

Portfolio Codebase Documentation

Complete Source Code Bundle (React 19, TanStack Start, Tailwind, Nitro)

Total Included Source Files: 121

INCLUDED FILE DIRECTORY:

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- public\google694108a7e5405cf5.html
- public\llms.txt
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- src\components\portfolio\About.tsx
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- src\components\portfolio\Intro.tsx
- src\components\portfolio\LiveStatusWidget.tsx
- src\components\portfolio\MagneticButton.tsx
- src\components\portfolio\Nav.tsx
- src\components\portfolio\Projects.tsx
- src\components\portfolio\Reveal.tsx
- src\components\portfolio\ScrollProgress.tsx
- src\components\portfolio\Stack.tsx
- src\components\portfolio\StatsBar.tsx
- src\components\portfolio\TextScramble.tsx

- src\components\portfolio\TiltCard.tsx
- src\components\portfolio\WorkspacePanel.tsx
- src\components\portfolio\project-visuals.tsx
- src\components\portfolio\projects\CarouselProjectCard.tsx
- src\components\portfolio\projects\WorkspacePanel.tsx
- src\components\portfolio\projects\widgets\ArchitectureNode.tsx
- src\components\portfolio\projects\widgets\ChallengeCard.tsx
- src\components\portfolio\projects\widgets\LighthouseDial.tsx
- src\components\portfolio\projects\widgets\ScenarioTimeline.tsx
- src\components\portfolio\projects\widgets\useSpotlight.ts
- src\components\ui\accordion.tsx
- src\components\ui\alert-dialog.tsx
- src\components\ui\alert.tsx
- src\components\ui\aspect-ratio.tsx
- src\components\ui\avatar.tsx
- src\components\ui\badge.tsx
- src\components\ui\breadcrumb.tsx
- src\components\ui\button.tsx
- src\components\ui\calendar.tsx
- src\components\ui\card.tsx
- src\components\ui\carousel.tsx
- src\components\ui\chart.tsx
- src\components\ui\checkbox.tsx
- src\components\ui\collapsible.tsx
- src\components\ui\command.tsx
- src\components\ui\context-menu.tsx
- src\components\ui\dialog.tsx
- src\components\ui\drawer.tsx
- src\components\ui\dropdown-menu.tsx
- src\components\ui\form.tsx
- src\components\ui\hover-card.tsx
- src\components\ui\input-otp.tsx
- src\components\ui\input.tsx
- src\components\ui\label.tsx
- src\components\ui\menubar.tsx
- src\components\ui\navigation-menu.tsx
- src\components\ui\pagination.tsx
- src\components\ui\popover.tsx
- src\components\ui\progress.tsx
- src\components\ui\radio-group.tsx
- src\components\ui\resizable.tsx
- src\components\ui\scroll-area.tsx
- src\components\ui\select.tsx
- src\components\ui\separator.tsx
- src\components\ui\sheet.tsx
- src\components\ui\sidebar.tsx
- src\components\ui\skeleton.tsx
- src\components\ui\slider.tsx
- src\components\ui\sonner.tsx
- src\components\ui\switch.tsx
- src\components\ui\table.tsx
- src\components\ui\tabs.tsx

- src\components\ui\textarea.tsx
- src\components\ui\toggle-group.tsx
- src\components\ui\toggle.tsx
- src\components\ui\tooltip.tsx
- src\hooks\use-mobile.tsx
- src\lib\error-capture.ts
- src\lib\error-page.ts
- src\lib\lovable-error-reporting.ts
- src\lib\motion.ts
- src\lib\project-details.ts
- src\lib\site-data.ts
- src\lib\sound.ts
- src\lib\utils.ts
- src\routeTree.gen.ts
- src\router.tsx
- src\routes\README.md
- src\routes__root.tsx
- src\routes\index.tsx
- src\routes\projects\vcentre.tsx
- src\routes\projects\veltrix.tsx
- src\routes\projects\vurlo.tsx
- src\server.ts
- src\start.ts
- src\styles.css
- tsconfig.json
- tunnel.log
- vite.config.ts

FILE #1: .codex\hooks.json

```
1 | {
2 |   "hooks": {
3 |     "PostToolUse": [
4 |       {
5 |         "matcher": "Edit|Write|apply_patch",
6 |         "hooks": [
7 |           {
8 |             "type": "command",
9 |             "command": "node \"agents/skills/impeccable/scripts/hook.mjs\"",
10 |            "timeout": 5,
11 |            "statusMessage": "Checking UI changes"
12 |          }
13 |        ]
14 |      }
15 |    ]
16 |  }
17 | }
```

FILE #2: .dev.vars

```
1 | GROQ_API_KEY=gsk_wj1YYWUI6KuTRh31HVx3WGdyb3FYUMe0gsw4iD1TPtgMbLcR10L
```

FILE #3: .env

```
1 | GROQ_API_KEY=gsk_wj1YYWUI6KuTRh31HVx3WGdyb3FYUMe0gsw4iD1TPtgMbLcR10L
```

FILE #4: .gitignore

```
1 | # Logs
2 | logs
3 | *.log
4 | npm-debug.log*
5 | yarn-debug.log*
6 | yarn-error.log*
7 | pnpm-debug.log*
8 | lerna-debug.log*
9 |
10 | node_modules
11 | dist
12 | dist-ssr
13 | .output
14 | .vinxi
15 | .tanstack/**
16 | .nitro
17 | *.local
18 |
19 | # Environment variables
20 | .env
21 | .env.local
22 | .env.*.local
23 | .env.*
24 |
25 |
26 | # Wrangler / Cloudflare
27 | .wrangler/
```

```
28 | .dev.vars
29 |
30 | # Editor directories and files
31 | .vscode/*
32 | !.vscode/extensions.json
33 | .idea
34 | .DS_Store
35 | *.suo
36 | *.ntvs*
37 | *.njsproj
38 | *.sln
39 | *.sw?
40 |
41 | # Agent files
42 | .workspace/
43 | .agents/
44 | .gemini/
45 |
46 | # Archives and build bundles
47 | *.zip
48 | *.tar.gz
49 | *.tgz
50 |
51 |
```

FILE #5: .node-version

```
1 | 22
```

FILE #6: .prettierignore

```
1 | node_modules
2 | dist
3 | .output
4 | .vinxi
5 | pnpm-lock.yaml
6 | package-lock.json
7 | bun.lock
8 | routeTree.gen.ts
```

FILE #7: .prettierrc

```
1 | {
2 |   "printWidth": 100,
3 |   "semi": true,
4 |   "singleQuote": false,
5 |   "trailingComma": "all"
6 | }
```

FILE #8: AGENTS.md

```
1 | <!-- LOVABLE-BEGIN -->
2 |
3 | > [!IMPORTANT]
4 | > This project is connected to [Lovable](https://lovable.dev). Avoid rewriting
5 | > published git history ? force pushing, or rebasing/amending/squashing commits
```

```
6 | > that are already pushed ? as it rewrites history on Lovable's side and the
7 | > user will likely lose their project history.
8 | >
9 | > Commits you push to the connected branch sync back to Lovable and show up in
10 | > the editor, so keep the branch in a working state.
11 |
12 | <!-- LOVABLE:END -->
```

FILE #9: DESIGN.md

```
1 | ---
2 | name: Cosmic Obsidian Developer Desk
3 | description: Sleek dark cosmic developer portfolio with glowing cyan and violet accent highlights and res...
4 | colors:
5 |   primary: "#8b5cf6"
6 |   secondary: "#06b6d4"
7 |   neutral-bg: "#07050f"
8 |   neutral-surface: "#0d0a18"
9 |   neutral-card: "#131020"
10 |   neutral-text: "#fafafa"
11 |   neutral-muted: "#a1a1aa"
12 |   neutral-subtle: "#71717a"
13 |   border: "rgba(255, 255, 255, 0.08)"
14 | typography:
15 |   display:
16 |     fontFamily: "Inter Tight, Inter, sans-serif"
17 |     fontSize: "clamp(2.5rem, 7vw, 5.25rem)"
18 |     fontWeight: 600
19 |     lineHeight: 1.02
20 |     letterSpacing: "-0.02em"
21 |   body:
22 |     fontFamily: "Inter, sans-serif"
23 |     fontSize: "15px"
24 |     fontWeight: 400
25 |     lineHeight: 1.6
26 |   mono:
27 |     fontFamily: "JetBrains Mono, monospace"
28 |     fontSize: "11px"
29 |     fontWeight: 400
30 |   rounded:
31 |     sm: "8px"
32 |     md: "10px"
33 |     lg: "12px"
34 |     xl: "16px"
35 |     2xl: "20px"
36 |   components:
37 |     button-primary:
38 |       backgroundColor: "{colors.primary}"
39 |       textColor: "{colors.neutral-text}"
40 |       rounded: "{rounded.lg}"
41 |       padding: "10px 20px"
42 |     card:
43 |       backgroundColor: "{colors.neutral-card}"
44 |       rounded: "{rounded.lg}"
45 |       padding: "24px"
46 | ---
47 |
48 | # Design System: Cosmic Obsidian Developer Desk
49 |
50 | ## 1. Overview
51 |
52 | **Creative North Star: "Cosmic Obsidian Developer Desk"**
53 |
```

```
54 | A high-fidelity developer showcase aesthetic built on a dark, layered cosmic color space. The design comb...
55 |
56 | The system rejects default beige "AI cream" palettes, plain uninspired grids, and generic outline borders...
57 |
58 | **Key Characteristics:**
59 |
60 | - Deep dark background tinting (#07050f)
61 | - Glowing violet (#8b5cf6) and cyan (#06b6d4) interactive anchors
62 | - High-contrast typography pairing displaying tight sans-serif titles alongside monospace metadata labels
63 | - Responsive 3D parallax scroll reveals and mouse-movement reflections that feel physically tactical
64 |
65 | ## 2. Colors
66 |
67 | A dark, atmospheric color spectrum with vibrant cyberpunk glowing elements.
68 |
69 | ### Primary
70 |
71 | - **Cosmic Violet** (#8b5cf6): The primary brand identifier. Used for glowing borders, button backgrounds...
72 |
73 | ### Secondary
74 |
75 | - **Electric Cyan** (#06b6d4): The secondary highlight color. Used sparingly for status pulses, active ta...
76 |
77 | ### Neutral
78 |
79 | - **Cosmic Void** (#07050f): The base background color of the canvas.
80 | - **Obsidian Surface** (#0d0a18): Background color for panel containers and overlays.
81 | - **Tactical Card** (#131020): Background color for cards and secondary content block groupings.
82 | - **Pure Bright Ink** (#fafafa): High-contrast foreground text color.
83 | - **Dim Slate** (#a1a1aa): Color for body paragraphs and descriptive copy.
84 | - **Deep Slate** (#71717a): Color for secondary labels and timestamps.
85 |
86 | ### Named Rules
87 |
88 | **The Accent Economy Rule.** Accent colors must occupy less than 10% of any viewport. Their scarcity is w...
89 |
90 | **The Tinted Neutral Rule.** Dark neutrals are never absolute black (#000000); they are tinted with deep ...
91 |
92 | ## 3. Typography
93 |
94 | **Display Font:** Inter Tight (with Inter fallback)
95 | **Body Font:** Inter (with system-ui fallback)
96 | **Label/Mono Font:** JetBrains Mono
97 |
98 | A high-contrast editorial developer pairing: heavy geometric display headlines with structural, clean mon...
99 |
100 | ### Hierarchy
101 |
102 | - **Display** (600, clamp(2.5rem, 7vw, 5.25rem), 1.02): Used for primary page title headlines in the Hero...
103 | - **Headline** (600, 3xl/4xl, 1.1): Used for main section titles (e.g. "Selected Work", "About").
104 | - **Title** (600, xl/2xl, 1.2): Used for project names and card titles.
105 | - **Body** (400, 15px, 1.6): Used for bios, summary narratives, and project taglines. Cap line length at ...
106 | - **Label** (400, 11px, tracking-widest, uppercase): Used for eyebrows, metrics indicators, and tech stac...
107 |
108 | ### Named Rules
109 |
110 | **The Balance Wrap Rule.** Headline and Display headings always employ `text-wrap: balance` to prevent aw...
111 |
112 | ## 4. Elevation
113 |
114 | The system is flat by default, opting for crisp solid borders and subtle backdrop-filters instead of heav...
115 |
116 | ### Shadow Vocabulary
117 |
118 | - **Ambient Accent Glow** (`box-shadow: 0 0 30px rgba(139, 92, 246, 0.09)`): Used under highlighted cards...
```

```
119 |
120 | ### Named Rules
121 |
122 | **The Restrained Elevation Rule.** Components are flat at rest. Depth is strictly interactive and trigger...
123 |
124 | ## 5. Components
125 |
126 | ### Buttons
127 |
128 | - **Shape:** Rounded corners (12px radius, `var(--radius-lg)`).
129 | - **Primary:** Violet background with white text, micro-glow on hover, and smooth interactive scale trans...
130 | - **Ghost:** Borderless with slate text, fading to cyan outline highlights on hover.
131 |
132 | ### Cards / Containers
133 |
134 | - **Corner Style:** Rounded corners (12px radius, `var(--radius-lg)`).
135 | - **Background:** Dark Obsidian Card fill (#131020).
136 | - **Interactive State:** Hover zooms visual contents by 4% (`scale-[1.04]`), triggers cursor spotlight, a...
137 |
138 | ### Inputs / Fields
139 |
140 | - **Style:** Dark outline input field (#131020) with transparent border.
141 | - **Focus:** Violet border ring (#8b5cf6) with 2px offset.
142 |
143 | ### Navigation
144 |
145 | - **Style:** Sticky top glassmorphic navbar with backdrop-filter blur (12px). Link hover states trigger a...
146 |
147 | ## 6. Do's and Don'ts
148 |
149 | ### Do:
150 |
151 | - **Do** wrap project articles inside `` components to animate them using GPU-accelerated 3D scro...
152 | - **Do** restrict wide letter-spacing tracking strictly to labels using JetBrains Mono.
153 | - **Do** guard all browser-only globals (like `window` or `document`) with `typeof window !== "undefined"...
154 |
155 | ### Don't:
156 |
157 | - **Don't** use plain, flat layouts or generic template themes without micro-glows or cosmic neutral back...
158 | - **Don't** add side-stripe colored borders on cards as decorative highlights.
159 | - **Don't** include sketchy SVG illustrations or hand-drawn graphics.
160 | - **Don't** load heavy JS libraries for page scroll animations; use CSS Compositor properties (opacity, s...
```

FILE #10: PRODUCT.md

```
1 | # Product
2 |
3 | ## Register
4 |
5 | brand
6 |
7 | ## Platform
8 |
9 | web
10 |
11 | ## Users
12 |
13 | Recruiters, engineering managers, startup founders, and prospective clients evaluating candidates for hig...
14 |
15 | ## Product Purpose
16 |
17 | To serve as an elite personal developer showcase that demonstrates clean visual aesthetics, high-performa...
18 |
```



```
19 | ## Positioning
20 |
21 | An impeccably crafted portfolio demonstrating that modern web applications can be visually stunning, inte...
22 |
23 | ## Conversion & proof
24 |
25 | - Primary and secondary CTA: View Case Study / "Let's talk" (contact form)
26 | - The line a visitor remembers after 10 seconds: A master-level developer who constructs beautiful, highl...
27 | - Belief ladder:
28 |   1. The UI is exceptionally polished, responsive, and animated.
29 |   2. The developer values performance (0% JS-overhead on scroll, pure CSS parallax).
30 |   3. They have shipped real complex code (Vurlo SaaS, Veltrix Bot, Vcentre Scraper).
31 |   4. Engaging them brings both architectural reliability and UI/UX craftsmanship.
32 | - Proof on hand: Live showcase of functional applications (Firestore sync cart, automated LLM consensus l...
33 |
34 | ## Brand Personality
35 |
36 | Sleek, premium dark-mode, neon glowing accent fields, industrial developer aesthetic, expert confidence, ...
37 |
38 | ## Anti-references
39 |
40 | - Plain, flat generic layouts or uninspired bootstrap-style grids
41 | - Overly sketchy SVG doodles or low-fidelity illustrations
42 | - Heavy JS scroll libraries that trigger continuous Chrome browser layout shifts or stutters
43 |
44 | ## Design Principles
45 |
46 | 1. Performance as an Aesthetic: Visual motion must feel instant, running entirely on the CSS composit...
47 | 2. Interactive Delight: Leverage micro-interactions (like 3D card tilt on mouse move and directional ...
48 | 3. Show, Don't Tells: Let actual project details, metrics, and functional highlights carry the strate...
49 |
50 | ## Accessibility & Inclusion
51 |
52 | - Keep contrast ratios at or above WCAG AA (4.5:1 floor for body text).
53 | - Strictly respect user preferences by disabling transforms and transitions when `@media (prefers-reduced...
```

FILE #11: README.md

```
1 | # Ayush Agarwal ? Systems & Agent Engineer
2 |
3 | High-performance AI agent architectures, e-commerce SaaS engines, and interactive web platforms.
4 |
5 | ![Website](https://img.shields.io/badge/Website-theayush.pages.dev-blue?style=flat-square)(https://thea...
6 | ![Lighthouse SEO](https://img.shields.io/badge/Lighthouse--SEO-100%2F100-success?style=flat-square)(htt...
7 | ![Schema.org](https://img.shields.io/badge/Schema.org-Valid-success?style=flat-square)(https://validato...
8 |
9 | ---
10 |
11 | ## ? Technical Highlights of this Portfolio
12 |
13 | This repository isn't just a static resume page ? it is a production-grade TanStack Start (React 19) ...
14 |
15 | - Full Server-Side Rendering (SSR): Case study layouts are fully hydrated and persistent in the raw H...
16 |
17 | * Spec-Compliant Entity Graphs: 100% warnings-and-error-free Schema.org JSON-LD graph linking `Person...
18 | * AI Engine Optimization (GEO):
19 |   - Complete custom `/llms.txt` file configured to feed structured markdown data to training models.
20 |   - Explicit user-agent allow rules for `GPTBot`, `ClaudeBot`, `PerplexityBot`, and `Google-Extended` in ...
21 | * Automated Asset Pipelines: Dynamic sitemap (`sitemap.xml`) generation at compile time via node scri...
22 | * Premium UX/UI: Immersive glassmorphic components, SVG Lighthouse dials, mouse-tracking spotlights, ...
23 |
24 | ---
25 |
```

```
26 | ## ?? The Tech Stack
27 |
28 | - **Frontend & Routing**: React 19, TanStack Start (SSR), TanStack Router, TypeScript.
29 | - **Styling & Animation**: Tailwind CSS, Framer Motion, Vanilla CSS variables.
30 | - **Backend & AI Integrations**: Python, Go, Gemini API, Groq, Cerebras, Playwright, Jinja2, Firebase Adm...
31 |
32 | ---
33 |
34 | ## ? Shipped Projects Detailed Inside
35 |
36 | ### 1. [Vurlo (Full-Stack E-commerce SaaS)](https://theayush.pages.dev/projects/vurlo)
37 |
38 | - **Code Repository**: [theayushagarwal/vurlo-ecommerce](https://github.com/theayushagarwal/vurlo-ecommerce)
39 | - **Architecture**: Implements atomic Firestore transaction rules preventing inventory race conditions. I...
40 |
41 | ### 2. [Veltrix (Autonomous Social Media Engine)](https://theayush.pages.dev/projects/veltrix)
42 |
43 | - **Architecture**: A pipeline that generates, reviews, and schedules visual content. Gemini Flash drafts...
44 |
45 | ### 3. [Vcentre (Competitor Intelligence Scraper)](https://theayush.pages.dev/projects/vcentre)
46 |
47 | - **Architecture**: Scrapes competitor engagement data, normalizes outlier metrics against cohort-specifi...
48 |
49 | ---
50 |
51 | ## ? Getting Started
52 |
53 | ### Prerequisites
54 |
55 | Make sure you have [Node.js](https://nodejs.org/) installed (v18+ recommended).
56 |
57 | ### Install dependencies
58 |
59 | ```bash
60 | npm install
61 | # or
62 | bun install
63 | ```
64 |
65 | ### Development server
66 |
67 | ```bash
68 | npm run dev
69 | # or
70 | bun dev
71 | ```
72 |
73 | Open [http://localhost:3000](http://localhost:3000) to view the development server.
74 |
75 | ### Build for production
76 |
77 | ```bash
78 | npm run build
79 | ```
80 |
81 | The output will be built inside the `.output/` folder, configured out-of-the-box for Cloudflare Pages/Wor...
```

FILE #12: bunfig.toml

```
1 | [install]
2 | # 24h supply-chain guard: skip package versions published less than a day ago.
3 | minimumReleaseAge = 86400
4 | # Each entry bypasses the 24h guard for one package ? confirm with the user
```

```
5 | # before adding any.
6 | minimumReleaseAgeExcludes = ["@lovable.dev/vite-tanstack-config", "@lovable.dev/mcp-js", "@lovable.dev/vi...
```

FILE #13: components.json

```
1 | {
2 |   "$schema": "https://ui.shadcn.com/schema.json",
3 |   "style": "new-york",
4 |   "rsc": false,
5 |   "tsx": true,
6 |   "tailwind": {
7 |     "css": "src/styles.css",
8 |     "baseColor": "slate",
9 |     "cssVariables": true,
10 |    "prefix": ""
11 |  },
12 |   "iconLibrary": "lucide",
13 |   "rtl": false,
14 |   "aliases": {
15 |     "components": "@components",
16 |     "utils": "@lib/utils",
17 |     "ui": "@components/ui",
18 |     "lib": "@lib",
19 |     "hooks": "@hooks"
20 |   },
21 |   "registries": {}
22 | }
```

FILE #14: eslint.config.js

```
1 | import js from "@eslint/js";
2 | import eslintPluginPrettier from "eslint-plugin-prettier/recommended";
3 | import globals from "globals";
4 | import reactHooks from "eslint-plugin-react-hooks";
5 | import reactRefresh from "eslint-plugin-react-refresh";
6 | import tseslint from "typescript-eslint";
7 |
8 | export default tseslint.config(
9 |   { ignores: ["dist", ".output", ".vinxi", ".agents"] },
10 |   {
11 |     extends: [js.configs.recommended, ...tseslint.configs.recommended],
12 |     files: ["**/*.ts", "**/*.tsx"],
13 |     languageOptions: {
14 |       ecmaVersion: 2020,
15 |       globals: globals.browser,
16 |     },
17 |     plugins: {
18 |       "react-hooks": reactHooks,
19 |       "react-refresh": reactRefresh,
20 |     },
21 |     rules: {
22 |       ...reactHooks.configs.recommended.rules,
23 |       "no-restricted-imports": [
24 |         "error",
25 |         {
26 |           paths: [
27 |             {
28 |               name: "server-only",
29 |               message:
30 |                 "TanStack Start does not use the Next.js `server-only` package. Rename the module to `*.s...`",
31 |             }
32 |           ],
33 |         }
34 |       ],
35 |     },
36 |   }
37 | );
```

```
32 |         ],
33 |     },
34 | ],
35 |     "react-refresh/only-export-components": ["warn", { allowConstantExport: true }],
36 |     "@typescript-eslint/no-unused-vars": "off",
37 | },
38 | },
39 |     eslintPluginPrettier,
40 | );
```

FILE #15: generate_codebase_pdf.py

```
1 | import os
2 | import sys
3 | from fpdf import FPDF
4 |
5 | class CodebasePDF(FPDF):
6 |     def header(self):
7 |         self.set_font('Helvetica', 'B', 9)
8 |         self.set_text_color(100, 100, 100)
9 |         self.cell(0, 8, 'PORTFOLIO CODEBASE DOCUMENTATION | THEAYUSH.PAGES.DEV', border=False, align='L')
10 |        self.cell(0, 8, 'PAGE ' + str(self.page_no()), border=False, align='R')
11 |        self.ln(10)
12 |        self.set_draw_color(220, 220, 220)
13 |        self.line(10, 18, 200, 18)
14 |
15 |    def footer(self):
16 |        self.set_y(-15)
17 |        self.set_font('Helvetica', 'I', 8)
18 |        self.set_text_color(140, 140, 140)
19 |        self.cell(0, 10, 'Generated for Ayush Agarwal Portfolio Codebase (Excluding Binary & Build Artifa...',
20 |
21 | def generate_pdf():
22 |     project_root = r"c:\Users\ayush\Downloads\portfolio-redesigned (2)\portfolio"
23 |     output_pdf_path = os.path.join(project_root, "portfolio_codebase_complete.pdf")
24 |
25 |     # Excluded directories
26 |     exclude_dirs = {
27 |         'node_modules', '.git', '.output', 'dist', '.wrangler',
28 |         '.impeccable', '.agents', '.nitro', 'scratch', 'brain'
29 |     }
30 |
31 |     # Excluded extensions
32 |     exclude_exts = {
33 |         '.png', '.jpg', '.jpeg', '.webp', '.ico', '.woff', '.woff2',
34 |         '.ttf', '.eot', '.mp3', '.wav', '.pdf', '.zip', '.tar', '.gz',
35 |         '.lock', '.exe', '.bin', '.db', '.sqlite'
36 |     }
37 |
38 |     # Specific excluded files
39 |     exclude_files = {'package-lock.json', 'skills-lock.json', 'portfolio_codebase_complete.pdf'}
40 |
41 |     files_to_process = []
42 |
43 |     for root, dirs, files in os.walk(project_root):
44 |         # Prune excluded directories
45 |         dirs[:] = [d for d in dirs if d not in exclude_dirs]
46 |
47 |         for file in sorted(files):
48 |             if file in exclude_files:
49 |                 continue
50 |             ext = os.path.splitext(file)[1].lower()
51 |             if ext in exclude_exts:
```

```
52 |         continue
53 |
54 |         full_path = os.path.join(root, file)
55 |         rel_path = os.path.relpath(full_path, project_root)
56 |
57 |         # Skip if file size > 300KB (e.g. huge minified bundles)
58 |         if os.path.getsize(full_path) > 300 * 1024:
59 |             continue
60 |
61 |         files_to_process.append((rel_path, full_path))
62 |
63 |     files_to_process.sort(key=lambda x: x[0])
64 |
65 |     print(f"Total source files to compile into PDF: {len(files_to_process)}")
66 |
67 |     pdf = CodebasePDF(orientation='P', unit='mm', format='A4')
68 |     pdf.set_auto_page_break(auto=True, margin=15)
69 |     pdf.add_page()
70 |
71 |     # Cover / Title Section
72 |     pdf.set_font('Helvetica', 'B', 22)
73 |     pdf.set_text_color(15, 23, 42) # Slate 900
74 |     pdf.cell(0, 14, 'Portfolio Codebase Documentation', ln=True, align='C')
75 |
76 |     pdf.set_font('Helvetica', '', 12)
77 |     pdf.set_text_color(100, 116, 139) # Slate 500
78 |     pdf.cell(0, 8, 'Complete Source Code Bundle (React 19, TanStack Start, Tailwind, Nitro)', ln=True, al...
79 |     pdf.cell(0, 8, f'Total Included Source Files: {len(files_to_process)}', ln=True, align='C')
80 |     pdf.ln(10)
81 |
82 |     # Table of Contents Summary Box
83 |     pdf.set_fill_color(241, 245, 249) # Slate 100
84 |     pdf.set_draw_color(203, 213, 225)
85 |     pdf.set_font('Helvetica', 'B', 11)
86 |     pdf.set_text_color(30, 41, 59)
87 |     pdf.cell(0, 10, '    INCLUDED FILE DIRECTORY:', ln=True, fill=True, border=1)
88 |
89 |     pdf.set_font('Courier', '', 8)
90 |     pdf.set_text_color(51, 65, 85)
91 |
92 |     for rel_path, _ in files_to_process:
93 |         pdf.cell(0, 5, f'    - {rel_path}', ln=True)
94 |
95 |     pdf.add_page()
96 |
97 |     # Process each file
98 |     for idx, (rel_path, full_path) in enumerate(files_to_process, 1):
99 |         print(f"[{idx}/{len(files_to_process)}] Adding {rel_path}...")
100 |
101 |         pdf.set_font('Helvetica', 'B', 13)
102 |         pdf.set_text_color(15, 23, 42)
103 |         pdf.set_fill_color(226, 232, 240)
104 |         pdf.cell(0, 10, f'FILE #{idx}: {rel_path}', ln=True, fill=True, border=1)
105 |         pdf.ln(3)
106 |
107 |         try:
108 |             with open(full_path, 'r', encoding='utf-8', errors='replace') as f:
109 |                 content = f.read()
110 |
111 |                 lines = content.splitlines()
112 |                 pdf.set_font('Courier', '', 7.5)
113 |                 pdf.set_text_color(30, 41, 59)
114 |
115 |                 for line_num, line_text in enumerate(lines, 1):
116 |                     # Format line with line number
```

```
117 |             # Sanitize tabs
118 |             line_clean = line_text.replace('\t', '    ')
119 |             # Escape characters if needed (fpdf handles basic latin-1 / utf8)
120 |             formatted_line = f"{line_num:4d} | {line_clean}"
121 |
122 |             # Truncate very wide lines to avoid overflow
123 |             if len(formatted_line) > 115:
124 |                 formatted_line = formatted_line[:112] + "..."
125 |
126 |             # Encode to latin-1 safe string for fpdf
127 |             safe_line = formatted_line.encode('latin-1', 'replace').decode('latin-1')
128 |             pdf.cell(0, 4, safe_line, ln=True)
129 |
130 |         pdf.ln(8)
131 |
132 |         except Exception as e:
133 |             pdf.set_font('Helvetica', 'I', 9)
134 |             pdf.set_text_color(220, 38, 38)
135 |             pdf.cell(0, 6, f"[Error reading file: {str(e)}]", ln=True)
136 |             pdf.ln(5)
137 |
138 |     pdf.output(output_pdf_path)
139 |     print(f"\nSUCCESS! PDF generated successfully at:\n{output_pdf_path}")
140 |
141 | if __name__ == "__main__":
142 |     generate_pdf()
```

FILE #16: package.json

```
1 | {
2 |   "name": "tanstack_start_ts",
3 |   "private": true,
4 |   "sideEffects": false,
5 |   "type": "module",
6 |   "scripts": {
7 |     "dev": "vite dev --host",
8 |     "prebuild": "node scripts/update-sitemap.js",
9 |     "build": "vite build",
10 |     "build:dev": "vite build --mode development",
11 |     "preview": "vite preview",
12 |     "lint": "eslint .",
13 |     "format": "prettier --write ."
14 |   },
15 |   "dependencies": {
16 |     "@hookform/resolvers": "^5.2.2",
17 |     "@radix-ui/react-accordion": "^1.2.12",
18 |     "@radix-ui/react-alert-dialog": "^1.1.15",
19 |     "@radix-ui/react-aspect-ratio": "^1.1.8",
20 |     "@radix-ui/react-avatar": "^1.1.11",
21 |     "@radix-ui/react-checkbox": "^1.3.3",
22 |     "@radix-ui/react-collapsible": "^1.1.12",
23 |     "@radix-ui/react-context-menu": "^2.2.16",
24 |     "@radix-ui/react-dialog": "^1.1.15",
25 |     "@radix-ui/react-dropdown-menu": "^2.1.16",
26 |     "@radix-ui/react-hover-card": "^1.1.15",
27 |     "@radix-ui/react-label": "^2.1.8",
28 |     "@radix-ui/react-menubar": "^1.1.16",
29 |     "@radix-ui/react-navigation-menu": "^1.2.14",
30 |     "@radix-ui/react-popover": "^1.1.15",
31 |     "@radix-ui/react-progress": "^1.1.8",
32 |     "@radix-ui/react-radio-group": "^1.3.8",
33 |     "@radix-ui/react-scroll-area": "^1.2.10",
34 |     "@radix-ui/react-select": "^2.2.6",
```

```
35 |     "@radix-ui/react-separator": "^1.1.8",
36 |     "@radix-ui/react-slider": "^1.3.6",
37 |     "@radix-ui/react-slot": "^1.2.4",
38 |     "@radix-ui/react-switch": "^1.2.6",
39 |     "@radix-ui/react-tabs": "^1.1.13",
40 |     "@radix-ui/react-toggle": "^1.1.10",
41 |     "@radix-ui/react-toggle-group": "^1.1.11",
42 |     "@radix-ui/react-tooltip": "^1.2.8",
43 |     "@tailwindcss/vite": "^4.2.1",
44 |     "@tanstack/react-query": "^5.101.1",
45 |     "@tanstack/react-router": "^1.170.16",
46 |     "@tanstack/react-start": "^1.168.26",
47 |     "@tanstack/router-plugin": "^1.168.18",
48 |     "class-variance-authority": "^0.7.1",
49 |     "clsx": "^2.1.1",
50 |     "cmdk": "^1.1.1",
51 |     "date-fns": "^4.1.0",
52 |     "embla-carousel-react": "^8.6.0",
53 |     "input-otp": "^1.4.2",
54 |     "lucide-react": "^0.575.0",
55 |     "motion": "^12.42.2",
56 |     "react": "^19.2.0",
57 |     "react-day-picker": "^9.14.0",
58 |     "react-dom": "^19.2.0",
59 |     "react-hook-form": "^7.71.2",
60 |     "react-resizable-panels": "^4.6.5",
61 |     "recharts": "^2.15.4",
62 |     "sonner": "^2.0.7",
63 |     "tailwind-merge": "^3.5.0",
64 |     "tailwindcss": "^4.2.1",
65 |     "tw-animate-css": "^1.3.4",
66 |     "vaul": "^1.1.2",
67 |     "vite-tsconfig-paths": "^6.0.2",
68 |     "zod": "^3.24.2"
69 |   },
70 |   "devDependencies": {
71 |     "@eslint/js": "^9.32.0",
72 |     "@lovable.dev/vite-tanstack-config": "^2.6.4",
73 |     "@types/node": "^22.16.5",
74 |     "@types/react": "^19.2.0",
75 |     "@types/react-dom": "^19.2.0",
76 |     "@vitejs/plugin-react": "^5.2.0",
77 |     "eslint": "^9.32.0",
78 |     "eslint-config-prettier": "^10.1.1",
79 |     "eslint-plugin-prettier": "^5.2.6",
80 |     "eslint-plugin-react-hooks": "^5.2.0",
81 |     "eslint-plugin-react-refresh": "^0.4.20",
82 |     "globals": "^15.15.0",
83 |     "nitro": "3.0.260603-beta",
84 |     "prettier": "^3.7.3",
85 |     "typescript": "^5.8.3",
86 |     "typescript-eslint": "^8.56.1",
87 |     "vite": "^8.0.16"
88 |   }
89 | }
```

FILE #17: public_headers

```
1 | /sitemap.xml
2 |   Content-Type: application/xml; charset=utf-8
3 |
4 | /robots.txt
5 |   Content-Type: text/plain; charset=utf-8
```

```
6 |  
7 | /*  
8 | Content-Security-Policy: default-src 'self'; script-src 'self' 'unsafe-inline' 'unsafe-eval' https://fo...  
9 | X-Frame-Options: DENY  
10 | X-Content-Type-Options: nosniff  
11 | Referrer-Policy: strict-origin-when-cross-origin  
12 | X-XSS-Protection: 1; mode=block  
13 |  
14 | /assets/*  
15 | cache-control: public, max-age=31536000, immutable
```

FILE #18: public\google694108a7e5405cf5.html

```
1 | google-site-verification: google694108a7e5405cf5.html
```

FILE #19: public\llms.txt

```
1 | # Ayush Agarwal | Systems & Agent Engineer  
2 |  
3 | High-performance AI agent architectures, e-commerce SaaS engines, and interactive web platforms.  
4 |  
5 | ## Contact & Links  
6 | - Website: [theayush.pages.dev](https://theayush.pages.dev/)  
7 | - Email: theayush.codes@gmail.com  
8 | - GitHub: [github.com/theayushagarwal](https://github.com/theayushagarwal)  
9 | - LinkedIn: [linkedin.com/in/ayushagarwal17](https://linkedin.com/in/ayushagarwal17)  
10 |  
11 | ## Technical Profile  
12 | Specializes in production-grade LLM state machines, secure database schemas, high-velocity backend pipeli...  
13 |  
14 | ### Core Stack  
15 | - **AI & Automation**: LangGraph, Gemini API, Groq, Cerebras, Playwright, Jinja2, GitHub Actions.  
16 | - **Backend & DB**: Python, Go, FastAPI, SQLite, Supabase, Firebase Admin SDK.  
17 | - **Frontend & Systems**: TypeScript, React 19, TanStack Start / Router, Tailwind CSS, Framer Motion.  
18 | - **Availability**: Actively seeking global remote software engineering roles, high-velocity contracts, a...  
19 |  
20 | ---  
21 |  
22 | ## Key Projects  
23 |  
24 | ### [Vurlo](https://theayush.pages.dev/projects/vurlo) (Full-Stack E-commerce SaaS)  
25 | - **Case Study URL**: [theayush.pages.dev/projects/vurlo](https://theayush.pages.dev/projects/vurlo)  
26 | - **Role**: Founder & Lead Architect  
27 | - **Build Time**: 10 Days  
28 | - **Key Metrics**: 100/100 Lighthouse SEO, 0 inventory oversells, under 400ms loading latency.  
29 | - **Architecture**: Atomic inventory check-and-decrement transactions in Firestore preventing race condit...  
30 |  
31 | ### [Veltrix](https://theayush.pages.dev/projects/veltrix) (Autonomous Social Engine)  
32 | - **Case Study URL**: [theayush.pages.dev/projects/veltrix](https://theayush.pages.dev/projects/veltrix)  
33 | - **Role**: Solo Developer  
34 | - **Cadence**: 2 automated posts/day  
35 | - **Operational Cost**: ~$0.0002 per post  
36 | - **Architecture**: Multimodel adversarial loop. Gemini 2.0 Flash drafts captions; Groq (Llama) and Cereb...  
37 |  
38 | ### [Vcentre](https://theayush.pages.dev/projects/vcentre) (Competitor Intelligence Scraper)  
39 | - **Case Study URL**: [theayush.pages.dev/projects/vcentre](https://theayush.pages.dev/projects/vcentre)  
40 | - **Role**: Solo Developer  
41 | - **Cadence**: Daily @ 2:00 AM UTC  
42 | - **Stack**: Python, Apify, FastAPI, SQLite  
43 | - **Architecture**: Nightly scraping of target competitor channels. Separates Reels and Photos into cohor...
```


FILE #20: public\manifest.json

```
1 | {
2 |   "id": "/",
3 |   "short_name": "Ayush Portfolio",
4 |   "name": "Ayush Agarwal | Systems & Agent Engineer",
5 |   "icons": [
6 |     {
7 |       "src": "/favicon.png",
8 |       "type": "image/png",
9 |       "sizes": "192x192"
10 |    },
11 |    {
12 |      "src": "/favicon.png",
13 |      "type": "image/png",
14 |      "sizes": "512x512",
15 |      "purpose": "any maskable"
16 |    }
17 |  ],
18 |   "start_url": "/",
19 |   "scope": "/",
20 |   "lang": "en-US",
21 |   "description": "Systems & Agent Engineer Portfolio detailing e-commerce SaaS, AI agent pipelines, and h...",
22 |   "categories": ["portfolio", "technology"],
23 |   "background_color": "#09090b",
24 |   "theme_color": "#09090b",
25 |   "display": "standalone",
26 |   "orientation": "portrait"
27 | }
```

FILE #21: public\robots.txt

```
1 | # https://www.robotstxt.org/robotstxt.html
2 | User-agent: GPTBot
3 | Allow: /
4 |
5 | User-agent: ClaudeBot
6 | Allow: /
7 |
8 | User-agent: PerplexityBot
9 | Allow: /
10 |
11 | User-agent: Google-Extended
12 | Allow: /
13 |
14 | User-agent: *
15 | Allow: /
16 |
17 | Sitemap: https://theayush.pages.dev/sitemap.xml
```

FILE #22: public\sitemap.xml

```
1 | <?xml version="1.0" encoding="UTF-8"?>
2 | <urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
3 |   <url>
4 |     <loc>https://theayush.pages.dev/</loc>
5 |     <lastmod>2026-07-30</lastmod>
6 |     <changefreq>weekly</changefreq>
7 |     <priority>1.0</priority>
```

```
8 | </url>
9 | <url>
10 | <loc>https://theayush.pages.dev/projects/vurlo</loc>
11 | <lastmod>2026-07-30</lastmod>
12 | <changefreq>monthly</changefreq>
13 | <priority>0.8</priority>
14 | </url>
15 | <url>
16 | <loc>https://theayush.pages.dev/projects/veltrix</loc>
17 | <lastmod>2026-07-30</lastmod>
18 | <changefreq>monthly</changefreq>
19 | <priority>0.8</priority>
20 | </url>
21 | <url>
22 | <loc>https://theayush.pages.dev/projects/vcentre</loc>
23 | <lastmod>2026-07-30</lastmod>
24 | <changefreq>monthly</changefreq>
25 | <priority>0.8</priority>
26 | </url>
27 | </urlset>
```

FILE #23: scripts\update-sitemap.js

```
1 | import fs from "fs";
2 | import path from "path";
3 |
4 | const sitemapPath = path.resolve("public/sitemap.xml");
5 | const currentDate = new Date().toISOString().split("T")[0];
6 |
7 | const sitemapContent = `<?xml version="1.0" encoding="UTF-8"?>
8 | <urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
9 |   <url>
10 |     <loc>https://theayush.pages.dev/</loc>
11 |     <lastmod>${currentDate}</lastmod>
12 |     <changefreq>weekly</changefreq>
13 |     <priority>1.0</priority>
14 |   </url>
15 |   <url>
16 |     <loc>https://theayush.pages.dev/projects/vurlo</loc>
17 |     <lastmod>${currentDate}</lastmod>
18 |     <changefreq>monthly</changefreq>
19 |     <priority>0.8</priority>
20 |   </url>
21 |   <url>
22 |     <loc>https://theayush.pages.dev/projects/veltrix</loc>
23 |     <lastmod>${currentDate}</lastmod>
24 |     <changefreq>monthly</changefreq>
25 |     <priority>0.8</priority>
26 |   </url>
27 |   <url>
28 |     <loc>https://theayush.pages.dev/projects/vcentre</loc>
29 |     <lastmod>${currentDate}</lastmod>
30 |     <changefreq>monthly</changefreq>
31 |     <priority>0.8</priority>
32 |   </url>
33 | </urlset>
34 | `;
35 |
36 | fs.writeFileSync(sitemapPath, sitemapContent, "utf8");
37 | console.log(`Updated sitemap.xml with lastmod date: ${currentDate}`);
```

FILE #24: src\components\portfolio\About.tsx

```
1 | import { Reveal } from "../Reveal";
2 | import { PROFILE } from "@lib/site-data";
3 |
4 | const MILESTONES = [
5 |   {
6 |     date: "June 2026",
7 |     title: "Shipped Veltrix Bot",
8 |     desc: "Engineered multi-model LLM consensus engine auto-publishing twice daily.",
9 |   },
10 |  {
11 |    date: "June 2026",
12 |    title: "Initiated Vurlo SaaS",
13 |    desc: "Designed full-stack e-commerce architecture, Firestore rules, and Razorpay routing.",
14 |  },
15 |  {
16 |    date: "July 2026",
17 |    title: "Launched Vcentre Scraper",
18 |    desc: "Created 10-provider LLM scraping pipeline and feedback scoring loop.",
19 |  },
20 |  {
21 |    date: "July 2026",
22 |    title: "Crafted Portfolio Redesign",
23 |    desc: "Developed glassmorphic 3D developer showcase with fluid animations.",
24 |  },
25 | ];
26 |
27 | export function About() {
28 |   return (
29 |     <section id="about" className="relative py-32 md:py-40">
30 |       <div className="mx-auto max-w-6xl px-6">
31 |         <div className="grid gap-16 md:grid-cols-12">
32 |           <div className="md:col-span-4">
33 |             <Reveal direction="light">
34 |               <p className="eyebrow">About</p>
35 |               <h2 className="mt-4 font-display text-4xl font-semibold tracking-tight md:text-5xl">
36 |                 Built to ship. <br />
37 |                 Wired to scale.
38 |               </h2>
39 |             </Reveal>
40 |           </div>
41 |
42 |           <div className="space-y-6 md:col-span-7 md:col-start-6">
43 |             {PROFILE.bio.map((para, i) => (
44 |               <Reveal key={para.slice(0, 12)} delay={i} direction="light">
45 |                 <p className="text-lg leading-relaxed text-muted-foreground">{para}</p>
46 |               </Reveal>
47 |             ))}
48 |
49 |             <Reveal delay={2} direction="light">
50 |               <div className="mt-12 space-y-6">
51 |                 <p className="eyebrow">Milestones</p>
52 |                 <div className="relative pl-6 border-l border-border space-y-8 mt-4">
53 |                   {MILESTONES.map((m, idx) => (
54 |                     <div key={idx} className="relative">
55 |                       </* Timeline dot */>
56 |                       <span className="absolute -left-[29px] top-1.5 flex h-3 w-3 items-center justify-ce...
57 |                         <span className="h-1 w-1 rounded-full bg-primary" />
58 |                       </span>
59 |                       <div className="flex flex-col gap-1">
60 |                         <span className="font-mono text-[9px] uppercase tracking-widest text-primary font...
61 |                           {m.date}
62 |                         </span>
```

```

63 |                 <h4 className="font-display font-semibold text-foreground text-sm">
64 |                     {m.title}
65 |                 </h4>
66 |                 <p className="text-xs text-muted-foreground leading-relaxed">{m.desc}</p>
67 |             </div>
68 |         </div>
69 |     )}}
70 | </div>
71 | </div>
72 | </Reveal>
73 |
74 | <Reveal delay={3} direction="light">
75 |     <div className="mt-12 grid grid-cols-2 gap-8 border-t border-border pt-8">
76 |         <div>
77 |             <p className="eyebrow">Currently</p>
78 |             <p className="mt-2 text-sm text-foreground">{PROFILE.currently}</p>
79 |         </div>
80 |         <div>
81 |             <p className="eyebrow">Previously</p>
82 |             <p className="mt-2 text-sm text-foreground">{PROFILE.previously}</p>
83 |         </div>
84 |     </div>
85 | </Reveal>
86 | </div>
87 | </div>
88 | </div>
89 | </section>
90 | );
91 | }

```

FILE #25: src\components\portfolio\Chatbox.tsx

```

1 | import { useEffect, useRef, useState } from "react";
2 | import { Bot, X, Send, Loader2 } from "lucide-react";
3 | import { createServerFn } from "@tanstack/react-start";
4 | import { playTick } from "@lib/sound";
5 | import { motion, AnimatePresence } from "motion/react";
6 |
7 | // Server function calling the Groq API securely
8 | const askGroq = createServerFn({ method: "POST" }).handler(
9 |   async ({ data: prompt }: { data: string }) => {
10 |     const apiKey = process.env.GROQ_API_KEY;
11 |     if (!apiKey) {
12 |       return "__NO_API_KEY__";
13 |     }
14 |
15 |     try {
16 |       const response = await fetch("https://api.groq.com/openai/v1/chat/completions", {
17 |         method: "POST",
18 |         headers: {
19 |           "Content-Type": "application/json",
20 |           Authorization: `Bearer ${apiKey}`,
21 |         },
22 |         body: JSON.stringify({
23 |           model: "llama-3.1-8b-instant",
24 |           messages: [
25 |             {
26 |               role: "system",
27 |               content: `You are an AI assistant representing Ayush Agarwal on his portfolio website.
28 | Ayush is a 17-year-old Systems & Agent Engineer based in Vellore, India. He builds high-performance e-com...
29 |
30 | Availability:
31 | - Open to remote software engineering roles, high-velocity contract work, and freelance consulting.

```

```
32 | - Due to starting college in August 2026, he is actively seeking remote part-time positions, flexible fre...
33 |
34 | Engineering Philosophy:
35 | - Pragmatism over theoretical hype. He builds "systems that ship, agents that work, and products that sca...
36 | - Prioritizes engineering integrity: implementing proper outlier baselines, multi-model consensus, fallba...
37 |
38 | Key Stats & Outcomes:
39 | - Shipped 3 production AI/SaaS projects solo in 30 days total (June - July 2026).
40 | - Portfolio infra cost: ?0 (designed for maximum efficiency using free tiers).
41 | - Lighthouse performance & SEO score: 100.
42 |
43 | Key Projects:
44 | 1. Vurlo (Live at https://vurlo.store):
45 |   - Tech: React 19, TanStack Start, Firebase Auth & Firestore Rules, Razorpay checkout, Gemini API, Tail...
46 |   - Built solo in 10 days. Features in-memory caching, atomic database transaction stock-locking to prev...
47 | 2. Veltrix (Autonomous Instagram Growth Pipeline):
48 |   - Tech: Python, Gemini, Groq, Cerebras, OpenRouter, SiliconFlow, Hugging Face, Unsplash, NVIDIA, Logo....
49 |   - Scale: Orchestrates 18+ active API keys with a gated paid API model. Gemini Pro is invoked for creat...
50 |   - Fallbacks:
51 |     - Audit Chain: Groq (Llama 3.3 70B) ?? Cerebras (gpt-oss-120b) ?? OpenRouter (Gemma 31B free) ?? Raw...
52 |     - Image Gen: SiliconFlow (Flux/SD3) ?? Hugging Face ?? Gemini internal generator.
53 |     - Logo Assets: Logo.dev ?? Brandfetch API ?? Brandfetch CDN ?? Google Favicon API ?? text placeholder.
54 |     - Logs & DB: Supabase ?? local SQLite (veltrix.db) and JSON backup.
55 |     - Cost per post is only ~$0.0002. Runs 2 posts / day automatically.
56 | 3. Vcentre (Competitor Intelligence Scraper):
57 |   - Tech: Python, Apify, SQLite, Supabase, Groq, Cerebras, FastAPI, GitHub Actions.
58 |   - Scrapes competitor Instagram accounts nightly. Cohort-scores Reels/Photos separately, filters outlie...
59 |
60 | Tech Stack:
61 | - AI Agents & Pipelines: Python, Gemini API, Groq, Cerebras, LangGraph, GitHub Actions.
62 | - E-Commerce & SaaS: React 19, TanStack Start, Firebase Auth & Firestore Rules, Razorpay Checkout.
63 | - Data & Systems: SQLite, Supabase, FastAPI, Playwright, Jinja2, Cloudflare Pages.
64 | - Interfaces: TypeScript, Tailwind CSS, Framer Motion, Figma.
65 |
66 | Social Links:
67 | - GitHub: https://github.com/theayushagarwal
68 | - LinkedIn: https://linkedin.com/in/ayushagarwal17
69 | - Email: theayush.codes@gmail.com
70 | - Contact: Suggest filling out the contact form at the bottom of the page or emailing theayush.codes@gmai...
71 |
72 | Navigational & Page Context Hints:
73 | - You can direct users to scroll and explore sections of this page:
74 |   - Timeline: "Check out the chronological timeline under the About section."
75 |   - Stack: "Scroll down to see his full visual stack competencies."
76 |   - Projects: "Click any project card in the Selected Work section to open the live workspace code panel."
77 |
78 | Guardrails & Instructions:
79 | - Do not make up facts or project details.
80 | - If asked about topics outside Ayush's background, politely redirect them back to his portfolio work.
81 | - Format responses beautifully using bolding and inline code `like this` where appropriate.
82 | - When mentioning emails, keep them as theayush.codes@gmail.com.
83 |
84 | Tone & Style:
85 | - Engineering-focused, direct, and crisp.
86 | - Keep simple answers short (3-4 sentences). For deep technical/architectural queries, feel free to give ...
87 |     },
88 |     {
89 |       role: "user",
90 |       content: prompt,
91 |     },
92 |   ],
93 |   temperature: 0.5,
94 |   max_tokens: 500,
95 | },
96 | );
```

```
97 |  
98 |     if (!response.ok) {  
99 |         throw new Error(`Groq API returned status ${response.status}`);  
100 |     }  
101 |  
102 |     const resData = await response.json();  
103 |     return resData.choices[0].message.content || "";  
104 | } catch (err) {  
105 |     console.error("Groq API error:", err);  
106 |     return "__API_ERROR__";  
107 | }  
108 | },  
109 | );  
110 |  
111 | interface Message {  
112 |     sender: "user" | "bot";  
113 |     text: string;  
114 | }  
115 |  
116 | const QUICK_PROMPTS = [  
117 |     { label: "? Veltrix Consensus", query: "How does Veltrix's consensus system work?" },  
118 |     { label: "? Vurlo E-Commerce", query: "Tell me about Vurlo.store" },  
119 |     { label: "? Availability", query: "Are you open to freelance or full-time roles?" },  
120 |     { label: "? Stack", query: "What is your main technology stack?" },  
121 | ];  
122 |  
123 | function getLocalResponse(prompt: string): string {  
124 |     const lowercase = prompt.toLowerCase();  
125 |  
126 |     if (  
127 |         lowercase.includes("veltrix") ||  
128 |         lowercase.includes("consensus") ||  
129 |         lowercase.includes("instagram")  
130 |     ) {  
131 |         return "Veltrix is an autonomous Instagram account that designs slides and publishes posts twice a da...  
132 |     }  
133 |     if (  
134 |         lowercase.includes("vurlo") ||  
135 |         lowercase.includes("e-commerce") ||  
136 |         lowercase.includes("saas")  
137 |     ) {  
138 |         return "Vurlo (vurlo.store) is a full-stack e-commerce SaaS platform built solo in 10 days. It handle...  
139 |     }  
140 |     if (  
141 |         lowercase.includes("vcentre") ||  
142 |         lowercase.includes("scraper") ||  
143 |         lowercase.includes("competitor")  
144 |     ) {  
145 |         return "Vcentre scrapes competitor Instagram accounts nightly. It isolates Reel/Photo engagement base...  
146 |     }  
147 |     if (  
148 |         lowercase.includes("stack") ||  
149 |         lowercase.includes("tech") ||  
150 |         lowercase.includes("languages") ||  
151 |         lowercase.includes("framework")  
152 |     ) {  
153 |         return "Ayush's main stack is TypeScript, React 19, TanStack Start, Python, Supabase, SQLite, and Pla...  
154 |     }  
155 |     if (  
156 |         lowercase.includes("hire") ||  
157 |         lowercase.includes("work") ||  
158 |         lowercase.includes("contact") ||  
159 |         lowercase.includes("freelance") ||  
160 |         lowercase.includes("email")  
161 |     ) {
```

```
162 |     return "Ayush is available for hire and freelance work! You can contact him by filling out the form a...
163 | }
164 | if (
165 |     lowercase.includes("age") ||
166 |     lowercase.includes("years old") ||
167 |     lowercase.includes("college") ||
168 |     lowercase.includes("vit")
169 | ) {
170 |     return "Ayush is 17 years old and starts his Computer Science Engineering (CSE) degree at VIT (Vellor...
171 | }
172 | return "I'm Ayush's virtual agent. I can tell you about Vurlo, Veltrix, Vcentre, his tech stack, or his...
173 | }
174 |
175 | function formatMessageText(text: string) {
176 |     // Parse email addresses
177 |     const emailRegex = /([a-zA-Z0-9._-]+@[a-zA-Z0-9._-]+\.[a-zA-Z0-9._-]+)/g;
178 |
179 |     return text.split("\n").map((line, lineIdx) => {
180 |         // Split by bold (**bold**) and inline code (`code`)
181 |         const parts = line.split(/(\*\*.*?\*\*|\`.*?\`)/g);
182 |
183 |         return (
184 |             <div key={lineIdx} className={lineIdx > 0 ? "mt-1.5" : ""}>
185 |                 {parts.map((part, partIdx) => {
186 |                     // Bold formatting
187 |                     if (part.startsWith("**") && part.endsWith("**")) {
188 |                         return (
189 |                             <strong key={partIdx} className="font-semibold text-foreground">
190 |                                 {part.slice(2, -2)}
191 |                             </strong>
192 |                         );
193 |                     }
194 |                     // Code formatting
195 |                     if (part.startsWith("`") && part.endsWith("`")) {
196 |                         return (
197 |                             <code
198 |                                 key={partIdx}
199 |                                 className="font-mono text-[11px] bg-black/40 px-1 py-0.5 rounded border border-white/5 te...
200 |                             >
201 |                                 {part.slice(1, -1)}
202 |                             </code>
203 |                         );
204 |                     }
205 |
206 |                     // Email formatting within regular text
207 |                     const subParts = part.split(emailRegex);
208 |                     return subParts.map((subPart, subPartIdx) => {
209 |                         if (emailRegex.test(subPart)) {
210 |                             return (
211 |                                 <a
212 |                                     key={subPartIdx}
213 |                                     href={`mailto:${subPart}`}
214 |                                     className="underline text-primary hover:text-secondary transition-colors"
215 |                                 >
216 |                                     {subPart}
217 |                                 </a>
218 |                             );
219 |                         }
220 |                         return subPart;
221 |                     });
222 |                 })}
223 |             </div>
224 |         );
225 |     });
226 | }
```

```
227 |  
228 | export function Chatbox() {  
229 |   const [isOpen, setIsOpen] = useState(false);  
230 |   const [showTooltip, setShowTooltip] = useState(false);  
231 |   const [messages, setMessages] = useState<Message[]>([  
232 |     {  
233 |       sender: "bot",  
234 |       text: "Hey! I'm Ayush's virtual agent. Ask me anything about his projects, tech stack, or availabil...",  
235 |     },  
236 |   ]);  
237 |   const [inputValue, setInputValue] = useState("");  
238 |   const [isLoading, setIsLoading] = useState(false);  
239 |   const messagesEndRef = useRef<HTMLDivElement>(null);  
240 |  
241 |   const scrollToBottom = () => {  
242 |     messagesEndRef.current?.scrollIntoView({ behavior: "smooth" });  
243 |   };  
244 |  
245 |   useEffect(() => {  
246 |     if (isOpen) {  
247 |       scrollToBottom();  
248 |     }  
249 |   }, [messages, isOpen]);  
250 |  
251 |   const handleSend = async (text: string) => {  
252 |     if (!text.trim() || isLoading) return;  
253 |  
254 |     playTick(0.2); // Play sound on send  
255 |     setMessages((prev) => [...prev, { sender: "user", text }]);  
256 |     setInputValue("");  
257 |     setIsLoading(true);  
258 |  
259 |     // Normalize keys: lowercase, strip punctuation/emojis, collapse whitespace  
260 |     const cacheKey = text  
261 |       .trim()  
262 |       .toLowerCase()  
263 |       .replace(/[.,/#!$%^&*;:{}=\-_`~()]/g, "")  
264 |       .replace(/\s+/g, " ");  
265 |  
266 |     // Bounded eviction setter to stay within storage limits safely  
267 |     const safeSetItem = (key: string, value: string) => {  
268 |       try {  
269 |         if (typeof window !== "undefined") {  
270 |           const keysStr = sessionStorage.getItem("vagent_cache_keys");  
271 |           let keys: string[] = keysStr ? JSON.parse(keysStr) : [];  
272 |  
273 |           keys = keys.filter((k) => k !== key);  
274 |           keys.push(key);  
275 |  
276 |           // Enforce 50-item cache size ceiling (FIFO eviction)  
277 |           if (keys.length > 50) {  
278 |             const oldest = keys.shift();  
279 |             if (oldest) {  
280 |               sessionStorage.removeItem(`vagent_cache_${oldest}`);  
281 |             }  
282 |           }  
283 |  
284 |           sessionStorage.setItem(`vagent_cache_${key}`, value);  
285 |           sessionStorage.setItem("vagent_cache_keys", JSON.stringify(keys));  
286 |         }  
287 |       } catch (e) {  
288 |         console.warn("sessionStorage cache write bypassed:", e);  
289 |       }  
290 |     };  
291 |   };
```



```

292 |     try {
293 |         // Check client-side sessionStorage cache first
294 |         if (typeof window !== "undefined") {
295 |             const cachedResponse = sessionStorage.getItem(`vagent_cache_${cacheKey}`);
296 |             if (cachedResponse) {
297 |                 // Play instant responsive audio feedback + minor typing indicator delay
298 |                 await new Promise((resolve) => setTimeout(resolve, 300));
299 |                 playTick(-0.2);
300 |                 setMessages((prev) => [...prev, { sender: "bot", text: cachedResponse }]);
301 |                 setIsLoading(false);
302 |                 return;
303 |             }
304 |         }
305 |
306 |         // Call the server function which queries Groq
307 |         const response = await askGroq({ data: text });
308 |
309 |         if (response === "__NO_API_KEY__" || response === "__API_ERROR__") {
310 |             // Fallback to local heuristic responder with simulated thinking delay
311 |             await new Promise((resolve) => setTimeout(resolve, 800));
312 |             const localResponse = getLocalResponse(text);
313 |             safeSetItem(cacheKey, localResponse);
314 |             playTick(-0.2); // Play sound on receive
315 |             setMessages((prev) => [...prev, { sender: "bot", text: localResponse }]);
316 |         } else {
317 |             safeSetItem(cacheKey, response);
318 |             playTick(-0.2); // Play sound on receive
319 |             setMessages((prev) => [...prev, { sender: "bot", text: response }]);
320 |         }
321 |     } catch (e) {
322 |         console.error(e);
323 |         await new Promise((resolve) => setTimeout(resolve, 800));
324 |         const localResponse = getLocalResponse(text);
325 |         safeSetItem(cacheKey, localResponse);
326 |         playTick(-0.2);
327 |         setMessages((prev) => [...prev, { sender: "bot", text: localResponse }]);
328 |     } finally {
329 |         setIsLoading(false);
330 |     }
331 | };
332 |
333 | return (
334 |     <div className="fixed bottom-6 right-6 z-40 font-sans">
335 |         {/* Hover Tooltip */}
336 |         <div
337 |             className={`absolute right-14 top-1/2 -translate-y-1/2 whitespace-nowrap rounded-lg border border...
338 |             showTooltip && !isOpen ? "opacity-100 translate-x-0" : "opacity-0 translate-x-2"
339 |         `}
340 |         >
341 |             Ask my virtual agent
342 |         </div>
343 |
344 |         {/* Toggle button */}
345 |         <button
346 |             onClick={() => {
347 |                 playTick(0);
348 |                 setIsOpen(!isOpen);
349 |             }}
350 |             onMouseEnter={() => setShowTooltip(true)}
351 |             onMouseLeave={() => setShowTooltip(false)}
352 |             className="flex h-12 w-12 items-center justify-center rounded-full bg-primary text-white shadow-[...
353 |             aria-label="Open Chatbot"
354 |         >
355 |             {isOpen ? <X className="h-5 w-5" /> : <Bot className="h-5 w-5" />}
356 |         </button>

```

```

357 |
358 |     {/* Chat window */}
359 |     <AnimatePresence>
360 |         {isOpen && (
361 |             <motion.div
362 |                 initial={{ opacity: 0, scale: 0.95, y: 15, filter: "blur(6px)" }}
363 |                 animate={{ opacity: 1, scale: 1, y: 0, filter: "blur(0px)" }}
364 |                 exit={{ opacity: 0, scale: 0.95, y: 15, filter: "blur(6px)" }}
365 |                 transition={{ type: "spring", stiffness: 350, damping: 28 }}
366 |                 className="absolute bottom-16 right-0 w-[calc(100vw-3rem)] sm:w-96 h-[400px] sm:h-[460px] sur...
367 |             >
368 |                 {/* Header */}
369 |                 <div className="flex items-center justify-between border-b border-white/10 bg-black/40 px-4 p...
370 |                     <div className="flex items-center gap-2">
371 |                         <span className="relative flex h-2 w-2">
372 |                             <span className="absolute inline-flex h-full w-full animate-ping rounded-full bg-emeral...
373 |                             <span className="relative inline-flex h-2 w-2 rounded-full bg-emerald-500" />
374 |                         </span>
375 |                         <span className="font-mono text-[10px] uppercase tracking-widest text-foreground font-sem...
376 |                         <Bot className="h-3.5 w-3.5 text-primary animate-pulse" /> v-agent.sys initialized
377 |                     </span>
378 |                 </div>
379 |                 <button
380 |                     onClick={() => {
381 |                         playTick(0);
382 |                         setIsOpen(false);
383 |                     }}
384 |                     className="text-subtle transition-colors hover:text-foreground cursor-pointer"
385 |                 >
386 |                     <X className="h-4 w-4" />
387 |                 </button>
388 |             </div>
389 |
390 |             {/* Messages list */}
391 |             <div className="flex-1 overflow-y-auto p-4 space-y-4 scrollbar-thin">
392 |                 {messages.map((m, idx) => (
393 |                     <div
394 |                         key={idx}
395 |                         className={`flex w-full ${m.sender === "user" ? "justify-end" : "justify-start"}`}
396 |                     >
397 |                         <div
398 |                             className={`max-w-[80%] rounded-2xl px-4 py-2.5 text-xs leading-relaxed ${
399 |                                 m.sender === "user"
400 |                                     ? "bg-primary text-white rounded-tr-none shadow-[0_2px_8px_rgba(139,92,246,0.2)]"
401 |                                     : "bg-surface border border-white/10 text-muted-foreground rounded-tl-none"
402 |                             }`}
403 |                         >
404 |                             {formatMessageText(m.text)}
405 |                         </div>
406 |                     </div>
407 |                 )))
408 |
409 |             {isLoading && (
410 |                 <div className="flex w-full justify-start">
411 |                     <div className="bg-surface border border-white/10 text-subtle rounded-2xl rounded-tl-no...
412 |                         <Loader2 className="h-3 w-3 animate-spin text-primary" />
413 |                         <span>Agent is searching logs...</span>
414 |                     </div>
415 |                 </div>
416 |             )}
417 |             <div ref={messagesEndRef} />
418 |         </div>
419 |
420 |         {/* Quick prompt chips */}
421 |         {messages.length === 1 && (

```

```

422 |         <div className="px-4 py-3 border-t border-white/5 bg-black/10 select-none">
423 |             <p className="text-[9px] uppercase tracking-widest text-subtle mb-2 font-mono">
424 |                 Suggested Prompts
425 |             </p>
426 |             <div className="flex flex-wrap gap-1.5">
427 |                 {QUICK_PROMPTS.map((qp, idx) => (
428 |                     <button
429 |                         key={idx}
430 |                         onClick={() => handleSend(qp.query)}
431 |                         className="text-[10px] text-muted-foreground bg-surface border border-white/5 hover...
432 |                     >
433 |                         {qp.label}
434 |                     </button>
435 |                 ))}
436 |             </div>
437 |         </div>
438 |     )}
439 |
440 |     {/* Input field */}
441 |     <form
442 |         onSubmit={(e) => {
443 |             e.preventDefault();
444 |             handleSend(inputValue);
445 |         }}
446 |         className="border-t border-white/10 bg-black/30 p-3 flex gap-2 select-none"
447 |     >
448 |         <input
449 |             type="text"
450 |             value={inputValue}
451 |             onChange={(e) => setInputValue(e.target.value)}
452 |             placeholder="Ask v-agent..."
453 |             className="flex-1 bg-surface border border-white/10 focus:border-primary/60 outline-none ..."
454 |             disabled={isLoading}
455 |         />
456 |         <button
457 |             type="submit"
458 |             disabled={isLoading || !inputValue.trim()}
459 |             className="flex h-8 w-8 items-center justify-center rounded-xl bg-primary text-white disa...
460 |         >
461 |             <Send className="h-3.5 w-3.5" />
462 |         </button>
463 |     </form>
464 | </motion.div>
465 | )}
466 | </AnimatePresence>
467 | </div>
468 | );
469 | }

```

FILE #26: src\components\portfolio\CommandPalette.tsx

```

1 | import { useEffect, useState } from "react";
2 | import { Command, Search, Copy, ExternalLink, ArrowRight, Code, Cpu, Database, User, Mail, Sparkles, X } ...
3 | import { toast } from "sonner";
4 | import { PROFILE } from "@lib/site-data";
5 |
6 | interface CommandItem {
7 |     id: string;
8 |     title: string;
9 |     category: "Projects" | "Navigation" | "Contact & Socials";
10 |    icon: any;
11 |    action: () => void;
12 |    badge?: string;

```

```
13 | }
14 |
15 | export function CommandPalette() {
16 |   const [open, setOpen] = useState(false);
17 |   const [search, setSearch] = useState("");
18 |   const [selectedIndex, setSelectedIndex] = useState(0);
19 |
20 |   // Global keydown listener for Cmd+K / Ctrl+K
21 |   useEffect(() => {
22 |     const handleKeyDown = (e: KeyboardEvent) => {
23 |       if ((e.metaKey || e.ctrlKey) && e.key.toLowerCase() === "k") {
24 |         e.preventDefault();
25 |         setOpen((prev) => !prev);
26 |       }
27 |       if (e.key === "Escape" && open) {
28 |         setOpen(false);
29 |       }
30 |     };
31 |
32 |     window.addEventListener("keydown", handleKeyDown);
33 |     return () => window.removeEventListener("keydown", handleKeyDown);
34 |   }, [open]);
35 |
36 |   // Listen to custom window trigger event
37 |   useEffect(() => {
38 |     const handleCustomTrigger = () => setOpen(true);
39 |     window.addEventListener("open-command-palette", handleCustomTrigger);
40 |     return () => window.removeEventListener("open-command-palette", handleCustomTrigger);
41 |   }, []);
42 |
43 |   const items: CommandItem[] = [
44 |     {
45 |       id: "proj-vurlo",
46 |       title: "Vurlo ? E-commerce SaaS Storefront",
47 |       category: "Projects",
48 |       icon: Code,
49 |       badge: "Razorpay + Firebase",
50 |       action: () => {
51 |         window.location.href = "/projects/vurlo";
52 |       },
53 |     },
54 |     {
55 |       id: "proj-veltrix",
56 |       title: "Veltrix ? Autonomous Instagram AI Pipeline",
57 |       category: "Projects",
58 |       icon: Cpu,
59 |       badge: "Gemini + Adversarial AI",
60 |       action: () => {
61 |         window.location.href = "/projects/veltrix";
62 |       },
63 |     },
64 |     {
65 |       id: "proj-vcentre",
66 |       title: "Vcentre ? Competitor Intelligence Scraper",
67 |       category: "Projects",
68 |       icon: Database,
69 |       badge: "Nightly Analytics",
70 |       action: () => {
71 |         window.location.href = "/projects/vcentre";
72 |       },
73 |     },
74 |     {
75 |       id: "copy-email",
76 |       title: `Copy Email (${PROFILE.email})`,
77 |       category: "Contact & Socials",
```

```
78 |     icon: Copy,
79 |     badge: "1-Click Copy",
80 |     action: () => {
81 |         navigator.clipboard.writeText(PROFILE.email);
82 |         toast.success("Email copied to clipboard!", {
83 |             description: PROFILE.email,
84 |         });
85 |     },
86 | },
87 | {
88 |     id: "nav-stack",
89 |     title: "Jump to Tech Stack & Infrastructure",
90 |     category: "Navigation",
91 |     icon: ArrowRight,
92 |     action: () => {
93 |         document.getElementById("stack")?.scrollIntoView({ behavior: "smooth" });
94 |     },
95 | },
96 | {
97 |     id: "nav-about",
98 |     title: "Jump to About & Bio",
99 |     category: "Navigation",
100 |     icon: User,
101 |     action: () => {
102 |         document.getElementById("about")?.scrollIntoView({ behavior: "smooth" });
103 |     },
104 | },
105 | {
106 |     id: "nav-contact",
107 |     title: "Jump to Contact Form",
108 |     category: "Navigation",
109 |     icon: Mail,
110 |     action: () => {
111 |         document.getElementById("contact")?.scrollIntoView({ behavior: "smooth" });
112 |     },
113 | },
114 | {
115 |     id: "open-github",
116 |     title: "Open GitHub Profile",
117 |     category: "Contact & Socials",
118 |     icon: ExternalLink,
119 |     badge: "GitHub",
120 |     action: () => {
121 |         window.open(PROFILE.socials.github, "_blank");
122 |     },
123 | },
124 | {
125 |     id: "open-linkedin",
126 |     title: "Open LinkedIn Profile",
127 |     category: "Contact & Socials",
128 |     icon: ExternalLink,
129 |     badge: "LinkedIn",
130 |     action: () => {
131 |         window.open(PROFILE.socials.linkedin, "_blank");
132 |     },
133 | },
134 | ];
135 |
136 | const filteredItems = items.filter((item) =>
137 |     item.title.toLowerCase().includes(search.toLowerCase()) ||
138 |     item.category.toLowerCase().includes(search.toLowerCase())
139 | );
140 |
141 | // Reset selected index when search changes
142 | useEffect(() => {
```

```
143 |     setSelectedIndex(0);
144 | }, [search]);
145 |
146 | // Handle arrow key navigation inside modal
147 | const handleModalKeyDown = (e: React.KeyboardEvent) => {
148 |     if (e.key === "ArrowDown") {
149 |         e.preventDefault();
150 |         setSelectedIndex((prev) => (prev + 1) % Math.max(1, filteredItems.length));
151 |     } else if (e.key === "ArrowUp") {
152 |         e.preventDefault();
153 |         setSelectedIndex((prev) => (prev - 1 + filteredItems.length) % Math.max(1, filteredItems.length));
154 |     } else if (e.key === "Enter" && filteredItems[selectedIndex]) {
155 |         e.preventDefault();
156 |         filteredItems[selectedIndex].action();
157 |         setOpen(false);
158 |     }
159 | };
160 |
161 | if (!open) return null;
162 |
163 | return (
164 |     <div
165 |         className="fixed inset-0 z-50 flex items-start justify-center pt-20 px-4 bg-slate-950/70 backdrop-b...
166 |         onClick={() => setOpen(false)}
167 |     >
168 |         <div
169 |             className="w-full max-w-xl overflow-hidden rounded-2xl border border-slate-800 bg-slate-900/95 sh...
170 |             onClick={(e) => e.stopPropagation()}
171 |             onKeyDown={handleModalKeyDown}
172 |             tabIndex={0}
173 |         >
174 |             {/* Search Header */}
175 |             <div className="flex items-center border-b border-slate-800 px-4 py-3">
176 |                 <Search className="h-5 w-5 text-cyan-400 mr-3 shrink-0" />
177 |                 <input
178 |                     type="text"
179 |                     placeholder="Type a command or search projects, tech stack..."
180 |                     value={search}
181 |                     onChange={(e) => setSearch(e.target.value)}
182 |                     className="w-full bg-transparent text-sm text-slate-100 placeholder-slate-500 focus:outline-n...
183 |                     autoFocus
184 |                 />
185 |                 <button
186 |                     onClick={() => setOpen(false)}
187 |                     className="rounded-lg p-1 text-slate-400 hover:bg-slate-800 hover:text-slate-200"
188 |                 >
189 |                     <X className="h-4 w-4" />
190 |                 </button>
191 |             </div>
192 |
193 |             {/* Results List */}
194 |             <div className="max-h-80 overflow-y-auto p-2">
195 |                 {filteredItems.length === 0 ? (
196 |                     <div className="py-8 text-center text-sm text-slate-500">
197 |                         No matching commands found.
198 |                     </div>
199 |                 ) : (
200 |                     filteredItems.map((item, idx) => {
201 |                         const Icon = item.icon;
202 |                         const isSelected = idx === selectedIndex;
203 |
204 |                         return (
205 |                             <div
206 |                                 key={item.id}
207 |                                 onClick={() => {
```

```
208 |         item.action();
209 |         setOpen(false);
210 |     }}
211 |     onMouseEnter={() => setSelectedIndex(idx)}
212 |     className={`flex cursor-pointer items-center justify-between rounded-xl px-3.5 py-2.5 t...
213 |         isSelected ? "bg-cyan-500/10 text-cyan-300 border border-cyan-500/30" : "text-slate-3...
214 |     }`}
215 | >
216 | <div className="flex items-center gap-3">
217 |   <Icon className={`h-4 w-4 ${isSelected ? "text-cyan-400" : "text-slate-400"}`} />
218 |   <span className="font-medium">{item.title}</span>
219 | </div>
220 |
221 | {item.badge && (
222 |   <span className="rounded-md border border-slate-700 bg-slate-800/80 px-2 py-0.5 font-...
223 |     {item.badge}
224 |   </span>
225 | )}
226 | </div>
227 | );
228 | })
229 | })
230 | </div>
231 |
232 | {/* Footer shortcuts */}
233 | <div className="flex items-center justify-between border-t border-slate-800/80 bg-slate-950/60 px...
234 |   <div className="flex items-center gap-3">
235 |     <span><kbd className="rounded bg-slate-800 px-1.5 py-0.5 text-[10px] text-slate-300">??</kbd>...
236 |     <span><kbd className="rounded bg-slate-800 px-1.5 py-0.5 text-[10px] text-slate-300">?</kbd> ...
237 |   </div>
238 |   <span><kbd className="rounded bg-slate-800 px-1.5 py-0.5 text-[10px] text-slate-300">ESC</kbd> ...
239 | </div>
240 | </div>
241 | </div>
242 | );
243 | }
244 |
245 | export function CommandPaletteTrigger() {
246 |   const trigger = () => {
247 |     window.dispatchEvent(new CustomEvent("open-command-palette"));
248 |   };
249 |
250 |   return (
251 |     <button
252 |       onClick={trigger}
253 |       className="inline-flex items-center gap-1.5 rounded-full border border-slate-800 bg-slate-900/80 px...
254 |       title="Open Command Palette (Cmd+K)"
255 |     >
256 |       <Command className="h-3.5 w-3.5 text-cyan-400" />
257 |       <span className="hidden sm:inline font-mono text-[11px]">?K</span>
258 |     </button>
259 |   );
260 | }
```

FILE #27: src\components\portfolio\Contact.tsx

```
1 | import { ArrowUpRight, FileText, Github, LinkedIn, Mail, Twitter } from "lucide-react";
2 | import { Reveal } from "../Reveal";
3 | import { PROFILE } from "@lib/site-data";
4 |
5 | const SOCIAL_LINKS = [
6 |   { icon: Mail, label: "Email", href: `mailto:${PROFILE.email}` },
7 |   { icon: Github, label: "Git Hub", href: PROFILE.socials.github },
```

```
8 | { icon: Linkedin, label: "LinkedIn", href: PROFILE.socials.linkedin },
9 | { icon: Twitter, label: "Twitter", href: PROFILE.socials.twitter },
10 | { icon: FileText, label: "Résumé", href: PROFILE.socials.resume },
11 | ];
12 |
13 | export function Contact() {
14 |   return (
15 |     <section id="contact" className="relative py-24 sm:py-32 md:py-48 overflow-hidden">
16 |       <div className="mx-auto max-w-6xl px-4 sm:px-6">
17 |         <div className="grid gap-12 md:gap-16 md:grid-cols-12 md:items-end">
18 |           <div className="md:col-span-7 min-w-0">
19 |             <Reveal>
20 |               <p className="eyebrow">Contact</p>
21 |               <h2 className="mt-4 sm:mt-6 font-display text-3xl sm:text-5xl font-semibold tracking-tight ...>
22 |                 Have something <br className="hidden sm:inline" />
23 |                 <span className="font-serif-accent font-normal italic text-muted-foreground">
24 |                   worth building?
25 |                 </span>
26 |               </h2>
27 |             </Reveal>
28 |
29 |             <Reveal delay={1}>
30 |               <a
31 |                 href={`mailto:${PROFILE.email}`}
32 |                 className="link-underline mt-6 sm:mt-10 inline-flex max-w-full items-center gap-2 font-di...
33 |               >
34 |                 <span>{PROFILE.email}</span>
35 |                 <ArrowUpRight className="h-4 w-4 sm:h-6 sm:w-6 shrink-0" />
36 |               </a>
37 |             </Reveal>
38 |           </div>
39 |
40 |           <div className="md:col-span-5">
41 |             <Reveal delay={2}>
42 |               <p className="text-sm text-muted-foreground">
43 |                 I take on a small number of engagements each quarter ? usually founding-engineer
44 |                 contracts, technical due diligence, or targeted six-week builds. Replies within 24h.
45 |               </p>
46 |
47 |               <ul className="mt-8 grid grid-cols-1 gap-px overflow-hidden rounded-xl border border-border...>
48 |                 {SOCIAL_LINKS.map((s) => (
49 |                   <li key={s.label}>
50 |                     <a
51 |                       href={s.href}
52 |                       aria-label={s.label}
53 |                       {...(s.label !== "Email"
54 |                         ? { target: "_blank", rel: "noopener noreferrer" }
55 |                         : {})}
56 |                       className="flex items-center justify-between bg-background px-5 py-4 text-sm text-m...
57 |                     >
58 |                       <span className="flex items-center gap-3">
59 |                         <s.icon className="h-4 w-4" aria-hidden="true" />
60 |                         {s.label}
61 |                       </span>
62 |                       <ArrowUpRight className="h-4 w-4" aria-hidden="true" />
63 |                     </a>
64 |                   </li>
65 |                 )))
66 |               </ul>
67 |             </Reveal>
68 |           </div>
69 |         </div>
70 |       </div>
71 |     </section>
72 |   );
```


73 | }

FILE #28: src\components\portfolio\CursorFollower.tsx

```
1 | import { useEffect, useRef } from "react";
2 |
3 | export function CursorFollower() {
4 |   const ringRef = useRef<HTMLDivElement>(null);
5 |   const dotRef = useRef<HTMLDivElement>(null);
6 |   // All mutable state lives in refs ? never triggers a React re-render
7 |   const mouse = useRef({ x: -200, y: -200 });
8 |   const pos = useRef({ x: -200, y: -200 });
9 |   const hovered = useRef(false);
10 |   const visible = useRef(false);
11 |
12 |   useEffect(() => {
13 |     // Hide entirely on touch screens ? no cursor to follow
14 |     const isTouchDevice = "ontouchstart" in window || navigator.maxTouchPoints > 0;
15 |     if (isTouchDevice) return;
16 |
17 |     const handleMouseMove = (e: MouseEvent) => {
18 |       mouse.current.x = e.clientX;
19 |       mouse.current.y = e.clientY;
20 |
21 |       if (!visible.current) {
22 |         visible.current = true;
23 |         // Snap elements into view without waiting for RAF
24 |         if (ringRef.current) ringRef.current.style.opacity = "1";
25 |         if (dotRef.current) dotRef.current.style.opacity = "1";
26 |       }
27 |     };
28 |
29 |     const handleMouseOver = (e: MouseEvent) => {
30 |       const target = e.target as HTMLElement;
31 |       if (!target) return;
32 |       // Avoid expensive getComputedStyle ? check tag and closest only
33 |       const isClickable =
34 |         target.tagName === "A" ||
35 |         target.tagName === "BUTTON" ||
36 |         !!target.closest("a") ||
37 |         !!target.closest("button");
38 |       hovered.current = isClickable;
39 |     };
40 |
41 |     let rafId: number;
42 |     const tick = () => {
43 |       // Lerp the ring toward the cursor
44 |       pos.current.x += (mouse.current.x - pos.current.x) * 0.12;
45 |       pos.current.y += (mouse.current.y - pos.current.y) * 0.12;
46 |
47 |       const scale = hovered.current ? 1.5 : 1;
48 |       const dotScale = hovered.current ? 0.4 : 1;
49 |
50 |       if (ringRef.current) {
51 |         ringRef.current.style.transform = `translate3d(${pos.current.x}px,${pos.current.y}px,0) translate...
52 |       }
53 |       if (dotRef.current) {
54 |         dotRef.current.style.transform = `translate3d(${mouse.current.x}px,${mouse.current.y}px,0) transl...
55 |       }
56 |
57 |       rafId = requestAnimationFrame(tick);
58 |     };
59 |
```

```

60 |     window.addEventListener("mousemove", handleMouseMove, { passive: true });
61 |     window.addEventListener("mouseover", handleMouseOver, { passive: true });
62 |     rafId = requestAnimationFrame(tick);
63 |
64 |     return () => {
65 |         window.removeEventListener("mousemove", handleMouseMove);
66 |         window.removeEventListener("mouseover", handleMouseOver);
67 |         cancelAnimationFrame(rafId);
68 |     };
69 | }, []); // ? empty deps: effect runs ONCE, never torn down
70 |
71 | return (
72 |     <>
73 |     { /* Outer spring-lagged ring ? hidden until first mouse move */}
74 |     <div
75 |         ref={ringRef}
76 |         style={{ opacity: 0 }}
77 |         className="pointer-events-none fixed left-0 top-0 z-[99999] h-5 w-5 rounded-full border border-pr...
78 |     />
79 |     { /* Inner instant dot */}
80 |     <div
81 |         ref={dotRef}
82 |         style={{ opacity: 0 }}
83 |         className="pointer-events-none fixed left-0 top-0 z-[99999] h-1.5 w-1.5 rounded-full bg-primary s...
84 |     />
85 | </>
86 | );
87 | }

```

FILE #29: src\components\portfolio\Faq.tsx

```

1 | import { useState } from "react";
2 | import { motion, AnimatePresence } from "motion/react";
3 | import { Plus, Minus } from "lucide-react";
4 | import { Reveal } from "../Reveal";
5 | import { playTick } from "@lib/sound";
6 |
7 | interface FAQItem {
8 |     question: string;
9 |     answer: string;
10 | }
11 |
12 | const FAQS: FAQItem[] = [
13 |     {
14 |         question: "How does Veltrix's adversarial AI consensus loop work?",
15 |         answer:
16 |             "Veltrix is an autonomous pipeline that orchestrates 18+ active APIs (content channels, competitor ...
17 |     },
18 |     {
19 |         question: "What is Vcentre and how does it analyze competitor engagement?",
20 |         answer:
21 |             "Vcentre scrapes competitor accounts nightly using isolated Reels and Photos baseline calculations ...
22 |     },
23 |     {
24 |         question: "What technology stack does Ayush Agarwal specialize in?",
25 |         answer:
26 |             "For AI and pipelines, the primary languages are Python and TypeScript, deploying models with Gemin...
27 |     },
28 |     {
29 |         question: "Is Ayush available for remote internships, freelance, or contract work?",
30 |         answer:
31 |             "Yes. Ayush starts his Computer Science Engineering (CSE) degree at VIT in August 2026 and is activ...
32 |     },

```

```

33 |   {
34 |     question: "How was the Vurlo e-commerce SaaS platform optimized for performance?",
35 |     answer:
36 |       "Vurlo was built solo in 10 days and achieved a 100/100 Lighthouse SEO score. It integrates server-...
37 |   },
38 | ];
39 |
40 | export function Faq() {
41 |   const [openIndex, setOpenIndex] = useState<number | null>(null);
42 |
43 |   const toggleOpen = (index: number) => {
44 |     playTick(0.1);
45 |     setOpenIndex(openIndex === index ? null : index);
46 |   };
47 |
48 |   return (
49 |     <section id="faq" className="relative py-32 md:py-40">
50 |       <div className="mx-auto max-w-4xl px-6">
51 |         <Reveal>
52 |           <p className="eyebrow">FAQ</p>
53 |           <h2 className="mt-4 font-display text-4xl font-semibold tracking-tight md:text-5xl">
54 |             Frequently Asked Questions
55 |           </h2>
56 |           <p className="mt-4 text-muted-foreground">
57 |             A quick reference for AI search engines, crawlers, and recruiters seeking details on
58 |             engineering philosophy, projects, and availability.
59 |           </p>
60 |         </Reveal>
61 |
62 |         <div className="mt-16 space-y-4">
63 |           {FAQS.map((faq, index) => {
64 |             const isOpen = openIndex === index;
65 |             return (
66 |               <Reveal key={index} delay={index * 0.1}>
67 |                 <div className="rounded-xl border border-border bg-card/40 backdrop-blur-sm overflow-hidd...
68 |                   <button
69 |                     onClick={() => toggleOpen(index)}
70 |                     className="flex w-full items-center justify-between p-6 text-left"
71 |                     aria-expanded={isOpen}
72 |                   >
73 |                     <span className="font-display font-medium text-foreground md:text-lg">
74 |                       {faq.question}
75 |                     </span>
76 |                     <span className="ml-4 shrink-0 rounded-full border border-border p-1 text-muted-foreg...
77 |                       {isOpen ? <Minus className="h-4 w-4" /> : <Plus className="h-4 w-4" />}
78 |                     </span>
79 |                   </button>
80 |                   <AnimatePresence initial={false}>
81 |                     {isOpen && (
82 |                       <motion.div
83 |                         initial={{ height: 0, opacity: 0 }}
84 |                         animate={{ height: "auto", opacity: 1 }}
85 |                         exit={{ height: 0, opacity: 0 }}
86 |                         transition={{ duration: 0.25, ease: "easeInOut" }}
87 |                       >
88 |                         <div className="border-t border-border/50 px-6 pb-6 pt-4 text-sm leading-relaxed ...
89 |                           {faq.answer}
90 |                         </div>
91 |                       </motion.div>
92 |                     )}
93 |                   </AnimatePresence>
94 |                 </div>
95 |               </Reveal>
96 |             );
97 |           )}

```

```

98 |         </div>
99 |     </div>
100 | </section>
101 | );
102 | }

```

FILE #30: src\components\portfolio\Footer.tsx

```

1 | import { ArrowUp } from "lucide-react";
2 |
3 | export function Footer() {
4 |     return (
5 |         <footer className="border-t border-border bg-background/30 backdrop-blur-md">
6 |             <div className="mx-auto flex max-w-6xl flex-col gap-6 px-6 py-10 font-mono text-xs text-subtle md:f...
7 |                 <div className="flex flex-col gap-1.5">
8 |                     <p className="text-foreground font-semibold">Ayush Agarwal</p>
9 |                     <p>© 2026 · Crafted in Vellore, India</p>
10 |                 </div>
11 |
12 |                 <div className="flex flex-wrap items-center gap-4 text-[11px] uppercase tracking-wider">
13 |                     <a
14 |                         href="https://github.com/theayushagarwal"
15 |                         target="_blank"
16 |                         rel="noopener noreferrer"
17 |                         className="hover:text-primary transition-colors"
18 |                     >
19 |                         GitHub
20 |                     </a>
21 |                     <span>.</span>
22 |                     <a
23 |                         href="https://linkedin.com/in/ayushagarwal17"
24 |                         target="_blank"
25 |                         rel="noopener noreferrer"
26 |                         className="hover:text-primary transition-colors"
27 |                     >
28 |                         LinkedIn
29 |                     </a>
30 |                     <span>.</span>
31 |                     <a
32 |                         href="mailto:theayush.codes@gmail.com"
33 |                         className="hover:text-primary transition-colors"
34 |                     >
35 |                         Email
36 |                     </a>
37 |                 </div>
38 |
39 |                 <div className="flex items-center gap-4">
40 |                     <p className="hidden md:block text-[10px] text-subtle/80">
41 |                         Built with TanStack Start · Motion
42 |                     </p>
43 |                     <a
44 |                         href="#top"
45 |                         className="inline-flex items-center gap-1.5 transition-colors hover:text-foreground border bo...
46 |                     >
47 |                         <span>Back to top</span>
48 |                         <ArrowUp className="h-3 w-3" aria-hidden="true" />
49 |                     </a>
50 |                 </div>
51 |             </div>
52 |         </footer>
53 |     );
54 | }

```

FILE #31: src\components\portfolio\Hero.tsx

```
1 | import { ArrowUpRight } from "lucide-react";
2 | import { useRef } from "react";
3 | import { motion, useReducedMotion, useScroll, useTransform } from "motion/react";
4 | import { MagneticButton } from "../MagneticButton";
5 | import { TextScramble } from "../TextScramble";
6 | import { EASE } from "@lib/motion";
7 | import { PROFILE } from "@lib/site-data";
8 | import { HeroCanvas } from "../HeroCanvas";
9 |
10 | export function Hero({ revealed }: { revealed: boolean }) {
11 |   const ref = useRef<HTMLDivElement>(null);
12 |   const reduceMotion = useReducedMotion();
13 |   const { scrollYProgress } = useScroll({
14 |     target: ref,
15 |     offset: ["start start", "end start"],
16 |   });
17 |   const y = useTransform(scrollYProgress, [0, 1], [0, 60]);
18 |   const opacity = useTransform(scrollYProgress, [0, 0.8], [1, 0]);
19 |
20 |   // Bold 3D Kinetic Entrance Stagger
21 |   const containerVariants = {
22 |     hidden: { opacity: 0 },
23 |     visible: {
24 |       opacity: 1,
25 |       transition: {
26 |         staggerChildren: 0.14,
27 |         delayChildren: 0.1,
28 |       },
29 |     },
30 |   };
31 |
32 |   const nameVariants = {
33 |     hidden: reduceMotion
34 |       ? { opacity: 0 }
35 |       : { opacity: 0, y: 20, filter: "blur(8px)" },
36 |     visible: {
37 |       opacity: 1,
38 |       y: 0,
39 |       filter: "blur(0px)",
40 |       transition: { duration: 0.8, ease: EASE },
41 |     },
42 |   };
43 |
44 |   const headlineVariants = {
45 |     hidden: reduceMotion
46 |       ? { opacity: 0 }
47 |       : { opacity: 0, y: 45, rotateX: 18, filter: "blur(14px)", transformOrigin: "bottom left" },
48 |     visible: {
49 |       opacity: 1,
50 |       y: 0,
51 |       rotateX: 0,
52 |       filter: "blur(0px)",
53 |       transition: { duration: 1.05, ease: EASE },
54 |     },
55 |   };
56 |
57 |   const subVariants = {
58 |     hidden: reduceMotion
59 |       ? { opacity: 0 }
60 |       : { opacity: 0, y: 30, filter: "blur(10px)" },
61 |     visible: {
62 |       opacity: 1,
```

```
63 |     y: 0,
64 |     filter: "blur(0px)",
65 |     transition: { duration: 0.9, ease: EASE },
66 |   },
67 | };
68 |
69 | const buttonVariants = {
70 |   hidden: reduceMotion
71 |     ? { opacity: 0 }
72 |     : { opacity: 0, y: 24, scale: 0.92, filter: "blur(6px)" },
73 |   visible: {
74 |     opacity: 1,
75 |     y: 0,
76 |     scale: 1,
77 |     filter: "blur(0px)",
78 |     transition: { duration: 0.75, ease: EASE },
79 |   },
80 | };
81 |
82 | return (
83 |   <section id="top" ref={ref} className="relative overflow-hidden pt-28 pb-16 md:pt-36 md:pb-24 perspec...
84 |     /* 3D Interactive Particle Mesh Canvas */
85 |     <HeroCanvas />
86 |
87 |     /* Shifting Gradient Mesh */
88 |     <div
89 |       className="pointer-events-none absolute inset-0 -z-20 overflow-hidden mesh-container scroll-paral...
90 |       aria-hidden
91 |     >
92 |       <div className="mesh-blob mesh-blob-1" />
93 |       <div className="mesh-blob mesh-blob-2" />
94 |       <div className="mesh-blob mesh-blob-3" />
95 |     </div>
96 |
97 |     /* Fine dot-grid ? structure, not glow. Fades toward the edges. */
98 |     <div
99 |       aria-hidden
100 |       className="pointer-events-none absolute inset-0 -z-10 dot-grid"
101 |       style={{
102 |         maskImage: "radial-gradient(60% 55% at 50% 30%, black, transparent)",
103 |         WebkitMaskImage: "radial-gradient(60% 55% at 50% 30%, black, transparent)",
104 |       }}
105 |     />
106 |
107 |     <motion.div style={{ y, opacity }} className="mx-auto max-w-6xl px-6">
108 |       <motion.div
109 |         variants={containerVariants}
110 |         initial="hidden"
111 |         animate={revealed ? "visible" : "hidden"}
112 |       >
113 |         /* Name + Tagline */
114 |         <motion.div variants={nameVariants}>
115 |           <p className="shiny-text font-display text-lg font-semibold tracking-tight md:text-xl">
116 |             <TextScramble text={PROFILE.name} delay={0.2} trigger={revealed} />
117 |           </p>
118 |           <p className="mt-1.5 max-w-md text-sm text-muted-foreground md:text-base">
119 |             {PROFILE.tagline}
120 |           </p>
121 |         </motion.div>
122 |
123 |         /* Status Eyebrow */
124 |         <motion.div variants={subVariants} className="mt-5 flex items-center gap-3">
125 |           <span className="relative flex h-2 w-2">
126 |             <span className="absolute inline-flex h-full w-full animate-ping rounded-full bg-secondary ...
127 |             <span className="relative inline-flex h-2 w-2 rounded-full bg-secondary" />
```

```

128 |         </span>
129 |         <p className="eyebrow">{PROFILE.status}</p>
130 |     </motion.div>
131 |
132 |     {/* 3D Kinetic Headline */}
133 |     <motion.div variants={headlineVariants} className="perspective-1000">
134 |         <h1 className="mt-5 max-w-4xl text-3xl sm:text-5xl font-semibold leading-[1.05] tracking-tigh...
135 |             {PROFILE.headline.line1}
136 |             <br />
137 |             <span className="text-muted-foreground">{PROFILE.headline.line2}</span>{" "}
138 |             <span className="font-serif-accent font-normal italic text-transparent bg-clip-text bg-grad...
139 |                 {PROFILE.headline.accent}
140 |             </span>
141 |         </h1>
142 |     </motion.div>
143 |
144 |     {/* Pitch Paragraph */}
145 |     <motion.div variants={subVariants}>
146 |         <p className="mt-5 max-w-xl text-base leading-relaxed text-muted-foreground md:text-lg">
147 |             {PROFILE.pitch}
148 |         </p>
149 |     </motion.div>
150 |
151 |     {/* CTA Buttons */}
152 |     <motion.div variants={buttonVariants} className="mt-7 flex flex-wrap items-center gap-3">
153 |         <MagneticButton href="#work" variant="primary">
154 |             View selected work
155 |             <ArrowUpRight className="h-4 w-4 transition-transform duration-300 group-hover:translate-x-...
156 |         </MagneticButton>
157 |         <MagneticButton href="#contact" variant="ghost">
158 |             Let's talk
159 |         </MagneticButton>
160 |     </motion.div>
161 | </motion.div>
162 | </motion.div>
163 | </section>
164 | );
165 | }

```

FILE #32: src\components\portfolio\HeroCanvas.tsx

```

1 | import { useEffect, useRef } from "react";
2 |
3 | interface Node {
4 |   x: number;
5 |   y: number;
6 |   z: number;
7 |   vx: number;
8 |   vy: number;
9 |   radius: number;
10 |   color: string;
11 |   alpha: number;
12 |   label?: string;
13 |   isSquare?: boolean;
14 | }
15 |
16 | interface Pulse {
17 |   fromIndex: number;
18 |   toIndex: number;
19 |   progress: number;
20 |   speed: number;
21 |   color: string;
22 | }

```

```
23 |  
24 | interface Wave {  
25 |   x: number;  
26 |   y: number;  
27 |   radius: number;  
28 |   maxRadius: number;  
29 |   alpha: number;  
30 | }  
31 |  
32 | const TECH_GLYPHS = ["VURLO", "VELTRIX", "VCENTRE", "18+", "400ms", "AI", "v2.0", "SYS_OK", "99.9%"];  
33 |  
34 | export function HeroCanvas() {  
35 |   const canvasRef = useRef<HTMLCanvasElement>(null);  
36 |  
37 |   useEffect(() => {  
38 |     const canvas = canvasRef.current;  
39 |     if (!canvas) return;  
40 |  
41 |     const ctx = canvas.getContext("2d");  
42 |     if (!ctx) return;  
43 |  
44 |     let animationFrameId: number;  
45 |     let isVisible = true;  
46 |     let dpr = 1;  
47 |  
48 |     const updateSize = () => {  
49 |       const parent = canvas.parentElement;  
50 |       const w = parent ? parent.getBoundingClientRect().width : window.innerWidth;  
51 |       const h = parent ? parent.getBoundingClientRect().height : window.innerHeight;  
52 |       dpr = Math.min(window.devicePixelRatio || 1, 2);  
53 |  
54 |       canvas.width = (w || window.innerWidth) * dpr;  
55 |       canvas.height = (h || Math.max(window.innerHeight, 600)) * dpr;  
56 |       canvas.style.width = `${w || window.innerWidth}px`;  
57 |       canvas.style.height = `${h || Math.max(window.innerHeight, 600)}px`;  
58 |  
59 |       ctx.scale(dpr, dpr);  
60 |     };  
61 |  
62 |     updateSize();  
63 |  
64 |     const resizeObserver = new ResizeObserver(() => {  
65 |       updateSize();  
66 |     });  
67 |  
68 |     if (canvas.parentElement) {  
69 |       resizeObserver.observe(canvas.parentElement);  
70 |     }  
71 |  
72 |     const intersectionObserver = new IntersectionObserver(  
73 |       ([entry]) => {  
74 |         isVisible = entry.isIntersecting;  
75 |       },  
76 |       { threshold: 0.05 }  
77 |     );  
78 |     intersectionObserver.observe(canvas);  
79 |  
80 |     const handleVisibilityChange = () => {  
81 |       if (document.hidden) isVisible = false;  
82 |     };  
83 |     document.addEventListener("visibilitychange", handleVisibilityChange);  
84 |  
85 |     let cssWidth = canvas.width / dpr;  
86 |     let cssHeight = canvas.height / dpr;  
87 |
```



```
88 |     const mouse = { x: cssWidth / 2, y: cssHeight / 2, targetX: cssWidth / 2, targetY: cssHeight / 2 };
89 |
90 |     const handleMouseMove = (e: MouseEvent) => {
91 |         const rect = canvas.getBoundingClientRect();
92 |         mouse.targetX = e.clientX - rect.left;
93 |         mouse.targetY = e.clientY - rect.top;
94 |     };
95 |
96 |     const waves: Wave[] = [];
97 |
98 |     const handleClick = (e: MouseEvent) => {
99 |         const rect = canvas.getBoundingClientRect();
100 |         const clickX = e.clientX - rect.left;
101 |         const clickY = e.clientY - rect.top;
102 |         waves.push({
103 |             x: clickX,
104 |             y: clickY,
105 |             radius: 5,
106 |             maxRadius: Math.min(cssWidth, cssHeight) * 0.48,
107 |             alpha: 0.95,
108 |         });
109 |     };
110 |
111 |     window.addEventListener("mousemove", handleMouseMove);
112 |     window.addEventListener("click", handleClick);
113 |
114 |     // Cyan & Violet Dual-Tone Palette
115 |     const COLORS = [
116 |         "rgba(0, 240, 255, ", // Laser Cyan
117 |         "rgba(168, 85, 247, ", // Deep Violet
118 |         "rgba(56, 189, 248, ", // Electric Cyan-Blue
119 |         "rgba(192, 132, 252, ", // Neon Violet Flare
120 |         "rgba(255, 255, 255, ", // Starlight White
121 |     ];
122 |
123 |     const isMobile = (cssWidth || window.innerWidth) < 768;
124 |     const nodeCount = isMobile ? 28 : 68;
125 |     const nodes: Node[] = [];
126 |
127 |     for (let i = 0; i < nodeCount; i++) {
128 |         const hasLabel = i < TECH_GLYPHS.length;
129 |         nodes.push({
130 |             x: Math.random() * (cssWidth || 1200),
131 |             y: Math.random() * (cssHeight || 800),
132 |             z: Math.random() * 2 + 0.8,
133 |             vx: (Math.random() - 0.5) * (isMobile ? 0.4 : 0.65),
134 |             vy: (Math.random() - 0.5) * (isMobile ? 0.4 : 0.65),
135 |             radius: Math.random() * 3.2 + 1.8,
136 |             color: COLORS[Math.floor(Math.random() * COLORS.length)],
137 |             alpha: Math.random() * 0.5 + 0.45,
138 |             label: hasLabel ? TECH_GLYPHS[i] : undefined,
139 |             isSquare: i % 4 === 0,
140 |         });
141 |     }
142 |
143 |     const pulses: Pulse[] = [];
144 |
145 |     const spawnPulse = () => {
146 |         if (nodes.length < 2) return;
147 |         const fromIndex = Math.floor(Math.random() * nodes.length);
148 |         let toIndex = -1;
149 |         let minDistance = isMobile ? 130 : 170;
150 |
151 |         for (let j = 0; j < nodes.length; j++) {
152 |             if (j === fromIndex) continue;
```

```
153 |     const dx = nodes[fromIndex].x - nodes[j].x;
154 |     const dy = nodes[fromIndex].y - nodes[j].y;
155 |     const dist = Math.sqrt(dx * dx + dy * dy);
156 |     if (dist < minDistance && Math.random() > 0.25) {
157 |         minDistance = dist;
158 |         toIndex = j;
159 |     }
160 | }
161 |
162 | if (toIndex !== -1) {
163 |     pulses.push({
164 |         fromIndex,
165 |         toIndex,
166 |         progress: 0,
167 |         speed: 0.022 + Math.random() * 0.025,
168 |         color: Math.random() > 0.5 ? "rgba(0, 240, 255, " : "rgba(192, 132, 252, ",
169 |     });
170 | }
171 | };
172 |
173 | let time = 0;
174 |
175 | const render = () => {
176 |     animationFrameId = requestAnimationFrame(render);
177 |     if (!isVisible) return;
178 |
179 |     time += 0.03;
180 |     cssWidth = canvas.width / dpr;
181 |     cssHeight = canvas.height / dpr;
182 |
183 |     if (Math.random() < 0.11 && pulses.length < (isMobile ? 6 : 14)) {
184 |         spawnPulse();
185 |     }
186 |
187 |     mouse.x += (mouse.targetX - mouse.x) * 0.08;
188 |     mouse.y += (mouse.targetY - mouse.y) * 0.08;
189 |
190 |     ctx.clearRect(0, 0, cssWidth, cssHeight);
191 |
192 |     // --- DUAL-TONE CYAN & VIOLET AMBIENT NEBULA ORBS ---
193 |     // Left Cyan Spotlight
194 |     const orb1X = cssWidth * 0.22 + Math.sin(time * 0.4) * 40;
195 |     const orb1Y = cssHeight * 0.35 + Math.cos(time * 0.4) * 30;
196 |     const grad1 = ctx.createRadialGradient(orb1X, orb1Y, 0, orb1X, orb1Y, 380);
197 |     grad1.addColorStop(0, "rgba(0, 240, 255, 0.20)");
198 |     grad1.addColorStop(0.6, "rgba(56, 189, 248, 0.08)");
199 |     grad1.addColorStop(1, "rgba(0, 0, 0, 0)");
200 |     ctx.fillStyle = grad1;
201 |     ctx.fillRect(0, 0, cssWidth, cssHeight);
202 |
203 |     // Right Violet Spotlight
204 |     const orb2X = cssWidth * 0.78 + Math.cos(time * 0.35) * 40;
205 |     const orb2Y = cssHeight * 0.45 + Math.sin(time * 0.35) * 30;
206 |     const grad2 = ctx.createRadialGradient(orb2X, orb2Y, 0, orb2X, orb2Y, 400);
207 |     grad2.addColorStop(0, "rgba(168, 85, 247, 0.22)");
208 |     grad2.addColorStop(0.6, "rgba(192, 132, 252, 0.09)");
209 |     grad2.addColorStop(1, "rgba(0, 0, 0, 0)");
210 |     ctx.fillStyle = grad2;
211 |     ctx.fillRect(0, 0, cssWidth, cssHeight);
212 |
213 |     // Mouse Intersecting Dual Spotlight (Soft & Subtle)
214 |     const spotlightGrad = ctx.createRadialGradient(
215 |         mouse.x,
216 |         mouse.y,
217 |         0,
```

```
218 |         mouse.x,  
219 |         mouse.y,  
220 |         220  
221 |     );  
222 |     spotlightGrad.addColorStop(0, "rgba(168, 85, 247, 0.12)");  
223 |     spotlightGrad.addColorStop(0.5, "rgba(0, 240, 255, 0.05)");  
224 |     spotlightGrad.addColorStop(1, "rgba(0, 0, 0, 0)");  
225 |     ctx.fillStyle = spotlightGrad;  
226 |     ctx.fillRect(0, 0, cssWidth, cssHeight);  
227 |  
228 |     // --- CLICK SHOCKWAVE RIPPLES ---  
229 |     for (let i = waves.length - 1; i >= 0; i--) {  
230 |         const wave = waves[i];  
231 |         wave.radius += 5;  
232 |         wave.alpha *= 0.95;  
233 |  
234 |         ctx.beginPath();  
235 |         ctx.arc(wave.x, wave.y, wave.radius, 0, Math.PI * 2);  
236 |         ctx.strokeStyle = `rgba(0, 240, 255, ${wave.alpha})`;   
237 |         ctx.lineWidth = 2;  
238 |         ctx.stroke();  
239 |  
240 |         ctx.beginPath();  
241 |         ctx.arc(wave.x, wave.y, wave.radius * 0.8, 0, Math.PI * 2);  
242 |         ctx.strokeStyle = `rgba(192, 132, 252, ${wave.alpha * 0.7})`;   
243 |         ctx.lineWidth = 1.2;  
244 |         ctx.stroke();  
245 |  
246 |         for (let j = 0; j < nodes.length; j++) {  
247 |             const n = nodes[j];  
248 |             const dx = n.x - wave.x;  
249 |             const dy = n.y - wave.y;  
250 |             const dist = Math.sqrt(dx * dx + dy * dy);  
251 |             if (Math.abs(dist - wave.radius) < 32) {  
252 |                 n.x += (dx / (dist || 1)) * 3.5;  
253 |                 n.y += (dy / (dist || 1)) * 3.5;  
254 |             }  
255 |         }  
256 |  
257 |         if (wave.alpha < 0.02 || wave.radius > wave.maxRadius) {  
258 |             waves.splice(i, 1);  
259 |         }  
260 |     }  
261 |  
262 |     // --- GEOMETRIC CONSTELLATION LINES ---  
263 |     const maxDist = isMobile ? 115 : 150;  
264 |     for (let i = 0; i < nodes.length; i++) {  
265 |         for (let j = i + 1; j < nodes.length; j++) {  
266 |             const dx = nodes[i].x - nodes[j].x;  
267 |             const dy = nodes[i].y - nodes[j].y;  
268 |             const dist = Math.sqrt(dx * dx + dy * dy);  
269 |  
270 |             if (dist < maxDist) {  
271 |                 const lineAlpha = (1 - dist / maxDist) * 0.35;  
272 |                 ctx.beginPath();  
273 |                 ctx.strokeStyle = `rgba(192, 132, 252, ${lineAlpha})`;   
274 |                 ctx.lineWidth = 1.1;  
275 |                 ctx.moveTo(nodes[i].x, nodes[i].y);  
276 |                 ctx.lineTo(nodes[j].x, nodes[j].y);  
277 |                 ctx.stroke();  
278 |             }  
279 |         }  
280 |     }  
281 |  
282 |     // --- SIGNAL PULSES ---
```

```
283 |     for (let i = pulses.length - 1; i >= 0; i--) {
284 |         const pulse = pulses[i];
285 |         pulse.progress += pulse.speed;
286 |
287 |         const from = nodes[pulse.fromIndex];
288 |         const to = nodes[pulse.toIndex];
289 |
290 |         if (from && to) {
291 |             const px = from.x + (to.x - from.x) * pulse.progress;
292 |             const py = from.y + (to.y - from.y) * pulse.progress;
293 |
294 |             ctx.beginPath();
295 |             const pulseGrad = ctx.createRadialGradient(px, py, 0, px, py, 7);
296 |             pulseGrad.addColorStop(0, "rgba(255, 255, 255, 1)");
297 |             pulseGrad.addColorStop(0.5, `${pulse.color}0.95`);
298 |             pulseGrad.addColorStop(1, `${pulse.color}0`);
299 |             ctx.fillStyle = pulseGrad;
300 |             ctx.arc(px, py, 7, 0, Math.PI * 2);
301 |             ctx.fill();
302 |         }
303 |
304 |         if (pulse.progress >= 1) {
305 |             pulses.splice(i, 1);
306 |         }
307 |     }
308 |
309 |     // --- CYAN & VIOLET NODES & GLYPHS ---
310 |     for (let i = 0; i < nodes.length; i++) {
311 |         const n = nodes[i];
312 |
313 |         const mdx = mouse.x - n.x;
314 |         const mdy = mouse.y - n.y;
315 |         const mdist = Math.sqrt(mdx * mdx + mdy * mdy);
316 |         if (mdist < 180) {
317 |             const force = (180 - mdist) / 180;
318 |             n.x -= (mdx / mdist) * force * 1.3 * n.z;
319 |             n.y -= (mdy / mdist) * force * 1.3 * n.z;
320 |         }
321 |
322 |         n.x += n.vx * n.z;
323 |         n.y += n.vy * n.z;
324 |
325 |         if (n.x < 0) n.x = cssWidth;
326 |         if (n.x > cssWidth) n.x = 0;
327 |         if (n.y < 0) n.y = cssHeight;
328 |         if (n.y > cssHeight) n.y = 0;
329 |
330 |         const currentRadius = n.radius + Math.sin(time * 3 + i) * 1.0;
331 |
332 |         if (n.isSquare) {
333 |             ctx.fillStyle = `${n.color}${n.alpha}`;
334 |             ctx.fillRect(n.x - currentRadius, n.y - currentRadius, currentRadius * 2, currentRadius * 2);
335 |         } else {
336 |             ctx.beginPath();
337 |             const pGrad = ctx.createRadialGradient(
338 |                 n.x,
339 |                 n.y,
340 |                 0,
341 |                 n.x,
342 |                 n.y,
343 |                 currentRadius * 4
344 |             );
345 |             pGrad.addColorStop(0, `${n.color}${n.alpha}`);
346 |             pGrad.addColorStop(0.4, `${n.color}${n.alpha * 0.5}`);
347 |             pGrad.addColorStop(1, `${n.color}0`);
```

```

348 |
349 |         ctx.fillStyle = pGrad;
350 |         ctx.arc(n.x, n.y, currentRadius * 4, 0, Math.PI * 2);
351 |         ctx.fill();
352 |     }
353 |
354 |     if (n.label) {
355 |         ctx.font = '10px "JetBrains Mono", monospace';
356 |         ctx.fillStyle = `rgba(0, 240, 255, ${n.alpha * 0.9})`;
357 |         ctx.fillText(n.label, n.x + 8, n.y + 3);
358 |     }
359 | }
360 | };
361 |
362 | render();
363 |
364 | return () => {
365 |     resizeObserver.disconnect();
366 |     intersectionObserver.disconnect();
367 |     document.removeEventListener("visibilitychange", handleVisibilityChange);
368 |     window.removeEventListener("mousemove", handleMouseMove);
369 |     window.removeEventListener("click", handleClick);
370 |     cancelAnimationFrame(animationFrameId);
371 | };
372 | }, []);
373 |
374 | return (
375 |     <canvas
376 |         ref={canvasRef}
377 |         className="pointer-events-none absolute inset-0 z-0 h-full w-full opacity-100"
378 |         aria-hidden
379 |     />
380 | );
381 | }

```

FILE #33: src\components\portfolio\Intro.tsx

```

1 | import { motion, useReducedMotion } from "motion/react";
2 | import { useEffect } from "react";
3 | import { EASE } from "@lib/motion";
4 |
5 | const WORDS = [
6 |   { text: "Ideas.", weight: "font-semibold text-foreground" },
7 |   {
8 |     text: "Into.",
9 |     weight: "font-normal italic text-muted-foreground font-serif-accent",
10 |   },
11 |   { text: "Systems.", weight: "font-semibold text-foreground" },
12 | ];
13 |
14 | export function Intro({ onDone }: { onDone: () => void }) {
15 |   const reduceMotion = useReducedMotion();
16 |
17 |   useEffect(() => {
18 |     // Last word finishes animating in at ~1.39s (2 × 0.32s stagger + 0.75s
19 |     // duration). Hold on the completed phrase for ~550ms before exiting ?
20 |     // long enough to actually read as a pause, not a stutter.
21 |     const holdDelay = reduceMotion ? 500 : 1950;
22 |     const t = setTimeout(onDone, holdDelay);
23 |     return () => clearTimeout(t);
24 |   }, [onDone, reduceMotion]);
25 |
26 |   return (

```

```

27 |     <motion.div
28 |       key="intro"
29 |       exit={{
30 |         opacity: 0,
31 |         filter: reduceMotion ? "blur(0px)" : "blur(18px)",
32 |         scale: 1.015,
33 |       }}
34 |       transition={{ duration: reduceMotion ? 0.3 : 0.65, ease: EASE }}
35 |       className="fixed inset-0 z-50 flex items-center justify-center bg-background grain"
36 |       aria-hidden="true"
37 |     >
38 |       <div className="flex flex-wrap items-baseline justify-center gap-x-4 gap-y-2 px-6 text-center">
39 |         {WORDS.map((w, i) => (
40 |           <motion.span
41 |             key={w.text}
42 |             initial={reduceMotion ? { opacity: 0 } : { opacity: 0, y: 18, filter: "blur(12px)" }}
43 |             animate={reduceMotion ? { opacity: 1 } : { opacity: 1, y: 0, filter: "blur(0px)" }}
44 |             transition={{
45 |               duration: reduceMotion ? 0.4 : 0.75,
46 |               ease: EASE,
47 |               delay: reduceMotion ? 0 : i * 0.32,
48 |             }}
49 |             className={`font-display text-5xl tracking-tight sm:text-6xl md:text-8xl ${w.weight}`}
50 |           >
51 |             {w.text}
52 |           </motion.span>
53 |         ))}
54 |       </div>
55 |     </motion.div>
56 |   );
57 | }

```

FILE #34: src\components\portfolio\LiveStatusWidget.tsx

```

1 | import { useEffect, useState } from "react";
2 | import { Clock, MapPin } from "lucide-react";
3 |
4 | export function LiveStatusWidget() {
5 |   const [timeString, setTimeString] = useState("");
6 |
7 |   useEffect(() => {
8 |     const updateTime = () => {
9 |       const now = new Date();
10 |      // Formatted in IST / India Standard Time
11 |      const formatted = new Intl.DateTimeFormat("en-US", {
12 |        timeZone: "Asia/Kolkata",
13 |        hour: "2-digit",
14 |        minute: "2-digit",
15 |        second: "2-digit",
16 |        hour12: true,
17 |      }).format(now);
18 |      setTimeString(formatted);
19 |    };
20 |
21 |    updateTime();
22 |    const interval = setInterval(updateTime, 1000);
23 |    return () => clearInterval(interval);
24 |  }, []);
25 |
26 |   return (
27 |     <div className="inline-flex flex-wrap items-center gap-2.5 rounded-full border border-slate-800/80 bg...
28 |       { /* Open for roles indicator */ }
29 |     <span className="flex items-center gap-1.5 font-medium text-emerald-400">

```

```

30 |         <span className="relative flex h-2 w-2">
31 |             <span className="absolute inline-flex h-full w-full animate-ping rounded-full bg-emerald-400 op...
32 |             <span className="relative inline-flex h-2 w-2 rounded-full bg-emerald-500"></span>
33 |         </span>
34 |         AVAILABLE FOR ROLES
35 |     </span>
36 |
37 |     <span className="text-slate-700">?</span>
38 |
39 |     {/* Clock */}
40 |     <span className="flex items-center gap-1 font-mono text-slate-300">
41 |         <Clock className="h-3 w-3 text-cyan-400" />
42 |         {timeString || "07:12:00 PM"} IST
43 |     </span>
44 |
45 |     <span className="hidden sm:inline text-slate-700">?</span>
46 |
47 |     {/* Location */}
48 |     <span className="hidden sm:flex items-center gap-1 text-slate-400">
49 |         <MapPin className="h-3 w-3 text-purple-400" />
50 |         Vellore, IN (UTC+5:30)
51 |     </span>
52 | </div>
53 | );
54 | }

```

FILE #35: src\components\portfolio\MagneticButton.tsx

```

1 | import { motion, useMotionValue, useReducedMotion, useSpring, useTransform } from "motion/react";
2 | import { type ReactNode, useRef } from "react";
3 | import { cn } from "@lib/utils";
4 |
5 | /**
6 |  * A button/link that nudges toward the cursor on hover ? a few px, never
7 |  * more. Falls back to a static element for touch devices and when the user
8 |  * prefers reduced motion.
9 |  */
10 | export function MagneticButton({
11 |   href,
12 |   children,
13 |   className,
14 |   variant = "primary",
15 |   onClick,
16 | }: {
17 |   href?: string;
18 |   children: ReactNode;
19 |   className?: string;
20 |   variant?: "primary" | "ghost";
21 |   onClick?: () => void;
22 | }): {
23 |   const ref = useRef<HTMLAnchorElement>(null);
24 |   const reduceMotion = useReducedMotion();
25 |   const x = useMotionValue(0);
26 |   const y = useMotionValue(0);
27 |   const springX = useSpring(x, { stiffness: 300, damping: 20, mass: 0.4 });
28 |   const springY = useSpring(y, { stiffness: 300, damping: 20, mass: 0.4 });
29 |
30 |   // Create secondary springs and transforms for the inner text to create an elastic parallax drift
31 |   const textX = useSpring(x, { stiffness: 240, damping: 18, mass: 0.5 });
32 |   const textY = useSpring(y, { stiffness: 240, damping: 18, mass: 0.5 });
33 |   const innerX = useTransform(textX, (v) => v * 0.3);
34 |   const innerY = useTransform(textY, (v) => v * 0.3);
35 |

```

```

36 |   const handleMove = (e: React.MouseEvent<HTMLAnchorElement>) => {
37 |     if (reduceMotion) return;
38 |     const rect = ref.current?.getBoundingClientRect();
39 |     if (!rect) return;
40 |     const relX = e.clientX - (rect.left + rect.width / 2);
41 |     const relY = e.clientY - (rect.top + rect.height / 2);
42 |     x.set(relX * 0.25);
43 |     y.set(relY * 0.35);
44 |   };
45 |
46 |   const handleLeave = () => {
47 |     x.set(0);
48 |     y.set(0);
49 |   };
50 |
51 |   return (
52 |     <motion.a
53 |       ref={ref}
54 |       href={href}
55 |       onClick={onClick}
56 |       onMouseMove={handleMove}
57 |       onMouseLeave={handleLeave}
58 |       style={{ x: springX, y: springY }}
59 |       className={cn(
60 |         "group inline-flex items-center gap-2 rounded-full px-6 py-3 text-sm font-medium",
61 |         variant === "primary" ? "btn-primary" : "btn-ghost",
62 |         className,
63 |       )}
64 |     >
65 |       <motion.span
66 |         style={reduceMotion ? {} : { x: innerX, y: innerY }}
67 |         className="inline-flex items-center gap-2"
68 |       >
69 |         {children}
70 |       </motion.span>
71 |     </motion.a>
72 |   );
73 | }

```

FILE #36: src\components\portfolio\Nav.tsx

```

1 | import { AnimatePresence, motion, useScroll } from "motion/react";
2 | import { useEffect, useState } from "react";
3 | import { Menu, X } from "lucide-react";
4 | import { EASE } from "@lib/motion";
5 | import { NAV_ITEMS, PROFILE } from "@lib/site-data";
6 | import { playTick } from "@lib/sound";
7 | import { CommandPaletteTrigger } from "../CommandPalette";
8 |
9 | export function Nav() {
10 |   const { scrollY } = useScroll();
11 |   const [scrolled, setScrolled] = useState(false);
12 |   const [open, setOpen] = useState(false);
13 |   const [activeSection, setActiveSection] = useState("top");
14 |
15 |   useEffect(() => scrollY.on("change", (y) => setScrolled(y > 20)), [scrollY]);
16 |
17 |   // Lock body scroll while the mobile menu is open.
18 |   useEffect(() => {
19 |     document.body.style.overflow = open ? "hidden" : "";
20 |     return () => {
21 |       document.body.style.overflow = "";
22 |     };

```



```

23 |   }, [open]);
24 |
25 |   // High-performance IntersectionObserver Scrollspy
26 |   useEffect(() => {
27 |     const sections = ["top", "work", "about", "stack", "contact"];
28 |
29 |     // rootMargin offset simulates the previous 250px offset earlier detection beautifully
30 |     const observerOptions = {
31 |       root: null,
32 |       rootMargin: "-20% 0px -60% 0px",
33 |       threshold: 0,
34 |     };
35 |
36 |     const observerCallback = (entries: IntersectionObserverEntry[]) => {
37 |       entries.forEach((entry) => {
38 |         if (entry.isIntersecting) {
39 |           setActiveSection(entry.target.id);
40 |         }
41 |       });
42 |     };
43 |
44 |     const observer = new IntersectionObserver(observerCallback, observerOptions);
45 |
46 |     sections.forEach((sectionId) => {
47 |       const el = document.getElementById(sectionId);
48 |       if (el) observer.observe(el);
49 |     });
50 |
51 |     return () => observer.disconnect();
52 |   }, []);
53 |
54 |   return (
55 |     <>
56 |       <motion.header
57 |         initial={{ y: -20, opacity: 0 }}
58 |         animate={{ y: 0, opacity: 1 }}
59 |         transition={{ duration: 0.6, delay: 0.15, ease: EASE }}
60 |         className={`fixed inset-x-0 top-0 z-40 transition-colors duration-500 ${
61 |           scrolled || open ? "border-b border-border bg-background/70 backdrop-blur-xl" : ""
62 |         }`
63 |       >
64 |       <div className="mx-auto flex max-w-6xl items-center justify-between px-6 py-4">
65 |         <a href="#top" onMouseEnter={playTick} className="group flex items-center gap-2">
66 |           <span className="h-1.5 w-1.5 rounded-full bg-primary transition-colors group-hover:bg-seconda...
67 |           <span className="font-display text-sm font-semibold tracking-tight">
68 |             {PROFILE.name}
69 |           </span>
70 |         </a>
71 |
72 |         <nav className="hidden items-center gap-8 md:flex">
73 |           {NAV_ITEMS.map((item, i) => {
74 |             const pan = (i / (NAV_ITEMS.length - 1)) * 2 - 1;
75 |             const isActive = activeSection === item.href.slice(1);
76 |             return (
77 |               <a
78 |                 key={item.href}
79 |                 href={item.href}
80 |                 onMouseEnter={() => playTick(pan)}
81 |                 className={`relative text-sm font-medium transition-colors py-1 ${
82 |                   isActive ? "text-primary" : "text-muted-foreground hover:text-foreground"
83 |                 }`}
84 |               >
85 |                 {item.label}
86 |                 {isActive && (
87 |                   <motion.span

```

```
88 |         layoutId="activeNavIndicator"
89 |         className="absolute -bottom-1 left-0 right-0 h-0.5 rounded-full bg-primary"
90 |         transition={{ type: "spring", stiffness: 380, damping: 30 }}
91 |       />
92 |     )}
93 |   </a>
94 | );
95 |   )}}
96 | </nav>
97 |
98 |   <div className="flex items-center gap-3">
99 |     <CommandPaletteTrigger />
100 |     <a
101 |       href="#contact"
102 |       onMouseEnter={playTick}
103 |       className="btn-ghost hidden rounded-full px-4 py-1.5 text-xs font-medium md:inline-flex"
104 |     >
105 |       Get in touch
106 |     </a>
107 |
108 |     <button
109 |       type="button"
110 |       onClick={() => setOpen((v) => !v)}
111 |       onMouseEnter={playTick}
112 |       aria-expanded={open}
113 |       aria-label={open ? "Close menu" : "Open menu"}
114 |       className="flex h-9 w-9 items-center justify-center rounded-full text-foreground transition...
115 |     >
116 |       {open ? <X className="h-4 w-4" /> : <Menu className="h-4 w-4" />}
117 |     </button>
118 |   </div>
119 | </div>
120 | </motion.header>
121 |
122 | <AnimatePresence>
123 |   {open && (
124 |     <motion.div
125 |       initial={{ opacity: 0 }}
126 |       animate={{ opacity: 1 }}
127 |       exit={{ opacity: 0 }}
128 |       transition={{ duration: 0.3, ease: EASE }}
129 |       className="fixed inset-0 z-30 bg-background/95 backdrop-blur-2xl md:hidden"
130 |     >
131 |       <nav className="flex h-full flex-col items-center justify-center gap-8 px-6">
132 |         {NAV_ITEMS.map((item, i) => {
133 |           const pan = (i / (NAV_ITEMS.length - 1)) * 2 - 1;
134 |           const isActive = activeSection === item.href.slice(1);
135 |           return (
136 |             <motion.a
137 |               key={item.href}
138 |               href={item.href}
139 |               onMouseEnter={() => playTick(pan)}
140 |               onClick={() => setOpen(false)}
141 |               initial={{ opacity: 0, y: 12 }}
142 |               animate={{ opacity: 1, y: 0 }}
143 |               transition={{ duration: 0.4, delay: 0.08 * i, ease: EASE }}
144 |               className={`font-display text-3xl font-semibold tracking-tight transition-colors ${
145 |                 isActive ? "text-primary" : "text-foreground"
146 |               }`}
147 |             >
148 |               {item.label}
149 |             </motion.a>
150 |           );
151 |         })}
152 |       </nav>
```

```

153 |             href="#contact"
154 |             onMouseEnter={playTick}
155 |             onClick={() => setOpen(false)}
156 |             initial={{ opacity: 0, y: 12 }}
157 |             animate={{ opacity: 1, y: 0 }}
158 |             transition={{
159 |               duration: 0.4,
160 |               delay: 0.08 * NAV_ITEMS.length,
161 |               ease: EASE,
162 |             }}
163 |             className="btn-primary mt-4 rounded-full px-6 py-3 text-sm font-medium"
164 |           >
165 |             Get in touch
166 |           </motion.a>
167 |         </nav>
168 |       </motion.div>
169 |     )}
170 |   </AnimatePresence>
171 | </>
172 | );
173 | }

```

FILE #37: src\components\portfolio\Projects.tsx

```

1 | import { lazy, Suspense } from "react";
2 | import { useRef, useState, useEffect } from "react";
3 | import { motion, useMotionValue, animate } from "motion/react";
4 | import { Reveal } from "../Reveal";
5 | import { PROJECTS, type Project } from "@lib/site-data";
6 | import { playTick } from "@lib/sound";
7 | import { CarouselProjectCard } from "../projects/CarouselProjectCard";
8 |
9 | const ACTIVE_THEMES = [
10 |   { glow: "rgba(34, 211, 238, 0.025)", border: "rgba(34, 211, 238, 0.35)" }, // Vurlo (Cyan)
11 |   { glow: "rgba(16, 185, 129, 0.025)", border: "rgba(16, 185, 129, 0.35)" }, // Veltrix (Emerald)
12 |   { glow: "rgba(99, 102, 241, 0.025)", border: "rgba(99, 102, 241, 0.35)" }, // Vcentre (Indigo)
13 | ];
14 |
15 | const WorkspacePanel = lazy(() =>
16 |   import("../projects/WorkspacePanel").then((module) => ({
17 |     default: module.WorkspacePanel,
18 |   })),
19 | );
20 |
21 | export function Projects() {
22 |   const [activeProject, setActiveProject] = useState<Project | null>(null);
23 |   const carouselContainerRef = useRef<HTMLDivElement>(null);
24 |   const [activeCarouselIndex, setActiveCarouselIndex] = useState(0);
25 |   const scrollTimeoutRef = useRef<number | null>(null);
26 |
27 |   // 3D Carousel Motion values for smooth compositor animations per card
28 |   const cardRotations = [useMotionValue(0), useMotionValue(-25), useMotionValue(-25)];
29 |   const cardRotateXs = [useMotionValue(0), useMotionValue(-8), useMotionValue(-8)];
30 |   const cardScales = [useMotionValue(1), useMotionValue(0.85), useMotionValue(0.85)];
31 |   const cardOpacities = [useMotionValue(1), useMotionValue(0.55), useMotionValue(0.55)];
32 |   const cardZs = [useMotionValue(0), useMotionValue(-120), useMotionValue(-120)];
33 |   const cardShines = [useMotionValue(0), useMotionValue(-150), useMotionValue(-150)];
34 |
35 |   // Caching card centers to avoid reading layouts on scroll frames
36 |   const cardCentersRef = useRef<number[]>([]);
37 |
38 |   // Scrollspy & scroll progress handler
39 |   useEffect(() => {

```

```
40 |     const container = carouselContainerRef.current;
41 |     if (!container) return;
42 |
43 |     // Cache card centers and total scroll limit
44 |     const updateLayout = () => {
45 |         const cardElements = container.getElementsByClassName("snap-center");
46 |         const centers = [];
47 |         for (let i = 0; i < cardElements.length; i++) {
48 |             const el = cardElements[i] as HTMLElement;
49 |             centers.push(el.offsetLeft + el.clientWidth / 2);
50 |         }
51 |         cardCentersRef.current = centers;
52 |     };
53 |
54 |     const handleScroll = () => {
55 |         // Instantly kill all active spotlight glows by firing synthetic mouseleave on card articles.
56 |         const articles = container.querySelectorAll("article");
57 |         articles.forEach((el) => el.dispatchEvent(new MouseEvent("mouseleave", { bubbles: false })));
58 |
59 |         // Toggle carousel-scrolling class to disable mouse events and fade out glows on scroll
60 |         container.classList.add("carousel-scrolling");
61 |         if (scrollTimeoutRef.current) {
62 |             window.clearTimeout(scrollTimeoutRef.current);
63 |         }
64 |         scrollTimeoutRef.current = window.setTimeout(() => {
65 |             container.classList.remove("carousel-scrolling");
66 |         }, 150);
67 |
68 |         // Recalculate on the fly if not set yet
69 |         if (cardCentersRef.current.length === 0) {
70 |             updateLayout();
71 |         }
72 |
73 |         const scrollLeft = container.scrollLeft;
74 |         const clientWidth = container.clientWidth;
75 |         const containerCenter = scrollLeft + clientWidth / 2;
76 |         const centers = cardCentersRef.current;
77 |
78 |         // Update each card's raw transforms locally relative to the viewport center
79 |         for (let i = 0; i < cardRotations.length; i++) {
80 |             const center = centers[i] || 0;
81 |             const distance = center - containerCenter;
82 |             const maxDist = clientWidth * 0.65;
83 |             const ratio = Math.max(-1, Math.min(1, distance / maxDist)); // ranges -1 to 1
84 |
85 |             cardRotations[i].set(ratio * -25); // rotateY
86 |             cardRotateXs[i].set(Math.abs(ratio) * -8); // rotateX
87 |             cardScales[i].set(1 - Math.abs(ratio) * 0.15); // scale
88 |             cardOpacities[i].set(1 - Math.abs(ratio) * 0.45); // opacity
89 |             cardZs[i].set(Math.abs(ratio) * -120); // translateZ
90 |             cardShines[i].set(ratio * 150); // shine X percentage shift
91 |         }
92 |
93 |         // Update active dots index using robust closest-to-center element detection
94 |         let closestIndex = 0;
95 |         let minDistance = Infinity;
96 |         for (let i = 0; i < centers.length; i++) {
97 |             const center = centers[i] || 0;
98 |             const distance = Math.abs(center - containerCenter);
99 |             if (distance < minDistance) {
100 |                 minDistance = distance;
101 |                 closestIndex = i;
102 |             }
103 |         }
104 |         setActiveCarouselIndex((prev) => {
```

```
105 |         if (prev !== closestIndex && closestIndex >= 0 && closestIndex < PROJECTS.length) {
106 |             return closestIndex;
107 |         }
108 |         return prev;
109 |     });
110 | };
111 |
112 | // Initial calculations
113 | updateLayout();
114 | handleScroll();
115 |
116 | // Recalculate after mount tick intervals to ensure rendering positions resolve correctly
117 | const renderTimeouts = [150, 400, 1000].map((delay) =>
118 |     window.setTimeout(() => {
119 |         updateLayout();
120 |         handleScroll();
121 |     }, delay),
122 | );
123 |
124 | const handleResize = () => {
125 |     updateLayout();
126 |     handleScroll();
127 | };
128 |
129 | // Listen to scroll for card changes & motion values
130 | container.addEventListener("scroll", handleScroll, { passive: true });
131 |
132 | // Only read layout dimensions on resize, NOT on every scroll frame
133 | window.addEventListener("resize", handleResize, { passive: true });
134 |
135 | return () => {
136 |     container.removeEventListener("scroll", handleScroll);
137 |     window.removeEventListener("resize", handleResize);
138 |     renderTimeouts.forEach((t) => window.clearTimeout(t));
139 |     if (scrollTimeoutRef.current) {
140 |         window.clearTimeout(scrollTimeoutRef.current);
141 |     }
142 | };
143 | }, []);
144 |
145 | // Keyboard navigation
146 | useEffect(() => {
147 |     const handleKeyDown = (e: KeyboardEvent) => {
148 |         if (activeProject) return;
149 |
150 |         const container = carouselContainerRef.current;
151 |         if (!container) return;
152 |
153 |         const cardElements = container.getElementsByClassName("snap-center");
154 |         if (cardElements.length === 0) return;
155 |
156 |         if (e.key === "ArrowRight") {
157 |             const nextIndex = Math.min(activeCarouselIndex + 1, PROJECTS.length - 1);
158 |             const nextEl = cardElements[nextIndex] as HTMLElement;
159 |             if (nextEl) {
160 |                 const targetScrollLeft =
161 |                     nextEl.offsetLeft - (container.clientWidth - nextEl.clientWidth) / 2;
162 |                 animate(container.scrollLeft, targetScrollLeft, {
163 |                     type: "tween",
164 |                     ease: [0.22, 1, 0.36, 1],
165 |                     duration: 0.5,
166 |                     onUpdate: (latest) => {
167 |                         container.scrollLeft = latest;
168 |                     },
169 |                 });

```

```
170 |         playTick(0.1);
171 |     }
172 | } else if (e.key === "ArrowLeft") {
173 |     const prevIndex = Math.max(activeCarouselIndex - 1, 0);
174 |     const prevEl = cardElements[prevIndex] as HTMLElement;
175 |     if (prevEl) {
176 |         const targetScrollLeft =
177 |             prevEl.offsetLeft - (container.clientWidth - prevEl.clientWidth) / 2;
178 |         animate(container.scrollLeft, targetScrollLeft, {
179 |             type: "tween",
180 |             ease: [0.22, 1, 0.36, 1],
181 |             duration: 0.5,
182 |             onUpdate: (latest) => {
183 |                 container.scrollLeft = latest;
184 |             },
185 |         });
186 |         playTick(-0.1);
187 |     }
188 | }
189 | };
190 |
191 | window.addEventListener("keydown", handleKeyDown);
192 | return () => window.removeEventListener("keydown", handleKeyDown);
193 | }, [activeCarouselIndex, activeProject]);
194 |
195 | const handleViewCaseStudy = (p: Project) => {
196 |     setActiveProject(p);
197 | };
198 |
199 | return (
200 |     <section id="work" className="relative py-32 md:py-40">
201 |         {/* Ambient background glow bleed */}
202 |         <div
203 |             className="pointer-events-none absolute inset-0 -z-10 transition-colors duration-1000 ease-out"
204 |             style={{
205 |                 background: `radial-gradient(600px circle at 50% 50%, ${ACTIVE_THEMES[activeCarouselIndex]?.glo...
206 |             }}
207 |         />
208 |
209 |         <div className="mx-auto max-w-6xl px-6">
210 |             <Reveal>
211 |                 <div className="flex items-end justify-between gap-8">
212 |                     <div>
213 |                         <p className="eyebrow">Selected work · 2026</p>
214 |                         <h2 className="mt-4 font-display text-4xl font-semibold tracking-tight md:text-5xl">
215 |                             Systems shipped, <br className="hidden md:block" />
216 |                             not just experiments.
217 |                         </h2>
218 |                     </div>
219 |
220 |                     {/* Editorial Sliding Card Counter */}
221 |                     <div className="hidden md:flex flex-col items-end gap-1 font-mono">
222 |                         <span className="text-[9px] uppercase tracking-widest text-muted-foreground/60">
223 |                             Active Case Study
224 |                         </span>
225 |                         <div className="flex items-baseline gap-2 select-none">
226 |                             <div className="h-[48px] overflow-hidden text-5xl font-bold tracking-tight text-foreground...
227 |                                 <div
228 |                                     className="transition-transform duration-500 ease-[cubic-bezier(0.22,1,0.36,1)] flex ...
229 |                                     style={{ transform: `translateY(-${activeCarouselIndex * 48}px)` }}
230 |                                 >
231 |                                     <span className="h-[48px] flex items-center justify-end leading-none">01</span>
232 |                                     <span className="h-[48px] flex items-center justify-end leading-none">02</span>
233 |                                     <span className="h-[48px] flex items-center justify-end leading-none">03</span>
234 |                                 </div>
```

```

235 |         </div>
236 |         <span className="text-muted-foreground/20 text-2xl font-light">/</span>
237 |         <span className="text-muted-foreground/50 text-xl font-medium">03</span>
238 |     </div>
239 | </div>
240 | </div>
241 | </Reveal>
242 |
243 | <Reveal delay={2} direction="up" className="relative mt-12">
244 |     {/* Left/Right Fading Overlays */}
245 |     <div className="pointer-events-none absolute inset-y-0 left-0 w-[8vw] z-20 bg-gradient-to-r fro...
246 |     <div className="pointer-events-none absolute inset-y-0 right-0 w-[8vw] z-20 bg-gradient-to-l fr...
247 |
248 |     {/* Scroll Container */}
249 |     <div
250 |         ref={carouselContainerRef}
251 |         className="flex gap-6 md:gap-12 overflow-x-auto snap-x snap-mandatory scrollbar-none py-8 px-...
252 |         style={{ touchAction: "pan-x pan-y" }}
253 |     >
254 |         {PROJECTS.map((p, i) => (
255 |             <CarouselProjectCard
256 |                 key={p.index}
257 |                 p={p}
258 |                 i={i}
259 |                 onViewCaseStudy={handleViewCaseStudy}
260 |                 rotateX={cardRotateXs[i]}
261 |                 rotateY={cardRotations[i]}
262 |                 scale={cardScales[i]}
263 |                 opacity={cardOpacities[i]}
264 |                 z={cardZs[i]}
265 |                 shineX={cardShines[i]}
266 |                 isActive={activeCarouselIndex === i}
267 |             />
268 |         ))}
269 |     </div>
270 | </Reveal>
271 |
272 | {/* Navigation Dots */}
273 | <div className="mt-6 flex justify-center gap-2 relative z-20">
274 |     {PROJECTS.map((p, idx) => (
275 |         <button
276 |             key={p.index}
277 |             onClick={() => {
278 |                 const container = carouselContainerRef.current;
279 |                 if (!container) return;
280 |                 const cardElements = container.getElementsByClassName("snap-center");
281 |                 const targetEl = cardElements[idx] as HTMLElement;
282 |                 if (targetEl) {
283 |                     const targetScrollLeft =
284 |                         targetEl.offsetLeft - (container.clientWidth - targetEl.clientWidth) / 2;
285 |                     animate(container.scrollLeft, targetScrollLeft, {
286 |                         type: "tween",
287 |                         ease: [0.22, 1, 0.36, 1],
288 |                         duration: 0.5,
289 |                         onUpdate: (latest) => {
290 |                             container.scrollLeft = latest;
291 |                         },
292 |                     });
293 |                     playTick((idx / (PROJECTS.length - 1)) * 2 - 1);
294 |                 }
295 |             })}
296 |     <div className={`h-2.5 rounded-full transition-all duration-300 cursor-pointer ${
297 |         activeCarouselIndex === idx
298 |         ? "w-8 bg-primary"
299 |         : "w-2.5 bg-white/20 hover:bg-white/40"

```

```
300 |         `}``  
301 |         aria-label={`Go to project ${idx + 1}`}`  
302 |     />  
303 |     )}}  
304 | </div>  
305 | </div>  
306 |  
307 |     { /* Workspace Portal Backdrop Overlay */  
308 |     <Suspense fallback={null}>  
309 |         {activeProject && (  
310 |             <WorkspacePanel project={activeProject} onClose={() => setActiveProject(null)} />  
311 |         )}  
312 |     </Suspense>  
313 | </section>  
314 | );  
315 | }
```

FILE #38: src\components\portfolio\Reveal.tsx

```
1 | import { useEffect, useRef, type ReactNode, type ElementType } from "react";  
2 |  
3 | interface RevealProps {  
4 |     children: ReactNode;  
5 |     className?: string;  
6 |     delay?: number; // Accepts a number (in steps) to stagger animations  
7 |     as?: ElementType;  
8 |     direction?: "up" | "left" | "right" | "light"; // Added light direction  
9 | }  
10 |  
11 | export function Reveal({  
12 |     children,  
13 |     className = "",  
14 |     delay = 0,  
15 |     as: Tag = "div",  
16 |     direction = "up",  
17 | }: RevealProps) {  
18 |     const ref = useRef<HTMLElement>(null);  
19 |  
20 |     useEffect(() => {  
21 |         const el = ref.current;  
22 |         if (!el) return;  
23 |  
24 |         const observer = new IntersectionObserver(  
25 |             (entries) => {  
26 |                 entries.forEach((entry) => {  
27 |                     if (entry.isIntersecting) {  
28 |                         // Add a small timeout if delay is passed, allowing CSS to handle the stagger  
29 |                         if (delay > 0) {  
30 |                             setTimeout(() => el.classList.add("in-view"), delay * 120);  
31 |                         } else {  
32 |                             el.classList.add("in-view");  
33 |                         }  
34 |                         observer.unobserve(el); // Run once and destroy observer for 0 overhead  
35 |                     }  
36 |                 });  
37 |             },  
38 |             {  
39 |                 threshold: 0.1,  
40 |                 rootMargin: "0px 0px -10% 0px",  
41 |             },  
42 |         );  
43 |  
44 |         observer.observe(el);
```



```
45 |     return () => observer.disconnect();
46 |   }, [delay]);
47 |
48 |   // Determine which CSS class to use based on direction
49 |   const revealClass =
50 |     direction === "left"
51 |       ? "reveal-3d-left"
52 |       : direction === "right"
53 |         ? "reveal-3d-right"
54 |         : direction === "light"
55 |           ? "reveal-light"
56 |           : "reveal-3d";
57 |
58 |   return (
59 |     <Tag ref={ref as React.Ref<HTMLElement>} className={` ${revealClass} ${className}`} >
60 |       {children}
61 |     </Tag>
62 |   );
63 | }
```

FILE #39: src\components\portfolio\ScrollProgress.tsx

```
1 | import { motion, useScroll, useSpring } from "motion/react";
2 |
3 | export function ScrollProgress() {
4 |   const { scrollYProgress } = useScroll();
5 |   const scaleX = useSpring(scrollYProgress, {
6 |     stiffness: 120,
7 |     damping: 30,
8 |     mass: 0.2,
9 |   });
10 |   return (
11 |     <motion.div
12 |       style={{
13 |         scaleX,
14 |         boxShadow: "0 0 10px rgba(139, 92, 246, 0.6)",
15 |       }}
16 |       className="fixed left-0 top-0 z-50 h-0.5 w-full origin-left bg-primary"
17 |     </>
18 |   );
19 | }
```

FILE #40: src\components\portfolio\Stack.tsx

```
1 | import { Reveal } from "../Reveal";
2 | import { STACK } from "@lib/site-data";
3 |
4 | export function Stack() {
5 |   return (
6 |     <section id="stack" className="relative py-32 md:py-40">
7 |       <div className="mx-auto max-w-6xl px-6">
8 |         <Reveal>
9 |           <p className="eyebrow">Stack</p>
10 |          <h2 className="mt-4 font-display text-4xl font-semibold tracking-tight md:text-5xl">
11 |            Tools I reach for.
12 |          </h2>
13 |          <p className="mt-4 max-w-lg text-muted-foreground">
14 |            Boring where it should be boring, sharp where it matters. I choose latency and clarity
15 |            over hype.
16 |          </p>
17 |        </Reveal>

```

```
18 |  
19 |     <div className="mt-16 grid gap-px overflow-hidden rounded-2xl border border-border bg-border md:g...  
20 |       {STACK.map((g, i) => (  
21 |         <Reveal key={g.group} delay={i}>  
22 |           <div className="h-full bg-background p-8">  
23 |             <p className="eyebrow">{g.group}</p>  
24 |             <p className="mt-2 text-xs text-muted-foreground/80 leading-relaxed font-normal">  
25 |               {g.desc}  
26 |             </p>  
27 |             <ul className="mt-5 flex flex-wrap gap-2">  
28 |               {g.items.map((it) => (  
29 |                 <li  
30 |                   key={it}  
31 |                   className="rounded-md border border-border bg-card px-3 py-1.5 text-sm text-muted-f...  
32 |                 >  
33 |                   {it}  
34 |                 </li>  
35 |               )}}  
36 |             </ul>  
37 |           </div>  
38 |         </Reveal>  
39 |       )}}  
40 |     </div>  
41 |   </div>  
42 | </section>  
43 | );  
44 | }
```

FILE #41: src\components\portfolio\StatsBar.tsx

```
1 | import { HERO_STATS } from "@lib/site-data";  
2 | import { Reveal } from "../Reveal";  
3 |  
4 | export function StatsBar() {  
5 |   return (  
6 |     <section className="border-y border-border/60 bg-background/25 py-10 backdrop-blur-sm select-none rel...  
7 |       <div className="mx-auto max-w-6xl px-6">  
8 |         <Reveal>  
9 |           <dl className="grid grid-cols-2 gap-y-8 gap-x-6 sm:grid-cols-4 text-center sm:text-left">  
10 |             {HERO_STATS.map((s) => (  
11 |               <div key={s.k} className="flex flex-col gap-1.5">  
12 |                 <dd className="font-display text-3xl font-semibold tabular-nums text-foreground">  
13 |                   {s.v}  
14 |                 </dd>  
15 |                 <dt className="text-[10px] uppercase tracking-widest text-subtle font-mono">  
16 |                   {s.k}  
17 |                 </dt>  
18 |               </div>  
19 |             )}}  
20 |           </dl>  
21 |         </Reveal>  
22 |       </div>  
23 |     </section>  
24 |   );  
25 | }
```

FILE #42: src\components\portfolio\TextScramble.tsx

```
1 | import { useState, useEffect } from "react";  
2 |  
3 | export function TextScramble({
```

```
4 |   text,
5 |   delay = 0,
6 |   trigger = true,
7 | }: {
8 |   text: string;
9 |   delay?: number;
10 |   trigger?: boolean;
11 | }) {
12 |   const [displayText, setDisplayText] = useState(text);
13 |
14 |   useEffect(() => {
15 |     if (!trigger) return;
16 |     let isMounted = true;
17 |     const chars = "!<>_\\/[ ]{}?+=*^?#_abcdefghijklmnopqrstuvwxyz";
18 |     let frame = 0;
19 |     const queue: { from: string; to: string; start: number; end: number; char?: string }[] = [];
20 |
21 |     // Initialize queue with target chars and randomized start/end frames
22 |     for (let i = 0; i < text.length; i++) {
23 |       const from = chars[Math.floor(Math.random() * chars.length)];
24 |       const to = text[i];
25 |       const start = Math.floor(Math.random() * 25);
26 |       const end = start + Math.floor(Math.random() * 25) + 8;
27 |       queue.push({ from, to, start, end });
28 |     }
29 |
30 |     let timer: number;
31 |     const update = () => {
32 |       let output = "";
33 |       let complete = 0;
34 |       for (let i = 0; i < queue.length; i++) {
35 |         const item = queue[i];
36 |         if (frame >= item.end) {
37 |           complete++;
38 |           output += item.to;
39 |         } else if (frame >= item.start) {
40 |           if (!item.char || Math.random() < 0.28) {
41 |             item.char = chars[Math.floor(Math.random() * chars.length)];
42 |           }
43 |           output += item.char;
44 |         } else {
45 |           output += item.from;
46 |         }
47 |       }
48 |
49 |       if (isMounted) {
50 |         setDisplayText(output);
51 |       }
52 |
53 |       if (complete === queue.length) {
54 |         return;
55 |       }
56 |
57 |       frame++;
58 |       timer = requestAnimationFrame(update);
59 |     };
60 |
61 |     const timeout = setTimeout(() => {
62 |       update();
63 |     }, delay * 1000);
64 |
65 |     return () => {
66 |       isMounted = false;
67 |       clearTimeout(timeout);
68 |       cancelAnimationFrame(timer);
```

```
69 |     };
70 |   }, [text, delay, trigger]);
71 |
72 |   return <span>{displayText}</span>;
73 | }
```

FILE #43: src\components\portfolio\TiltCard.tsx

```
1 | import { motion, useMotionValue, useSpring, useTransform } from "motion/react";
2 | import { type ReactNode, useRef } from "react";
3 |
4 | export function TiltCard({ children, className }: { children: ReactNode; className?: string }) {
5 |   const x = useMotionValue(0);
6 |   const y = useMotionValue(0);
7 |
8 |   // Smooth out coordinate tracking using low-damping springs for buttery motion
9 |   const mouseX = useSpring(x, { damping: 25, stiffness: 180, mass: 0.6 });
10 |  const mouseY = useSpring(y, { damping: 25, stiffness: 180, mass: 0.6 });
11 |
12 |  // Map mouse coordinate ratios (-0.5 to 0.5) to subtle tilt degrees
13 |  const rotateX = useTransform(mouseY, [-0.5, 0.5], [6, -6]);
14 |  const rotateY = useTransform(mouseX, [-0.5, 0.5], [-6, 6]);
15 |
16 |  // 1. Cache the bounding rect to prevent layout thrashing
17 |  const rectRef = useRef<DOMRect | null>(null);
18 |
19 |  const handleMouseEnter = (e: React.MouseEvent<HTMLDivElement>) => {
20 |    // Calculate layout ONCE on enter
21 |    rectRef.current = e.currentTarget.getBoundingClientRect();
22 |  };
23 |
24 |  const handleMouseMove = (e: React.MouseEvent<HTMLDivElement>) => {
25 |    // 2. If we don't have the rect yet (e.g. scrolled into view while hovering), get it
26 |    if (!rectRef.current) {
27 |      rectRef.current = e.currentTarget.getBoundingClientRect();
28 |    }
29 |
30 |    const rect = rectRef.current;
31 |
32 |    // 3. Pure math on cached values (no DOM reading = zero lag)
33 |    const relativeX = (e.clientX - rect.left) / rect.width - 0.5;
34 |    const relativeY = (e.clientY - rect.top) / rect.height - 0.5;
35 |
36 |    x.set(relativeX);
37 |    y.set(relativeY);
38 |  };
39 |
40 |  const handleMouseLeave = () => {
41 |    // 4. Clear cache on exit so it recalculates next time
42 |    rectRef.current = null;
43 |
44 |    // Reset positions smoothly on exit
45 |    x.set(0);
46 |    y.set(0);
47 |  };
48 |
49 |  return (
50 |    <div style={{ perspective: "1000px" }} className={className}>
51 |      <motion.div
52 |        onMouseEnter={handleMouseEnter}
53 |        onMouseMove={handleMouseMove}
54 |        onMouseLeave={handleMouseLeave}
55 |        style={{
```

```
56 |         rotateX,
57 |         rotateY,
58 |         transformStyle: "preserve-3d",
59 |         willChange: "transform", // 5. Hint browser to keep this on a dedicated GPU layer
60 |     }}
61 |     className="w-full h-full"
62 |   >
63 |     {children}
64 |   </motion.div>
65 | </div>
66 | );
67 | }
```

FILE #44: src\components\portfolio\WorkspacePanel.tsx

```
1 | export { WorkspacePanel } from "../projects/WorkspacePanel";
```

FILE #45: src\components\portfolio\project-visuals.tsx

```
1 | import type { SVGProps } from "react";
2 | import { cn } from "@lib/utils";
3 | import { motion } from "motion/react";
4 |
5 | /**
6 |  * Frames a project visual like a dev tool window: a thin chrome bar with a
7 |  * mono label (not macOS traffic lights ? this is an engineering artifact,
8 |  * not an app icon). No blur, no translucency ? a flat surface with a border,
9 |  * same as every other card on the site.
10 |  */
11 | export function DeviceFrame({
12 |   label,
13 |   className,
14 |   children,
15 |   noPadding = false,
16 |   aspectClass = "aspect-[16/10]",
17 |   fillHeight = false,
18 | }): {
19 |   label: string;
20 |   className?: string;
21 |   children: React.ReactNode;
22 |   noPadding?: boolean;
23 |   aspectClass?: string;
24 |   fillHeight?: boolean;
25 | }) {
26 |   return (
27 |     <div
28 |       className={cn(
29 |         "overflow-hidden rounded-xl border border-border bg-surface",
30 |         fillHeight && "flex flex-col h-full",
31 |         className,
32 |       )}
33 |     >
34 |       <div className="flex items-center gap-2 border-b border-border px-4 py-2.5 shrink-0">
35 |         <div className="flex gap-1.5 aria-hidden="true">
36 |           <span className="h-1.5 w-1.5 rounded-full bg-white/15" />
37 |           <span className="h-1.5 w-1.5 rounded-full bg-white/15" />
38 |           <span className="h-1.5 w-1.5 rounded-full bg-white/15" />
39 |         </div>
40 |         <span className="font-mono text-[10px] tracking-wide text-subtle">{label}</span>
41 |       </div>
42 |     </div>
```

```
43 |         className={cn(
44 |             "relative flex items-center justify-center w-full",
45 |             fillHeight ? "flex-1 min-h-0" : aspectClass,
46 |             noPadding ? "p-0" : "p-6",
47 |         )}
48 |     >
49 |         {children}
50 |     </div>
51 | </div>
52 | );
53 | }
54 |
55 | const svgBase: SVGProps<SVGSVGElement> = {
56 |     viewBox: "0 0 320 180",
57 |     className: "h-full w-full",
58 |     fill: "none",
59 | };
60 |
61 | /** Halcyon ? step rail + streaming citations. */
62 | export function AgentTraceVisual() {
63 |     return (
64 |         <svg {...svgBase} aria-hidden="true">
65 |             <motion.line
66 |                 x1="24"
67 |                 y1="18"
68 |                 x2="24"
69 |                 y2="162"
70 |                 className="stroke-border"
71 |                 strokeWidth={1}
72 |                 variants={{
73 |                     hidden: { pathLength: 0 },
74 |                     visible: {
75 |                         pathLength: 1,
76 |                         transition: { duration: 0.8, ease: "easeOut" },
77 |                     },
78 |                 }}
79 |             />
80 |             {[18, 78, 138].map((y, i) => (
81 |                 <motion.circle
82 |                     key={y}
83 |                     cx="24"
84 |                     cy={y}
85 |                     r="4"
86 |                     className={i === 2 ? "fill-secondary" : "fill-subtle"}
87 |                     variants={{
88 |                         hidden: { scale: 0, opacity: 0 },
89 |                         visible: {
90 |                             scale: 1,
91 |                             opacity: 1,
92 |                             transition: { delay: i * 0.15, duration: 0.3, type: "spring", stiffness: 200 },
93 |                         },
94 |                     }}
95 |                 />
96 |             ))}
97 |             {[
98 |                 { y: 24, w: 210 },
99 |                 { y: 84, w: 170 },
100 |                 { y: 144, w: 130 },
101 |             ].map((row, i) => (
102 |                 <motion.rect
103 |                     key={row.y}
104 |                     x="44"
105 |                     y={row.y}
106 |                     width={row.w}
107 |                     height="6"
```

```
108 |         rx="3"
109 |         className="fill-white/10"
110 |         variants={{
111 |             hidden: { scaleX: 0, originX: 0 },
112 |             visible: {
113 |                 scaleX: 1,
114 |                 transition: { duration: 0.6, delay: i * 0.12 + 0.1, ease: "easeOut" },
115 |             },
116 |             hover: {
117 |                 fill: "rgba(255, 255, 255, 0.18)",
118 |                 x: 48,
119 |                 transition: { duration: 0.3, ease: [0.22, 1, 0.36, 1], delay: i * 0.04 },
120 |             },
121 |         }}
122 |     />
123 | ))}
124 | <motion.rect
125 |     x="44"
126 |     y="38"
127 |     width="150"
128 |     height="4"
129 |     rx="2"
130 |     className="fill-white/[0.06]"
131 |     variants={{
132 |         hidden: { scaleX: 0, originX: 0 },
133 |         visible: { scaleX: 1, transition: { duration: 0.5, delay: 0.25 } },
134 |         hover: { x: 48, transition: { duration: 0.3, delay: 0.04 } },
135 |     }}
136 | />
137 | <motion.rect
138 |     x="44"
139 |     y="98"
140 |     width="120"
141 |     height="4"
142 |     rx="2"
143 |     className="fill-white/[0.06]"
144 |     variants={{
145 |         hidden: { scaleX: 0, originX: 0 },
146 |         visible: { scaleX: 1, transition: { duration: 0.5, delay: 0.37 } },
147 |         hover: { x: 48, transition: { duration: 0.3, delay: 0.08 } },
148 |     }}
149 | />
150 | <motion.rect
151 |     x="44"
152 |     y="158"
153 |     width="90"
154 |     height="4"
155 |     rx="2"
156 |     className="fill-white/[0.06]"
157 |     variants={{
158 |         hidden: { scaleX: 0, originX: 0 },
159 |         visible: { scaleX: 1, transition: { duration: 0.5, delay: 0.49 } },
160 |         hover: { x: 48, transition: { duration: 0.3, delay: 0.12 } },
161 |     }}
162 | />
163 | {[44, 92, 132].map((x, i) => (
164 |     <motion.rect
165 |         key={x}
166 |         x={x}
167 |         y="168"
168 |         width="34"
169 |         height="10"
170 |         rx="5"
171 |         className="fill-secondary/10 stroke-secondary/30"
172 |         strokeWidth={1}
```

```
173 |         variants={{
174 |             hidden: { scale: 0, opacity: 0 },
175 |             visible: {
176 |                 scale: 1,
177 |                 opacity: 1,
178 |                 transition: {
179 |                     type: "spring",
180 |                     stiffness: 260,
181 |                     damping: 20,
182 |                     delay: 0.4 + i * 0.1,
183 |                 },
184 |             },
185 |             hover: {
186 |                 scale: 1.06,
187 |                 stroke: "rgba(6, 182, 212, 0.6)",
188 |                 fill: "rgba(6, 182, 212, 0.15)",
189 |                 transition: { duration: 0.2 },
190 |             },
191 |         }}
192 |     />
193 | ))}
194 | <circle cx="150" cy="144" r="2.5" className="fill-secondary">
195 |   <animate attributeName="opacity" values="1;0.2;1" dur="1.6s" repeatCount="indefinite" />
196 | </circle>
197 | </svg>
198 | );
199 | }
200 |
201 | /** Parity ? eval rows with pass marks and a fill bar. */
202 | export function TerminalEvalsVisual() {
203 |   const rows = [0, 1, 2, 3, 4];
204 |   return (
205 |     <svg {...svgBase} aria-hidden="true">
206 |       {rows.map((i) => {
207 |         const y = 20 + i * 26;
208 |         const pass = i !== 3;
209 |         return (
210 |           <g key={i}>
211 |             <motion.circle
212 |               cx="22"
213 |               cy={y}
214 |               r="6"
215 |               className={pass ? "stroke-secondary" : "stroke-primary"}
216 |               strokeWidth="1.4"
217 |               variants={{
218 |                 hidden: { scale: 0, opacity: 0 },
219 |                 visible: {
220 |                   scale: 1,
221 |                   opacity: 1,
222 |                   transition: { type: "spring", stiffness: 200, delay: i * 0.08 },
223 |                 },
224 |               }}
225 |             />
226 |             {pass ? (
227 |               <motion.path
228 |                 d={`M19 ${y} l2 2.5 l4 -5`}
229 |                 className="stroke-secondary"
230 |                 strokeWidth="1.4"
231 |                 strokeLinecap="round"
232 |                 strokeLinejoin="round"
233 |                 variants={{
234 |                   hidden: { pathLength: 0 },
235 |                   visible: {
236 |                     pathLength: 1,
237 |                     transition: { duration: 0.25, delay: i * 0.08 + 0.2 },
```



```

238 |         },
239 |       }}
240 |     />
241 |   ) : (
242 |     <motion.path
243 |       d={`M19.5 ${y - 2.5} l5 5 M24.5 ${y - 2.5} l-5 5`}
244 |       className="stroke-primary"
245 |       strokeWidth="1.4"
246 |       strokeLinecap="round"
247 |       variants={{
248 |         hidden: { pathLength: 0 },
249 |         visible: {
250 |           pathLength: 1,
251 |           transition: { duration: 0.25, delay: i * 0.08 + 0.2 },
252 |         },
253 |       }}
254 |     />
255 |   )}
256 |   <motion.rect
257 |     x="42"
258 |     y={y - 3}
259 |     width={140 - i * 12}
260 |     height="6"
261 |     rx="3"
262 |     className="fill-white/10"
263 |     variants={{
264 |       hidden: { scaleX: 0, originX: 0 },
265 |       visible: {
266 |         scaleX: 1,
267 |         transition: { duration: 0.5, delay: i * 0.08 + 0.1 },
268 |       },
269 |       hover: {
270 |         fill: "rgba(255, 255, 255, 0.18)",
271 |         width: 140 - i * 12 + 10,
272 |         transition: { duration: 0.3, ease: [0.22, 1, 0.36, 1] },
273 |       },
274 |     }}
275 |   />
276 | </g>
277 | );
278 | }}}
279 | <rect x="0" y="156" width="320" height="4" rx="2" className="fill-white/[0.06]" />
280 | <motion.rect
281 |   x="0"
282 |   y="156"
283 |   width="248"
284 |   height="4"
285 |   rx="2"
286 |   className="fill-secondary/70"
287 |   variants={{
288 |     hidden: { width: 0 },
289 |     visible: {
290 |       width: 248,
291 |       transition: { duration: 1.0, ease: [0.22, 1, 0.36, 1], delay: 0.4 },
292 |     },
293 |   }}
294 | />
295 | <text x="0" y="176" className="fill-subtle font-mono text-[9px]">
296 |   120/152 evals · 79%
297 | </text>
298 | </svg>
299 | );
300 | }
301 |
302 | /** Cinder ? token waveform + a small chip motif. */

```

```

303 | export function WaveformVisual() {
304 |   const bars = [8, 18, 30, 44, 60, 78, 96, 74, 52, 38, 26, 16, 22, 34, 50, 68, 84, 58, 30, 14];
305 |   return (
306 |     <svg {...svgBase} aria-hidden="true">
307 |       <g transform="translate(0, 40)">
308 |         {bars.map((h, i) => (
309 |           <motion.rect
310 |             key={i}
311 |             x={i * 15 + 6}
312 |             y={50 - h / 2}
313 |             width="6"
314 |             height={h}
315 |             rx="2"
316 |             className={i % 5 === 0 ? "fill-secondary/80" : "fill-white/12"}
317 |             variants={{
318 |               hidden: { scaleY: 0, originY: 0.5 },
319 |               visible: {
320 |                 scaleY: 1,
321 |                 transition: {
322 |                   delay: i * 0.02,
323 |                   duration: 0.4,
324 |                   ease: "easeOut",
325 |                 },
326 |               },
327 |               hover: {
328 |                 scaleY: 1.25,
329 |                 opacity: 1,
330 |                 fill: i % 5 === 0 ? "rgba(6, 182, 212, 1)" : "rgba(255, 255, 255, 0.3)",
331 |                 transition: {
332 |                   delay: i * 0.015,
333 |                   duration: 0.3,
334 |                   ease: [0.22, 1, 0.36, 1],
335 |                 },
336 |               },
337 |             }}
338 |           />
339 |         )}}
340 |       </g>
341 |       <motion.g
342 |         transform="translate(128, 128)"
343 |         className="stroke-border"
344 |         strokeWidth="1"
345 |         variants={{
346 |           hover: {
347 |             stroke: "rgba(6, 182, 212, 0.4)",
348 |             transition: { duration: 0.3 },
349 |           },
350 |         }}
351 |       >
352 |         <rect x="0" y="0" width="64" height="34" rx="4" className="fill-transparent" />
353 |         {[10, 24, 38, 52].map((x) => (
354 |           <line key={`t${x}`} x1={x} y1="0" x2={x} y2="-6" />
355 |         ))}
356 |         {[10, 24, 38, 52].map((x) => (
357 |           <line key={`b${x}`} x1={x} y1="34" x2={x} y2="40" />
358 |         ))}
359 |       </motion.g>
360 |       <text x="128" y="150" className="fill-subtle font-mono text-[9px]">
361 |         M2 · 62 tok/s
362 |       </text>
363 |     </svg>
364 |   );
365 | }
366 |
367 | /** Loomstate ? unstructured points resolving into a connected graph. */

```

```
368 | export function GraphNetworkVisual() {
369 |   const nodes = [
370 |     { x: 46, y: 40 },
371 |     { x: 100, y: 24 },
372 |     { x: 160, y: 46 },
373 |     { x: 220, y: 28 },
374 |     { x: 270, y: 54 },
375 |     { x: 70, y: 100 },
376 |     { x: 140, y: 92 },
377 |     { x: 200, y: 108 },
378 |     { x: 250, y: 128 },
379 |     { x: 110, y: 150 },
380 |     { x: 180, y: 150 },
381 |   ];
382 |   const edges: [number, number][] = [
383 |     [0, 5],
384 |     [1, 5],
385 |     [1, 6],
386 |     [2, 6],
387 |     [2, 7],
388 |     [3, 7],
389 |     [4, 7],
390 |     [4, 8],
391 |     [5, 9],
392 |     [6, 9],
393 |     [6, 10],
394 |     [7, 10],
395 |     [8, 10],
396 |   ];
397 |   return (
398 |     <svg {...svgBase} aria-hidden="true">
399 |       {edges.map(([a, b], i) => (
400 |         <motion.line
401 |           key={i}
402 |           x1={nodes[a].x}
403 |           y1={nodes[a].y}
404 |           x2={nodes[b].x}
405 |           y2={nodes[b].y}
406 |           className="stroke-white/10"
407 |           strokeWidth={1}
408 |           variants={{
409 |             hidden: { pathLength: 0 },
410 |             visible: {
411 |               pathLength: 1,
412 |               transition: { duration: 0.6, delay: i * 0.03 + 0.1, ease: "easeOut" },
413 |             },
414 |             hover: {
415 |               stroke: "rgba(6, 182, 212, 0.25)",
416 |               strokeWidth: 1.2,
417 |               transition: { duration: 0.4, delay: i * 0.01 },
418 |             },
419 |           }}
420 |         />
421 |       ))}
422 |     {nodes.map((n, i) => (
423 |       <motion.circle
424 |         key={i}
425 |         cx={n.x}
426 |         cy={n.y}
427 |         r={i > 8 ? 5 : 3.5}
428 |         className={i > 8 ? "fill-primary/80" : "fill-white/25"}
429 |         variants={{
430 |           hidden: { scale: 0, opacity: 0 },
431 |           visible: {
432 |             scale: 1,
```

```
433 |         opacity: 1,
434 |         transition: {
435 |             type: "spring",
436 |             stiffness: 260,
437 |             damping: 20,
438 |             delay: i * 0.03 + 0.2,
439 |         },
440 |     },
441 |     hover: {
442 |         scale: i > 8 ? 1.25 : 1.3,
443 |         fill: i > 8 ? "rgba(139, 92, 246, 1)" : "rgba(255, 255, 255, 0.7)",
444 |         transition: {
445 |             type: "spring",
446 |             stiffness: 300,
447 |             damping: 15,
448 |             delay: i * 0.005,
449 |         },
450 |     },
451 | }}
452 | />
453 | )))
454 | </svg>
455 | );
456 | }
457 |
458 | export const PROJECT_VISUALS = {
459 |   "agent-trace": AgentTraceVisual,
460 |   "terminal-evals": TerminalEvalsVisual,
461 |   waveform: WaveformVisual,
462 |   "graph-network": GraphNetworkVisual,
463 | } as const;
```

FILE #46: src\components\portfolio\projects\CarouselProjectCard.tsx

```
1 | import { ArrowUpRight } from "lucide-react";
2 | import { useEffect, useRef, useState } from "react";
3 | import { motion, useTransform, type MotionValue } from "motion/react";
4 | import { DeviceFrame } from "../project-visuals";
5 | import { PROJECT_VISUALS } from "../project-visuals";
6 | import { revealVariants } from "@lib/motion";
7 | import { type Project } from "@lib/site-data";
8 | import { playTick } from "@lib/sound";
9 | import { PROJECT_DETAILS } from "@lib/project-details";
10 | import { useSpotlight } from "../widgets/useSpotlight";
11 |
12 | const ACTIVE_THEMES = [
13 |   { glow: "rgba(34, 211, 238, 0.025)", border: "rgba(34, 211, 238, 0.35)" }, // Vurlo (Cyan)
14 |   { glow: "rgba(16, 185, 129, 0.025)", border: "rgba(16, 185, 129, 0.35)" }, // Veltrix (Emerald)
15 |   { glow: "rgba(99, 102, 241, 0.025)", border: "rgba(99, 102, 241, 0.35)" }, // Vcentre (Indigo)
16 | ];
17 |
18 | function Metrics({ metrics }: { metrics: Project["metrics"] }) {
19 |   return (
20 |     <dl className="grid grid-cols-2 gap-x-4 gap-y-3 sm:grid-cols-4">
21 |       {metrics.map((m) => (
22 |         <div key={m.k}>
23 |           <dt className="font-mono text-[9px] uppercase tracking-widest text-subtle/80">{m.k}</dt>
24 |           <dd className="mt-0.5 font-display text-lg font-semibold tabular-nums text-foreground">
25 |             {m.v}
26 |           </dd>
27 |         </div>
28 |       ))}
29 |     </dl>
```

```
30 |   );
31 | }
32 |
33 | export function CarouselProjectCard({
34 |   p,
35 |   i,
36 |   onViewCaseStudy,
37 |   rotateX,
38 |   rotateY,
39 |   scale,
40 |   opacity,
41 |   z,
42 |   shineX,
43 |   isActive,
44 | }): {
45 |   p: Project;
46 |   i: number;
47 |   onViewCaseStudy: (p: Project) => void;
48 |   rotateX: MotionValue<number>;
49 |   rotateY: MotionValue<number>;
50 |   scale: MotionValue<number>;
51 |   opacity: MotionValue<number>;
52 |   z: MotionValue<number>;
53 |   shineX: MotionValue<number>;
54 |   isActive: boolean;
55 | }) {
56 |   const cardRef = useRef<HTMLDivElement>(null);
57 |
58 |   // Interpolate numerical shine percentage to absolute left positioning
59 |   const shineLeft = useTransform(shineX, (val) => `${val - 100}%`);
60 |
61 |   const details = PROJECT_DETAILS[p.name];
62 |   const glowColor = details?.theme.glow
63 |     ? details.theme.glow.replace("0.15", "0.09")
64 |     : "rgba(6, 182, 212, 0.09)";
65 |   const { ref: spotlightRef, onMove } = useSpotlight();
66 |   const Visual = PROJECT_VISUALS[p.visual];
67 |   const [isLightboxOpen, setIsLightboxOpen] = useState(false);
68 |
69 |   const tapStartRef = useRef<{ x: number; y: number; t: number } | null>(null);
70 |   const suppressNextClickRef = useRef(false);
71 |
72 |   useEffect(() => {
73 |     if (!isLightboxOpen) return;
74 |     const handleKeyDown = (e: KeyboardEvent) => {
75 |       if (e.key === "Escape") {
76 |         setIsLightboxOpen(false);
77 |       }
78 |     };
79 |     window.addEventListener("keydown", handleKeyDown);
80 |     return () => window.removeEventListener("keydown", handleKeyDown);
81 |   }, [isLightboxOpen]);
82 |
83 |   return (
84 |     <>
85 |     <div
86 |       ref={cardRef}
87 |       className={`w-[280px] sm:w-[480px] md:w-[580px] flex-shrink-0 snap-center py-6 ${
88 |         isActive ? "pointer-events-auto" : "pointer-events-none"
89 |       }`}
90 |     >
91 |     <div
92 |       style={{ perspective: "1200px", transformStyle: "preserve-3d" }}
93 |       className="pointer-events-none"
94 |     >
```

```

95 |         <motion.div
96 |             style={{
97 |                 rotateX,
98 |                 rotateY,
99 |                 scale,
100 |                 opacity,
101 |                 z,
102 |                 transformStyle: "preserve-3d",
103 |                 willChange: "transform, opacity",
104 |             }}
105 |             className={`h-full ${isActive ? "pointer-events-auto" : "pointer-events-none"}`}
106 |         >
107 |         <motion.article
108 |             ref={spotlightRef as React.RefObject<HTMLDivElement>}
109 |             onMouseMove={onMove}
110 |             variants={revealVariants}
111 |             whileHover={
112 |                 typeof window !== "undefined" && window.matchMedia("(hover: hover)").matches
113 |                 ? "hover"
114 |                 : undefined
115 |             }
116 |             style={
117 |                 {
118 |                     transformStyle: "preserve-3d",
119 |                     "--glow-color": glowColor,
120 |                     borderColor: isActive ? ACTIVE_THEMES[i].border : undefined,
121 |                     boxShadow: isActive ? `0 10px 30px -10px ${ACTIVE_THEMES[i].border}` : undefined,
122 |                 } as React.CSSProperties
123 |             }
124 |             className="surface-card group relative flex flex-col overflow-hidden p-6 md:p-8 h-full tran...
125 |         >
126 |         {/* Metallic Specular Shine Overlay */}
127 |         <motion.div
128 |             style={{
129 |                 left: shineLeft,
130 |                 background:
131 |                     "linear-gradient(115deg, transparent 0%, rgba(255, 255, 255, 0.005) 30%, rgba(255, 25...
132 |             }}
133 |             className="pointer-events-none absolute inset-y-0 w-[300%] z-10"
134 |         />
135 |
136 |         <motion.div
137 |             aria-hidden
138 |             className="pointer-events-none absolute -inset-px opacity-0 transition-opacity duration-5...
139 |             style={{
140 |                 background:
141 |                     "radial-gradient(320px circle at var(--gx, -999px) var(--gy, -999px), var(--glow-colo...
142 |             }}
143 |         />
144 |
145 |         {/* Mockup visual area */}
146 |         <div style={{ transform: "translateZ(40px)", transformStyle: "preserve-3d" }}>
147 |             <DeviceFrame
148 |                 label={p.visualLabel}
149 |                 className="relative overflow-hidden"
150 |                 noPadding={!!p.image}
151 |                 aspectClass="aspect-[16/10] bg-slate-900/60"
152 |             >
153 |                 {p.image ? (
154 |                     <button
155 |                         onClick={() => setIsLightboxOpen(true)}
156 |                         aria-label={`View full ${p.name} screenshot`}
157 |                         className="w-full h-full block text-left outline-none cursor-zoom-in pointer-events...
158 |                     >
159 |                         <img

```

```
160 |             src={p.image}
161 |             alt={p.name}
162 |             loading="eager"
163 |             decoding="async"
164 |             className="w-full h-full object-cover object-top block transition-transform durat...
165 |         />
166 |     </button>
167 | ) : (
168 |     <Visual />
169 | )}
170 | </DeviceFrame>
171 | </div>
172 |
173 | {/* Text metadata and specs */}
174 | <div
175 |     style={{ transform: "translateZ(20px)" }}
176 |     className="relative mt-6 flex flex-1 flex-col justify-between"
177 | >
178 |     <div>
179 |         <div className="flex items-center gap-3 font-mono text-xs text-subtle">
180 |             <span>{p.index}</span>
181 |             <span className="h-px w-5 bg-border" />
182 |             <span>{p.year}</span>
183 |         </div>
184 |
185 |         <h3 className="mt-3 font-display text-2xl font-semibold tracking-tight">
186 |             {p.name}
187 |         </h3>
188 |         <p className="mt-1 text-sm text-muted-foreground">{p.tagline}</p>
189 |         <p className="mt-4 text-xs leading-relaxed text-subtle/90">{p.summary}</p>
190 |     </div>
191 |
192 |     <div className="mt-6 pt-4 border-t border-border flex items-center justify-between">
193 |         <span className="font-display text-sm font-semibold tabular-nums text-foreground">
194 |             {p.metrics[0].v}
195 |             <span className="ml-1.5 font-mono text-[9px] font-normal uppercase tracking-widest te...
196 |                 {p.metrics[0].k}
197 |             </span>
198 |         </span>
199 |         <button
200 |             onPointerDown={(e) => {
201 |                 if (e.pointerType === "touch") {
202 |                     tapStartRef.current = { x: e.clientX, y: e.clientY, t: Date.now() };
203 |                 }
204 |             }}
205 |             onPointerUp={(e) => {
206 |                 if (e.pointerType !== "touch" || !tapStartRef.current) return;
207 |                 const { x, y, t } = tapStartRef.current;
208 |                 tapStartRef.current = null;
209 |                 const movedTooFar = Math.hypot(e.clientX - x, e.clientY - y) > 10;
210 |                 const heldTooLong = Date.now() - t > 500;
211 |                 if (!movedTooFar && !heldTooLong) {
212 |                     suppressNextClickRef.current = true;
213 |                     onViewCaseStudy(p);
214 |                     window.setTimeout(() => {
215 |                         suppressNextClickRef.current = false;
216 |                     }, 400);
217 |                 }
218 |             }}
219 |             onClick={() => {
220 |                 if (suppressNextClickRef.current) return;
221 |                 onViewCaseStudy(p);
222 |             }}
223 |             onMouseEnter={() => playTick(0)}
224 |             aria-label={`View ${p.name} case study`}
```

```

225 |         style={{ touchAction: "manipulation" }}
226 |         className={`relative z-30 inline-flex items-center gap-1.5 text-xs text-foreground fo...
227 |     >
228 |         <span>View Workspace</span>
229 |         <ArrowUpRight className="h-3.5 w-3.5 transition-transform duration-300 group-hover:tr...
230 |     </button>
231 | </div>
232 | </div>
233 | </motion.article>
234 | </motion.div>
235 | </div>
236 | </div>
237 |
238 | {isLightboxOpen && p.image && (
239 |     <div
240 |         className="fixed inset-0 z-[10000] flex items-center justify-center bg-black/95 p-4 backdrop-bl...
241 |         onClick={() => setIsLightboxOpen(false)}
242 |     >
243 |         <button
244 |             onClick={(e) => {
245 |                 e.stopPropagation();
246 |                 setIsLightboxOpen(false);
247 |             }}
248 |             aria-label="Close"
249 |             className="absolute top-4 right-4 rounded-full bg-white/10 p-2 text-white transition-colors h...
250 |         >
251 |             <svg
252 |                 xmlns="http://www.w3.org/2000/svg"
253 |                 className="h-6 w-6"
254 |                 fill="none"
255 |                 viewBox="0 0 24 24"
256 |                 stroke="currentColor"
257 |             >
258 |                 <path
259 |                     strokeLinecap="round"
260 |                     strokeLinejoin="round"
261 |                     strokeWidth={2}
262 |                     d="M6 18L18 6M6 6L18 18"
263 |                 />
264 |             </svg>
265 |         </button>
266 |
267 |         <img
268 |             src={` ${p.image}?v=1.2`}
269 |             alt={` ${p.name} screenshot`}
270 |             className="max-h-[90vh] max-w-[90vw] md:max-w-[1024px] w-full rounded-lg border border-border...
271 |             onClick={(e) => e.stopPropagation()}
272 |         />
273 |     </div>
274 | )}
275 | </>
276 | );
277 | }

```

FILE #47: src\components\portfolio\projects\WorkspacePanel.tsx

```

1 | import { ArrowUpRight, Link2, Minus, Plus } from "lucide-react";
2 | import { useRef, useState, useEffect } from "react";
3 | import { motion, AnimatePresence } from "motion/react";
4 | import { Link, useNavigate } from "@tanstack/react-router";
5 | import { toast } from "sonner";
6 |
7 | import { DeviceFrame } from "../project-visuals";

```



```
8 | import { PROJECT_DETAILS } from "@lib/project-details";
9 | import { playTick, playSwell, playExit, resumeAudio } from "@lib/sound";
10 | import { type Project } from "@lib/site-data";
11 | import { LighthouseDial } from "../widgets/LighthouseDial";
12 | import { ArchitectureNode } from "../widgets/ArchitectureNode";
13 | import { ScenarioTimeline } from "../widgets/ScenarioTimeline";
14 | import { StepSequence, ChallengeCard } from "../widgets/ChallengeCard";
15 |
16 | export function WorkspacePanel({ project, onClose }: { project: Project; onClose?: () => void }) {
17 |   const navigate = useNavigate();
18 |   const [isTransitioning, setIsTransitioning] = useState(true);
19 |   const [transitionStage, setTransitionStage] = useState<"entering" | "ready" | "exiting">(
20 |     "entering",
21 |   );
22 |   const [statusText, setStatusText] = useState("Initializing Session...");
23 |   const [progressBarWidth, setProgressbarWidth] = useState(0);
24 |   const [showWorkspaceContent, setShowWorkspaceContent] = useState(false);
25 |
26 |   // Workspace Scroll States
27 |   const [activeSection, setActiveSection] = useState("overview");
28 |   const [scrollProgress, setScrollProgress] = useState(0);
29 |   const [activeChallenge, setActiveChallenge] = useState<number | null>(0); // Default open the first cha...
30 |   const workspaceScrollRef = useRef<HTMLDivElement>(null);
31 |
32 |   const TABS = [
33 |     { id: "overview", label: "Overview" },
34 |     { id: "why-built", label: "Why Built" },
35 |     { id: "architecture", label: "Architecture" },
36 |     { id: "features", label: "Features" },
37 |     { id: "challenges", label: "Challenges" },
38 |     { id: "journey", label: "Journey" },
39 |     { id: "performance", label: "Performance" },
40 |     { id: "decisions", label: "Decisions" },
41 |     { id: "gallery", label: "Gallery" },
42 |   ];
43 |
44 |   const details = PROJECT_DETAILS[project.name];
45 |
46 |   // Scrollspy & scroll progress handler
47 |   const handleWorkspaceScroll = (e: React.UIEvent<HTMLDivElement>) => {
48 |     const target = e.currentTarget;
49 |     const progress = (target.scrollTop / (target.scrollHeight - target.clientHeight)) * 100;
50 |     setScrollProgress(progress);
51 |
52 |     const scrollTop = target.scrollTop;
53 |     const offsets = TABS.map((tab) => {
54 |       const el = document.getElementById(`ws-sec-${tab.id}`);
55 |       return { id: tab.id, offsetTop: el ? el.offsetTop - 140 : 0 };
56 |     });
57 |
58 |     const current = offsets.reduce((acc, curr) => {
59 |       if (scrollTop >= curr.offsetTop) {
60 |         return curr.id;
61 |       }
62 |       return acc;
63 |     }, TABS[0].id);
64 |
65 |     setActiveSection(current);
66 |   };
67 |
68 |   // Click smooth scroll jump
69 |   const scrollToSection = (id: string, index: number) => {
70 |     const el = document.getElementById(`ws-sec-${id}`);
71 |     if (el && workspaceScrollRef.current) {
72 |       workspaceScrollRef.current.scrollTo({
```

```
73 |         top: el.offsetTop - 110,  
74 |         behavior: "smooth",  
75 |     });  
76 |     const pan = (index / (TABS.length - 1)) * 2 - 1;  
77 |     playTick(pan);  
78 |   }  
79 | };  
80 |  
81 | // Run the entrance sequence on mount  
82 | useEffect(() => {  
83 |   playSwell();  
84 |   setProgressBarWidth(30);  
85 |   setStatusText("Opening Workspace...");  
86 |  
87 |   const t1 = setTimeout(() => {  
88 |     setStatusText("Loading Project...");  
89 |     setProgressBarWidth(60);  
90 |   }, 400);  
91 |  
92 |   const t2 = setTimeout(() => {  
93 |     setStatusText("Preparing Architecture...");  
94 |     setProgressBarWidth(85);  
95 |   }, 800);  
96 |  
97 |   const t3 = setTimeout(() => {  
98 |     setStatusText("Workspace Ready.");  
99 |     setProgressBarWidth(100);  
100 |   }, 1400);  
101 |  
102 |   const t4 = setTimeout(() => {  
103 |     setTransitionStage("ready");  
104 |     setShowWorkspaceContent(true);  
105 |     setIsTransitioning(false);  
106 |   }, 1800);  
107 |  
108 |   return () => {  
109 |     clearTimeout(t1);  
110 |     clearTimeout(t2);  
111 |     clearTimeout(t3);  
112 |     clearTimeout(t4);  
113 |   };  
114 | }, [project.name]);  
115 |  
116 | const handleExitWorkspace = () => {  
117 |   if (isTransitioning) return;  
118 |   setIsTransitioning(true);  
119 |   playExit();  
120 |   setShowWorkspaceContent(false);  
121 |   setProgressBarWidth(100);  
122 |  
123 |   setTimeout(() => {  
124 |     setTransitionStage("exiting");  
125 |     setStatusText("Closing Workspace...");  
126 |     setProgressBarWidth(0);  
127 |  
128 |     setTimeout(() => {  
129 |       setStatusText("Returning to Portfolio...");  
130 |     }, 750);  
131 |  
132 |     setTimeout(() => {  
133 |       if (onClose) {  
134 |         onClose();  
135 |       } else {  
136 |         navigate({ to: "/" });  
137 |       }  
138 |     });  
139 |   });  
140 | }
```

```

138 |     }, 1500);
139 |   }, 300);
140 | };
141 |
142 | // Escape key handler to close
143 | useEffect(() => {
144 |   const handleKeyDown = (e: KeyboardEvent) => {
145 |     if (e.key === "Escape" && !isTransitioning) {
146 |       handleExitWorkspace();
147 |     }
148 |   };
149 |   window.addEventListener("keydown", handleKeyDown);
150 |   return () => window.removeEventListener("keydown", handleKeyDown);
151 | }, [isTransitioning]);
152 |
153 | if (!details) return null;
154 |
155 | return (
156 |   <div
157 |     role="dialog"
158 |     aria-modal="true"
159 |     aria-label="Workspace Portal"
160 |     className="fixed inset-0 z-[9999] flex items-center justify-center overflow-hidden bg-[#09090b] p-4..."
161 |   >
162 |     /* Grid background layer */
163 |     <div className="absolute inset-0 workspace-grid pointer-events-none" />
164 |
165 |     /* Ambient project theme color fields */
166 |     <div
167 |       aria-hidden
168 |       style={{ backgroundColor: details.theme.glow, filter: "blur(140px)" }}
169 |       className="pointer-events-none absolute -left-40 -top-40 h-[520px] w-[520px] rounded-full opacity..."
170 |     />
171 |     <div
172 |       aria-hidden
173 |       style={{ backgroundColor: details.theme.glow, filter: "blur(140px)" }}
174 |       className="pointer-events-none absolute -bottom-40 -right-40 h-[520px] w-[520px] rounded-full opa..."
175 |     />
176 |
177 |     /* Loading stage - rendered as absolute overlay when entering/exiting */
178 |     {(transitionStage === "entering" || transitionStage === "exiting") && (
179 |       <div className="absolute inset-0 z-50 flex flex-col items-center justify-center bg-[#09090b] px-6..."
180 |         <div className="flex flex-wrap items-baseline justify-center gap-x-4 gap-y-2 text-center">
181 |           {(transitionStage === "entering"
182 |             ? [
183 |               { text: "Opening.", weight: "font-semibold text-foreground" },
184 |               {
185 |                 text: "the",
186 |                 weight: "font-normal italic text-muted-foreground font-serif-accent",
187 |               },
188 |               { text: `${project.name}.`, weight: "font-semibold text-foreground" },
189 |             ]
190 |             : [
191 |               { text: "Closing.", weight: "font-semibold text-foreground" },
192 |               {
193 |                 text: "the",
194 |                 weight: "font-normal italic text-muted-foreground font-serif-accent",
195 |               },
196 |               { text: "Workspace.", weight: "font-semibold text-foreground" },
197 |             ]
198 |           )}.map((w, idx) => (
199 |             <motion.span
200 |               key={`${transitionStage}-${w.text}-${idx}`}
201 |               initial={{ opacity: 0, y: 18, filter: "blur(12px)" }}
202 |               animate={{ opacity: 1, y: 0, filter: "blur(0px)" }}

```

```

203 |         transition={{
204 |             type: "spring",
205 |             stiffness: 120,
206 |             damping: 18,
207 |             delay: idx * 0.28,
208 |         }}
209 |         className={`font-display text-5xl tracking-tight sm:text-6xl md:text-8xl ${w.weight}`}
210 |     >
211 |         {w.text}
212 |     </motion.span>
213 |   )}}
214 | </div>
215 |
216 |   {/* Shiny-text status line */}
217 |   <div
218 |     className="mt-8 min-h-[16px] select-none"
219 |     style={{
220 |       animation: "bootCardIn 0.5s cubic-bezier(0.16, 1, 0.3, 1) both",
221 |       animationDelay: "0.2s",
222 |     }}
223 |   >
224 |     <p className="font-mono text-[10px] tracking-[0.3em] uppercase text-subtle shiny-text">
225 |       &gt; {statusText}
226 |     </p>
227 |   </div>
228 |
229 |   {/* Thin, premium progress bar */}
230 |   <div
231 |     className="mt-14 flex flex-col items-center"
232 |     style={{
233 |       animation: "bootCardIn 0.5s cubic-bezier(0.16, 1, 0.3, 1) both",
234 |       animationDelay: "0.4s",
235 |     }}
236 |   >
237 |     <div className="h-[1.5px] w-64 rounded-full overflow-hidden bg-white/10 relative">
238 |       <div
239 |         className={`h-full bg-gradient-to-r ${details.theme.primary} ${details.theme.secondary}`}
240 |         style={{
241 |           width: `${progressBarWidth}%`,
242 |           boxShadow: `0 0 8px ${details.theme.glow}`,
243 |           transition:
244 |             transitionStage === "exiting"
245 |               ? "width 1500ms linear"
246 |               : "width 400ms cubic-bezier(0.16, 1, 0.3, 1)",
247 |         }}
248 |       />
249 |     </div>
250 |     <div className="mt-4 font-mono text-[9px] uppercase tracking-[0.25em] text-subtle/85 flex ite...
251 |       <span className="relative flex h-1.5 w-1.5">
252 |         <span className="absolute inline-flex h-full w-full animate-ping rounded-full bg-secondar...
253 |         <span className="relative inline-flex h-1.5 w-1.5 rounded-full bg-secondary shadow-[0_0_4...
254 |       </span>
255 |     <span>
256 |       {transitionStage === "entering" ? "INITIALIZING_WORKSPACE" : "CLOSING_SESSION"}
257 |     </span>
258 |     <span>·</span>
259 |     <span className="tabular-nums font-bold text-foreground">{progressBarWidth}%</span>
260 |   </div>
261 | </div>
262 | </div>
263 | )}
264 |
265 |   {/* Active Case Study Workspace Content - Rendered always for SSR indexing */}
266 |   <div
267 |     ref={workspaceScrollRef}

```

```

268 |         onScroll={handleWorkspaceScroll}
269 |         style={{
270 |             opacity: transitionStage === "ready" ? 1 : 0,
271 |             pointerEvents: transitionStage === "ready" ? "auto" : "none",
272 |         }}
273 |         className={`relative z-10 flex h-full w-full max-w-6xl flex-col gap-10 overflow-y-auto px-4 pb-16...
274 |             transitionStage === "ready" && showWorkspaceContent ? "animate-workspace-in" : ""
275 |         `}
276 |     >
277 |     {/* Sticky header with documentation links and scroll progress */}
278 |     <div className="fixed inset-x-4 top-4 z-50 mx-auto max-w-6xl md:inset-x-8">
279 |         <div className="relative flex items-center justify-between rounded-2xl border border-white/15 b...
280 |             <motion.div
281 |                 aria-hidden
282 |                 className="pointer-events-none absolute -inset-[1px] rounded-2xl opacity-40"
283 |                 style={{
284 |                     background: `conic-gradient(from 0deg, transparent, ${details.theme.glow}, transparent 30...
285 |                     maskImage: "linear-gradient(#000, #000)",
286 |                     WebkitMaskImage: "linear-gradient(#000, #000)",
287 |                 }}
288 |                 animate={{ rotate: 360 }}
289 |                 transition={{ duration: 8, repeat: Infinity, ease: "linear" }}
290 |             />
291 |
292 |             {/* Thin Scroll Progress Bar across the top of the header border */}
293 |             <div className="absolute top-0 inset-x-0 h-[1.5px] rounded-t-2xl overflow-hidden bg-white/5 z...
294 |                 <div
295 |                     className={`h-full bg-gradient-to-r ${details.theme.primary} ${details.theme.secondary}`}
296 |                     style={{ width: `${scrollProgress}%` }}
297 |                 />
298 |             </div>
299 |
300 |             <span className="pointer-events-none absolute left-0 top-0 h-3 w-3 rounded-tl-md border-l bor...
301 |             <span className="pointer-events-none absolute right-0 top-0 h-3 w-3 rounded-tr-md border-r bo...
302 |
303 |             {/* Left elements: Exit workspace link and title */}
304 |             <div className="flex items-center gap-3 relative z-10">
305 |                 <button
306 |                     onClick={handleExitWorkspace}
307 |                     onMouseEnter={() => playTick(0)}
308 |                     className="group flex items-center gap-1 font-mono text-[10px] uppercase tracking-wider t...
309 |                 >
310 |                     <span className="transition-transform group-hover:-translate-x-0.5">?</span> Exit
311 |                     Workspace
312 |                 </button>
313 |                 <span className="hidden h-3 w-px bg-white/10 sm:block" />
314 |                 <span className="hidden font-mono text-[10px] uppercase tracking-[0.2em] text-foreground/80...
315 |                     {project.index}.{project.name}
316 |                 </span>
317 |             </div>
318 |
319 |             {/* Document Navigation Tabs - scrollable on mobile, static on desktop */}
320 |             <nav className="flex items-center gap-1 overflow-x-auto scrollbar-none py-1 max-w-[40%] sm:ma...
321 |                 {TABS.map((tab, idx) => (
322 |                     <button
323 |                         key={tab.id}
324 |                         onClick={() => scrollToSection(tab.id, idx)}
325 |                         className={`relative rounded-lg px-2.5 py-1.5 font-mono text-[9px] uppercase tracking-w...
326 |                         activeSection === tab.id
327 |                             ? "text-foreground font-semibold"
328 |                             : "text-muted-foreground/80 hover:text-foreground"
329 |                     `}
330 |                 >
331 |                     {activeSection === tab.id && (
332 |                         <motion.span

```

```
333 |         layoutId="ws-tab-pill"
334 |         className="absolute inset-0 rounded-lg bg-white/5 border border-white/10 shadow-inner"
335 |         transition={{ type: "spring", stiffness: 500, damping: 40 }}
336 |     />
337 |   }}
338 |   <span className="relative z-10">{tab.label}</span>
339 | </button>
340 | )}}
341 | </nav>
342 |
343 | {/ * Status Indicator & Copy Link */}
344 | <div className="flex items-center gap-4 relative z-10">
345 |   <button
346 |     onClick={() => {
347 |       navigator.clipboard.writeText(
348 |         `${window.location.href.split("#")[0]}#ws-sec-${activeSection}`,
349 |       );
350 |       toast.success("Section link copied");
351 |       playTick(0);
352 |     }}
353 |     className="flex items-center gap-1.5 font-mono text-[9px] uppercase tracking-widest text-...
354 |   >
355 |     <Link2 className="h-3 w-3" /> <span className="hidden sm:inline">Copy Link</span>
356 |   </button>
357 |   <span className="flex items-center gap-1.5 font-mono text-[9px] uppercase tracking-widest t...
358 |     <span className="relative flex h-1.5 w-1.5">
359 |       <span className="absolute inline-flex h-full w-full animate-ping rounded-full bg-emeral...
360 |       <span className="relative inline-flex h-1.5 w-1.5 rounded-full bg-emerald-400" />
361 |     </span>
362 |     <span className="hidden sm:inline">Active Session</span>
363 |   </span>
364 | </div>
365 | </div>
366 | </div>
367 |
368 | {/ * Section 1: Overview */}
369 | <section id="ws-sec-overview" className="scroll-mt-32 space-y-12">
370 |   <div className="border-b border-white/5 pb-6">
371 |     <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
372 |       01 . Project Overview
373 |     </span>
374 |     <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
375 |       {project.name}
376 |     </h3>
377 |     <p className="mt-2 text-xs text-subtle">
378 |       {project.name === "Vurlo"
379 |         ? "Built in 10 Days ? React 19 ? TanStack Start"
380 |         : project.name === "Veltrix"
381 |         ? "Autonomous Run ? Python ? Gemini API"
382 |         : "Competitor Scan ? FastAPI ? Groq Fallbacks"}
383 |     </p>
384 |   </div>
385 |
386 |   <div className="grid gap-8 md:grid-cols-3">
387 |     {/ * Left Column: Hero screenshot */}
388 |     <div className="md:col-span-2 space-y-6">
389 |       <DeviceFrame
390 |         label={` ${project.name.toLowerCase()}/hero.webp`}
391 |         noPadding
392 |         className="border-white/10 shadow-2xl overflow-hidden bg-black/40"
393 |       >
394 |         {project.image ? (
395 |           <img
396 |             src={` ${project.image}?v=1.2`}
397 |             alt="Hero view"
```

```
398 |         className="w-full object-cover object-top max-h-[360px]"
399 |       />
400 |     ) : (
401 |       <div className="aspect-[16/10] bg-black/30 flex items-center justify-center font-mono t...
402 |         Visual Screen Offline
403 |       </div>
404 |     )}
405 |   </DeviceFrame>
406 | </div>
407 |
408 | {/* Right Column: Project stats card */}
409 | <div className="rounded-xl border border-white/10 bg-card p-6 flex flex-col justify-between g...
410 |   <div className="space-y-4">
411 |     <div>
412 |       <h4 className="font-mono text-[9px] uppercase tracking-widest text-subtle">
413 |         Description
414 |       </h4>
415 |       <p className="mt-2 text-xs text-foreground/80 leading-relaxed font-medium">
416 |         {project.summary}
417 |       </p>
418 |     </div>
419 |     <div className="grid grid-cols-2 gap-4 border-t border-white/5 pt-4">
420 |       <div>
421 |         <h5 className="font-mono text-[9px] uppercase tracking-widest text-subtle">
422 |           Timeline
423 |         </h5>
424 |         <p className="mt-0.5 text-xs text-foreground/90 font-semibold font-mono">
425 |           {project.metrics.find(
426 |             (m) =>
427 |               m.k.toLowerCase().includes("build") || m.k.toLowerCase().includes("time"),
428 |             )?.v || "10 Days"}
429 |         </p>
430 |       </div>
431 |       <div>
432 |         <h5 className="font-mono text-[9px] uppercase tracking-widest text-subtle">
433 |           Role
434 |         </h5>
435 |         <p className="mt-0.5 text-xs text-foreground/90 font-semibold truncate">
436 |           {project.role}
437 |         </p>
438 |       </div>
439 |       <div>
440 |         <h5 className="font-mono text-[9px] uppercase tracking-widest text-subtle">
441 |           Status
442 |         </h5>
443 |         <p className="mt-0.5 text-xs text-foreground/90 font-semibold">Production</p>
444 |       </div>
445 |       <div>
446 |         <h5 className="font-mono text-[9px] uppercase tracking-widest text-subtle">
447 |           Year
448 |         </h5>
449 |         <p className="mt-0.5 text-xs text-foreground/90 font-semibold">
450 |           {project.year}
451 |         </p>
452 |       </div>
453 |       <div>
454 |         <h5 className="font-mono text-[9px] uppercase tracking-widest text-subtle">
455 |           Deployment
456 |         </h5>
457 |         <p className="mt-0.5 text-xs text-foreground/90 font-semibold">
458 |           {project.name === "Vurlo"
459 |             ? "Production SaaS"
460 |             : project.name === "Veltrix"
461 |             ? "2 posts / day (auto)"
462 |             : "Daily scan @ 2AM UTC"}
```

```

463 |         </p>
464 |     </div>
465 | <div>
466 |     <h5 className="font-mono text-[9px] uppercase tracking-widest text-subtle">
467 |         Core Stack
468 |     </h5>
469 |     <p className="mt-0.5 text-xs text-foreground/90 font-semibold truncate">
470 |         {project.name === "Vurlo"
471 |           ? "React 19 / Firebase"
472 |           : project.name === "Veltrix"
473 |           ? "Python / Playwright"
474 |           : "FastAPI / SQLite"}
475 |     </p>
476 | </div>
477 | </div>
478 | </div>
479 | <div className="flex flex-col gap-2 pt-2 border-t border-white/5">
480 |     {project.href && project.href !== "#" && (
481 |       <a
482 |         href={project.href}
483 |         target="_blank"
484 |         rel="noopener noreferrer"
485 |         className="flex items-center justify-center gap-1.5 rounded-lg bg-secondary/10 border...
486 |       >
487 |         Visit Live Site <ArrowUpRight className="h-3 w-3" />
488 |     </a>
489 |   )}
490 |   {project.name === "Vurlo" ? (
491 |     <a
492 |       href="https://github.com/theayushagarwal/vurlo-ecommerce"
493 |       target="_blank"
494 |       rel="noopener noreferrer"
495 |       className="flex items-center justify-center gap-1.5 rounded-lg bg-white/5 border bord...
496 |     >
497 |       View Source Code
498 |     </a>
499 |   ) : (
500 |     <div className="flex items-center justify-center gap-1.5 rounded-lg bg-white/5 border b...
501 |       Private Repository
502 |     </div>
503 |   )}
504 | </div>
505 | </div>
506 | </div>
507 | </section>
508 |
509 | {/* Section 2: Why I Built It */}
510 | <section id="ws-sec-why-built" className="scroll-mt-32 space-y-12">
511 |   <div className="border-b border-white/5 pb-6">
512 |     <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
513 |       02 . Core Rationale
514 |     </span>
515 |     <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
516 |       Why I Built It
517 |     </h3>
518 |     <p className="mt-2 text-sm text-subtle">
519 |       Understanding the core problems, proposed solutions, and realized product
520 |       achievements.
521 |     </p>
522 |   </div>
523 |
524 |   <div className="max-w-3xl">
525 |     <span className="font-mono text-[9px] uppercase tracking-widest text-red-400/80 font-bold">
526 |       The problem
527 |     </span>

```



```
528 |         <p className="mt-3 font-display text-xl md:text-2xl font-medium text-foreground/95 leading-sn...
529 |           {details.whyBuilt.problem}
530 |         </p>
531 |       </div>
532 |
533 |       {details.whyBuilt.scenario && <ScenarioTimeline events={details.whyBuilt.scenario} />}
534 |
535 |       <div className="rounded-xl border border-white/10 bg-card p-6 shadow-lg">
536 |         <span className="mb-6 block font-mono text-[9px] uppercase tracking-widest text-subtle">
537 |           The fix, step by step
538 |         </span>
539 |         {details.whyBuilt.solutionSteps ? (
540 |           <StepSequence steps={details.whyBuilt.solutionSteps} accent={details.theme.glow} />
541 |         ) : (
542 |           <p className="text-xs leading-relaxed text-subtle">{details.whyBuilt.solution}</p>
543 |         )}
544 |       </div>
545 |
546 |       <div>
547 |         <span className="font-mono text-[9px] uppercase tracking-widest text-emerald-400/80 font-bold">
548 |           Result
549 |         </span>
550 |         <p className="mt-2 max-w-2xl text-sm text-foreground/90">{details.whyBuilt.result}</p>
551 |         {details.whyBuilt.resultStats && (
552 |           <dl className="mt-5 grid max-w-2xl grid-cols-2 gap-x-6 gap-y-4 sm:grid-cols-4">
553 |             {details.whyBuilt.resultStats.map((s) => (
554 |               <div key={s.label}>
555 |                 <dt className="font-mono text-[9px] uppercase tracking-widest text-subtle/80">
556 |                   {s.label}
557 |                 </dt>
558 |                 <dd className="mt-0.5 font-display text-lg font-semibold tabular-nums text-foreground">
559 |                   {s.value}
560 |                 </dd>
561 |               </div>
562 |             ))}
563 |             </dl>
564 |           )}
565 |         </div>
566 |       </section>
567 |
568 |       {/* Section 3: Architecture */}
569 |       <section id="ws-sec-architecture" className="scroll-mt-32 space-y-12">
570 |         <div className="border-b border-white/5 pb-6">
571 |           <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
572 |             03 . System Pipeline
573 |           </span>
574 |           <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
575 |             Architecture Pipelines
576 |           </h3>
577 |           <p className="mt-2 text-sm text-subtle">
578 |             Visual system flow diagrams representing data pipelines and structured checks.
579 |           </p>
580 |         </div>
581 |
582 |         <div className="rounded-xl border border-white/10 bg-card p-6 shadow-lg">
583 |           <span className="mb-6 block font-mono text-[9px] uppercase tracking-widest text-subtle">
584 |             Pipeline Data flow nodes (Glow on Hover)
585 |           </span>
586 |           <div className="grid gap-4 sm:grid-cols-2 md:grid-cols-3">
587 |             {details.architecture.map((node, idx) => (
588 |               <ArchitectureNode key={node.id} node={node} idx={idx} glow={details.theme.glow} />
589 |             ))}
590 |           </div>
591 |         </div>
592 |       </section>
```

```
593 |
594 |     { /* Section 4: Features */ }
595 |     <section id="ws-sec-features" className="scroll-mt-32 space-y-12">
596 |         <div className="border-b border-white/5 pb-6">
597 |             <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
598 |                 04 . Core Capabilities
599 |             </span>
600 |             <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
601 |                 Features & Capabilities
602 |             </h3>
603 |             <p className="mt-2 text-sm text-subtle">
604 |                 Key features and built-in components implemented inside the codebase.
605 |             </p>
606 |         </div>
607 |
608 |         <div className="rounded-xl border border-white/10 bg-card p-6 shadow-lg">
609 |             <span className="mb-6 block font-mono text-[9px] uppercase tracking-widest text-subtle">
610 |                 Feature Checklist
611 |             </span>
612 |             <ul className="grid gap-4 sm:grid-cols-2 lg:grid-cols-3">
613 |                 {details.features.map((feature) => (
614 |                     <li
615 |                         key={feature}
616 |                         className="flex items-center gap-3 rounded-lg border border-white/5 bg-surface p-3 tran...
617 |                     >
618 |                         <span className="flex h-5 w-5 shrink-0 items-center justify-center rounded-full bg-emer...
619 |                             ?
620 |                         </span>
621 |                         <span className="text-foreground/90 font-medium">{feature}</span>
622 |                     </li>
623 |                 ) ) }
624 |             </ul>
625 |         </div>
626 |     </section>
627 |
628 |     { /* Section 5: Engineering Challenges */ }
629 |     <section id="ws-sec-challenges" className="scroll-mt-32 space-y-12">
630 |         <div className="border-b border-white/5 pb-6">
631 |             <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
632 |                 03 . Technical Barriers
633 |             </span>
634 |             <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
635 |                 Solving Complex Bottlenecks
636 |             </h3>
637 |             <p className="mt-2 text-sm text-subtle">
638 |                 A deep dive into security rule architectures, race conditions, and verification
639 |                 solutions.
640 |             </p>
641 |         </div>
642 |
643 |         <div className="space-y-4 max-w-4xl">
644 |             {details.challenges.map((challenge, idx) => (
645 |                 <ChallengeCard
646 |                     key={challenge.title}
647 |                     index={idx}
648 |                     challenge={challenge}
649 |                     active={activeChallenge === idx}
650 |                     onClick={() => setActiveChallenge(activeChallenge === idx ? null : idx)}
651 |                     accentGlow={details.theme.glow}
652 |                 />
653 |             ) ) }
654 |         </div>
655 |     </section>
656 |
657 |     { /* Section 6: Journey */ }
```

```
658 |         <section id="ws-sec-journey" className="scroll-mt-32 space-y-12">
659 |             <div className="border-b border-white/5 pb-6">
660 |                 <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
661 |                     06 . Build Log Timeline
662 |                 </span>
663 |                 <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
664 |                     Development Journey
665 |                 </h3>
666 |                 <p className="mt-2 text-sm text-subtle">
667 |                     A chronological day-by-day log detailing core milestones and features added.
668 |                 </p>
669 |             </div>
670 |
671 |             <div className="rounded-xl border border-white/10 bg-card p-6 shadow-lg">
672 |                 <span className="mb-6 block font-mono text-[9px] uppercase tracking-widest text-subtle">
673 |                     Milestone timeline
674 |                 </span>
675 |                 <div className="relative pl-6 border-l border-white/10 space-y-8 py-2">
676 |                     {details.journey.map((step) => (
677 |                         <div key={step.day} className="relative group">
678 |                             <span className="absolute -left-[31px] top-1.5 flex h-4.5 w-4.5 items-center justify-ce...
679 |                                 <span className="h-1.5 w-1.5 rounded-full bg-secondary transition-transform group-hov...
680 |                             </span>
681 |                             <div>
682 |                                 <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-b...
683 |                                     {step.day}
684 |                                 </span>
685 |                                 <h4 className="font-display font-semibold text-foreground text-sm mt-0.5">
686 |                                     {step.milestone}
687 |                                 </h4>
688 |                                 <p className="text-xs text-subtle mt-1">{step.details}</p>
689 |                             </div>
690 |                         </div>
691 |                     )))
692 |                 </div>
693 |             </div>
694 |         </section>
695 |
696 |         {/* Section 7: Performance */}
697 |         <section id="ws-sec-performance" className="scroll-mt-32 space-y-12">
698 |             {details.lighthouse ? (
699 |                 <>
700 |                     <div className="border-b border-white/5 pb-6">
701 |                         <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
702 |                             07 . Technical Audits
703 |                         </span>
704 |                         <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
705 |                             Performance & Audits
706 |                         </h3>
707 |                         <p className="mt-2 text-sm text-subtle">
708 |                             Audited system lighthouse stats for performance, accessibility, best practices,
709 |                             and SEO.
710 |                         </p>
711 |                     </div>
712 |
713 |                     <div className="rounded-xl border border-white/10 bg-card p-6 shadow-lg flex flex-col justi...
714 |                         <span className="mb-6 block font-mono text-[9px] uppercase tracking-widest text-subtle">
715 |                             Lighthouse Audit (Active SVG Gauges)
716 |                         </span>
717 |                         <div className="grid grid-cols-2 gap-4 sm:grid-cols-4">
718 |                             <LighthouseDial score={details.lighthouse.performance} label="Performance" />
719 |                             <LighthouseDial score={details.lighthouse.accessibility} label="Accessibility" />
720 |                             <LighthouseDial score={details.lighthouse.bestPractices} label="Best Practices" />
721 |                             <LighthouseDial score={details.lighthouse.seo} label="SEO" />
722 |                         </div>
```

```
723 |         </div>
724 |     </>
725 | ) : details.reliability ? (
726 |     <>
727 |         <div className="border-b border-white/5 pb-6">
728 |             <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
729 |                 07 . System Reliability
730 |             </span>
731 |             <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
732 |                 Operational Metrics
733 |             </h3>
734 |             <p className="mt-2 text-sm text-subtle">
735 |                 No public-facing site to audit ? these are the safeguards that keep an unattended
736 |                 pipeline honest.
737 |             </p>
738 |         </div>
739 |
740 |         <div className="grid grid-cols-1 gap-4 sm:grid-cols-2">
741 |             {details.reliability.map((stat) => (
742 |                 <div
743 |                     key={stat.label}
744 |                     className="rounded-xl border border-white/10 bg-card p-5 shadow-lg flex flex-col gap-2"
745 |                 >
746 |                     <span className="font-mono text-[9px] uppercase tracking-widest text-subtle">
747 |                         {stat.label}
748 |                     </span>
749 |                     <span className="font-display text-3xl font-semibold text-foreground">
750 |                         {stat.value}
751 |                     </span>
752 |                     <p className="text-xs text-subtle leading-relaxed">{stat.detail}</p>
753 |                 </div>
754 |             )))
755 |         </div>
756 |     </>
757 | ) : null}
758 | </section>
759 |
760 | {/* Section 8: Tech Decisions & Lessons */}
761 | <section id="ws-sec-decisions" className="scroll-mt-32 space-y-12">
762 |     <div className="border-b border-white/5 pb-6">
763 |         <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
764 |             05 . Architecture Decisions
765 |         </span>
766 |         <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
767 |             Trade-off Explanations
768 |         </h3>
769 |         <p className="mt-2 text-sm text-subtle">
770 |             Concise rationales for technology selections and future roadmaps.
771 |         </p>
772 |     </div>
773 |
774 |     <div className="grid gap-8 md:grid-cols-2">
775 |         <div className="space-y-4">
776 |             <span className="mb-2 block font-mono text-[9px] uppercase tracking-widest text-subtle">
777 |                 Design Choices
778 |             </span>
779 |             {details.decisions.map((dec) => (
780 |                 <div
781 |                     key={dec.tech}
782 |                     className="rounded-xl border border-white/10 bg-surface p-5 shadow-sm"
783 |                 >
784 |                     <div className="flex items-center justify-between gap-4">
785 |                         <h4 className="font-display font-semibold text-foreground text-sm">
786 |                             {dec.title}
787 |                         </h4>
```

```
788 |         <span className="rounded bg-white/5 px-2 py-0.5 font-mono text-[9px] text-secondary">
789 |             {dec.tech}
790 |         </span>
791 |     </div>
792 |     <p className="mt-2 text-xs leading-relaxed text-subtle">{dec.explanation}</p>
793 | </div>
794 |   )}}
795 | </div>
796 |
797 | <div className="rounded-xl border border-white/10 bg-card p-6 shadow-lg">
798 |   <span className="mb-6 block font-mono text-[9px] uppercase tracking-widest text-subtle">
799 |     If I built V2...
800 |   </span>
801 |   <div className="space-y-4">
802 |     <p className="text-xs text-subtle leading-relaxed font-medium italic mb-4">
803 |       "Evaluating scaling adjustments and cache controls shows true engineering
804 |       maturity."
805 |     </p>
806 |     <ul className="grid gap-3 font-sans text-[13px] text-muted-foreground/90 leading-relaxed">
807 |       {details.lessons.map((lesson) => (
808 |         <li
809 |           key={lesson}
810 |           className="flex items-start gap-3 rounded-lg border border-white/5 bg-surface p-3 t...
811 |         >
812 |           <span className="flex h-5 w-5 shrink-0 items-center justify-center rounded bg-indig...
813 |             ?
814 |           </span>
815 |           <span className="text-foreground/90 font-medium">{lesson}</span>
816 |         </li>
817 |       )}}
818 |     </ul>
819 |   </div>
820 | </div>
821 | </div>
822 | </section>
823 |
824 | {/* Section 10: Gallery */}
825 | <section id="ws-sec-gallery" className="scroll-mt-32 space-y-12">
826 |   <div className="border-b border-white/5 pb-6">
827 |     <span className="font-mono text-[9px] uppercase tracking-widest text-secondary font-bold">
828 |       06 . Screenshots
829 |     </span>
830 |     <h3 className="font-display text-4xl font-semibold text-foreground mt-2">
831 |       Visual Gallery
832 |     </h3>
833 |     <p className="mt-2 text-sm text-subtle">
834 |       Scrollable preview tracks showcasing panel responsive structures.
835 |     </p>
836 |   </div>
837 |
838 |   <div className="relative rounded-xl border border-white/10 bg-card p-6 shadow-lg">
839 |     <span className="mb-6 block font-mono text-[9px] uppercase tracking-widest text-subtle">
840 |       Horizontal Screenshot Scroll
841 |     </span>
842 |     <div className="flex gap-6 overflow-x-auto pb-4 scrollbar-thin select-none">
843 |       {details.gallery.map((item, idx) => (
844 |         <div key={`_${item.label}-${idx}`} className="flex-none w-[85vw] sm:w-[480px]">
845 |           <DeviceFrame
846 |             label={item.label}
847 |             noPadding
848 |             className="border-white/10 overflow-hidden shadow-md"
849 |           >
850 |             <img
851 |               src={`_${item.img}?v=1.2`}
852 |               alt={item.label}
```

```

853 |         loading="lazy"
854 |         className="w-full object-cover object-top max-h-[300px]"
855 |     />
856 |     </DeviceFrame>
857 | </div>
858 |   )})
859 |   {project.secondaryImage && (
860 |     <div className="flex-none w-[85vw] sm:w-[480px]">
861 |       <DeviceFrame
862 |         label={project.secondaryLabel || "Admin console"}
863 |         noPadding
864 |         className="border-white/10 overflow-hidden shadow-md"
865 |       >
866 |         <img
867 |           src={`_${project.secondaryImage}?v=1.3`}
868 |           alt="Secondary screenshot"
869 |           loading="lazy"
870 |           className="w-full object-cover object-top max-h-[300px]"
871 |         />
872 |       </DeviceFrame>
873 |     </div>
874 |   )}
875 | </div>
876 | </div>
877 | </section>
878 |
879 | {/* Section 11: End Workspace CTA */}
880 | <section className="border-t border-white/10 pt-16 pb-32 flex flex-col items-center text-center m...
881 |   <h4 className="font-display text-2xl font-semibold text-foreground">
882 |     Finished exploring this session?
883 |   </h4>
884 |   <p className="text-xs text-subtle leading-relaxed">
885 |     End the workspace session to release resources and return back to the primary portfolio
886 |     index.
887 |   </p>
888 |   <button
889 |     onClick={handleExitWorkspace}
890 |     onMouseEnter={() => playTick(0)}
891 |     className="group inline-flex items-center gap-2 rounded-xl bg-gradient-to-r from-red-500 to-a...
892 |   >
893 |     End Workspace Session
894 |     <span className="text-[14px] leading-none transition-transform group-hover:translate-x-1">
895 |       ?
896 |     </span>
897 |   </button>
898 | </section>
899 | </div>
900 | </div>
901 | );
902 | }

```

FILE #48: src\components\portfolio\projects\widgets\ArchitectureNode.tsx

```

1 | import { motion } from "motion/react";
2 | import { useSpotlight } from "../useSpotlight";
3 |
4 | /** Architecture pipeline node with mouse-tracking spotlight + staggered entrance */
5 | export function ArchitectureNode({
6 |   node,
7 |   idx,
8 |   glow,
9 | }): {
10 |   node: { id: string; label: string; desc: string };

```

```

11 |   idx: number;
12 |   glow: string;
13 | }) {
14 |   const { ref, onMove, background } = useSpotlight();
15 |   return (
16 |     <motion.div
17 |       ref={ref as React.RefObject<HTMLDivElement>}
18 |       onMouseMove={onMove}
19 |       initial={{ opacity: 0, y: 14 }}
20 |       whileInView={{ opacity: 1, y: 0 }}
21 |       viewport={{ once: true, margin: "-40px" }}
22 |       transition={{ type: "spring", stiffness: 260, damping: 26, delay: idx * 0.06 }}
23 |       style={{ "--glow-color": glow } as React.CSSProperties}
24 |       className="group/node relative overflow-hidden rounded-xl border border-white/10 bg-surface p-4 tra...
25 |     >
26 |       <motion.div
27 |         aria-hidden
28 |         className="pointer-events-none absolute -inset-px opacity-0 transition-opacity duration-500 group...
29 |         style={{ background }}
30 |       />
31 |       <div className="relative z-10">
32 |         <span className="font-mono text-[9px] text-secondary font-bold uppercase tracking-wider block mb-1">
33 |           0{idx + 1} // {node.id}
34 |         </span>
35 |         <h4 className="font-display font-semibold text-foreground text-sm">{node.label}</h4>
36 |         <p className="mt-2 text-xs text-subtle leading-relaxed">{node.desc}</p>
37 |       </div>
38 |     </motion.div>
39 |   );
40 | }

```

FILE #49: src\components\portfolio\projects\widgets\ChallengeCard.tsx

```

1 | import { ChevronDown } from "lucide-react";
2 | import { motion, AnimatePresence } from "motion/react";
3 |
4 | /** Numbered step sequence with optional file chips ? used for both `solutionSteps` and `fixSteps` */
5 | export function StepSequence({
6 |   steps,
7 |   accent,
8 | }): {
9 |   steps: { step: string; detail?: string; file?: string }[];
10 |   accent: string;
11 | }) {
12 |   return (
13 |     <div className="relative pl-7 border-l border-white/10 space-y-6">
14 |       {steps.map((s, i) => (
15 |         <motion.div
16 |           key={i}
17 |           initial={{ opacity: 0, y: 10 }}
18 |           whileInView={{ opacity: 1, y: 0 }}
19 |           viewport={{ once: true, margin: "-20px" }}
20 |           transition={{ delay: i * 0.07, type: "spring", stiffness: 300, damping: 28 }}
21 |           className="relative"
22 |         >
23 |           <span
24 |             style={{ "--accent": accent } as React.CSSProperties}
25 |             className="absolute -left-[37px] top-0 flex h-6 w-6 items-center justify-center rounded-full ...
26 |           >
27 |             {i + 1}
28 |           </span>
29 |           <div className="flex flex-wrap items-center gap-2">
30 |             <h5 className="font-display font-semibold text-foreground text-[13px]">{s.step}</h5>

```

```
31 |         {s.file && (  
32 |             <code className="rounded bg-white/5 border border-white/10 px-1.5 py-0.5 font-mono text-[10...  
33 |                 {s.file}  
34 |             </code>  
35 |         )}  
36 |     </div>  
37 |     {s.detail && (  
38 |         <p className="mt-1 text-xs leading-relaxed text-foreground/70">{s.detail}</p>  
39 |     )}  
40 | </motion.div>  
41 | )})  
42 | </div>  
43 | );  
44 | }  
45 |  
46 | /** Expandable accordion challenge cards */  
47 | export function ChallengeCard({  
48 |     index,  
49 |     challenge,  
50 |     active,  
51 |     onClick,  
52 |     accentGlow,  
53 | }): {  
54 |     index: number;  
55 |     challenge: {  
56 |         title: string;  
57 |         problem: string;  
58 |         difficulty: string;  
59 |         solution: string;  
60 |         fixSteps?: { step: string; detail?: string; file?: string }[];  
61 |         learned: string;  
62 |     };  
63 |     active: boolean;  
64 |     onClick: () => void;  
65 |     accentGlow: string;  
66 | }) {  
67 |     return (  
68 |         <div className="rounded-xl border border-white/10 bg-surface overflow-hidden transition-all duration-...  
69 |             <button  
70 |                 onClick={onClick}  
71 |                 className="flex w-full items-center justify-between p-5 text-left font-display font-semibold text...  
72 |             >  
73 |                 <span className="flex items-center gap-3">  
74 |                     <span className="font-mono text-xs text-subtle">0{index + 1}.</span>  
75 |                     {challenge.title}  
76 |                 </span>  
77 |                 <ChevronDown  
78 |                     className={`h-4 w-4 text-subtle transition-transform duration-300 ${active ? "rotate-180" : ""}`}  
79 |                 />  
80 |             </button>  
81 |             <AnimatePresence initial={false}>  
82 |                 {active && (  
83 |                     <motion.div  
84 |                         initial={{ height: 0, opacity: 0 }}  
85 |                         animate={{ height: "auto", opacity: 1 }}  
86 |                         exit={{ height: 0, opacity: 0 }}  
87 |                         transition={{ type: "spring", stiffness: 300, damping: 34, mass: 1 }}  
88 |                     >  
89 |                         <motion.div  
90 |                             variants={{  
91 |                                 hidden: {},  
92 |                                 show: { transition: { staggerChildren: 0.08, delayChildren: 0.05 } },  
93 |                             }}  
94 |                             initial="hidden"  
95 |                             animate="show"
```



```

96 |         style={{ "--accent-glow": accentGlow } as React.CSSProperties}
97 |         className="border-t border-white/5 bg-white/[0.03] backdrop-blur-md p-5 space-y-4 text-xs l...
98 |     >
99 |     <motion.div
100 |         variants={{ hidden: { opacity: 0, y: 8 }, show: { opacity: 1, y: 0 } }}
101 |         transition={{ type: "spring", stiffness: 320, damping: 28 }}
102 |         className="relative rounded-lg border-l-2 pl-3"
103 |         style={{ borderColor: accentGlow }}
104 |     >
105 |         <span className="block font-mono text-[9px] uppercase tracking-widest text-subtle mb-1">
106 |             Problem
107 |         </span>
108 |         <p className="text-foreground/90 font-medium text-[13px]">{challenge.problem}</p>
109 |     </motion.div>
110 |     <div className="grid gap-4 sm:grid-cols-2">
111 |         <motion.div
112 |             variants={{ hidden: { opacity: 0, y: 8 }, show: { opacity: 1, y: 0 } }}
113 |             transition={{ type: "spring", stiffness: 320, damping: 28 }}
114 |         >
115 |             <span className="block font-mono text-[9px] uppercase tracking-widest text-subtle mb-1">
116 |                 Why it was difficult
117 |             </span>
118 |             <p>{challenge.difficulty}</p>
119 |         </motion.div>
120 |         <motion.div
121 |             variants={{ hidden: { opacity: 0, y: 8 }, show: { opacity: 1, y: 0 } }}
122 |             transition={{ type: "spring", stiffness: 320, damping: 28 }}
123 |             className="sm:col-span-2"
124 |         >
125 |             <span className="block font-mono text-[9px] uppercase tracking-widest text-subtle mb-3">
126 |                 The fix
127 |             </span>
128 |             {challenge.fixSteps ? (
129 |                 <StepSequence steps={challenge.fixSteps} accent={accentGlow} />
130 |             ) : (
131 |                 <p>{challenge.solution}</p>
132 |             )}
133 |         </motion.div>
134 |     </div>
135 |     <motion.div
136 |         variants={{ hidden: { opacity: 0, y: 8 }, show: { opacity: 1, y: 0 } }}
137 |         transition={{ type: "spring", stiffness: 320, damping: 28 }}
138 |         className="pt-2 border-t border-white/5"
139 |     >
140 |         <span className="block font-mono text-[9px] uppercase tracking-widest text-secondary/80 m...
141 |             What I learned
142 |         </span>
143 |         <p className="text-secondary/90 italic">"{challenge.learned}"</p>
144 |     </motion.div>
145 | </motion.div>
146 | </motion.div>
147 |     )}
148 | </AnimatePresence>
149 | </div>
150 | );
151 | }

```

FILE #50: src\components\portfolio\projects\widgets\LighthouseDial.tsx

```

1 | import { useEffect, useState } from "react";
2 | import { animate } from "motion/react";
3 |
4 | /** Animated circular SVG gauges for performance scores with count-up sync and completion glow */

```

```
5 | export function LighthouseDial({ score, label }: { score: number; label: string }) {
6 |   const [animatedScore, setAnimatedScore] = useState(0);
7 |   const [justFinished, setJustFinished] = useState(false);
8 |
9 |   const radius = 34;
10 |   const circumference = 2 * Math.PI * radius;
11 |   const strokeDashoffset = circumference - (animatedScore / 100) * circumference;
12 |   const strokeColor = score >= 90 ? "#10b981" : score >= 50 ? "#f59e0b" : "#ef4444";
13 |
14 |   useEffect(() => {
15 |     const controls = animate(0, score, {
16 |       duration: 1.1,
17 |       delay: 0.35,
18 |       ease: [0.16, 1, 0.3, 1],
19 |       onUpdate: (v) => setAnimatedScore(Math.round(v)),
20 |       onComplete: () => {
21 |         setJustFinished(true);
22 |         setTimeout(() => setJustFinished(false), 550);
23 |       },
24 |     });
25 |     return () => controls.stop();
26 |   }, [score]);
27 |
28 |   return (
29 |     <div
30 |       style={{ "--dial-glow": `${strokeColor}66` } as React.CSSProperties}
31 |       className={`flex flex-col items-center gap-3 rounded-xl border border-white/10 bg-white/[0.03] back...
32 |       justFinished ? "shadow-[0_0_26px_var(--dial-glow)]" : ""
33 |     >
34 |     <div className="relative h-20 w-20">
35 |       <svg className="h-full w-full -rotate-90" viewBox="0 0 80 80">
36 |         <circle
37 |           cx="40"
38 |           cy="40"
39 |           r={radius}
40 |           className="stroke-white/5"
41 |           strokeWidth="5.5"
42 |           fill="transparent"
43 |         />
44 |         <circle
45 |           cx="40"
46 |           cy="40"
47 |           r={radius}
48 |           style={{
49 |             stroke: strokeColor,
50 |             strokeDasharray: circumference,
51 |             strokeDashoffset: strokeDashoffset,
52 |           }}
53 |           strokeWidth="5.5"
54 |           fill="transparent"
55 |           strokeLinecap="round"
56 |         />
57 |       </svg>
58 |       <div className="absolute inset-0 flex items-center justify-center font-display text-lg font-bold ...
59 |         {animatedScore}
60 |       </div>
61 |     </div>
62 |     <span className="font-mono text-[9px] uppercase tracking-widest text-subtle">{label}</span>
63 |   </div>
64 | );
65 | }
66 | }
```

FILE #51: src\components\portfolio\projects\widgets\ScenarioTimeline.tsx

```
1 | import { motion } from "motion/react";
2 |
3 | export function ScenarioTimeline({
4 |   events,
5 | }): {
6 |   events: { t: string; event: string; danger?: boolean }[];
7 | }) {
8 |   return (
9 |     <div className="rounded-xl border border-white/10 bg-card p-6 shadow-lg">
10 |       <span className="mb-6 block font-mono text-[9px] uppercase tracking-widest text-subtle">
11 |         What actually happens
12 |       </span>
13 |       <div className="relative pl-6 border-l border-white/10 space-y-5">
14 |         {events.map((e, i) => (
15 |           <motion.div
16 |             key={i}
17 |             initial={{ opacity: 0, x: -10 }}
18 |             whileInView={{ opacity: 1, x: 0 }}
19 |             viewport={{ once: true, margin: "-20px" }}
20 |             transition={{ delay: i * 0.08, type: "spring", stiffness: 300, damping: 30 }}
21 |             className="relative"
22 |           >
23 |             <span
24 |               className={`absolute -left-[27px] top-1 flex h-3.5 w-3.5 items-center justify-center rounded...
25 |                 e.danger ? "bg-red-500/20 border-red-500/50" : "bg-card border-white/15"
26 |               `}
27 |             >
28 |               <span
29 |                 className={`h-1.5 w-1.5 rounded-full ${e.danger ? "bg-red-400 animate-pulse" : "bg-second...
30 |               />
31 |             </span>
32 |             <div className="flex flex-wrap items-baseline gap-2">
33 |               <span
34 |                 className={`font-mono text-[10px] font-bold uppercase tracking-widest ${
35 |                   e.danger ? "text-red-400" : "text-secondary"
36 |                 `}
37 |               >
38 |                 {e.t}
39 |               </span>
40 |               <p
41 |                 className={`text-xs leading-relaxed ${e.danger ? "text-red-200/95 font-medium" : "text-fo...
42 |               >
43 |                 {e.event}
44 |               </p>
45 |             </div>
46 |           </motion.div>
47 |         ))}
48 |       </div>
49 |     </div>
50 |   );
51 | }
```

FILE #52: src\components\portfolio\projects\widgets\useSpotlight.ts

```
1 | import { useEffect, useRef } from "react";
2 | import { useMotionValue, useTransform } from "motion/react";
3 |
4 | export function useSpotlight() {
5 |   const ref = useRef<HTMLElement>(null);
6 |   const rectRef = useRef<DOMRect | null>(null);
```

```
7 |  
8 |   useEffect(() => {  
9 |     const el = ref.current;  
10 |     if (!el) return;  
11 |  
12 |     const handleMouseEnter = () => {  
13 |       rectRef.current = el.getBoundingClientRect();  
14 |     };  
15 |  
16 |     const handleMouseMove = (e: MouseEvent) => {  
17 |       if (!rectRef.current) return;  
18 |       const x = e.clientX - rectRef.current.left;  
19 |       const y = e.clientY - rectRef.current.top;  
20 |       el.style.setProperty("--mouse-x", `${x}px`);  
21 |       el.style.setProperty("--mouse-y", `${y}px`);  
22 |     };  
23 |  
24 |     el.addEventListener("mouseenter", handleMouseEnter);  
25 |     el.addEventListener("mousemove", handleMouseMove);  
26 |  
27 |     return () => {  
28 |       el.removeEventListener("mouseenter", handleMouseEnter);  
29 |       el.removeEventListener("mousemove", handleMouseMove);  
30 |     };  
31 |   }, []);  
32 |  
33 |   const mouseX = useMotionValue(0);  
34 |   const mouseY = useMotionValue(0);  
35 |  
36 |   function onMove({ currentTarget, clientX, clientY }: React.MouseEvent) {  
37 |     const { left, top } = currentTarget.getBoundingClientRect();  
38 |     mouseX.set(clientX - left);  
39 |     mouseY.set(clientY - top);  
40 |   }  
41 |  
42 |   const background = useTransform(  
43 |     [mouseX, mouseY],  
44 |     ([x, y]) =>  
45 |       `radial-gradient(180px circle at ${x}px ${y}px, var(--glow-color, rgba(139, 92, 246, 0.15)), transp...  
46 |   );  
47 |  
48 |   return { ref, onMove, background };  
49 | }
```

FILE #53: src\components\uilaccordion.tsx

```
1 | import * as React from "react";  
2 | import * as AccordionPrimitive from "@radix-ui/react-accordion";  
3 | import { ChevronDown } from "lucide-react";  
4 |  
5 | import { cn } from "@lib/utlis";  
6 |  
7 | const Accordion = AccordionPrimitive.Root;  
8 |  
9 | const AccordionItem = React.forwardRef<  
10 |   React.ElementRef<typeof AccordionPrimitive.Item>,  
11 |   React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Item>  
12 | >(({ className, ...props }, ref) => (  
13 |   <AccordionPrimitive.Item ref={ref} className={cn("border-b", className)} {...props} />  
14 | ));  
15 | AccordionItem.displayName = "AccordionItem";  
16 |  
17 | const AccordionTrigger = React.forwardRef<
```

```
18 |   React.ElementRef<typeof AccordionPrimitive.Trigger>,
19 |   React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Trigger>
20 | >(({ className, children, ...props }, ref) => (
21 |   <AccordionPrimitive.Header className="flex">
22 |     <AccordionPrimitive.Trigger
23 |       ref={ref}
24 |       className={cn(
25 |         "flex flex-1 items-center justify-between py-4 text-sm font-medium cursor-pointer transition-all ...
26 |         className,
27 |       )}
28 |       {...props}
29 |     >
30 |       {children}
31 |       <ChevronDown className="h-4 w-4 shrink-0 text-muted-foreground transition-transform duration-200" />
32 |     </AccordionPrimitive.Trigger>
33 |   </AccordionPrimitive.Header>
34 | ));
35 | AccordionTrigger.displayName = AccordionPrimitive.Trigger.displayName;
36 |
37 | const AccordionContent = React.forwardRef<
38 |   React.ElementRef<typeof AccordionPrimitive.Content>,
39 |   React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Content>
40 | >(({ className, children, ...props }, ref) => (
41 |   <AccordionPrimitive.Content
42 |     ref={ref}
43 |     className="overflow-hidden text-sm data-[state=closed]:animate-accordion-up data-[state=open]:animate...
44 |     {...props}
45 |   >
46 |     <div className={cn("pb-4 pt-0", className)}>{children}</div>
47 |   </AccordionPrimitive.Content>
48 | ));
49 | AccordionContent.displayName = AccordionPrimitive.Content.displayName;
50 |
51 | export { Accordion, AccordionItem, AccordionTrigger, AccordionContent };
```

FILE #54: src\components\ui>alert-dialog.tsx

```
1 | import * as React from "react";
2 | import * as AlertDialogPrimitive from "@radix-ui/react-alert-dialog";
3 |
4 | import { cn } from "@lib/utils";
5 | import { buttonVariants } from "@components/ui/button";
6 |
7 | const AlertDialog = AlertDialogPrimitive.Root;
8 |
9 | const AlertDialogTrigger = AlertDialogPrimitive.Trigger;
10 |
11 | const AlertDialogPortal = AlertDialogPrimitive.Portal;
12 |
13 | const AlertDialogOverlay = React.forwardRef<
14 |   React.ElementRef<typeof AlertDialogPrimitive.Overlay>,
15 |   React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Overlay>
16 | >(({ className, ...props }, ref) => (
17 |   <AlertDialogPrimitive.Overlay
18 |     className={cn(
19 |       "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-[...
20 |     className,
21 |   )}
22 |   {...props}
23 |   ref={ref}
24 | />
25 | ));
26 | AlertDialogOverlay.displayName = AlertDialogPrimitive.Overlay.displayName;
```

```
27 |
28 | const AlertDialogContent = React.forwardRef<
29 |   React.ElementRef<typeof AlertDialogPrimitive.Content>,
30 |   React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Content>
31 | >(({ className, ...props }, ref) => (
32 |   <AlertDialogPortal>
33 |     <AlertDialogOverlay />
34 |     <AlertDialogPrimitive.Content
35 |       ref={ref}
36 |       className={cn(
37 |         "fixed left-[50%] top-[50%] z-50 grid w-full max-w-lg translate-x-[-50%] translate-y-[-50%] gap-4...
38 |         className,
39 |       )}
40 |       {...props}
41 |     />
42 |   </AlertDialogPortal>
43 | ));
44 | AlertDialogContent.displayName = AlertDialogPrimitive.Content.displayName;
45 |
46 | const AlertDialogHeader = ({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) => (
47 |   <div className={cn("flex flex-col space-y-2 text-center sm:text-left", className)} {...props} />
48 | );
49 | AlertDialogHeader.displayName = "AlertDialogHeader";
50 |
51 | const AlertDialogFooter = ({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) => (
52 |   <div
53 |     className={cn("flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2", className)}
54 |     {...props}
55 |   />
56 | );
57 | AlertDialogFooter.displayName = "AlertDialogFooter";
58 |
59 | const AlertDialogTitle = React.forwardRef<
60 |   React.ElementRef<typeof AlertDialogPrimitive.Title>,
61 |   React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Title>
62 | >(({ className, ...props }, ref) => (
63 |   <AlertDialogPrimitive.Title
64 |     ref={ref}
65 |     className={cn("text-lg font-semibold", className)}
66 |     {...props}
67 |   />
68 | ));
69 | AlertDialogTitle.displayName = AlertDialogPrimitive.Title.displayName;
70 |
71 | const AlertDialogDescription = React.forwardRef<
72 |   React.ElementRef<typeof AlertDialogPrimitive.Description>,
73 |   React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Description>
74 | >(({ className, ...props }, ref) => (
75 |   <AlertDialogPrimitive.Description
76 |     ref={ref}
77 |     className={cn("text-sm text-muted-foreground", className)}
78 |     {...props}
79 |   />
80 | ));
81 | AlertDialogDescription.displayName = AlertDialogPrimitive.Description.displayName;
82 |
83 | const AlertDialogAction = React.forwardRef<
84 |   React.ElementRef<typeof AlertDialogPrimitive.Action>,
85 |   React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Action>
86 | >(({ className, ...props }, ref) => (
87 |   <AlertDialogPrimitive.Action ref={ref} className={cn(buttonVariants(), className)} {...props} />
88 | ));
89 | AlertDialogAction.displayName = AlertDialogPrimitive.Action.displayName;
90 |
91 | const AlertDialogCancel = React.forwardRef<
```

```

92 |   React.ElementRef<typeof AlertDialogPrimitive.Cancel>,
93 |   React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Cancel>
94 | >(({ className, ...props }, ref) => (
95 |   <AlertDialogPrimitive.Cancel
96 |     ref={ref}
97 |     className={cn(buttonVariants({ variant: "outline" }), "mt-2 sm:mt-0", className)}
98 |     {...props}
99 |   />
100 | ));
101 | AlertDialogCancel.displayName = AlertDialogPrimitive.Cancel.displayName;
102 |
103 | export {
104 |   AlertDialog,
105 |   AlertDialogPortal,
106 |   AlertDialogOverlay,
107 |   AlertDialogTrigger,
108 |   AlertDialogContent,
109 |   AlertDialogHeader,
110 |   AlertDialogFooter,
111 |   AlertDialogTitle,
112 |   AlertDialogDescription,
113 |   AlertDialogAction,
114 |   AlertDialogCancel,
115 | };

```

FILE #55: src\components\ui\alert.tsx

```

1 | import * as React from "react";
2 | import { cva, type VariantProps } from "class-variance-authority";
3 |
4 | import { cn } from "@lib/utlis";
5 |
6 | const alertVariants = cva(
7 |   "relative w-full rounded-lg border px-4 py-3 text-sm [&svg+div]:translate-y-[-3px] [&svg]:absolute [&...
8 |   {
9 |     variants: {
10 |       variant: {
11 |         default: "bg-background text-foreground",
12 |         destructive:
13 |           "border-destructive/50 text-destructive dark:border-destructive [&svg]:text-destructive",
14 |       },
15 |     },
16 |     defaultVariants: {
17 |       variant: "default",
18 |     },
19 |   },
20 | );
21 |
22 | const Alert = React.forwardRef<
23 |   HTMLDivElement,
24 |   React.HTMLAttributes<HTMLDivElement> & VariantProps<typeof alertVariants>
25 | >(({ className, variant, ...props }, ref) => (
26 |   <div ref={ref} role="alert" className={cn(alertVariants({ variant }), className)} {...props} />
27 | ));
28 | Alert.displayName = "Alert";
29 |
30 | const AlertTitle = React.forwardRef<HTMLParagraphElement, React.HTMLAttributes<HTMLHeadingElement>>(
31 |   ({ className, ...props }, ref) => (
32 |     <h5
33 |       ref={ref}
34 |       className={cn("mb-1 font-medium leading-none tracking-tight", className)}
35 |       {...props}
36 |     />

```

```
37 |   ),
38 | );
39 | AlertTitle.displayName = "AlertTitle";
40 |
41 | const AlertDescription = React.forwardRef<
42 |   HTMLParagraphElement,
43 |   React.HTMLAttributes<HTMLParagraphElement>
44 | >(({ className, ...props }, ref) => (
45 |   <div ref={ref} className={cn("text-sm [&p]:leading-relaxed", className)} {...props} />
46 | ));
47 | AlertDescription.displayName = "AlertDescription";
48 |
49 | export { Alert, AlertTitle, AlertDescription };
```

FILE #56: src\components\uilaspect-ratio.tsx

```
1 | import * as AspectRatioPrimitive from "@radix-ui/react-aspect-ratio";
2 |
3 | const AspectRatio = AspectRatioPrimitive.Root;
4 |
5 | export { AspectRatio };
```

FILE #57: src\components\uilavatar.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import * as AvatarPrimitive from "@radix-ui/react-avatar";
5 |
6 | import { cn } from "@/lib/utils";
7 |
8 | const Avatar = React.forwardRef<
9 |   React.ElementRef<typeof AvatarPrimitive.Root>,
10 |   React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Root>
11 | >(({ className, ...props }, ref) => (
12 |   <AvatarPrimitive.Root
13 |     ref={ref}
14 |     className={cn("relative flex h-10 w-10 shrink-0 overflow-hidden rounded-full", className)}
15 |     {...props}
16 |   />
17 | ));
18 | Avatar.displayName = AvatarPrimitive.Root.displayName;
19 |
20 | const AvatarImage = React.forwardRef<
21 |   React.ElementRef<typeof AvatarPrimitive.Image>,
22 |   React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Image>
23 | >(({ className, ...props }, ref) => (
24 |   <AvatarPrimitive.Image
25 |     ref={ref}
26 |     className={cn("aspect-square h-full w-full", className)}
27 |     {...props}
28 |   />
29 | ));
30 | AvatarImage.displayName = AvatarPrimitive.Image.displayName;
31 |
32 | const AvatarFallback = React.forwardRef<
33 |   React.ElementRef<typeof AvatarPrimitive.Fallback>,
34 |   React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Fallback>
35 | >(({ className, ...props }, ref) => (
36 |   <AvatarPrimitive.Fallback
37 |     ref={ref}
```



```
38 |     className={cn(
39 |       "flex h-full w-full items-center justify-center rounded-full bg-muted",
40 |       className,
41 |     )}
42 |     {...props}
43 |   />
44 | ));
45 | AvatarFallback.displayName = AvatarPrimitive.Fallback.displayName;
46 |
47 | export { Avatar, AvatarImage, AvatarFallback };
```

FILE #58: src\components\uilbadge.tsx

```
1 | import * as React from "react";
2 | import { cva, type VariantProps } from "class-variance-authority";
3 |
4 | import { cn } from "@lib/utils";
5 |
6 | const badgeVariants = cva(
7 |   "inline-flex items-center rounded-md border px-2.5 py-0.5 text-xs font-semibold transition-colors focus...",
8 |   {
9 |     variants: {
10 |       variant: {
11 |         default: "border-transparent bg-primary text-primary-foreground shadow hover:bg-primary/80",
12 |         secondary:
13 |           "border-transparent bg-secondary text-secondary-foreground hover:bg-secondary/80",
14 |         destructive:
15 |           "border-transparent bg-destructive text-destructive-foreground shadow hover:bg-destructive/80",
16 |         outline: "text-foreground",
17 |       },
18 |     },
19 |     defaultVariants: {
20 |       variant: "default",
21 |     },
22 |   },
23 | );
24 |
25 | export interface BadgeProps
26 |   extends React.HTMLAttributes<HTMLDivElement>, VariantProps<typeof badgeVariants> {}
27 |
28 | function Badge({ className, variant, ...props }: BadgeProps) {
29 |   return <div className={cn(badgeVariants({ variant }), className)} {...props} />;
30 | }
31 |
32 | export { Badge, badgeVariants };
```

FILE #59: src\components\uilbreadcrumb.tsx

```
1 | import * as React from "react";
2 | import { Slot } from "@radix-ui/react-slot";
3 | import { ChevronRight, MoreHorizontal } from "lucide-react";
4 |
5 | import { cn } from "@lib/utils";
6 |
7 | const Breadcrumb = React.forwardRef<
8 |   HTMLDivElement,
9 |   React.ComponentPropsWithoutRef<"nav"> & {
10 |     separator?: React.ReactNode;
11 |   }
12 | >(({ ...props }, ref) => <nav ref={ref} aria-label="breadcrumb" {...props} />);
13 | Breadcrumb.displayName = "Breadcrumb";
```

```
14 |
15 | const BreadcrumbList = React.forwardRef<HTMLListElement, React.ComponentPropsWithoutRef<"ol">>(
16 |   ({ className, ...props }, ref) => (
17 |     <ol
18 |       ref={ref}
19 |       className={cn(
20 |         "flex flex-wrap items-center gap-1.5 break-words text-sm text-muted-foreground sm:gap-2.5",
21 |         className,
22 |       )}
23 |       {...props}
24 |     />
25 |   ),
26 | );
27 | BreadcrumbList.displayName = "BreadcrumbList";
28 |
29 | const BreadcrumbItem = React.forwardRef<HTMLLIElement, React.ComponentPropsWithoutRef<"li">>(
30 |   ({ className, ...props }, ref) => (
31 |     <li ref={ref} className={cn("inline-flex items-center gap-1.5", className)} {...props} />
32 |   ),
33 | );
34 | BreadcrumbItem.displayName = "BreadcrumbItem";
35 |
36 | const BreadcrumbLink = React.forwardRef<
37 |   HTMLAnchorElement,
38 |   React.ComponentPropsWithoutRef<"a"> & {
39 |     asChild?: boolean;
40 |   }
41 | >(({ asChild, className, ...props }, ref) => {
42 |   const Comp = asChild ? Slot : "a";
43 |
44 |   return (
45 |     <Comp
46 |       ref={ref}
47 |       className={cn("transition-colors hover:text-foreground", className)}
48 |       {...props}
49 |     />
50 |   );
51 | });
52 | BreadcrumbLink.displayName = "BreadcrumbLink";
53 |
54 | const BreadcrumbPage = React.forwardRef<HTMLSpanElement, React.ComponentPropsWithoutRef<"span">>(
55 |   ({ className, ...props }, ref) => (
56 |     <span
57 |       ref={ref}
58 |       role="link"
59 |       aria-disabled="true"
60 |       aria-current="page"
61 |       className={cn("font-normal text-foreground", className)}
62 |       {...props}
63 |     />
64 |   ),
65 | );
66 | BreadcrumbPage.displayName = "BreadcrumbPage";
67 |
68 | const BreadcrumbSeparator = ({ children, className, ...props }: React.ComponentProps<"li">) => (
69 |   <li
70 |     role="presentation"
71 |     aria-hidden="true"
72 |     className={cn("[&svg]:w-3.5 [&svg]:h-3.5", className)}
73 |     {...props}
74 |   >
75 |     {children ?? <ChevronRight />}
76 |   </li>
77 | );
78 | BreadcrumbSeparator.displayName = "BreadcrumbSeparator";
```

```

79 |
80 | const BreadcrumbEllipsis = ({ className, ...props }: React.ComponentProps<"span">) => (
81 |   <span
82 |     role="presentation"
83 |     aria-hidden="true"
84 |     className={cn("flex h-9 w-9 items-center justify-center", className)}
85 |     {...props}
86 |   >
87 |     <MoreHorizontal className="h-4 w-4" />
88 |     <span className="sr-only">More</span>
89 |   </span>
90 | );
91 | BreadcrumbEllipsis.displayName = "BreadcrumbElipssis";
92 |
93 | export {
94 |   Breadcrumb,
95 |   BreadcrumbList,
96 |   BreadcrumbItem,
97 |   BreadcrumbLink,
98 |   BreadcrumbPage,
99 |   BreadcrumbSeparator,
100 |   BreadcrumbEllipsis,
101 | };

```

FILE #60: src\components\ui\button.tsx

```

1 | import * as React from "react";
2 | import { Slot } from "@radix-ui/react-slot";
3 | import { cva, type VariantProps } from "class-variance-authority";
4 |
5 | import { cn } from "@lib/utlis";
6 |
7 | const buttonVariants = cva(
8 |   "inline-flex items-center justify-center gap-2 whitespace-nowrap rounded-md text-sm font-medium cursor-...
9 |   {
10 |     variants: {
11 |       variant: {
12 |         default: "bg-primary text-primary-foreground shadow hover:bg-primary/90",
13 |         destructive: "bg-destructive text-destructive-foreground shadow-sm hover:bg-destructive/90",
14 |         outline:
15 |           "border border-input bg-background shadow-sm hover:bg-accent hover:text-accent-foreground",
16 |         secondary: "bg-secondary text-secondary-foreground shadow-sm hover:bg-secondary/80",
17 |         ghost: "hover:bg-accent hover:text-accent-foreground",
18 |         link: "text-primary underline-offset-4 hover:underline",
19 |       },
20 |       size: {
21 |         default: "h-9 px-4 py-2",
22 |         sm: "h-8 rounded-md px-3 text-xs",
23 |         lg: "h-10 rounded-md px-8",
24 |         icon: "h-9 w-9",
25 |       },
26 |     },
27 |     defaultVariants: {
28 |       variant: "default",
29 |       size: "default",
30 |     },
31 |   },
32 | );
33 |
34 | export interface ButtonProps
35 |   extends React.ButtonHTMLAttributes<HTMLButtonElement>, VariantProps<typeof buttonVariants> {
36 |   asChild?: boolean;
37 | }

```

```
38 |  
39 | const Button = React.forwardRef<HTMLButtonElement, ButtonProps>(  
40 |   ({ className, variant, size, asChild = false, ...props }, ref) => {  
41 |     const Comp = asChild ? Slot : "button";  
42 |     return (  
43 |       <Comp className={cn(buttonVariants({ variant, size, className }))} ref={ref} {...props} />  
44 |     );  
45 |   },  
46 | );  
47 | Button.displayName = "Button";  
48 |  
49 | export { Button, buttonVariants };
```

FILE #61: src\components\uilcalendar.tsx

```
1 | "use client";  
2 |  
3 | import * as React from "react";  
4 | import { ChevronDownIcon, ChevronLeftIcon, ChevronRightIcon } from "lucide-react";  
5 | import { DayButton, DayPicker, getDefaultClassNames } from "react-day-picker";  
6 |  
7 | import { cn } from "@lib/utils";  
8 | import { Button, buttonVariants } from "@components/ui/button";  
9 |  
10 | function Calendar({  
11 |   className,  
12 |   classNames,  
13 |   showOutsideDays = true,  
14 |   captionLayout = "label",  
15 |   buttonVariant = "ghost",  
16 |   formatters,  
17 |   components,  
18 |   ...props  
19 | }: React.ComponentProps<typeof DayPicker> & {  
20 |   buttonVariant?: React.ComponentProps<typeof Button>["variant"];  
21 | }) {  
22 |   const defaultClassNames = getDefaultClassNames();  
23 |  
24 |   return (  
25 |     <DayPicker  
26 |       showOutsideDays={showOutsideDays}  
27 |       className={cn(  
28 |         "bg-background group/calendar p-3 [--cell-size:2rem] [[data-slot=card-content]_&]:bg-transparent ...  
29 |         String.raw`rtl:**:[.rdp-button\_next>svg]:rotate-180`,  
30 |         String.raw`rtl:**:[.rdp-button\_previous>svg]:rotate-180`,  
31 |         className,  
32 |       )}  
33 |       captionLayout={captionLayout}  
34 |       formatters={{  
35 |         formatMonthDropdown: (date) => date.toLocaleString("default", { month: "short" }},  
36 |         ...formatters,  
37 |       }}  
38 |       classNames={{  
39 |         root: cn("w-fit", defaultClassNames.root),  
40 |         months: cn("relative flex flex-col gap-4 md:flex-row", defaultClassNames.months),  
41 |         month: cn("flex w-full flex-col gap-4", defaultClassNames.month),  
42 |         nav: cn(  
43 |           "absolute inset-x-0 top-0 flex w-full items-center justify-between gap-1",  
44 |           defaultClassNames.nav,  
45 |         ),  
46 |         button_previous: cn(  
47 |           buttonVariants({ variant: buttonVariant }),  
48 |           "h-[--cell-size] w-[--cell-size] select-none p-0 aria-disabled:opacity-50",
```

```

49 |         defaultClassNames.button_previous,
50 |     ),
51 |     button_next: cn(
52 |         buttonVariants({ variant: buttonVariant }),
53 |         "h(--cell-size) w(--cell-size) select-none p-0 aria-disabled:opacity-50",
54 |         defaultClassNames.button_next,
55 |     ),
56 |     month_caption: cn(
57 |         "flex h(--cell-size) w-full items-center justify-center px(--cell-size)",
58 |         defaultClassNames.month_caption,
59 |     ),
60 |     dropdowns: cn(
61 |         "flex h(--cell-size) w-full items-center justify-center gap-1.5 text-sm font-medium",
62 |         defaultClassNames.dropdowns,
63 |     ),
64 |     dropdown_root: cn(
65 |         "has-focus:border-ring border-input shadow-xs has-focus:ring-ring/50 has-focus:ring-[3px] relat...",
66 |         defaultClassNames.dropdown_root,
67 |     ),
68 |     dropdown: cn("bg-popover absolute inset-0 opacity-0", defaultClassNames.dropdown),
69 |     caption_label: cn(
70 |         "select-none font-medium",
71 |         captionLayout === "label"
72 |             ? "text-sm"
73 |             : "[&>svg]:text-muted-foreground flex h-8 items-center gap-1 rounded-md pl-2 pr-1 text-sm [&>...",
74 |         defaultClassNames.caption_label,
75 |     ),
76 |     table: "w-full border-collapse",
77 |     weekdays: cn("flex", defaultClassNames.weekdays),
78 |     weekday: cn(
79 |         "text-muted-foreground flex-1 select-none rounded-md text-[0.8rem] font-normal",
80 |         defaultClassNames.weekday,
81 |     ),
82 |     week: cn("mt-2 flex w-full", defaultClassNames.week),
83 |     week_number_header: cn("w(--cell-size) select-none", defaultClassNames.week_number_header),
84 |     week_number: cn(
85 |         "text-muted-foreground select-none text-[0.8rem]",
86 |         defaultClassNames.week_number,
87 |     ),
88 |     day: cn(
89 |         "group/day relative aspect-square h-full w-full select-none p-0 text-center [&:first-child[data...",
90 |         defaultClassNames.day,
91 |     ),
92 |     range_start: cn("bg-accent rounded-l-md", defaultClassNames.range_start),
93 |     range_middle: cn("rounded-none", defaultClassNames.range_middle),
94 |     range_end: cn("bg-accent rounded-r-md", defaultClassNames.range_end),
95 |     today: cn(
96 |         "bg-accent text-accent-foreground rounded-md data-[selected=true]:rounded-none",
97 |         defaultClassNames.today,
98 |     ),
99 |     outside: cn(
100 |         "text-muted-foreground aria-selected:text-muted-foreground",
101 |         defaultClassNames.outside,
102 |     ),
103 |     disabled: cn("text-muted-foreground opacity-50", defaultClassNames.disabled),
104 |     hidden: cn("invisible", defaultClassNames.hidden),
105 |     ...classNames,
106 | })
107 | components={{
108 |     Root: ({ className, rootRef, ...props }) => {
109 |         return <div data-slot="calendar" ref={rootRef} className={cn(className)} {...props} />;
110 |     },
111 |     Chevron: ({ className, orientation, ...props }) => {
112 |         if (orientation === "left") {
113 |             return <ChevronLeftIcon className={cn("size-4", className)} {...props} />;

```

```
114 |     }
115 |
116 |     if (orientation === "right") {
117 |       return <ChevronRightIcon className={cn("size-4", className)} {...props} />;
118 |     }
119 |
120 |     return <ChevronDownIcon className={cn("size-4", className)} {...props} />;
121 |   },
122 |   DayButton: CalendarDayButton,
123 |   WeekNumber: ({ children, ...props }) => {
124 |     return (
125 |       <td {...props}>
126 |         <div className="flex size-(--cell-size) items-center justify-center text-center">
127 |           {children}
128 |         </div>
129 |       </td>
130 |     );
131 |   },
132 |   ...components,
133 | })
134 | {...props}
135 | />
136 | );
137 | }
138 |
139 | function CalendarDayButton({
140 |   className,
141 |   day,
142 |   modifiers,
143 |   ...props
144 | }: React.ComponentProps<typeof DayButton>) {
145 |   const defaultClassNames = getDefaultClassNames();
146 |
147 |   const ref = React.useRef<HTMLButtonElement>(null);
148 |   React.useEffect(() => {
149 |     if (modifiers.focused) ref.current?.focus();
150 |   }, [modifiers.focused]);
151 |
152 |   return (
153 |     <Button
154 |       ref={ref}
155 |       variant="ghost"
156 |       size="icon"
157 |       data-day={day.date.toLocaleDateString()}
158 |       data-selected-single={
159 |         modifiers.selected &&
160 |         !modifiers.range_start &&
161 |         !modifiers.range_end &&
162 |         !modifiers.range_middle
163 |       }
164 |       data-range-start={modifiers.range_start}
165 |       data-range-end={modifiers.range_end}
166 |       data-range-middle={modifiers.range_middle}
167 |       className={cn(
168 |         "data-[selected-single=true]:bg-primary data-[selected-single=true]:text-primary-foreground data-...
169 |         defaultClassNames.day,
170 |         className,
171 |       )}
172 |       {...props}
173 |     />
174 |   );
175 | }
176 |
177 | export { Calendar, CalendarDayButton };
```

FILE #62: src\components\uilcard.tsx

```
1 | import * as React from "react";
2 |
3 | import { cn } from "@lib/utils";
4 |
5 | const Card = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(<
6 |   ({ className, ...props }, ref) => (
7 |     <div
8 |       ref={ref}
9 |       className={cn("rounded-xl border bg-card text-card-foreground shadow", className)}
10 |       {...props}
11 |     />
12 |   ),
13 | );
14 | Card.displayName = "Card";
15 |
16 | const CardHeader = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(<
17 |   ({ className, ...props }, ref) => (
18 |     <div ref={ref} className={cn("flex flex-col space-y-1.5 p-6", className)} {...props} />
19 |   ),
20 | );
21 | CardHeader.displayName = "CardHeader";
22 |
23 | const CardTitle = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(<
24 |   ({ className, ...props }, ref) => (
25 |     <div
26 |       ref={ref}
27 |       className={cn("font-semibold leading-none tracking-tight", className)}
28 |       {...props}
29 |     />
30 |   ),
31 | );
32 | CardTitle.displayName = "CardTitle";
33 |
34 | const CardDescription = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(<
35 |   ({ className, ...props }, ref) => (
36 |     <div ref={ref} className={cn("text-sm text-muted-foreground", className)} {...props} />
37 |   ),
38 | );
39 | CardDescription.displayName = "CardDescription";
40 |
41 | const CardContent = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(<
42 |   ({ className, ...props }, ref) => (
43 |     <div ref={ref} className={cn("p-6 pt-0", className)} {...props} />
44 |   ),
45 | );
46 | CardContent.displayName = "CardContent";
47 |
48 | const CardFooter = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(<
49 |   ({ className, ...props }, ref) => (
50 |     <div ref={ref} className={cn("flex items-center p-6 pt-0", className)} {...props} />
51 |   ),
52 | );
53 | CardFooter.displayName = "CardFooter";
54 |
55 | export { Card, CardHeader, CardFooter, CardTitle, CardDescription, CardContent };
```

FILE #63: src\components\uilcarousel.tsx

```
1 | import * as React from "react";
2 | import useEmblaCarousel, { type UseEmblaCarouselType } from "embla-carousel-react";
```

```
3 | import { ArrowLeft, ArrowRight } from "lucide-react";
4 |
5 | import { cn } from "@lib/utils";
6 | import { Button } from "@components/ui/button";
7 |
8 | type CarouselApi = UseEmblaCarouselType[1];
9 | type UseCarouselParameters = Parameters<typeof useEmblaCarousel>;
10 | type CarouselOptions = UseCarouselParameters[0];
11 | type CarouselPlugin = UseCarouselParameters[1];
12 |
13 | type CarouselProps = {
14 |   opts?: CarouselOptions;
15 |   plugins?: CarouselPlugin;
16 |   orientation?: "horizontal" | "vertical";
17 |   setApi?: (api: CarouselApi) => void;
18 | };
19 |
20 | type CarouselContextProps = {
21 |   carouselRef: ReturnType<typeof useEmblaCarousel>[0];
22 |   api: ReturnType<typeof useEmblaCarousel>[1];
23 |   scrollPrev: () => void;
24 |   scrollNext: () => void;
25 |   canScrollPrev: boolean;
26 |   canScrollNext: boolean;
27 | } & CarouselProps;
28 |
29 | const CarouselContext = React.createContext<CarouselContextProps | null>(null);
30 |
31 | function useCarousel() {
32 |   const context = React.useContext(CarouselContext);
33 |
34 |   if (!context) {
35 |     throw new Error("useCarousel must be used within a <Carousel />");
36 |   }
37 |
38 |   return context;
39 | }
40 |
41 | const Carousel = React.forwardRef<
42 |   HTMLDivElement,
43 |   React.HTMLAttributes<HTMLDivElement> & CarouselProps
44 | >(({ orientation = "horizontal", opts, setApi, plugins, className, children, ...props }, ref) => {
45 |   const [carouselRef, api] = useEmblaCarousel(
46 |     {
47 |       ...opts,
48 |       axis: orientation === "horizontal" ? "x" : "y",
49 |     },
50 |     plugins,
51 |   );
52 |   const [canScrollPrev, setCanScrollPrev] = React.useState(false);
53 |   const [canScrollNext, setCanScrollNext] = React.useState(false);
54 |
55 |   const onSelect = React.useCallback((api: CarouselApi) => {
56 |     if (!api) {
57 |       return;
58 |     }
59 |
60 |     setCanScrollPrev(api.canScrollPrev());
61 |     setCanScrollNext(api.canScrollNext());
62 |   }, []);
63 |
64 |   const scrollPrev = React.useCallback(() => {
65 |     api?.scrollPrev();
66 |   }, [api]);
67 |
```



```
68 |   const scrollNext = React.useCallback(() => {
69 |     api?.scrollNext();
70 |   }, [api]);
71 |
72 |   const handleKeyDown = React.useCallback(
73 |     (event: React.KeyboardEvent<HTMLDivElement>) => {
74 |       if (event.key === "ArrowLeft") {
75 |         event.preventDefault();
76 |         scrollPrev();
77 |       } else if (event.key === "ArrowRight") {
78 |         event.preventDefault();
79 |         scrollNext();
80 |       }
81 |     },
82 |     [scrollPrev, scrollNext],
83 |   );
84 |
85 |   React.useEffect(() => {
86 |     if (!api || !setApi) {
87 |       return;
88 |     }
89 |
90 |     setApi(api);
91 |   }, [api, setApi]);
92 |
93 |   React.useEffect(() => {
94 |     if (!api) {
95 |       return;
96 |     }
97 |
98 |     onSelect(api);
99 |     api.on("reInit", onSelect);
100 |    api.on("select", onSelect);
101 |
102 |    return () => {
103 |      api?.off("select", onSelect);
104 |    };
105 |   }, [api, onSelect]);
106 |
107 |   return (
108 |     <CarouselContext.Provider
109 |       value={{
110 |         carouselRef,
111 |         api: api,
112 |         opts,
113 |         orientation: orientation || (opts?.axis === "y" ? "vertical" : "horizontal"),
114 |         scrollPrev,
115 |         scrollNext,
116 |         canScrollPrev,
117 |         canScrollNext,
118 |       }}
119 |     >
120 |       <div
121 |         ref={ref}
122 |         onKeyDownCapture={handleKeyDown}
123 |         className={cn("relative", className)}
124 |         role="region"
125 |         aria-roledescription="carousel"
126 |         {...props}
127 |       >
128 |         {children}
129 |       </div>
130 |     </CarouselContext.Provider>
131 |   );
132 | });
```

```
133 | Carousel.displayName = "Carousel";
134 |
135 | const CarouselContent = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(
136 |   ({ className, ...props }, ref) => {
137 |     const { carouselRef, orientation } = useCarousel();
138 |
139 |     return (
140 |       <div ref={carouselRef} className="overflow-hidden">
141 |         <div
142 |           ref={ref}
143 |           className={cn(
144 |             "flex",
145 |             orientation === "horizontal" ? "-ml-4" : "-mt-4 flex-col",
146 |             className,
147 |           )}
148 |           {...props}
149 |         />
150 |       </div>
151 |     );
152 |   },
153 | );
154 | CarouselContent.displayName = "CarouselContent";
155 |
156 | const CarouselItem = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(
157 |   ({ className, ...props }, ref) => {
158 |     const { orientation } = useCarousel();
159 |
160 |     return (
161 |       <div
162 |         ref={ref}
163 |         role="group"
164 |         aria-roledescription="slide"
165 |         className={cn(
166 |           "min-w-0 shrink-0 grow-0 basis-full",
167 |           orientation === "horizontal" ? "pl-4" : "pt-4",
168 |           className,
169 |         )}
170 |         {...props}
171 |       />
172 |     );
173 |   },
174 | );
175 | CarouselItem.displayName = "CarouselItem";
176 |
177 | const CarouselPrevious = React.forwardRef<HTMLButtonElement, React.ComponentProps<typeof Button>>(
178 |   ({ className, variant = "outline", size = "icon", ...props }, ref) => {
179 |     const { orientation, scrollPrev, canScrollPrev } = useCarousel();
180 |
181 |     return (
182 |       <Button
183 |         ref={ref}
184 |         variant={variant}
185 |         size={size}
186 |         className={cn(
187 |           "absolute h-8 w-8 rounded-full",
188 |           orientation === "horizontal"
189 |             ? "-left-12 top-1/2 -translate-y-1/2"
190 |             : "-top-12 left-1/2 -translate-x-1/2 rotate-90",
191 |           className,
192 |         )}
193 |         disabled={!canScrollPrev}
194 |         onClick={scrollPrev}
195 |         {...props}
196 |       >
197 |         <ArrowLeft className="h-4 w-4" />
```

```

198 |         <span className="sr-only">Previous slide</span>
199 |     </Button>
200 | );
201 | },
202 | );
203 | CarouselPrevious.displayName = "CarouselPrevious";
204 |
205 | const CarouselNext = React.forwardRef<HTMLButtonElement, React.ComponentProps<typeof Button>>(
206 |   ({ className, variant = "outline", size = "icon", ...props }, ref) => {
207 |     const { orientation, scrollNext, canScrollNext } = useCarousel();
208 |
209 |     return (
210 |       <Button
211 |         ref={ref}
212 |         variant={variant}
213 |         size={size}
214 |         className={cn(
215 |           "absolute h-8 w-8 rounded-full",
216 |           orientation === "horizontal"
217 |             ? "-right-12 top-1/2 -translate-y-1/2"
218 |             : "-bottom-12 left-1/2 -translate-x-1/2 rotate-90",
219 |           className,
220 |         )}
221 |         disabled={!canScrollNext}
222 |         onClick={scrollNext}
223 |         {...props}
224 |       >
225 |         <ArrowRight className="h-4 w-4" />
226 |         <span className="sr-only">Next slide</span>
227 |       </Button>
228 |     );
229 |   },
230 | );
231 | CarouselNext.displayName = "CarouselNext";
232 |
233 | export {
234 |   type CarouselApi,
235 |   Carousel,
236 |   CarouselContent,
237 |   CarouselItem,
238 |   CarouselPrevious,
239 |   CarouselNext,
240 | };

```

FILE #64: src\components\uilchart.tsx

```

1 | import * as React from "react";
2 | import * as RechartsPrimitive from "recharts";
3 |
4 | import { cn } from "@lib/utils";
5 |
6 | // Format: { THEME_NAME: CSS_SELECTOR }
7 | const THEMES = { light: "", dark: ".dark" } as const;
8 |
9 | export type ChartConfig = {
10 |   [k in string]: {
11 |     label?: React.ReactNode;
12 |     icon?: React.ComponentType;
13 |   } & (
14 |     | { color?: string; theme?: never }
15 |     | { color?: never; theme: Record<keyof typeof THEMES, string> }
16 |   );
17 | };

```

```
18 |  
19 | type ChartContextProps = {  
20 |   config: ChartConfig;  
21 | };  
22 |  
23 | const ChartContext = React.createContext<ChartContextProps | null>(null);  
24 |  
25 | function useChart() {  
26 |   const context = React.useContext(ChartContext);  
27 |  
28 |   if (!context) {  
29 |     throw new Error("useChart must be used within a <ChartContainer />");  
30 |   }  
31 |  
32 |   return context;  
33 | }  
34 |  
35 | const ChartContainer = React.forwardRef<  
36 |   HTMLDivElement,  
37 |   React.ComponentProps<"div"> & {  
38 |     config: ChartConfig;  
39 |     children: React.ComponentProps<typeof RechartsPrimitive.ResponsiveContainer>["children"];  
40 |   }  
41 | >(({ id, className, children, config, ...props }, ref) => {  
42 |   const uniqueId = React.useId();  
43 |   const chartId = `chart-${id || uniqueId.replace(/:/g, "")}`;  
44 |  
45 |   return (  
46 |     <ChartContext.Provider value={{ config }}>  
47 |       <div  
48 |         data-chart={chartId}  
49 |         ref={ref}  
50 |         className={cn(  
51 |           "flex aspect-video justify-center text-xs [&_.recharts-cartesian-axis-tick_text]:fill-muted-for...  
52 |           className,  
53 |         )}  
54 |         {...props}  
55 |       >  
56 |         <ChartStyle id={chartId} config={config} />  
57 |         <RechartsPrimitive.ResponsiveContainer>{children}</RechartsPrimitive.ResponsiveContainer>  
58 |       </div>  
59 |     </ChartContext.Provider>  
60 |   );  
61 | });  
62 | ChartContainer.displayName = "Chart";  
63 |  
64 | const ChartStyle = ({ id, config }: { id: string; config: ChartConfig }) => {  
65 |   const colorConfig = Object.entries(config).filter(([, config]) => config.theme || config.color);  
66 |  
67 |   if (!colorConfig.length) {  
68 |     return null;  
69 |   }  
70 |  
71 |   return (  
72 |     <style  
73 |       dangerouslySetInnerHTML={{  
74 |         __html: Object.entries(THEMES)  
75 |           .map(  
76 |             ([theme, prefix]) => `  
77 | ${prefix} [data-chart=${id}] {  
78 | ${colorConfig  
79 |   .map(([key, itemConfig]) => {  
80 |     const color = itemConfig.theme?.[theme as keyof typeof itemConfig.theme] || itemConfig.color;  
81 |     return color ? `--color-${key}: ${color};` : null;  
82 |   })
```

```
83 |   .join("\n"))
84 | }
85 | `,
86 |   )
87 |   .join("\n"),
88 |   }}
89 |   />
90 | );
91 | };
92 |
93 | const ChartTooltip = RechartsPrimitive.Tooltip;
94 |
95 | const ChartTooltipContent = React.forwardRef<
96 |   HTMLDivElement,
97 |   React.ComponentProps<typeof RechartsPrimitive.Tooltip> &
98 |     React.ComponentProps<"div"> & {
99 |       hideLabel?: boolean;
100 |       hideIndicator?: boolean;
101 |       indicator?: "line" | "dot" | "dashed";
102 |       nameKey?: string;
103 |       labelKey?: string;
104 |     }
105 | > (
106 |   (
107 |     {
108 |       active,
109 |       payload,
110 |       className,
111 |       indicator = "dot",
112 |       hideLabel = false,
113 |       hideIndicator = false,
114 |       label,
115 |       labelFormatter,
116 |       labelClassName,
117 |       formatter,
118 |       color,
119 |       nameKey,
120 |       labelKey,
121 |     },
122 |     ref,
123 |   ) => {
124 |     const { config } = useChart();
125 |
126 |     const tooltipLabel = React.useMemo(() => {
127 |       if (hideLabel || !payload?.length) {
128 |         return null;
129 |       }
130 |
131 |       const [item] = payload;
132 |       const key = `${labelKey || item?.dataKey || item?.name || "value"}`;
133 |       const itemConfig = getPayloadConfigFromPayload(config, item, key);
134 |       const value =
135 |         !labelKey && typeof label === "string"
136 |           ? config[label as keyof typeof config]?.label || label
137 |           : itemConfig?.label;
138 |
139 |       if (labelFormatter) {
140 |         return (
141 |           <div className={cn("font-medium", labelClassName)}>{labelFormatter(value, payload)}</div>
142 |         );
143 |       }
144 |
145 |       if (!value) {
146 |         return null;
147 |       }
```

```

148 |
149 |     return <div className={cn("font-medium", labelClassName)}>{value}</div>;
150 |   }, [label, labelFormatter, payload, hideLabel, labelClassName, config, labelKey]);
151 |
152 |   if (!active || !payload?.length) {
153 |     return null;
154 |   }
155 |
156 |   const nestLabel = payload.length === 1 && indicator !== "dot";
157 |
158 |   return (
159 |     <div
160 |       ref={ref}
161 |       className={cn(
162 |         "grid min-w-[8rem] items-start gap-1.5 rounded-lg border border-border/50 bg-background px-2.5 ...
163 |         className,
164 |       )}
165 |     >
166 |       {!nestLabel ? tooltipLabel : null}
167 |       <div className="grid gap-1.5">
168 |         {payload
169 |           .filter((item) => item.type !== "none")
170 |           .map((item, index) => {
171 |             const key = `${nameKey} || item.name || item.dataKey || "value"`;
172 |             const itemConfig = getPayloadConfigFromPayload(config, item, key);
173 |             const indicatorColor = color || item.payload.fill || item.color;
174 |
175 |             return (
176 |               <div
177 |                 key={item.dataKey}
178 |                 className={cn(
179 |                   "flex w-full flex-wrap items-stretch gap-2 [&svg]:h-2.5 [&svg]:w-2.5 [&svg]:text-m...
180 |                   indicator === "dot" && "items-center",
181 |                 )}
182 |               >
183 |                 {formatter && item?.value !== undefined && item.name ? (
184 |                   formatter(item.value, item.name, item, index, item.payload)
185 |                 ) : (
186 |                   <>
187 |                     {itemConfig?.icon ? (
188 |                       <itemConfig.icon />
189 |                     ) : (
190 |                       !hideIndicator && (
191 |                         <div
192 |                           className={cn(
193 |                             "shrink-0 rounded-[2px] border(--color-border) bg(--color-bg)",
194 |                             {
195 |                               "h-2.5 w-2.5": indicator === "dot",
196 |                               "w-1": indicator === "line",
197 |                               "w-0 border-[1.5px] border-dashed bg-transparent":
198 |                                 indicator === "dashed",
199 |                               "my-0.5": nestLabel && indicator === "dashed",
200 |                             },
201 |                         )}
202 |                         style={
203 |                           {
204 |                             "--color-bg": indicatorColor,
205 |                             "--color-border": indicatorColor,
206 |                           } as React.CSSProperties
207 |                         }
208 |                       />
209 |                     )
210 |                   }
211 |                 <div
212 |                   className={cn(

```

```
213 |         "flex flex-1 justify-between leading-none",
214 |         nestLabel ? "items-end" : "items-center",
215 |     )}
216 |     >
217 |         <div className="grid gap-1.5">
218 |             {nestLabel ? tooltipLabel : null}
219 |             <span className="text-muted-foreground">
220 |                 {itemConfig?.label || item.name}
221 |             </span>
222 |         </div>
223 |         {item.value && (
224 |             <span className="font-mono font-medium tabular-nums text-foreground">
225 |                 {item.value.toLocaleString()}
226 |             </span>
227 |         )}
228 |     </div>
229 | </>
230 |     )}
231 | </div>
232 | );
233 | }}}}
234 | </div>
235 | </div>
236 | );
237 | },
238 | );
239 | ChartTooltipContent.displayName = "ChartTooltip";
240 |
241 | const ChartLegend = RechartsPrimitive.Legend;
242 |
243 | const ChartLegendContent = React.forwardRef<
244 |   HTMLDivElement,
245 |   React.ComponentProps<"div"> &
246 |     Pick<RechartsPrimitive.LegendProps, "payload" | "verticalAlign"> & {
247 |       hideIcon?: boolean;
248 |       nameKey?: string;
249 |     }
250 | >(({ className, hideIcon = false, payload, verticalAlign = "bottom", nameKey }, ref) => {
251 |   const { config } = useChart();
252 |
253 |   if (!payload?.length) {
254 |     return null;
255 |   }
256 |
257 |   return (
258 |     <div
259 |       ref={ref}
260 |       className={cn(
261 |         "flex items-center justify-center gap-4",
262 |         verticalAlign === "top" ? "pb-3" : "pt-3",
263 |         className,
264 |       )}
265 |     >
266 |       {payload
267 |         .filter((item) => item.type !== "none")
268 |         .map((item) => {
269 |           const key = `${nameKey} || item.dataKey || "value"`;
270 |           const itemConfig = getPayloadConfigFromPayload(config, item, key);
271 |
272 |           return (
273 |             <div
274 |               key={item.value}
275 |               className={cn(
276 |                 "flex items-center gap-1.5 [>svg]:h-3 [>svg]:w-3 [>svg]:text-muted-foreground",
277 |               )}
```

```
278 |         >
279 |         {itemConfig?.icon && !hideIcon ? (
280 |             <itemConfig.icon />
281 |         ) : (
282 |             <div
283 |                 className="h-2 w-2 shrink-0 rounded-[2px]"
284 |                 style={{
285 |                     backgroundColor: item.color,
286 |                 }}
287 |             />
288 |         )}
289 |         {itemConfig?.label}
290 |     </div>
291 | );
292 | }}}
293 | </div>
294 | );
295 | });
296 | ChartLegendContent.displayName = "ChartLegend";
297 |
298 | // Helper to extract item config from a payload.
299 | function getPayloadConfigFromPayload(config: ChartConfig, payload: unknown, key: string) {
300 |     if (typeof payload !== "object" || payload === null) {
301 |         return undefined;
302 |     }
303 |
304 |     const payloadPayload =
305 |         "payload" in payload && typeof payload.payload === "object" && payload.payload !== null
306 |         ? payload.payload
307 |         : undefined;
308 |
309 |     let configLabelKey: string = key;
310 |
311 |     if (key in payload && typeof payload[key as keyof typeof payload] === "string") {
312 |         configLabelKey = payload[key as keyof typeof payload] as string;
313 |     } else if (
314 |         payloadPayload &&
315 |         key in payloadPayload &&
316 |         typeof payloadPayload[key as keyof typeof payloadPayload] === "string"
317 |     ) {
318 |         configLabelKey = payloadPayload[key as keyof typeof payloadPayload] as string;
319 |     }
320 |
321 |     return configLabelKey in config ? config[configLabelKey] : config[key as keyof typeof config];
322 | }
323 |
324 | export {
325 |     ChartContainer,
326 |     ChartTooltip,
327 |     ChartTooltipContent,
328 |     ChartLegend,
329 |     ChartLegendContent,
330 |     ChartStyle,
331 | };
```

FILE #65: src\components\uilcheckbox.tsx

```
1 | import * as React from "react";
2 | import * as CheckboxPrimitive from "@radix-ui/react-checkbox";
3 | import { Check } from "lucide-react";
4 |
5 | import { cn } from "@lib/utlis";
6 |
```



```
7 | const Checkbox = React.forwardRef<
8 |   React.ElementRef<typeof CheckboxPrimitive.Root>,
9 |   React.ComponentPropsWithoutRef<typeof CheckboxPrimitive.Root>
10 | >(({ className, ...props }, ref) => (
11 |   <CheckboxPrimitive.Root
12 |     ref={ref}
13 |     className={cn(
14 |       "grid place-content-center peer h-4 w-4 shrink-0 rounded-sm border border-primary shadow cursor-poi...
15 |       className,
16 |     )}
17 |     {...props}
18 |   >
19 |     <CheckboxPrimitive.Indicator className={cn("grid place-content-center text-current")}>
20 |       <Check className="h-4 w-4" />
21 |     </CheckboxPrimitive.Indicator>
22 |   </CheckboxPrimitive.Root>
23 | ));
24 | Checkbox.displayName = CheckboxPrimitive.Root.displayName;
25 |
26 | export { Checkbox };
```

FILE #66: src\components\uilcollapsible.tsx

```
1 | "use client";
2 |
3 | import * as CollapsiblePrimitive from "@radix-ui/react-collapsible";
4 |
5 | const Collapsible = CollapsiblePrimitive.Root;
6 |
7 | const CollapsibleTrigger = CollapsiblePrimitive.CollapsibleTrigger;
8 |
9 | const CollapsibleContent = CollapsiblePrimitive.CollapsibleContent;
10 |
11 | export { Collapsible, CollapsibleTrigger, CollapsibleContent };
```

FILE #67: src\components\uilcommand.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import { type DialogProps } from "@radix-ui/react-dialog";
5 | import { Command as CommandPrimitive } from "cmdk";
6 | import { Search } from "lucide-react";
7 |
8 | import { cn } from "@lib/utils";
9 | import { Dialog, DialogContent } from "@components/ui/dialog";
10 |
11 | const Command = React.forwardRef<
12 |   React.ElementRef<typeof CommandPrimitive>,
13 |   React.ComponentPropsWithoutRef<typeof CommandPrimitive>
14 | >(({ className, ...props }, ref) => (
15 |   <CommandPrimitive
16 |     ref={ref}
17 |     className={cn(
18 |       "flex h-full w-full flex-col overflow-hidden rounded-md bg-popover text-popover-foreground",
19 |       className,
20 |     )}
21 |     {...props}
22 |   />
23 | ));
24 | Command.displayName = CommandPrimitive.displayName;
```

```
25 |
26 | const CommandDialog = ({ children, ...props }: DialogProps) => {
27 |   return (
28 |     <Dialog {...props}>
29 |       <DialogContent className="overflow-hidden p-0">
30 |         <Command className="[&_[cmdk-group-heading]]:px-2 [&_[cmdk-group-heading]]:font-medium [&_[cmdk-g...
31 |           {children}
32 |         </Command>
33 |       </DialogContent>
34 |     </Dialog>
35 |   );
36 | };
37 |
38 | const CommandInput = React.forwardRef<
39 |   React.ElementRef<typeof CommandPrimitive.Input>,
40 |   React.ComponentPropsWithoutRef<typeof CommandPrimitive.Input>
41 | >(({ className, ...props }, ref) => (
42 |   <div className="flex items-center border-b px-3" cmdk-input-wrapper="">
43 |     <Search className="mr-2 h-4 w-4 shrink-0 opacity-50" />
44 |     <CommandPrimitive.Input
45 |       ref={ref}
46 |       className={cn(
47 |         "flex h-10 w-full rounded-md bg-transparent py-3 text-sm outline-none placeholder:text-muted-fore...
48 |         className,
49 |       )}
50 |       {...props}
51 |     />
52 |   </div>
53 | ));
54 |
55 | CommandInput.displayName = CommandPrimitive.Input.displayName;
56 |
57 | const CommandList = React.forwardRef<
58 |   React.ElementRef<typeof CommandPrimitive.List>,
59 |   React.ComponentPropsWithoutRef<typeof CommandPrimitive.List>
60 | >(({ className, ...props }, ref) => (
61 |   <CommandPrimitive.List
62 |     ref={ref}
63 |     className={cn("max-h-[300px] overflow-y-auto overflow-x-hidden", className)}
64 |     {...props}
65 |   />
66 | ));
67 |
68 | CommandList.displayName = CommandPrimitive.List.displayName;
69 |
70 | const CommandEmpty = React.forwardRef<
71 |   React.ElementRef<typeof CommandPrimitive.Empty>,
72 |   React.ComponentPropsWithoutRef<typeof CommandPrimitive.Empty>
73 | >((props, ref) => (
74 |   <CommandPrimitive.Empty ref={ref} className="py-6 text-center text-sm" {...props} />
75 | ));
76 |
77 | CommandEmpty.displayName = CommandPrimitive.Empty.displayName;
78 |
79 | const CommandGroup = React.forwardRef<
80 |   React.ElementRef<typeof CommandPrimitive.Group>,
81 |   React.ComponentPropsWithoutRef<typeof CommandPrimitive.Group>
82 | >(({ className, ...props }, ref) => (
83 |   <CommandPrimitive.Group
84 |     ref={ref}
85 |     className={cn(
86 |       "overflow-hidden p-1 text-foreground [&_[cmdk-group-heading]]:px-2 [&_[cmdk-group-heading]]:py-1.5 ...
87 |       className,
88 |     )}
89 |     {...props}
```

```
90 |   />
91 | });
92 |
93 | CommandGroup.displayName = CommandPrimitive.Group.displayName;
94 |
95 | const CommandSeparator = React.forwardRef<
96 |   React.ElementRef<typeof CommandPrimitive.Separator>,
97 |   React.ComponentPropsWithoutRef<typeof CommandPrimitive.Separator>
98 | >(({ className, ...props }, ref) => (
99 |   <CommandPrimitive.Separator
100 |     ref={ref}
101 |     className={cn("-mx-1 h-px bg-border", className)}
102 |     {...props}
103 |   />
104 | ));
105 | CommandSeparator.displayName = CommandPrimitive.Separator.displayName;
106 |
107 | const CommandItem = React.forwardRef<
108 |   React.ElementRef<typeof CommandPrimitive.Item>,
109 |   React.ComponentPropsWithoutRef<typeof CommandPrimitive.Item>
110 | >(({ className, ...props }, ref) => (
111 |   <CommandPrimitive.Item
112 |     ref={ref}
113 |     className={cn(
114 |       "relative flex cursor-default gap-2 select-none items-center rounded-sm px-2 py-1.5 text-sm outline...
115 |       className,
116 |     )}
117 |     {...props}
118 |   />
119 | ));
120 |
121 | CommandItem.displayName = CommandPrimitive.Item.displayName;
122 |
123 | const CommandShortcut = ({ className, ...props }: React.HTMLAttributes<HTMLSpanElement>) => {
124 |   return (
125 |     <span
126 |       className={cn("ml-auto text-xs tracking-widest text-muted-foreground", className)}
127 |       {...props}
128 |     />
129 |   );
130 | };
131 | CommandShortcut.displayName = "CommandShortcut";
132 |
133 | export {
134 |   Command,
135 |   CommandDialog,
136 |   CommandInput,
137 |   CommandList,
138 |   CommandEmpty,
139 |   CommandGroup,
140 |   CommandItem,
141 |   CommandShortcut,
142 |   CommandSeparator,
143 | };
```

FILE #68: src\components\uilcontext-menu.tsx

```
1 | import * as React from "react";
2 | import * as ContextMenuPrimitive from "@radix-ui/react-context-menu";
3 | import { Check, ChevronRight, Circle } from "lucide-react";
4 |
5 | import { cn } from "@/lib/utils";
6 |
```

```
7 | const ContextMenu = ContextMenuPrimitive.Root;
8 |
9 | const ContextMenuTrigger = ContextMenuPrimitive.Trigger;
10 |
11 | const ContextMenuGroup = ContextMenuPrimitive.Group;
12 |
13 | const ContextMenuPortal = ContextMenuPrimitive.Portal;
14 |
15 | const ContextMenuSub = ContextMenuPrimitive.Sub;
16 |
17 | const ContextMenuRadioGroup = ContextMenuPrimitive.RadioGroup;
18 |
19 | const ContextMenuSubTrigger = React.forwardRef<
20 |   React.ElementRef<typeof ContextMenuPrimitive.SubTrigger>,
21 |   React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.SubTrigger> & {
22 |     inset?: boolean;
23 |   }
24 | >(({ className, inset, children, ...props }, ref) => (
25 |   <ContextMenuPrimitive.SubTrigger
26 |     ref={ref}
27 |     className={cn(
28 |       "flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-...
29 |       inset && "pl-8",
30 |       className,
31 |     )}
32 |     {...props}
33 |   >
34 |     {children}
35 |     <ChevronRight className="ml-auto h-4 w-4" />
36 |   </ContextMenuPrimitive.SubTrigger>
37 | ));
38 | ContextMenuSubTrigger.displayName = ContextMenuPrimitive.SubTrigger.displayName;
39 |
40 | const ContextMenuSubContent = React.forwardRef<
41 |   React.ElementRef<typeof ContextMenuPrimitive.SubContent>,
42 |   React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.SubContent>
43 | >(({ className, ...props }, ref) => (
44 |   <ContextMenuPrimitive.SubContent
45 |     ref={ref}
46 |     className={cn(
47 |       "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-...
48 |       className,
49 |     )}
50 |     {...props}
51 |   />
52 | ));
53 | ContextMenuSubContent.displayName = ContextMenuPrimitive.SubContent.displayName;
54 |
55 | const ContextMenuContent = React.forwardRef<
56 |   React.ElementRef<typeof ContextMenuPrimitive.Content>,
57 |   React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Content>
58 | >(({ className, ...props }, ref) => (
59 |   <ContextMenuPrimitive.Portal>
60 |     <ContextMenuPrimitive.Content
61 |       ref={ref}
62 |       className={cn(
63 |         "z-50 max-h-(--radix-context-menu-content-available-height) min-w-[8rem] overflow-y-auto overflow-...
64 |         className,
65 |       )}
66 |       {...props}
67 |     />
68 |   </ContextMenuPrimitive.Portal>
69 | ));
70 | ContextMenuContent.displayName = ContextMenuPrimitive.Content.displayName;
71 |
```

```
72 | const ContextMenuItem = React.forwardRef<
73 |   React.ElementRef<typeof ContextMenuPrimitive.Item>,
74 |   React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Item> & {
75 |     inset?: boolean;
76 |   }
77 | >(({ className, inset, ...props }, ref) => (
78 |   <ContextMenuPrimitive.Item
79 |     ref={ref}
80 |     className={cn(
81 |       "relative flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none ...
82 |       inset && "pl-8",
83 |       className,
84 |     )}
85 |     {...props}
86 |   />
87 | ));
88 | ContextMenuItem.displayName = ContextMenuPrimitive.Item.displayName;
89 |
90 | const ContextMenuCheckboxItem = React.forwardRef<
91 |   React.ElementRef<typeof ContextMenuPrimitive.CheckboxItem>,
92 |   React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.CheckboxItem>
93 | >(({ className, children, checked, ...props }, ref) => (
94 |   <ContextMenuPrimitive.CheckboxItem
95 |     ref={ref}
96 |     className={cn(
97 |       "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-...
98 |       className,
99 |     )}
100 |     checked={checked}
101 |     {...props}
102 |   >
103 |     <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
104 |       <ContextMenuPrimitive.ItemIndicator>
105 |         <Check className="h-4 w-4" />
106 |       </ContextMenuPrimitive.ItemIndicator>
107 |     </span>
108 |     {children}
109 |   </ContextMenuPrimitive.CheckboxItem>
110 | ));
111 | ContextMenuCheckboxItem.displayName = ContextMenuPrimitive.CheckboxItem.displayName;
112 |
113 | const ContextMenuRadioItem = React.forwardRef<
114 |   React.ElementRef<typeof ContextMenuPrimitive.RadioItem>,
115 |   React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.RadioItem>
116 | >(({ className, children, ...props }, ref) => (
117 |   <ContextMenuPrimitive.RadioItem
118 |     ref={ref}
119 |     className={cn(
120 |       "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-...
121 |       className,
122 |     )}
123 |     {...props}
124 |   >
125 |     <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
126 |       <ContextMenuPrimitive.ItemIndicator>
127 |         <Circle className="h-4 w-4 fill-current" />
128 |       </ContextMenuPrimitive.ItemIndicator>
129 |     </span>
130 |     {children}
131 |   </ContextMenuPrimitive.RadioItem>
132 | ));
133 | ContextMenuRadioItem.displayName = ContextMenuPrimitive.RadioItem.displayName;
134 |
135 | const ContextMenuLabel = React.forwardRef<
136 |   React.ElementRef<typeof ContextMenuPrimitive.Label>,
```

```
137 |   React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Label> & {
138 |     inset?: boolean;
139 |   }
140 | >(({ className, inset, ...props }, ref) => (
141 |   <ContextMenuPrimitive.Label
142 |     ref={ref}
143 |     className={cn("px-2 py-1.5 text-sm font-semibold text-foreground", inset && "pl-8", className)}
144 |     {...props}
145 |   />
146 | ));
147 | ContextMenuLabel.displayName = ContextMenuPrimitive.Label.displayName;
148 |
149 | const ContextMenuSeparator = React.forwardRef<
150 |   React.ElementRef<typeof ContextMenuPrimitive.Separator>,
151 |   React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Separator>
152 | >(({ className, ...props }, ref) => (
153 |   <ContextMenuPrimitive.Separator
154 |     ref={ref}
155 |     className={cn("-mx-1 my-1 h-px bg-border", className)}
156 |     {...props}
157 |   />
158 | ));
159 | ContextMenuSeparator.displayName = ContextMenuPrimitive.Separator.displayName;
160 |
161 | const ContextMenuShortcut = ({ className, ...props }: React.HTMLAttributes<HTMLSpanElement>) => {
162 |   return (
163 |     <span
164 |       className={cn("ml-auto text-xs tracking-widest text-muted-foreground", className)}
165 |       {...props}
166 |     />
167 |   );
168 | };
169 | ContextMenuShortcut.displayName = "ContextMenuShortcut";
170 |
171 | export {
172 |   ContextMenu,
173 |   ContextMenuTrigger,
174 |   ContextMenuContent,
175 |   ContextMenuItem,
176 |   ContextMenuCheckboxItem,
177 |   ContextMenuRadioItem,
178 |   ContextMenuLabel,
179 |   ContextMenuSeparator,
180 |   ContextMenuShortcut,
181 |   ContextMenuGroup,
182 |   ContextMenuPortal,
183 |   ContextMenuSub,
184 |   ContextMenuSubContent,
185 |   ContextMenuSubTrigger,
186 |   ContextMenuRadioGroup,
187 | };
```

FILE #69: src\components\ui\dialog.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import * as DialogPrimitive from "@radix-ui/react-dialog";
5 | import { X } from "lucide-react";
6 |
7 | import { cn } from "@lib/utils";
8 |
9 | const Dialog = DialogPrimitive.Root;
```

```
10 |
11 | const DialogTrigger = DialogPrimitive.Trigger;
12 |
13 | const DialogPortal = DialogPrimitive.Portal;
14 |
15 | const DialogClose = DialogPrimitive.Close;
16 |
17 | const DialogOverlay = React.forwardRef<
18 |   React.ElementRef<typeof DialogPrimitive.Overlay>,
19 |   React.ComponentPropsWithoutRef<typeof DialogPrimitive.Overlay>
20 | >(({ className, ...props }, ref) => (
21 |   <DialogPrimitive.Overlay
22 |     ref={ref}
23 |     className={cn(
24 |       "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-...
25 |       className,
26 |     )}
27 |     {...props}
28 |   />
29 | ));
30 | DialogOverlay.displayName = DialogPrimitive.Overlay.displayName;
31 |
32 | const DialogContent = React.forwardRef<
33 |   React.ElementRef<typeof DialogPrimitive.Content>,
34 |   React.ComponentPropsWithoutRef<typeof DialogPrimitive.Content>
35 | >(({ className, children, ...props }, ref) => (
36 |   <DialogPortal>
37 |     <DialogOverlay />
38 |     <DialogPrimitive.Content
39 |       ref={ref}
40 |       className={cn(
41 |         "fixed left-[50%] top-[50%] z-50 grid w-full max-w-lg translate-x-[-50%] translate-y-[-50%] gap-4...
42 |         className,
43 |       )}
44 |       {...props}
45 |     >
46 |       {children}
47 |       <DialogPrimitive.Close className="absolute right-4 top-4 rounded-sm opacity-70 ring-offset-backgrou...
48 |         <X className="h-4 w-4" />
49 |         <span className="sr-only">Close</span>
50 |       </DialogPrimitive.Close>
51 |     </DialogPrimitive.Content>
52 |   </DialogPortal>
53 | ));
54 | DialogContent.displayName = DialogPrimitive.Content.displayName;
55 |
56 | const DialogHeader = ({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) => (
57 |   <div className={cn("flex flex-col space-y-1.5 text-center sm:text-left", className)} {...props} />
58 | );
59 | DialogHeader.displayName = "DialogHeader";
60 |
61 | const DialogFooter = ({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) => (
62 |   <div
63 |     className={cn("flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2", className)}
64 |     {...props}
65 |   />
66 | );
67 | DialogFooter.displayName = "DialogFooter";
68 |
69 | const DialogTitle = React.forwardRef<
70 |   React.ElementRef<typeof DialogPrimitive.Title>,
71 |   React.ComponentPropsWithoutRef<typeof DialogPrimitive.Title>
72 | >(({ className, ...props }, ref) => (
73 |   <DialogPrimitive.Title
74 |     ref={ref}
```

```
75 |     className={cn("text-lg font-semibold leading-none tracking-tight", className)}
76 |     {...props}
77 |   />
78 | ));
79 | DialogTitle.displayName = DialogPrimitive.Title.displayName;
80 |
81 | const DialogDescription = React.forwardRef<
82 |   React.ElementRef<typeof DialogPrimitive.Description>,
83 |   React.ComponentPropsWithoutRef<typeof DialogPrimitive.Description>
84 | >(({ className, ...props }, ref) => (
85 |   <DialogPrimitive.Description
86 |     ref={ref}
87 |     className={cn("text-sm text-muted-foreground", className)}
88 |     {...props}
89 |   />
90 | ));
91 | DialogDescription.displayName = DialogPrimitive.Description.displayName;
92 |
93 | export {
94 |   Dialog,
95 |   DialogPortal,
96 |   DialogOverlay,
97 |   DialogTrigger,
98 |   DialogClose,
99 |   DialogContent,
100 |   DialogHeader,
101 |   DialogFooter,
102 |   DialogTitle,
103 |   DialogDescription,
104 | };
```

FILE #70: src\components\uildrawer.tsx

```
1 | import * as React from "react";
2 | import { Drawer as DrawerPrimitive } from "vaul";
3 |
4 | import { cn } from "@lib/utils";
5 |
6 | const Drawer = ({
7 |   shouldScaleBackground = true,
8 |   ...props
9 | }: React.ComponentProps<typeof DrawerPrimitive.Root>) => (
10 |   <DrawerPrimitive.Root shouldScaleBackground={shouldScaleBackground} {...props} />
11 | );
12 | Drawer.displayName = "Drawer";
13 |
14 | const DrawerTrigger = DrawerPrimitive.Trigger;
15 |
16 | const DrawerPortal = DrawerPrimitive.Portal;
17 |
18 | const DrawerClose = DrawerPrimitive.Close;
19 |
20 | const DrawerOverlay = React.forwardRef<
21 |   React.ElementRef<typeof DrawerPrimitive.Overlay>,
22 |   React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Overlay>
23 | >(({ className, ...props }, ref) => (
24 |   <DrawerPrimitive.Overlay
25 |     ref={ref}
26 |     className={cn("fixed inset-0 z-50 bg-black/80", className)}
27 |     {...props}
28 |   />
29 | ));
30 | DrawerOverlay.displayName = DrawerPrimitive.Overlay.displayName;
```



```
31 |
32 | const DrawerContent = React.forwardRef<
33 |   React.ElementRef<typeof DrawerPrimitive.Content>,
34 |   React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Content>
35 | >(({ className, children, ...props }, ref) => (
36 |   <DrawerPortal>
37 |     <DrawerOverlay />
38 |     <DrawerPrimitive.Content
39 |       ref={ref}
40 |       className={cn(
41 |         "fixed inset-x-0 bottom-0 z-50 mt-24 flex h-auto flex-col rounded-t-[10px] border bg-background",
42 |         className,
43 |       )}
44 |       {...props}
45 |     >
46 |       <div className="mx-auto mt-4 h-2 w-[100px] rounded-full bg-muted" />
47 |       {children}
48 |     </DrawerPrimitive.Content>
49 |   </DrawerPortal>
50 | ));
51 | DrawerContent.displayName = "DrawerContent";
52 |
53 | const DrawerHeader = ({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) => (
54 |   <div className={cn("grid gap-1.5 p-4 text-center sm:text-left", className)} {...props} />
55 | );
56 | DrawerHeader.displayName = "DrawerHeader";
57 |
58 | const DrawerFooter = ({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) => (
59 |   <div className={cn("mt-auto flex flex-col gap-2 p-4", className)} {...props} />
60 | );
61 | DrawerFooter.displayName = "DrawerFooter";
62 |
63 | const DrawerTitle = React.forwardRef<
64 |   React.ElementRef<typeof DrawerPrimitive.Title>,
65 |   React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Title>
66 | >(({ className, ...props }, ref) => (
67 |   <DrawerPrimitive.Title
68 |     ref={ref}
69 |     className={cn("text-lg font-semibold leading-none tracking-tight", className)}
70 |     {...props}
71 |   />
72 | ));
73 | DrawerTitle.displayName = DrawerPrimitive.Title.displayName;
74 |
75 | const DrawerDescription = React.forwardRef<
76 |   React.ElementRef<typeof DrawerPrimitive.Description>,
77 |   React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Description>
78 | >(({ className, ...props }, ref) => (
79 |   <DrawerPrimitive.Description
80 |     ref={ref}
81 |     className={cn("text-sm text-muted-foreground", className)}
82 |     {...props}
83 |   />
84 | ));
85 | DrawerDescription.displayName = DrawerPrimitive.Description.displayName;
86 |
87 | export {
88 |   Drawer,
89 |   DrawerPortal,
90 |   DrawerOverlay,
91 |   DrawerTrigger,
92 |   DrawerClose,
93 |   DrawerContent,
94 |   DrawerHeader,
95 |   DrawerFooter,
```

```
96 |   DrawerTitle,  
97 |   DrawerDescription,  
98 | };
```

FILE #71: src\components\uildropdown-menu.tsx

```
1 | "use client";  
2 |  
3 | import * as React from "react";  
4 | import * as DropdownMenuPrimitive from "@radix-ui/react-dropdown-menu";  
5 | import { Check, ChevronRight, Circle } from "lucide-react";  
6 |  
7 | import { cn } from "@lib/utils";  
8 |  
9 | const DropdownMenu = DropdownMenuPrimitive.Root;  
10 |  
11 | const DropdownMenuTrigger = DropdownMenuPrimitive.Trigger;  
12 |  
13 | const DropdownMenuGroup = DropdownMenuPrimitive.Group;  
14 |  
15 | const DropdownMenuPortal = DropdownMenuPrimitive.Portal;  
16 |  
17 | const DropdownMenuSub = DropdownMenuPrimitive.Sub;  
18 |  
19 | const DropdownMenuRadioGroup = DropdownMenuPrimitive.RadioGroup;  
20 |  
21 | const DropdownMenuSubTrigger = React.forwardRef(  
22 |   React.ElementRef<typeof DropdownMenuPrimitive.SubTrigger>,  
23 |   React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.SubTrigger> & {  
24 |     inset?: boolean;  
25 |   }  
26 | >(({ className, inset, children, ...props }, ref) => (  
27 |   <DropdownMenuPrimitive.SubTrigger  
28 |     ref={ref}  
29 |     className={cn(  
30 |       "flex cursor-default select-none items-center gap-2 rounded-sm px-2 py-1.5 text-sm outline-none foc...  
31 |       inset && "pl-8",  
32 |       className,  
33 |     )}  
34 |     {...props}  
35 |   >  
36 |     {children}  
37 |     <ChevronRight className="ml-auto" />  
38 |   </DropdownMenuPrimitive.SubTrigger>  
39 | ));  
40 | DropdownMenuSubTrigger.displayName = DropdownMenuPrimitive.SubTrigger.displayName;  
41 |  
42 | const DropdownMenuSubContent = React.forwardRef(  
43 |   React.ElementRef<typeof DropdownMenuPrimitive.SubContent>,  
44 |   React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.SubContent>  
45 | >(({ className, ...props }, ref) => (  
46 |   <DropdownMenuPrimitive.SubContent  
47 |     ref={ref}  
48 |     className={cn(  
49 |       "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-...  
50 |       className,  
51 |     )}  
52 |     {...props}  
53 |   />  
54 | ));  
55 | DropdownMenuSubContent.displayName = DropdownMenuPrimitive.SubContent.displayName;  
56 |  
57 | const DropdownMenuContent = React.forwardRef(<
```

```
58 |   React.ElementRef<typeof DropdownMenuPrimitive.Content>,
59 |   React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Content>
60 | >(({ className, sideOffset = 4, ...props }, ref) => (
61 |   <DropdownMenuPrimitive.Portal>
62 |     <DropdownMenuPrimitive.Content
63 |       ref={ref}
64 |       sideOffset={sideOffset}
65 |       className={cn(
66 |         "z-50 max-h-[var(--radix-dropdown-menu-content-available-height)] min-w-[8rem] overflow-y-auto ov...
67 |         "data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data...
68 |         className,
69 |       )}
70 |       {...props}
71 |     />
72 |   </DropdownMenuPrimitive.Portal>
73 | ));
74 | DropdownMenuContent.displayName = DropdownMenuPrimitive.Content.displayName;
75 |
76 | const DropdownMenuItem = React.forwardRef<
77 |   React.ElementRef<typeof DropdownMenuPrimitive.Item>,
78 |   React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Item> & {
79 |     inset?: boolean;
80 |   }
81 | >(({ className, inset, ...props }, ref) => (
82 |   <DropdownMenuPrimitive.Item
83 |     ref={ref}
84 |     className={cn(
85 |       "relative flex cursor-default select-none items-center gap-2 rounded-sm px-2 py-1.5 text-sm outline...
86 |       inset && "pl-8",
87 |       className,
88 |     )}
89 |     {...props}
90 |   />
91 | ));
92 | DropdownMenuItem.displayName = DropdownMenuPrimitive.Item.displayName;
93 |
94 | const DropdownMenuCheckboxItem = React.forwardRef<
95 |   React.ElementRef<typeof DropdownMenuPrimitive.CheckboxItem>,
96 |   React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.CheckboxItem>
97 | >(({ className, children, checked, ...props }, ref) => (
98 |   <DropdownMenuPrimitive.CheckboxItem
99 |     ref={ref}
100 |     className={cn(
101 |       "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-...
102 |       className,
103 |     )}
104 |     checked={checked}
105 |     {...props}
106 |   >
107 |     <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
108 |       <DropdownMenuPrimitive.ItemIndicator>
109 |         <Check className="h-4 w-4" />
110 |       </DropdownMenuPrimitive.ItemIndicator>
111 |     </span>
112 |     {children}
113 |   </DropdownMenuPrimitive.CheckboxItem>
114 | ));
115 | DropdownMenuCheckboxItem.displayName = DropdownMenuPrimitive.CheckboxItem.displayName;
116 |
117 | const DropdownMenuRadioItem = React.forwardRef<
118 |   React.ElementRef<typeof DropdownMenuPrimitive.RadioItem>,
119 |   React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.RadioItem>
120 | >(({ className, children, ...props }, ref) => (
121 |   <DropdownMenuPrimitive.RadioItem
122 |     ref={ref}
```

```
123 |     className={cn(
124 |       "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-...
125 |       className,
126 |     )}
127 |     {...props}
128 |   >
129 |     <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
130 |       <DropdownMenuPrimitive.ItemIndicator>
131 |         <Circle className="h-2 w-2 fill-current" />
132 |       </DropdownMenuPrimitive.ItemIndicator>
133 |     </span>
134 |     {children}
135 |   </DropdownMenuPrimitive.RadioItem>
136 | ));
137 | DropdownMenuRadioItem.displayName = DropdownMenuPrimitive.RadioItem.displayName;
138 |
139 | const DropdownMenuLabel = React.forwardRef<
140 |   React.ElementRef<typeof DropdownMenuPrimitive.Label>,
141 |   React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Label> & {
142 |     inset?: boolean;
143 |   }
144 | >(({ className, inset, ...props }, ref) => (
145 |   <DropdownMenuPrimitive.Label
146 |     ref={ref}
147 |     className={cn("px-2 py-1.5 text-sm font-semibold", inset && "pl-8", className)}
148 |     {...props}
149 |   />
150 | ));
151 | DropdownMenuLabel.displayName = DropdownMenuPrimitive.Label.displayName;
152 |
153 | const DropdownMenuSeparator = React.forwardRef<
154 |   React.ElementRef<typeof DropdownMenuPrimitive.Separator>,
155 |   React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Separator>
156 | >(({ className, ...props }, ref) => (
157 |   <DropdownMenuPrimitive.Separator
158 |     ref={ref}
159 |     className={cn("-mx-1 my-1 h-px bg-muted", className)}
160 |     {...props}
161 |   />
162 | ));
163 | DropdownMenuSeparator.displayName = DropdownMenuPrimitive.Separator.displayName;
164 |
165 | const DropdownMenuShortcut = ({ className, ...props }: React.HTMLAttributes<HTMLSpanElement>) => {
166 |   return (
167 |     <span className={cn("ml-auto text-xs tracking-widest opacity-60", className)} {...props} />
168 |   );
169 | };
170 | DropdownMenuShortcut.displayName = "DropdownMenuShortcut";
171 |
172 | export {
173 |   DropdownMenu,
174 |   DropdownMenuTrigger,
175 |   DropdownMenuContent,
176 |   DropdownMenuItem,
177 |   DropdownMenuCheckboxItem,
178 |   DropdownMenuRadioItem,
179 |   DropdownMenuLabel,
180 |   DropdownMenuSeparator,
181 |   DropdownMenuShortcut,
182 |   DropdownMenuGroup,
183 |   DropdownMenuPortal,
184 |   DropdownMenuSub,
185 |   DropdownMenuSubContent,
186 |   DropdownMenuSubTrigger,
187 |   DropdownMenuRadioGroup,
```

```
188 | };
```

FILE #72: src\components\uilform.tsx

```
1 | import * as React from "react";
2 | import * as LabelPrimitive from "@radix-ui/react-label";
3 | import { Slot } from "@radix-ui/react-slot";
4 | import {
5 |   Controller,
6 |   FormProvider,
7 |   useFormContext,
8 |   type ControllerProps,
9 |   type FieldPath,
10 |   type FieldValues,
11 | } from "react-hook-form";
12 |
13 | import { cn } from "@lib/utils";
14 | import { Label } from "@components/ui/label";
15 |
16 | const Form = FormProvider;
17 |
18 | type FormFieldContextValue<
19 |   TFieldValues extends FieldValues = FieldValues,
20 |   TName extends FieldPath<TFieldValues> = FieldPath<TFieldValues>,
21 | > = {
22 |   name: TName;
23 | };
24 |
25 | const FormFieldContext = React.createContext<FormFieldContextValue | null>(null);
26 |
27 | const FormField = <
28 |   TFieldValues extends FieldValues = FieldValues,
29 |   TName extends FieldPath<TFieldValues> = FieldPath<TFieldValues>,
30 | >({
31 |   ...props
32 | }): ControllerProps<TFieldValues, TName> => {
33 |   return (
34 |     <FormFieldContext.Provider value={{ name: props.name }}>
35 |       <Controller {...props} />
36 |     </FormFieldContext.Provider>
37 |   );
38 | };
39 |
40 | const useFormField = () => {
41 |   const fieldContext = React.useContext(FormFieldContext);
42 |   const itemContext = React.useContext(FormItemContext);
43 |   const { getFieldState, formState } = useFormContext();
44 |
45 |   if (!fieldContext) {
46 |     throw new Error("useFormField should be used within <FormField>");
47 |   }
48 |
49 |   if (!itemContext) {
50 |     throw new Error("useFormField should be used within <FormItem>");
51 |   }
52 |
53 |   const fieldState = getFieldState(fieldContext.name, formState);
54 |
55 |   const { id } = itemContext;
56 |
57 |   return {
58 |     id,
59 |     name: fieldContext.name,
```

```
60 |     formItemId: `${id}-form-item`,
61 |     formDescriptionId: `${id}-form-item-description`,
62 |     formMessageId: `${id}-form-item-message`,
63 |     ...fieldState,
64 |   };
65 | };
66 |
67 | type FormItemContextValue = {
68 |   id: string;
69 | };
70 |
71 | const FormItemContext = React.createContext<FormItemContextValue | null>(null);
72 |
73 | const FormItem = React.forwardRef<HTMLDivElement, React.HTMLAttributes<HTMLDivElement>>(
74 |   ({ className, ...props }, ref) => {
75 |     const id = React.useId();
76 |
77 |     return (
78 |       <FormItemContext.Provider value={{ id }}>
79 |         <div ref={ref} className={cn("space-y-2", className)} {...props} />
80 |       </FormItemContext.Provider>
81 |     );
82 |   },
83 | );
84 | FormItem.displayName = "FormItem";
85 |
86 | const FormLabel = React.forwardRef<
87 |   React.ElementRef<typeof LabelPrimitive.Root>,
88 |   React.ComponentPropsWithoutRef<typeof LabelPrimitive.Root>
89 | >(({ className, ...props }, ref) => {
90 |   const { error, formItemId } = useFormField();
91 |
92 |   return (
93 |     <Label
94 |       ref={ref}
95 |       className={cn(error && "text-destructive", className)}
96 |       htmlFor={formItemId}
97 |       {...props}
98 |     />
99 |   );
100 | });
101 | FormLabel.displayName = "FormLabel";
102 |
103 | const FormControl = React.forwardRef<
104 |   React.ElementRef<typeof Slot>,
105 |   React.ComponentPropsWithoutRef<typeof Slot>
106 | >(({ ...props }, ref) => {
107 |   const { error, formItemId, formDescriptionId, formMessageId } = useFormField();
108 |
109 |   return (
110 |     <Slot
111 |       ref={ref}
112 |       id={formItemId}
113 |       aria-describedby={!error ? `${formDescriptionId}` : `${formDescriptionId} ${formMessageId}`}
114 |       aria-invalid={!!error}
115 |       {...props}
116 |     />
117 |   );
118 | });
119 | FormControl.displayName = "FormControl";
120 |
121 | const FormDescription = React.forwardRef<
122 |   HTMLParagraphElement,
123 |   React.HTMLAttributes<HTMLParagraphElement>
124 | >(({ className, ...props }, ref) => {
```

```
125 |   const { formDescriptionId } = useFormField();
126 |
127 |   return (
128 |     <p
129 |       ref={ref}
130 |       id={formDescriptionId}
131 |       className={cn("text-[0.8rem] text-muted-foreground", className)}
132 |       {...props}
133 |     />
134 |   );
135 | });
136 | FormDescription.displayName = "FormDescription";
137 |
138 | const FormMessage = React.forwardRef<
139 |   HTMLParagraphElement,
140 |   React.HTMLAttributes<HTMLParagraphElement>
141 | >(({ className, children, ...props }, ref) => {
142 |   const { error, formMessageId } = useFormField();
143 |   const body = error ? String(error?.message ?? "") : children;
144 |
145 |   if (!body) {
146 |     return null;
147 |   }
148 |
149 |   return (
150 |     <p
151 |       ref={ref}
152 |       id={formMessageId}
153 |       className={cn("text-[0.8rem] font-medium text-destructive", className)}
154 |       {...props}
155 |     >
156 |       {body}
157 |     </p>
158 |   );
159 | });
160 | FormMessage.displayName = "FormMessage";
161 |
162 | export {
163 |   useFormField,
164 |   Form,
165 |   FormItem,
166 |   FormLabel,
167 |   FormControl,
168 |   FormDescription,
169 |   FormMessage,
170 |   FormField,
171 | };
```

FILE #73: src\components\uilhover-card.tsx

```
1 | import * as React from "react";
2 | import * as HoverCardPrimitive from "@radix-ui/react-hover-card";
3 |
4 | import { cn } from "@/lib/utils";
5 |
6 | const HoverCard = HoverCardPrimitive.Root;
7 |
8 | const HoverCardTrigger = HoverCardPrimitive.Trigger;
9 |
10 | const HoverCardContent = React.forwardRef<
11 |   React.ElementRef<typeof HoverCardPrimitive.Content>,
12 |   React.ComponentPropsWithoutRef<typeof HoverCardPrimitive.Content>
13 | >(({ className, align = "center", sideOffset = 4, ...props }, ref) => {
```

```
14 | <HoverCardPrimitive.Content
15 |   ref={ref}
16 |   align={align}
17 |   sideOffset={sideOffset}
18 |   className={cn(
19 |     "z-50 w-64 rounded-md border bg-popover p-4 text-popover-foreground shadow-md outline-none data-[st...
20 |     className,
21 |   )}
22 |   {...props}
23 | />
24 | ));
25 | HoverCardContent.displayName = HoverCardPrimitive.Content.displayName;
26 |
27 | export { HoverCard, HoverCardTrigger, HoverCardContent };
```

FILE #74: src\components\ui\input-otp.tsx

```
1 | import * as React from "react";
2 | import { OTPInput, OTPInputContext } from "input-otp";
3 | import { Minus } from "lucide-react";
4 |
5 | import { cn } from "@lib/utils";
6 |
7 | const InputOTP = React.forwardRef<
8 |   React.ElementRef<typeof OTPInput>,
9 |   React.ComponentPropsWithoutRef<typeof OTPInput>
10 | >(({ className, containerClassName, ...props }, ref) => (
11 |   <OTPInput
12 |     ref={ref}
13 |     containerClassName={cn(
14 |       "flex items-center gap-2 has-[:disabled]:opacity-50",
15 |       containerClassName,
16 |     )}
17 |     className={cn("disabled:cursor-not-allowed", className)}
18 |     {...props}
19 |   />
20 | ));
21 | InputOTP.displayName = "InputOTP";
22 |
23 | const InputOTPGroup = React.forwardRef<
24 |   React.ElementRef<"div">,
25 |   React.ComponentPropsWithoutRef<"div">
26 | >(({ className, ...props }, ref) => (
27 |   <div ref={ref} className={cn("flex items-center", className)} {...props} />
28 | ));
29 | InputOTPGroup.displayName = "InputOTPGroup";
30 |
31 | const InputOTPSlot = React.forwardRef<
32 |   React.ElementRef<"div">,
33 |   React.ComponentPropsWithoutRef<"div"> & { index: number }
34 | >(({ index, className, ...props }, ref) => {
35 |   const inputOTPContext = React.useContext(OTPInputContext);
36 |   const { char, hasFakeCaret, isActive } = inputOTPContext.slots[index];
37 |
38 |   return (
39 |     <div
40 |       ref={ref}
41 |       className={cn(
42 |         "relative flex h-9 w-9 items-center justify-center border-y border-r border-input text-sm shadow-...
43 |         isActive && "z-10 ring-1 ring-ring",
44 |         className,
45 |       )}
46 |       {...props}
```



```
47 |     >
48 |     {char}
49 |     {hasFakeCaret && (
50 |       <div className="pointer-events-none absolute inset-0 flex items-center justify-center">
51 |         <div className="h-4 w-px animate-caret-blink bg-foreground duration-1000" />
52 |       </div>
53 |     )}
54 |   </div>
55 | );
56 | });
57 | InputOTPSlot.displayName = "InputOTPSlot";
58 |
59 | const InputOTPSeparator = React.forwardRef<
60 |   React.ElementRef<"div">,
61 |   React.ComponentPropsWithoutRef<"div">
62 | >(({ ...props }, ref) => (
63 |   <div ref={ref} role="separator" {...props}>
64 |     <Minus />
65 |   </div>
66 | ));
67 | InputOTPSeparator.displayName = "InputOTPSeparator";
68 |
69 | export { InputOTP, InputOTPGroup, InputOTPSlot, InputOTPSeparator };
```

FILE #75: src\components\ui\input.tsx

```
1 | import * as React from "react";
2 |
3 | import { cn } from "@lib/utils";
4 |
5 | const Input = React.forwardRef<HTMLInputElement, React.ComponentProps<"input">>(
6 |   ({ className, type, ...props }, ref) => {
7 |     return (
8 |       <input
9 |         type={type}
10 |        className={cn(
11 |          "flex h-9 w-full rounded-md border border-input bg-transparent px-3 py-1 text-base shadow-sm tr...
12 |        className,
13 |        )}
14 |        ref={ref}
15 |        {...props}
16 |      />
17 |    );
18 |   },
19 | );
20 | Input.displayName = "Input";
21 |
22 | export { Input };
```

FILE #76: src\components\ui\label.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import * as LabelPrimitive from "@radix-ui/react-label";
5 | import { cva, type VariantProps } from "class-variance-authority";
6 |
7 | import { cn } from "@lib/utils";
8 |
9 | const labelVariants = cva(
10 |   "text-sm font-medium leading-none peer-disabled:cursor-not-allowed peer-disabled:opacity-70",
```

```
11 | );
12 |
13 | const Label = React.forwardRef<
14 |   React.ElementRef<typeof LabelPrimitive.Root>,
15 |   React.ComponentPropsWithoutRef<typeof LabelPrimitive.Root> & VariantProps<typeof labelVariants>
16 | >(({ className, ...props }, ref) => (
17 |   <LabelPrimitive.Root ref={ref} className={cn(labelVariants(), className)} {...props} />
18 | ));
19 | Label.displayName = LabelPrimitive.Root.displayName;
20 |
21 | export { Label };
```

FILE #77: src\components\uilmenubar.tsx

```
1 | import * as React from "react";
2 | import * as MenubarPrimitive from "@radix-ui/react-menubar";
3 | import { Check, ChevronRight, Circle } from "lucide-react";
4 |
5 | import { cn } from "@lib/utils";
6 |
7 | function MenubarMenu({ ...props }: React.ComponentProps<typeof MenubarPrimitive.Menu>) {
8 |   return <MenubarPrimitive.Menu {...props} />;
9 | }
10 |
11 | function MenubarGroup({ ...props }: React.ComponentProps<typeof MenubarPrimitive.Group>) {
12 |   return <MenubarPrimitive.Group {...props} />;
13 | }
14 |
15 | function MenubarPortal({ ...props }: React.ComponentProps<typeof MenubarPrimitive.Portal>) {
16 |   return <MenubarPrimitive.Portal {...props} />;
17 | }
18 |
19 | function MenubarRadioGroup({ ...props }: React.ComponentProps<typeof MenubarPrimitive.RadioGroup>) {
20 |   return <MenubarPrimitive.RadioGroup {...props} />;
21 | }
22 |
23 | function MenubarSub({ ...props }: React.ComponentProps<typeof MenubarPrimitive.Sub>) {
24 |   return <MenubarPrimitive.Sub data-slot="menubar-sub" {...props} />;
25 | }
26 |
27 | const Menubar = React.forwardRef<
28 |   React.ElementRef<typeof MenubarPrimitive.Root>,
29 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Root>
30 | >(({ className, ...props }, ref) => (
31 |   <MenubarPrimitive.Root
32 |     ref={ref}
33 |     className={cn(
34 |       "flex h-9 items-center space-x-1 rounded-md border bg-background p-1 shadow-sm",
35 |       className,
36 |     )}
37 |     {...props}
38 |   />
39 | ));
40 | Menubar.displayName = MenubarPrimitive.Root.displayName;
41 |
42 | const MenubarTrigger = React.forwardRef<
43 |   React.ElementRef<typeof MenubarPrimitive.Trigger>,
44 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Trigger>
45 | >(({ className, ...props }, ref) => (
46 |   <MenubarPrimitive.Trigger
47 |     ref={ref}
48 |     className={cn(
49 |       "flex cursor-default select-none items-center rounded-sm px-3 py-1 text-sm font-medium outline-none...
```

```
50 |     className,
51 |   )}
52 |   {...props}
53 | />
54 | ));
55 | MenubarTrigger.displayName = MenubarPrimitive.Trigger.displayName;
56 |
57 | const MenubarSubTrigger = React.forwardRef<
58 |   React.ElementRef<typeof MenubarPrimitive.SubTrigger>,
59 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.SubTrigger> & {
60 |     inset?: boolean;
61 |   }
62 | >(({ className, inset, children, ...props }, ref) => (
63 |   <MenubarPrimitive.SubTrigger
64 |     ref={ref}
65 |     className={cn(
66 |       "flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-...
67 |       inset && "pl-8",
68 |       className,
69 |     )}
70 |     {...props}
71 |   >
72 |     {children}
73 |     <ChevronRight className="ml-auto h-4 w-4" />
74 |   </MenubarPrimitive.SubTrigger>
75 | ));
76 | MenubarSubTrigger.displayName = MenubarPrimitive.SubTrigger.displayName;
77 |
78 | const MenubarSubContent = React.forwardRef<
79 |   React.ElementRef<typeof MenubarPrimitive.SubContent>,
80 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.SubContent>
81 | >(({ className, ...props }, ref) => (
82 |   <MenubarPrimitive.SubContent
83 |     ref={ref}
84 |     className={cn(
85 |       "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-...
86 |       className,
87 |     )}
88 |     {...props}
89 |   />
90 | ));
91 | MenubarSubContent.displayName = MenubarPrimitive.SubContent.displayName;
92 |
93 | const MenubarContent = React.forwardRef<
94 |   React.ElementRef<typeof MenubarPrimitive.Content>,
95 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Content>
96 | >(({ className, align = "start", alignOffset = -4, sideOffset = 8, ...props }, ref) => (
97 |   <MenubarPrimitive.Portal>
98 |     <MenubarPrimitive.Content
99 |       ref={ref}
100 |       align={align}
101 |       alignOffset={alignOffset}
102 |       sideOffset={sideOffset}
103 |       className={cn(
104 |         "z-50 min-w-[12rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shad...
105 |         className,
106 |       )}
107 |       {...props}
108 |     />
109 |   </MenubarPrimitive.Portal>
110 | ));
111 | MenubarContent.displayName = MenubarPrimitive.Content.displayName;
112 |
113 | const MenubarItem = React.forwardRef<
114 |   React.ElementRef<typeof MenubarPrimitive.Item>,>
```

```
115 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Item> & {
116 |     inset?: boolean;
117 |   }
118 | >(({ className, inset, ...props }, ref) => (
119 |   <MenubarPrimitive.Item
120 |     ref={ref}
121 |     className={cn(
122 |       "relative flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none ...
123 |       inset && "pl-8",
124 |       className,
125 |     )}
126 |     {...props}
127 |   />
128 | ));
129 | MenubarItem.displayName = MenubarPrimitive.Item.displayName;
130 |
131 | const MenubarCheckboxItem = React.forwardRef<
132 |   React.ElementRef<typeof MenubarPrimitive.CheckboxItem>,
133 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.CheckboxItem>
134 | >(({ className, children, checked, ...props }, ref) => (
135 |   <MenubarPrimitive.CheckboxItem
136 |     ref={ref}
137 |     className={cn(
138 |       "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-...
139 |       className,
140 |     )}
141 |     checked={checked}
142 |     {...props}
143 |   >
144 |     <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
145 |       <MenubarPrimitive.ItemIndicator>
146 |         <Check className="h-4 w-4" />
147 |       </MenubarPrimitive.ItemIndicator>
148 |     </span>
149 |     {children}
150 |   </MenubarPrimitive.CheckboxItem>
151 | ));
152 | MenubarCheckboxItem.displayName = MenubarPrimitive.CheckboxItem.displayName;
153 |
154 | const MenubarRadioItem = React.forwardRef<
155 |   React.ElementRef<typeof MenubarPrimitive.RadioItem>,
156 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.RadioItem>
157 | >(({ className, children, ...props }, ref) => (
158 |   <MenubarPrimitive.RadioItem
159 |     ref={ref}
160 |     className={cn(
161 |       "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-...
162 |       className,
163 |     )}
164 |     {...props}
165 |   >
166 |     <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
167 |       <MenubarPrimitive.ItemIndicator>
168 |         <Circle className="h-4 w-4 fill-current" />
169 |       </MenubarPrimitive.ItemIndicator>
170 |     </span>
171 |     {children}
172 |   </MenubarPrimitive.RadioItem>
173 | ));
174 | MenubarRadioItem.displayName = MenubarPrimitive.RadioItem.displayName;
175 |
176 | const MenubarLabel = React.forwardRef<
177 |   React.ElementRef<typeof MenubarPrimitive.Label>,
178 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Label> & {
179 |     inset?: boolean;
```

```
180 |   }
181 | >(({ className, inset, ...props }, ref) => (
182 |   <MenubarPrimitive.Label
183 |     ref={ref}
184 |     className={cn("px-2 py-1.5 text-sm font-semibold", inset && "pl-8", className)}
185 |     {...props}
186 |   />
187 | ));
188 | MenubarLabel.displayName = MenubarPrimitive.Label.displayName;
189 |
190 | const MenubarSeparator = React.forwardRef<
191 |   React.ElementRef<typeof MenubarPrimitive.Separator>,
192 |   React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Separator>
193 | >(({ className, ...props }, ref) => (
194 |   <MenubarPrimitive.Separator
195 |     ref={ref}
196 |     className={cn("-mx-1 my-1 h-px bg-muted", className)}
197 |     {...props}
198 |   />
199 | ));
200 | MenubarSeparator.displayName = MenubarPrimitive.Separator.displayName;
201 |
202 | const MenubarShortcut = ({ className, ...props }: React.HTMLAttributes<HTMLSpanElement>) => {
203 |   return (
204 |     <span
205 |       className={cn("ml-auto text-xs tracking-widest text-muted-foreground", className)}
206 |       {...props}
207 |     />
208 |   );
209 | };
210 | MenubarShortcut.displayName = "MenubarShortcut";
211 |
212 | export {
213 |   Menubar,
214 |   MenubarMenu,
215 |   MenubarTrigger,
216 |   MenubarContent,
217 |   MenubarItem,
218 |   MenubarSeparator,
219 |   MenubarLabel,
220 |   MenubarCheckboxItem,
221 |   MenubarRadioGroup,
222 |   MenubarRadioItem,
223 |   MenubarPortal,
224 |   MenubarSubContent,
225 |   MenubarSubTrigger,
226 |   MenubarGroup,
227 |   MenubarSub,
228 |   MenubarShortcut,
229 | };
```

FILE #78: src\components\uil\navigation-menu.tsx

```
1 | import * as React from "react";
2 | import * as NavigationMenuPrimitive from "@radix-ui/react-navigation-menu";
3 | import { cva } from "class-variance-authority";
4 | import { ChevronDown } from "lucide-react";
5 |
6 | import { cn } from "@lib/utils";
7 |
8 | const NavigationMenu = React.forwardRef<
9 |   React.ElementRef<typeof NavigationMenuPrimitive.Root>,
10 |   React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Root>
```

```
11 | >(({ className, children, ...props }, ref) => (  
12 |   <NavigationMenuPrimitive.Root  
13 |     ref={ref}  
14 |     className={cn("relative z-10 flex max-w-max flex-1 items-center justify-center", className)}  
15 |     {...props}  
16 |   >  
17 |     {children}  
18 |     <NavigationMenuViewport />  
19 |   </NavigationMenuPrimitive.Root>  
20 | ));  
21 | NavigationMenu.displayName = NavigationMenuPrimitive.Root.displayName;  
22 |  
23 | const NavigationMenuList = React.forwardRef(  
24 |   React.ElementRef<typeof NavigationMenuPrimitive.List>,  
25 |   React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.List>  
26 | >(({ className, ...props }, ref) => (  
27 |   <NavigationMenuPrimitive.List  
28 |     ref={ref}  
29 |     className={cn("group flex flex-1 list-none items-center justify-center space-x-1", className)}  
30 |     {...props}  
31 |   />  
32 | ));  
33 | NavigationMenuList.displayName = NavigationMenuPrimitive.List.displayName;  
34 |  
35 | const NavigationMenuItem = NavigationMenuPrimitive.Item;  
36 |  
37 | const navigationMenuTriggerStyle = cva(  
38 |   "group inline-flex h-9 w-max items-center justify-center rounded-md bg-background px-4 py-2 text-sm fon...  
39 | );  
40 |  
41 | const NavigationMenuTrigger = React.forwardRef(  
42 |   React.ElementRef<typeof NavigationMenuPrimitive.Trigger>,  
43 |   React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Trigger>  
44 | >(({ className, children, ...props }, ref) => (  
45 |   <NavigationMenuPrimitive.Trigger  
46 |     ref={ref}  
47 |     className={cn(navigationMenuTriggerStyle(), "group", className)}  
48 |     {...props}  
49 |   >  
50 |     {children}{" "  
51 |     <ChevronDown  
52 |       className="relative top-[1px] ml-1 h-3 w-3 transition duration-300 group-data-[state=open]:rotate-180"  
53 |       aria-hidden="true"  
54 |     />  
55 |   </NavigationMenuPrimitive.Trigger>  
56 | ));  
57 | NavigationMenuTrigger.displayName = NavigationMenuPrimitive.Trigger.displayName;  
58 |  
59 | const NavigationMenuContent = React.forwardRef(  
60 |   React.ElementRef<typeof NavigationMenuPrimitive.Content>,  
61 |   React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Content>  
62 | >(({ className, ...props }, ref) => (  
63 |   <NavigationMenuPrimitive.Content  
64 |     ref={ref}  
65 |     className={cn(  
66 |       "left-0 top-0 w-full data-[motion^=from-]:animate-in data-[motion^=to-]:animate-out data-[motion^=f...  
67 |       className,  
68 |     )}  
69 |     {...props}  
70 |   />  
71 | ));  
72 | NavigationMenuContent.displayName = NavigationMenuPrimitive.Content.displayName;  
73 |  
74 | const NavigationMenuLink = NavigationMenuPrimitive.Link;  
75 |
```

```

76 | const NavigationMenuViewport = React.forwardRef<
77 |   React.ElementRef<typeof NavigationMenuPrimitive.Viewport>,
78 |   React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Viewport>
79 | >(({ className, ...props }, ref) => (
80 |   <div className={cn("absolute left-0 top-full flex justify-center")}>
81 |     <NavigationMenuPrimitive.Viewport
82 |       className={cn(
83 |         "origin-top-center relative mt-1.5 h-[var(--radix-navigation-menu-viewport-height)] w-full overfl...
84 |         className,
85 |       )}
86 |       ref={ref}
87 |       {...props}
88 |     />
89 |   </div>
90 | ));
91 | NavigationMenuViewport.displayName = NavigationMenuPrimitive.Viewport.displayName;
92 |
93 | const NavigationMenuIndicator = React.forwardRef<
94 |   React.ElementRef<typeof NavigationMenuPrimitive.Indicator>,
95 |   React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Indicator>
96 | >(({ className, ...props }, ref) => (
97 |   <NavigationMenuPrimitive.Indicator
98 |     ref={ref}
99 |     className={cn(
100 |       "top-full z-[1] flex h-1.5 items-end justify-center overflow-hidden data-[state=visible]:animate-in...
101 |       className,
102 |     )}
103 |     {...props}
104 |   >
105 |     <div className="relative top-[60%] h-2 w-2 rotate-45 rounded-tl-sm bg-border shadow-md" />
106 |   </NavigationMenuPrimitive.Indicator>
107 | ));
108 | NavigationMenuIndicator.displayName = NavigationMenuPrimitive.Indicator.displayName;
109 |
110 | export {
111 |   navigationMenuTriggerStyle,
112 |   NavigationMenu,
113 |   NavigationMenuList,
114 |   NavigationMenuItem,
115 |   NavigationMenuContent,
116 |   NavigationMenuTrigger,
117 |   NavigationMenuLink,
118 |   NavigationMenuIndicator,
119 |   NavigationMenuViewport,
120 | };

```

FILE #79: src\components\ui\pagination.tsx

```

1 | import * as React from "react";
2 | import { ChevronLeft, ChevronRight, MoreHorizontal } from "lucide-react";
3 |
4 | import { cn } from "@lib/utls";
5 | import { ButtonProps, buttonVariants } from "@components/ui/button";
6 |
7 | const Pagination = ({ className, ...props }: React.ComponentProps<"nav">) => (
8 |   <nav
9 |     role="navigation"
10 |     aria-label="pagination"
11 |     className={cn("mx-auto flex w-full justify-center", className)}
12 |     {...props}
13 |   />
14 | );
15 | Pagination.displayName = "Pagination";

```

```
16 |
17 | const PaginationContent = React.forwardRef<HTMLUListElement, React.ComponentProps<"ul">>(  
18 |   ({ className, ...props }, ref) => (  
19 |     <ul ref={ref} className={cn("flex flex-row items-center gap-1", className)} {...props} />  
20 |   ),  
21 | );  
22 | PaginationContent.displayName = "PaginationContent";  
23 |  
24 | const PaginationItem = React.forwardRef<HTMLLIElement, React.ComponentProps<"li">>(  
25 |   ({ className, ...props }, ref) => <li ref={ref} className={cn("", className)} {...props} />,  
26 | );  
27 | PaginationItem.displayName = "PaginationItem";  
28 |  
29 | type PaginationLinkProps = {  
30 |   isActive?: boolean;  
31 | } & Pick<ButtonProps, "size"> &  
32 |   React.ComponentProps<"a">;  
33 |  
34 | const PaginationLink = ({ className, isActive, size = "icon", ...props }: PaginationLinkProps) => (  
35 |   <a  
36 |     aria-current={isActive ? "page" : undefined}  
37 |     className={cn(  
38 |       buttonVariants({  
39 |         variant: isActive ? "outline" : "ghost",  
40 |         size,  
41 |       }),  
42 |       className,  
43 |     )}  
44 |     {...props}  
45 |   />  
46 | );  
47 | PaginationLink.displayName = "PaginationLink";  
48 |  
49 | const PaginationPrevious = ({  
50 |   className,  
51 |   ...props  
52 | }: React.ComponentProps<typeof PaginationLink>) => (  
53 |   <PaginationLink  
54 |     aria-label="Go to previous page"  
55 |     size="default"  
56 |     className={cn("gap-1 pl-2.5", className)}  
57 |     {...props}  
58 |   >  
59 |     <ChevronLeft className="h-4 w-4" />  
60 |     <span>Previous</span>  
61 |   </PaginationLink>  
62 | );  
63 | PaginationPrevious.displayName = "PaginationPrevious";  
64 |  
65 | const PaginationNext = ({ className, ...props }: React.ComponentProps<typeof PaginationLink>) => (  
66 |   <PaginationLink  
67 |     aria-label="Go to next page"  
68 |     size="default"  
69 |     className={cn("gap-1 pr-2.5", className)}  
70 |     {...props}  
71 |   >  
72 |     <span>Next</span>  
73 |     <ChevronRight className="h-4 w-4" />  
74 |   </PaginationLink>  
75 | );  
76 | PaginationNext.displayName = "PaginationNext";  
77 |  
78 | const PaginationEllipsis = ({ className, ...props }: React.ComponentProps<"span">) => (  
79 |   <span  
80 |     aria-hidden
```



```
81 |     className={cn("flex h-9 w-9 items-center justify-center", className)}
82 |     {...props}
83 |   >
84 |     <MoreHorizontal className="h-4 w-4" />
85 |     <span className="sr-only">More pages</span>
86 |   </span>
87 | );
88 | PaginationEllipsis.displayName = "PaginationEllipsis";
89 |
90 | export {
91 |   Pagination,
92 |   PaginationContent,
93 |   PaginationLink,
94 |   PaginationItem,
95 |   PaginationPrevious,
96 |   PaginationNext,
97 |   PaginationEllipsis,
98 | };
```

FILE #80: src\components\uilpopover.tsx

```
1 | import * as React from "react";
2 | import * as PopoverPrimitive from "@radix-ui/react-popover";
3 |
4 | import { cn } from "@lib/utils";
5 |
6 | const Popover = PopoverPrimitive.Root;
7 |
8 | const PopoverTrigger = PopoverPrimitive.Trigger;
9 |
10 | const PopoverAnchor = PopoverPrimitive.Anchor;
11 |
12 | const PopoverContent = React.forwardRef<
13 |   React.ElementRef<typeof PopoverPrimitive.Content>,
14 |   React.ComponentPropsWithoutRef<typeof PopoverPrimitive.Content>
15 | >(({ className, align = "center", sideOffset = 4, ...props }, ref) => (
16 |   <PopoverPrimitive.Portal>
17 |     <PopoverPrimitive.Content
18 |       ref={ref}
19 |       align={align}
20 |       sideOffset={sideOffset}
21 |       className={cn(
22 |         "z-50 w-72 rounded-md border bg-popover p-4 text-popover-foreground shadow-md outline-none data-[...
23 |         className,
24 |       )}
25 |       {...props}
26 |     />
27 |   </PopoverPrimitive.Portal>
28 | ));
29 | PopoverContent.displayName = PopoverPrimitive.Content.displayName;
30 |
31 | export { Popover, PopoverTrigger, PopoverContent, PopoverAnchor };
```

FILE #81: src\components\uilprogress.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import * as ProgressPrimitive from "@radix-ui/react-progress";
5 |
6 | import { cn } from "@lib/utils";
```

```
7 |  
8 | const Progress = React.forwardRef<  
9 |   React.ElementRef<typeof ProgressPrimitive.Root>,  
10 |   React.ComponentPropsWithoutRef<typeof ProgressPrimitive.Root>  
11 | >(({ className, value, ...props }, ref) => (  
12 |   <ProgressPrimitive.Root  
13 |     ref={ref}  
14 |     className={cn("relative h-2 w-full overflow-hidden rounded-full bg-primary/20", className)}  
15 |     {...props}  
16 |   >  
17 |     <ProgressPrimitive.Indicator  
18 |       className="h-full w-full flex-1 bg-primary transition-all"  
19 |       style={{ transform: `translateX(-${100 - (value || 0)}%)` }}  
20 |     />  
21 |   </ProgressPrimitive.Root>  
22 | ));  
23 | Progress.displayName = ProgressPrimitive.Root.displayName;  
24 |  
25 | export { Progress };
```

FILE #82: src\components\uilradio-group.tsx

```
1 | import * as React from "react";  
2 | import * as RadioGroupPrimitive from "@radix-ui/react-radio-group";  
3 | import { Circle } from "lucide-react";  
4 |  
5 | import { cn } from "@lib/utils";  
6 |  
7 | const RadioGroup = React.forwardRef<  
8 |   React.ElementRef<typeof RadioGroupPrimitive.Root>,  
9 |   React.ComponentPropsWithoutRef<typeof RadioGroupPrimitive.Root>  
10 | >(({ className, ...props }, ref) => {  
11 |   return <RadioGroupPrimitive.Root className={cn("grid gap-2", className)} {...props} ref={ref} />;  
12 | });  
13 | RadioGroup.displayName = RadioGroupPrimitive.Root.displayName;  
14 |  
15 | const RadioGroupItem = React.forwardRef<  
16 |   React.ElementRef<typeof RadioGroupPrimitive.Item>,  
17 |   React.ComponentPropsWithoutRef<typeof RadioGroupPrimitive.Item>  
18 | >(({ className, ...props }, ref) => {  
19 |   return (  
20 |     <RadioGroupPrimitive.Item  
21 |       ref={ref}  
22 |       className={cn(  
23 |         "aspect-square h-4 w-4 rounded-full border border-primary text-primary shadow cursor-pointer focu...  
24 |         className,  
25 |       )}  
26 |       {...props}  
27 |     >  
28 |       <RadioGroupPrimitive.Indicator className="flex items-center justify-center">  
29 |         <Circle className="h-3.5 w-3.5 fill-primary" />  
30 |       </RadioGroupPrimitive.Indicator>  
31 |     </RadioGroupPrimitive.Item>  
32 |   );  
33 | });  
34 | RadioGroupItem.displayName = RadioGroupPrimitive.Item.displayName;  
35 |  
36 | export { RadioGroup, RadioGroupItem };
```

FILE #83: src\components\uilresizable.tsx

```
1 | import { GripVertical } from "lucide-react";
2 | import { Group, Panel, Separator } from "react-resizable-panels";
3 |
4 | import { cn } from "@lib/utils";
5 |
6 | const ResizablePanelGroup = ({ className, ...props }: React.ComponentProps<typeof Group>) => (
7 |   <Group
8 |     className={cn("flex h-full w-full data-[panel-group-direction=vertical]:flex-col", className)}
9 |     {...props}
10 |   />
11 | );
12 |
13 | const ResizablePanel = Panel;
14 |
15 | const ResizableHandle = ({
16 |   withHandle,
17 |   className,
18 |   ...props
19 | }: React.ComponentProps<typeof Separator> & {
20 |   withHandle?: boolean;
21 | }) => (
22 |   <Separator
23 |     className={cn(
24 |       "relative flex w-px items-center justify-center bg-border after:absolute after:inset-y-0 after:left...
25 |       className,
26 |     )}
27 |     {...props}
28 |   >
29 |     {withHandle && (
30 |       <div className="z-10 flex h-4 w-3 items-center justify-center rounded-sm border bg-border">
31 |         <GripVertical className="h-2.5 w-2.5" />
32 |       </div>
33 |     )}
34 |   </Separator>
35 | );
36 |
37 | export { ResizablePanelGroup, ResizablePanel, ResizableHandle };
```

FILE #84: src\components\ui\scroll-area.tsx

```
1 | import * as React from "react";
2 | import * as ScrollAreaPrimitive from "@radix-ui/react-scroll-area";
3 |
4 | import { cn } from "@lib/utils";
5 |
6 | const ScrollArea = React.forwardRef<
7 |   React.ElementRef<typeof ScrollAreaPrimitive.Root>,
8 |   React.ComponentPropsWithoutRef<typeof ScrollAreaPrimitive.Root>
9 | >(({ className, children, ...props }, ref) => (
10 |   <ScrollAreaPrimitive.Root
11 |     ref={ref}
12 |     className={cn("relative overflow-hidden", className)}
13 |     {...props}
14 |   >
15 |     <ScrollAreaPrimitive.Viewport className="h-full w-full rounded-[inherit]">
16 |       {children}
17 |     </ScrollAreaPrimitive.Viewport>
18 |     <ScrollBar />
19 |     <ScrollAreaPrimitive.Corner />
20 |   </ScrollAreaPrimitive.Root>
21 | ));
22 | ScrollArea.displayName = ScrollAreaPrimitive.Root.displayName;
23 |
```

```
24 | const ScrollBar = React.forwardRef<
25 |   React.ElementRef<typeof ScrollAreaPrimitive.ScrollAreaScrollbar>,
26 |   React.ComponentPropsWithoutRef<typeof ScrollAreaPrimitive.ScrollAreaScrollbar>
27 | >(({ className, orientation = "vertical", ...props }, ref) => (
28 |   <ScrollAreaPrimitive.ScrollAreaScrollbar
29 |     ref={ref}
30 |     orientation={orientation}
31 |     className={cn(
32 |       "flex touch-none select-none transition-colors",
33 |       orientation === "vertical" && "h-full w-2.5 border-l border-l-transparent p-[1px]",
34 |       orientation === "horizontal" && "h-2.5 flex-col border-t border-t-transparent p-[1px]",
35 |       className,
36 |     )}
37 |     {...props}
38 |   >
39 |     <ScrollAreaPrimitive.ScrollAreaThumb className="relative flex-1 rounded-full bg-border" />
40 |   </ScrollAreaPrimitive.ScrollAreaScrollbar>
41 | ));
42 | ScrollBar.displayName = ScrollAreaPrimitive.ScrollAreaScrollbar.displayName;
43 |
44 | export { ScrollArea, ScrollBar };
```

FILE #85: src\components\ui\select.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import * as SelectPrimitive from "@radix-ui/react-select";
5 | import { Check, ChevronDown, ChevronUp } from "lucide-react";
6 |
7 | import { cn } from "@lib/utils";
8 |
9 | const Select = SelectPrimitive.Root;
10 |
11 | const SelectGroup = SelectPrimitive.Group;
12 |
13 | const SelectValue = SelectPrimitive.Value;
14 |
15 | const SelectTrigger = React.forwardRef<
16 |   React.ElementRef<typeof SelectPrimitive.Trigger>,
17 |   React.ComponentPropsWithoutRef<typeof SelectPrimitive.Trigger>
18 | >(({ className, children, ...props }, ref) => (
19 |   <SelectPrimitive.Trigger
20 |     ref={ref}
21 |     className={cn(
22 |       "flex h-9 w-full items-center justify-between whitespace-nowrap rounded-md border border-input bg-t...
23 |       className,
24 |     )}
25 |     {...props}
26 |   >
27 |     {children}
28 |     <SelectPrimitive.Icon asChild>
29 |       <ChevronDown className="h-4 w-4 opacity-50" />
30 |     </SelectPrimitive.Icon>
31 |   </SelectPrimitive.Trigger>
32 | ));
33 | SelectTrigger.displayName = SelectPrimitive.Trigger.displayName;
34 |
35 | const SelectScrollUpButton = React.forwardRef<
36 |   React.ElementRef<typeof SelectPrimitive.ScrollUpButton>,
37 |   React.ComponentPropsWithoutRef<typeof SelectPrimitive.ScrollUpButton>
38 | >(({ className, ...props }, ref) => (
39 |   <SelectPrimitive.ScrollUpButton
```

```
40 |     ref={ref}
41 |     className={cn("flex cursor-default items-center justify-center py-1", className)}
42 |     {...props}
43 |   >
44 |     <ChevronUp className="h-4 w-4" />
45 |   </SelectPrimitive.ScrollUpButton>
46 | ));
47 | SelectScrollUpButton.displayName = SelectPrimitive.ScrollUpButton.displayName;
48 |
49 | const SelectScrollDownButton = React.forwardRef<
50 |   React.ElementRef<typeof SelectPrimitive.ScrollDownButton>,
51 |   React.ComponentPropsWithoutRef<typeof SelectPrimitive.ScrollDownButton>
52 | >(({ className, ...props }, ref) => (
53 |   <SelectPrimitive.ScrollDownButton
54 |     ref={ref}
55 |     className={cn("flex cursor-default items-center justify-center py-1", className)}
56 |     {...props}
57 |   >
58 |     <ChevronDown className="h-4 w-4" />
59 |   </SelectPrimitive.ScrollDownButton>
60 | ));
61 | SelectScrollDownButton.displayName = SelectPrimitive.ScrollDownButton.displayName;
62 |
63 | const SelectContent = React.forwardRef<
64 |   React.ElementRef<typeof SelectPrimitive.Content>,
65 |   React.ComponentPropsWithoutRef<typeof SelectPrimitive.Content>
66 | >(({ className, children, position = "popper", ...props }, ref) => (
67 |   <SelectPrimitive.Portal>
68 |     <SelectPrimitive.Content
69 |       ref={ref}
70 |       className={cn(
71 |         "relative z-50 max-h-(--radix-select-content-available-height) min-w-[8rem] overflow-y-auto overf...
72 |         position === "popper" &&
73 |         "data-[side=bottom]:translate-y-1 data-[side=left]:-translate-x-1 data-[side=right]:translate-x...
74 |         className,
75 |       )}
76 |       position={position}
77 |       {...props}
78 |     >
79 |       <SelectScrollUpButton />
80 |       <SelectPrimitive.Viewport
81 |         className={cn(
82 |           "p-1",
83 |           position === "popper" &&
84 |           "h-[var(--radix-select-trigger-height)] w-full min-w-[var(--radix-select-trigger-width)]",
85 |         )}
86 |     >
87 |       {children}
88 |     </SelectPrimitive.Viewport>
89 |     <SelectScrollDownButton />
90 |   </SelectPrimitive.Content>
91 | </SelectPrimitive.Portal>
92 | ));
93 | SelectContent.displayName = SelectPrimitive.Content.displayName;
94 |
95 | const SelectLabel = React.forwardRef<
96 |   React.ElementRef<typeof SelectPrimitive.Label>,
97 |   React.ComponentPropsWithoutRef<typeof SelectPrimitive.Label>
98 | >(({ className, ...props }, ref) => (
99 |   <SelectPrimitive.Label
100 |     ref={ref}
101 |     className={cn("px-2 py-1.5 text-sm font-semibold", className)}
102 |     {...props}
103 |   />
104 | ));
```

```
105 | SelectLabel.displayName = SelectPrimitive.Label.displayName;
106 |
107 | const SelectItem = React.forwardRef<
108 |   React.ElementRef<typeof SelectPrimitive.Item>,
109 |   React.ComponentPropsWithoutRef<typeof SelectPrimitive.Item>
110 | >(({ className, children, ...props }, ref) => (
111 |   <SelectPrimitive.Item
112 |     ref={ref}
113 |     className={cn(
114 |       "relative flex w-full cursor-default select-none items-center rounded-sm py-1.5 pl-2 pr-8 text-sm o...
115 |       className,
116 |     )}
117 |     {...props}
118 |   >
119 |     <span className="absolute right-2 flex h-3.5 w-3.5 items-center justify-center">
120 |       <SelectPrimitive.ItemIndicator>
121 |         <Check className="h-4 w-4" />
122 |       </SelectPrimitive.ItemIndicator>
123 |     </span>
124 |     <SelectPrimitive.ItemText>{children}</SelectPrimitive.ItemText>
125 |   </SelectPrimitive.Item>
126 | ));
127 | SelectItem.displayName = SelectPrimitive.Item.displayName;
128 |
129 | const SelectSeparator = React.forwardRef<
130 |   React.ElementRef<typeof SelectPrimitive.Separator>,
131 |   React.ComponentPropsWithoutRef<typeof SelectPrimitive.Separator>
132 | >(({ className, ...props }, ref) => (
133 |   <SelectPrimitive.Separator
134 |     ref={ref}
135 |     className={cn("-mx-1 my-1 h-px bg-muted", className)}
136 |     {...props}
137 |   />
138 | ));
139 | SelectSeparator.displayName = SelectPrimitive.Separator.displayName;
140 |
141 | export {
142 |   Select,
143 |   SelectGroup,
144 |   SelectValue,
145 |   SelectTrigger,
146 |   SelectContent,
147 |   SelectLabel,
148 |   SelectItem,
149 |   SelectSeparator,
150 |   SelectScrollUpButton,
151 |   SelectScrollDownButton,
152 | };
```

FILE #86: src\components\uilseparator.tsx

```
1 | import * as React from "react";
2 | import * as SeparatorPrimitive from "@radix-ui/react-separator";
3 |
4 | import { cn } from "@lib/utils";
5 |
6 | const Separator = React.forwardRef<
7 |   React.ElementRef<typeof SeparatorPrimitive.Root>,
8 |   React.ComponentPropsWithoutRef<typeof SeparatorPrimitive.Root>
9 | >(({ className, orientation = "horizontal", decorative = true, ...props }, ref) => (
10 |   <SeparatorPrimitive.Root
11 |     ref={ref}
12 |     decorative={decorative}
```

```
13 |     orientation={orientation}
14 |     className={cn(
15 |       "shrink-0 bg-border",
16 |       orientation === "horizontal" ? "h-[1px] w-full" : "h-full w-[1px]",
17 |       className,
18 |     )}
19 |     {...props}
20 |   />
21 | ));
22 | Separator.displayName = SeparatorPrimitive.Root.displayName;
23 |
24 | export { Separator };
```

FILE #87: src\components\uilsheet.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import * as SheetPrimitive from "@radix-ui/react-dialog";
5 | import { cva, type VariantProps } from "class-variance-authority";
6 | import { X } from "lucide-react";
7 |
8 | import { cn } from "@lib/utlis";
9 |
10 | const Sheet = SheetPrimitive.Root;
11 |
12 | const SheetTrigger = SheetPrimitive.Trigger;
13 |
14 | const SheetClose = SheetPrimitive.Close;
15 |
16 | const SheetPortal = SheetPrimitive.Portal;
17 |
18 | const SheetOverlay = React.forwardRef<
19 |   React.ElementRef<typeof SheetPrimitive.Overlay>,
20 |   React.ComponentPropsWithoutRef<typeof SheetPrimitive.Overlay>
21 | >(({ className, ...props }, ref) => (
22 |   <SheetPrimitive.Overlay
23 |     className={cn(
24 |       "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-...
25 |     className,
26 |   )}
27 |   {...props}
28 |   ref={ref}
29 | />
30 | ));
31 | SheetOverlay.displayName = SheetPrimitive.Overlay.displayName;
32 |
33 | const sheetVariants = cva(
34 |   "fixed z-50 gap-4 bg-background p-6 shadow-lg transition ease-in-out data-[state=closed]:duration-300 d...
35 |   {
36 |     variants: {
37 |       side: {
38 |         top: "inset-x-0 top-0 border-b data-[state=closed]:slide-out-to-top data-[state=open]:slide-in-fr...
39 |         bottom:
40 |           "inset-x-0 bottom-0 border-t data-[state=closed]:slide-out-to-bottom data-[state=open]:slide-in...
41 |         left: "inset-y-0 left-0 h-full w-3/4 border-r data-[state=closed]:slide-out-to-left data-[state=o...
42 |         right:
43 |           "inset-y-0 right-0 h-full w-3/4 border-l data-[state=closed]:slide-out-to-right data-[state=ope...
44 |       },
45 |     },
46 |     defaultVariants: {
47 |       side: "right",
48 |     },
```

```
49 |   },
50 | );
51 |
52 | interface SheetContentProps
53 |   extends
54 |     React.ComponentPropsWithoutRef<typeof SheetPrimitive.Content>,
55 |     VariantProps<typeof sheetVariants> {}
56 |
57 | const SheetContent = React.forwardRef<
58 |   React.ElementRef<typeof SheetPrimitive.Content>,
59 |   SheetContentProps
60 | >(({ side = "right", className, children, ...props }, ref) => (
61 |   <SheetPortal>
62 |     <SheetOverlay />
63 |     <SheetPrimitive.Content ref={ref} className={cn(sheetVariants({ side }), className)} {...props}>
64 |       <SheetPrimitive.Close className="absolute right-4 top-4 rounded-sm opacity-70 ring-offset-background...
65 |         <X className="h-4 w-4" />
66 |         <span className="sr-only">Close</span>
67 |       </SheetPrimitive.Close>
68 |       {children}
69 |     </SheetPrimitive.Content>
70 |   </SheetPortal>
71 | ));
72 | SheetContent.displayName = SheetPrimitive.Content.displayName;
73 |
74 | const SheetHeader = ({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) => (
75 |   <div className={cn("flex flex-col space-y-2 text-center sm:text-left", className)} {...props} />
76 | );
77 | SheetHeader.displayName = "SheetHeader";
78 |
79 | const SheetFooter = ({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) => (
80 |   <div
81 |     className={cn("flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2", className)}
82 |     {...props}
83 |   />
84 | );
85 | SheetFooter.displayName = "SheetFooter";
86 |
87 | const SheetTitle = React.forwardRef<
88 |   React.ElementRef<typeof SheetPrimitive.Title>,
89 |   React.ComponentPropsWithoutRef<typeof SheetPrimitive.Title>
90 | >(({ className, ...props }, ref) => (
91 |   <SheetPrimitive.Title
92 |     ref={ref}
93 |     className={cn("text-lg font-semibold text-foreground", className)}
94 |     {...props}
95 |   />
96 | ));
97 | SheetTitle.displayName = SheetPrimitive.Title.displayName;
98 |
99 | const SheetDescription = React.forwardRef<
100 |   React.ElementRef<typeof SheetPrimitive.Description>,
101 |   React.ComponentPropsWithoutRef<typeof SheetPrimitive.Description>
102 | >(({ className, ...props }, ref) => (
103 |   <SheetPrimitive.Description
104 |     ref={ref}
105 |     className={cn("text-sm text-muted-foreground", className)}
106 |     {...props}
107 |   />
108 | ));
109 | SheetDescription.displayName = SheetPrimitive.Description.displayName;
110 |
111 | export {
112 |   Sheet,
113 |   SheetPortal,
```



```
114 | SheetOverlay,  
115 | SheetTrigger,  
116 | SheetClose,  
117 | SheetContent,  
118 | SheetHeader,  
119 | SheetFooter,  
120 | SheetTitle,  
121 | SheetDescription,  
122 | };
```

FILE #88: src\components\uilsidebar.tsx

```
1 | import * as React from "react";  
2 | import { Slot } from "@radix-ui/react-slot";  
3 | import { cva, type VariantProps } from "class-variance-authority";  
4 | import { PanelLeft } from "lucide-react";  
5 |  
6 | import { useIsMobile } from "@/hooks/use-mobile";  
7 | import { cn } from "@/lib/utils";  
8 | import { Button } from "@/components/ui/button";  
9 | import { Input } from "@/components/ui/input";  
10 | import { Separator } from "@/components/ui/separator";  
11 | import {  
12 |   Sheet,  
13 |   SheetContent,  
14 |   SheetDescription,  
15 |   SheetHeader,  
16 |   SheetTitle,  
17 | } from "@/components/ui/sheet";  
18 | import { Skeleton } from "@/components/ui/skeleton";  
19 | import { Tooltip, TooltipContent, TooltipProvider, TooltipTrigger } from "@/components/ui/tooltip";  
20 |  
21 | const SIDEBAR_COOKIE_NAME = "sidebar_state";  
22 | const SIDEBAR_COOKIE_MAX_AGE = 60 * 60 * 24 * 7;  
23 | const SIDEBAR_WIDTH = "16rem";  
24 | const SIDEBAR_WIDTH_MOBILE = "18rem";  
25 | const SIDEBAR_WIDTH_ICON = "3rem";  
26 | const SIDEBAR_KEYBOARD_SHORTCUT = "b";  
27 |  
28 | type SidebarContextProps = {  
29 |   state: "expanded" | "collapsed";  
30 |   open: boolean;  
31 |   setOpen: (open: boolean) => void;  
32 |   openMobile: boolean;  
33 |   setOpenMobile: (open: boolean) => void;  
34 |   isMobile: boolean;  
35 |   toggleSidebar: () => void;  
36 | };  
37 |  
38 | const SidebarContext = React.createContext<SidebarContextProps | null>(null);  
39 |  
40 | function useSidebar() {  
41 |   const context = React.useContext(SidebarContext);  
42 |   if (!context) {  
43 |     throw new Error("useSidebar must be used within a SidebarProvider.");  
44 |   }  
45 |  
46 |   return context;  
47 | }  
48 |  
49 | const SidebarProvider = React.forwardRef<  
50 |   HTMLDivElement,  
51 |   React.ComponentProps<"div"> & {
```

```
52 |     defaultOpen?: boolean;
53 |     open?: boolean;
54 |     onOpenChange?: (open: boolean) => void;
55 |   }
56 | >{
57 |   (
58 |     {
59 |       defaultOpen = true,
60 |       open: openProp,
61 |       onOpenChange: setOpenProp,
62 |       className,
63 |       style,
64 |       children,
65 |       ...props
66 |     },
67 |     ref,
68 |   ) => {
69 |     const isMobile = useIsMobile();
70 |     const [openMobile, setOpenMobile] = React.useState(false);
71 |
72 |     // This is the internal state of the sidebar.
73 |     // We use openProp and setOpenProp for control from outside the component.
74 |     const [_open, _setOpen] = React.useState(defaultOpen);
75 |     const open = openProp ?? _open;
76 |     const setOpen = React.useCallback(
77 |       (value: boolean | ((value: boolean) => boolean)) => {
78 |         const openState = typeof value === "function" ? value(open) : value;
79 |         if (setOpenProp) {
80 |           setOpenProp(openState);
81 |         } else {
82 |           _setOpen(openState);
83 |         }
84 |
85 |         // This sets the cookie to keep the sidebar state.
86 |         document.cookie = `${SIDEBAR_COOKIE_NAME}=${openState}; path=/; max-age=${SIDEBAR_COOKIE_MAX_AGE}`;
87 |       },
88 |       [setOpenProp, open],
89 |     );
90 |
91 |     // Helper to toggle the sidebar.
92 |     const toggleSidebar = React.useCallback(() => {
93 |       return isMobile ? setOpenMobile((open) => !open) : setOpen((open) => !open);
94 |     }, [isMobile, setOpen, setOpenMobile]);
95 |
96 |     // Adds a keyboard shortcut to toggle the sidebar.
97 |     React.useEffect(() => {
98 |       const handleKeyDown = (event: KeyboardEvent) => {
99 |         if (event.key === SIDEBAR_KEYBOARD_SHORTCUT && (event.metaKey || event.ctrlKey)) {
100 |           event.preventDefault();
101 |           toggleSidebar();
102 |         }
103 |       };
104 |
105 |       window.addEventListener("keydown", handleKeyDown);
106 |       return () => window.removeEventListener("keydown", handleKeyDown);
107 |     }, [toggleSidebar]);
108 |
109 |     // We add a state so that we can do data-state="expanded" or "collapsed".
110 |     // This makes it easier to style the sidebar with Tailwind classes.
111 |     const state = open ? "expanded" : "collapsed";
112 |
113 |     const contextValue = React.useMemo<SidebarContextProps>(
114 |       () => ({
115 |         state,
116 |         open,
```

```
117 |         setOpen,
118 |         isMobile,
119 |         openMobile,
120 |         setOpenMobile,
121 |         toggleSidebar,
122 |     })),
123 |     [state, open, setOpen, isMobile, openMobile, setOpenMobile, toggleSidebar],
124 | );
125 |
126 | return (
127 |     <SidebarContext.Provider value={contextValue}>
128 |         <TooltipProvider delayDuration={0}>
129 |             <div
130 |                 style={
131 |                     {
132 |                         "--sidebar-width": SIDEBAR_WIDTH,
133 |                         "--sidebar-width-icon": SIDEBAR_WIDTH_ICON,
134 |                         ...style,
135 |                     } as React.CSSProperties
136 |                 }
137 |                 className={cn(
138 |                     "group/sidebar-wrapper flex min-h-svh w-full has-[[data-variant=inset]]:bg-sidebar",
139 |                     className,
140 |                 )}
141 |                 ref={ref}
142 |                 {...props}
143 |             >
144 |                 {children}
145 |             </div>
146 |         </TooltipProvider>
147 |     </SidebarContext.Provider>
148 | );
149 | },
150 | );
151 | SidebarProvider.displayName = "SidebarProvider";
152 |
153 | const Sidebar = React.forwardRef<
154 |     HTMLDivElement,
155 |     React.ComponentProps<"div"> & {
156 |         side?: "left" | "right";
157 |         variant?: "sidebar" | "floating" | "inset";
158 |         collapsible?: "offcanvas" | "icon" | "none";
159 |     }
160 | >((
161 |     (
162 |         {
163 |             side = "left",
164 |             variant = "sidebar",
165 |             collapsible = "offcanvas",
166 |             className,
167 |             children,
168 |             ...props
169 |         },
170 |         ref,
171 |     ) => {
172 |         const { isMobile, state, openMobile, setOpenMobile } = useSidebar();
173 |
174 |         if (collapsible === "none") {
175 |             return (
176 |                 <div
177 |                     className={cn(
178 |                         "flex h-full w-(--sidebar-width) flex-col bg-sidebar text-sidebar-foreground",
179 |                         className,
180 |                     )}
181 |                     ref={ref}
```

```
182 |         {...props}
183 |       >
184 |         {children}
185 |       </div>
186 |     );
187 |   }
188 |
189 |   if (isMobile) {
190 |     return (
191 |       <Sheet open={openMobile} onOpenChange={setOpenMobile} {...props}>
192 |         <SheetContent
193 |           data-sidebar="sidebar"
194 |           data-mobile="true"
195 |           className="w-(--sidebar-width) bg-sidebar p-0 text-sidebar-foreground [&gt;button]:hidden"
196 |           style={
197 |             {
198 |               "--sidebar-width": SIDEBAR_WIDTH_MOBILE,
199 |             } as React.CSSProperties
200 |           }
201 |           side={side}
202 |         >
203 |           <SheetHeader className="sr-only">
204 |             <SheetTitle>Sidebar</SheetTitle>
205 |             <SheetDescription>Displays the mobile sidebar.</SheetDescription>
206 |           </SheetHeader>
207 |           <div className="flex h-full w-full flex-col">{children}</div>
208 |         </SheetContent>
209 |       </Sheet>
210 |     );
211 |   }
212 |
213 |   return (
214 |     <div
215 |       ref={ref}
216 |       className="group peer hidden text-sidebar-foreground md:block"
217 |       data-state={state}
218 |       data-collapsible={state === "collapsed" ? collapsible : ""}
219 |       data-variant={variant}
220 |       data-side={side}
221 |     >
222 |       /* This is what handles the sidebar gap on desktop */
223 |       <div
224 |         className={cn(
225 |           "relative w-(--sidebar-width) bg-transparent transition-[width] duration-200 ease-linear",
226 |           "group-data-[collapsible=offcanvas]:w-0",
227 |           "group-data-[side=right]:rotate-180",
228 |           variant === "floating" || variant === "inset"
229 |             ? "group-data-[collapsible=icon]:w-[calc(var(--sidebar-width-icon)+_theme(spacing.4))]"
230 |             : "group-data-[collapsible=icon]:w-(--sidebar-width-icon)",
231 |         )}
232 |       />
233 |       <div
234 |         className={cn(
235 |           "fixed inset-y-0 z-10 hidden h-svh w-(--sidebar-width) transition-[left,right,width] duration...",
236 |           side === "left"
237 |             ? "left-0 group-data-[collapsible=offcanvas]:left-[calc(var(--sidebar-width)*-1)]"
238 |             : "right-0 group-data-[collapsible=offcanvas]:right-[calc(var(--sidebar-width)*-1)]",
239 |           // Adjust the padding for floating and inset variants.
240 |           variant === "floating" || variant === "inset"
241 |             ? "p-2 group-data-[collapsible=icon]:w-[calc(var(--sidebar-width-icon)+_theme(spacing.4)+...
242 |               : "group-data-[collapsible=icon]:w-(--sidebar-width-icon) group-data-[side=left]:border-r g...
243 |           className,
244 |         )}
245 |       {...props}
246 |     >
```

```
247 |         <div
248 |             data-sidebar="sidebar"
249 |             className="flex h-full w-full flex-col bg-sidebar group-data-[variant=floating]:rounded-lg gr...
250 |         >
251 |             {children}
252 |         </div>
253 |     </div>
254 | </div>
255 | );
256 | },
257 | );
258 | Sidebar.displayName = "Sidebar";
259 |
260 | const SidebarTrigger = React.forwardRef<
261 |     React.ElementRef<typeof Button>,
262 |     React.ComponentProps<typeof Button>
263 | >(({ className, onClick, ...props }, ref) => {
264 |     const { toggleSidebar } = useSidebar();
265 |
266 |     return (
267 |         <Button
268 |             ref={ref}
269 |             data-sidebar="trigger"
270 |             variant="ghost"
271 |             size="icon"
272 |             className={cn("h-7 w-7", className)}
273 |             onClick={(event) => {
274 |                 onClick?.(event);
275 |                 toggleSidebar();
276 |             }}
277 |             {...props}
278 |         >
279 |             <PanelLeft />
280 |             <span className="sr-only">Toggle Sidebar</span>
281 |         </Button>
282 |     );
283 | });
284 | SidebarTrigger.displayName = "SidebarTrigger";
285 |
286 | const SidebarRail = React.forwardRef<HTMLButtonElement, React.ComponentProps<"button">>(
287 |     ({ className, ...props }, ref) => {
288 |         const { toggleSidebar } = useSidebar();
289 |
290 |         return (
291 |             <button
292 |                 ref={ref}
293 |                 data-sidebar="rail"
294 |                 aria-label="Toggle Sidebar"
295 |                 tabIndex={-1}
296 |                 onClick={toggleSidebar}
297 |                 title="Toggle Sidebar"
298 |                 className={cn(
299 |                     "absolute inset-y-0 z-20 hidden w-4 -translate-x-1/2 transition-all ease-linear after:absolute ...
300 |                     "[[data-side=left]_&]:cursor-w-resize [[data-side=right]_&]:cursor-e-resize",
301 |                     "[[data-side=left][data-state=collapsed]_&]:cursor-e-resize [[data-side=right][data-state=colla...
302 |                     "group-data-[collapsible=offcanvas]:translate-x-0 group-data-[collapsible=offcanvas]:after:left...
303 |                     "[[data-side=left][data-collapsible=offcanvas]_&]:-right-2",
304 |                     "[[data-side=right][data-collapsible=offcanvas]_&]:-left-2",
305 |                     className,
306 |                 )}
307 |                 {...props}
308 |             />
309 |         );
310 |     },
311 | );
```

```
312 | SidebarRail.displayName = "SidebarRail";
313 |
314 | const SidebarInset = React.forwardRef<HTMLDivElement, React.ComponentProps<"main">>(  
315 |   ({ className, ...props }, ref) => {  
316 |     return (  
317 |       <main  
318 |         ref={ref}  
319 |         className={cn(  
320 |           "relative flex w-full flex-1 flex-col bg-background",  
321 |           "md:peer-data-[variant=inset]:m-2 md:peer-data-[state=collapsed]:peer-data-[variant=inset]:ml-2...",  
322 |           className,  
323 |         )}  
324 |         {...props}  
325 |       />  
326 |     );  
327 |   },  
328 | );  
329 | SidebarInset.displayName = "SidebarInset";  
330 |  
331 | const SidebarInput = React.forwardRef<  
332 |   React.ElementRef<typeof Input>,  
333 |   React.ComponentProps<typeof Input>  
334 | >(({ className, ...props }, ref) => {  
335 |   return (  
336 |     <Input  
337 |       ref={ref}  
338 |       data-sidebar="input"  
339 |       className={cn(  
340 |         "h-8 w-full bg-background shadow-none focus-visible:ring-2 focus-visible:ring-sidebar-ring",  
341 |         className,  
342 |       )}  
343 |       {...props}  
344 |     />  
345 |   );  
346 | });  
347 | SidebarInput.displayName = "SidebarInput";  
348 |  
349 | const SidebarHeader = React.forwardRef<HTMLDivElement, React.ComponentProps<"div">>(  
350 |   ({ className, ...props }, ref) => {  
351 |     return (  
352 |       <div  
353 |         ref={ref}  
354 |         data-sidebar="header"  
355 |         className={cn("flex flex-col gap-2 p-2", className)}  
356 |         {...props}  
357 |       />  
358 |     );  
359 |   },  
360 | );  
361 | SidebarHeader.displayName = "SidebarHeader";  
362 |  
363 | const SidebarFooter = React.forwardRef<HTMLDivElement, React.ComponentProps<"div">>(  
364 |   ({ className, ...props }, ref) => {  
365 |     return (  
366 |       <div  
367 |         ref={ref}  
368 |         data-sidebar="footer"  
369 |         className={cn("flex flex-col gap-2 p-2", className)}  
370 |         {...props}  
371 |       />  
372 |     );  
373 |   },  
374 | );  
375 | SidebarFooter.displayName = "SidebarFooter";  
376 |
```

```
377 | const SidebarSeparator = React.forwardRef<
378 |   React.ElementRef<typeof Separator>,
379 |   React.ComponentProps<typeof Separator>
380 | >(({ className, ...props }, ref) => {
381 |   return (
382 |     <Separator
383 |       ref={ref}
384 |       data-sidebar="separator"
385 |       className={cn("mx-2 w-auto bg-sidebar-border", className)}
386 |       {...props}
387 |     />
388 |   );
389 | });
390 | SidebarSeparator.displayName = "SidebarSeparator";
391 |
392 | const SidebarContent = React.forwardRef<HTMLDivElement, React.ComponentProps<"div">>(
393 |   ({ className, ...props }, ref) => {
394 |     return (
395 |       <div
396 |         ref={ref}
397 |         data-sidebar="content"
398 |         className={cn(
399 |           "flex min-h-0 flex-1 flex-col gap-2 overflow-auto group-data-[collapsible=icon]:overflow-hidden",
400 |           className,
401 |         )}
402 |         {...props}
403 |       />
404 |     );
405 |   },
406 | );
407 | SidebarContent.displayName = "SidebarContent";
408 |
409 | const SidebarGroup = React.forwardRef<HTMLDivElement, React.ComponentProps<"div">>(
410 |   ({ className, ...props }, ref) => {
411 |     return (
412 |       <div
413 |         ref={ref}
414 |         data-sidebar="group"
415 |         className={cn("relative flex w-full min-w-0 flex-col p-2", className)}
416 |         {...props}
417 |       />
418 |     );
419 |   },
420 | );
421 | SidebarGroup.displayName = "SidebarGroup";
422 |
423 | const SidebarGroupLabel = React.forwardRef<
424 |   HTMLDivElement,
425 |   React.ComponentProps<"div"> & { asChild?: boolean }
426 | >(({ className, asChild = false, ...props }, ref) => {
427 |   const Comp = asChild ? Slot : "div";
428 |
429 |   return (
430 |     <Comp
431 |       ref={ref}
432 |       data-sidebar="group-label"
433 |       className={cn(
434 |         "flex h-8 shrink-0 items-center rounded-md px-2 text-xs font-medium text-sidebar-foreground/70 ou...",
435 |         "group-data-[collapsible=icon]:-mt-8 group-data-[collapsible=icon]:opacity-0",
436 |         className,
437 |       )}
438 |       {...props}
439 |     />
440 |   );
441 | });
```

```
442 | SidebarGroupLabel.displayName = "SidebarGroupLabel";
443 |
444 | const SidebarGroupAction = React.forwardRef<
445 |   HTMLButtonElement,
446 |   React.ComponentProps<"button"> & { asChild?: boolean }
447 | >(({ className, asChild = false, ...props }, ref) => {
448 |   const Comp = asChild ? Slot : "button";
449 |
450 |   return (
451 |     <Comp
452 |       ref={ref}
453 |       data-sidebar="group-action"
454 |       className={cn(
455 |         "absolute right-3 top-3.5 flex aspect-square w-5 items-center justify-center rounded-md p-0 text-...
456 |         // Increases the hit area of the button on mobile.
457 |         "after:absolute after:-inset-2 after:md:hidden",
458 |         "group-data-[collapsible=icon]:hidden",
459 |         className,
460 |       )}
461 |       {...props}
462 |     />
463 |   );
464 | });
465 | SidebarGroupAction.displayName = "SidebarGroupAction";
466 |
467 | const SidebarGroupContent = React.forwardRef<HTMLDivElement, React.ComponentProps<"div">>(
468 |   ({ className, ...props }, ref) => (
469 |     <div
470 |       ref={ref}
471 |       data-sidebar="group-content"
472 |       className={cn("w-full text-sm", className)}
473 |       {...props}
474 |     />
475 |   ),
476 | );
477 | SidebarGroupContent.displayName = "SidebarGroupContent";
478 |
479 | const SidebarMenu = React.forwardRef<HTMLUListElement, React.ComponentProps<"ul">>(
480 |   ({ className, ...props }, ref) => (
481 |     <ul
482 |       ref={ref}
483 |       data-sidebar="menu"
484 |       className={cn("flex w-full min-w-0 flex-col gap-1", className)}
485 |       {...props}
486 |     />
487 |   ),
488 | );
489 | SidebarMenu.displayName = "SidebarMenu";
490 |
491 | const SidebarMenuItem = React.forwardRef<HTMLLIElement, React.ComponentProps<"li">>(
492 |   ({ className, ...props }, ref) => (
493 |     <li
494 |       ref={ref}
495 |       data-sidebar="menu-item"
496 |       className={cn("group/menu-item relative", className)}
497 |       {...props}
498 |     />
499 |   ),
500 | );
501 | SidebarMenuItem.displayName = "SidebarMenuItem";
502 |
503 | const sidebarMenuButtonVariants = cva(
504 |   "peer/menu-button flex w-full items-center gap-2 overflow-hidden rounded-md p-2 text-left text-sm outli...
505 |   {
506 |     variants: {
```



```
507 |     variant: {
508 |       default: "hover:bg-sidebar-accent hover:text-sidebar-accent-foreground",
509 |       outline:
510 |         "bg-background shadow-[0_0_1px_var(--sidebar-border)] hover:bg-sidebar-accent hover:text-side...",
511 |     },
512 |     size: {
513 |       default: "h-8 text-sm",
514 |       sm: "h-7 text-xs",
515 |       lg: "h-12 text-sm group-data-[collapsible=icon]:!p-0",
516 |     },
517 |   },
518 |   defaultVariants: {
519 |     variant: "default",
520 |     size: "default",
521 |   },
522 | },
523 | );
524 |
525 | const SidebarMenuButton = React.forwardRef<
526 |   HTMLButtonElement,
527 |   React.ComponentProps<"button"> & {
528 |     asChild?: boolean;
529 |     isActive?: boolean;
530 |     tooltip?: string | React.ComponentProps<typeof TooltipContent>;
531 |   } & VariantProps<typeof sidebarMenuButtonVariants>
532 | >({
533 |   (
534 |     {
535 |       asChild = false,
536 |       isActive = false,
537 |       variant = "default",
538 |       size = "default",
539 |       tooltip,
540 |       className,
541 |       ...props
542 |     },
543 |     ref,
544 |   ) => {
545 |     const Comp = asChild ? Slot : "button";
546 |     const { isMobile, state } = useSidebar();
547 |
548 |     const button = (
549 |       <Comp
550 |         ref={ref}
551 |         data-sidebar="menu-button"
552 |         data-size={size}
553 |         data-active={isActive}
554 |         className={cn(sidebarMenuButtonVariants({ variant, size }), className)}
555 |         {...props}
556 |       />
557 |     );
558 |
559 |     if (!tooltip) {
560 |       return button;
561 |     }
562 |
563 |     if (typeof tooltip === "string") {
564 |       tooltip = {
565 |         children: tooltip,
566 |       };
567 |     }
568 |
569 |     return (
570 |       <Tooltip>
571 |         <TooltipTrigger asChild>{button}</TooltipTrigger>
```

```
572 |         <TooltipContent
573 |             side="right"
574 |             align="center"
575 |             hidden={state !== "collapsed" || isMobile}
576 |             {...tooltip}
577 |         />
578 |     </Tooltip>
579 | );
580 | },
581 | );
582 | SidebarMenuButton.displayName = "SidebarMenuButton";
583 |
584 | const SidebarMenuAction = React.forwardRef<
585 |     HTMLButtonElement,
586 |     React.ComponentProps<"button"> & {
587 |         asChild?: boolean;
588 |         showOnHover?: boolean;
589 |     }
590 | >(({ className, asChild = false, showOnHover = false, ...props }, ref) => {
591 |     const Comp = asChild ? Slot : "button";
592 |
593 |     return (
594 |         <Comp
595 |             ref={ref}
596 |             data-sidebar="menu-action"
597 |             className={cn(
598 |                 "absolute right-1 top-1.5 flex aspect-square w-5 items-center justify-center rounded-md p-0 text-...
599 |                 // Increases the hit area of the button on mobile.
600 |                 "after:absolute after:-inset-2 after:md:hidden",
601 |                 "peer-data-[size=sm]/menu-button:top-1",
602 |                 "peer-data-[size=default]/menu-button:top-1.5",
603 |                 "peer-data-[size=lg]/menu-button:top-2.5",
604 |                 "group-data-[collapsible=icon]:hidden",
605 |                 showOnHover &&
606 |                 "group-focus-within/menu-item:opacity-100 group-hover/menu-item:opacity-100 data-[state=open]:o...
607 |                 className,
608 |             )}
609 |             {...props}
610 |         />
611 |     );
612 | });
613 | SidebarMenuAction.displayName = "SidebarMenuAction";
614 |
615 | const SidebarMenuBadge = React.forwardRef<HTMLDivElement, React.ComponentProps<"div">>(>
616 |     ({ className, ...props }, ref) => (
617 |         <div
618 |             ref={ref}
619 |             data-sidebar="menu-badge"
620 |             className={cn(
621 |                 "pointer-events-none absolute right-1 flex h-5 min-w-5 select-none items-center justify-center ro...
622 |                 "peer-hover/menu-button:text-sidebar-accent-foreground peer-data-[active=true]/menu-button:text-s...
623 |                 "peer-data-[size=sm]/menu-button:top-1",
624 |                 "peer-data-[size=default]/menu-button:top-1.5",
625 |                 "peer-data-[size=lg]/menu-button:top-2.5",
626 |                 "group-data-[collapsible=icon]:hidden",
627 |                 className,
628 |             )}
629 |             {...props}
630 |         />
631 |     ),
632 | );
633 | SidebarMenuBadge.displayName = "SidebarMenuBadge";
634 |
635 | const SidebarMenuSkeleton = React.forwardRef<
636 |     HTMLDivElement,
```

```
637 |   React.ComponentProps<"div"> & {
638 |     showIcon?: boolean;
639 |   }
640 | >(({ className, showIcon = false, ...props }, ref) => {
641 |   // Random width between 50 to 90%.
642 |   const width = React.useMemo(() => {
643 |     return `${Math.floor(Math.random() * 40) + 50}%`;
644 |   }, []);
645 |
646 |   return (
647 |     <div
648 |       ref={ref}
649 |       data-sidebar="menu-skeleton"
650 |       className={cn("flex h-8 items-center gap-2 rounded-md px-2", className)}
651 |       {...props}
652 |     >
653 |       {showIcon && <Skeleton className="size-4 rounded-md" data-sidebar="menu-skeleton-icon" />}
654 |       <Skeleton
655 |         className="h-4 max-w(--skeleton-width) flex-1"
656 |         data-sidebar="menu-skeleton-text"
657 |         style={
658 |           {
659 |             "--skeleton-width": width,
660 |           } as React.CSSProperties
661 |         }
662 |       />
663 |     </div>
664 |   );
665 | });
666 | SidebarMenuSkeleton.displayName = "SidebarMenuSkeleton";
667 |
668 | const SidebarMenuSub = React.forwardRef<HTMLUListElement, React.ComponentProps<"ul">>(>
669 |   ({ className, ...props }, ref) => (
670 |     <ul
671 |       ref={ref}
672 |       data-sidebar="menu-sub"
673 |       className={cn(
674 |         "mx-3.5 flex min-w-0 translate-x-py flex-col gap-1 border-l border-sidebar-border px-2.5 py-0.5",
675 |         "group-data-[collapsible=icon]:hidden",
676 |         className,
677 |       )}
678 |       {...props}
679 |     />
680 |   ),
681 | );
682 | SidebarMenuSub.displayName = "SidebarMenuSub";
683 |
684 | const SidebarMenuSubItem = React.forwardRef<HTMListItemElement, React.ComponentProps<"li">>(>
685 |   ({ ...props }, ref) => <li ref={ref} {...props} />,
686 | );
687 | SidebarMenuSubItem.displayName = "SidebarMenuSubItem";
688 |
689 | const SidebarMenuSubButton = React.forwardRef<
690 |   HTMLAnchorElement,
691 |   React.ComponentProps<"a"> & {
692 |     asChild?: boolean;
693 |     size?: "sm" | "md";
694 |     isActive?: boolean;
695 |   }
696 | >(({ asChild = false, size = "md", isActive, className, ...props }, ref) => {
697 |   const Comp = asChild ? Slot : "a";
698 |
699 |   return (
700 |     <Comp
701 |       ref={ref}
```

```
702 |     data-sidebar="menu-sub-button"
703 |     data-size={size}
704 |     data-active={isActive}
705 |     className={cn(
706 |       "flex h-7 min-w-0 -translate-x-px items-center gap-2 overflow-hidden rounded-md px-2 text-sidebar...
707 |       "data-[active=true]:bg-sidebar-accent data-[active=true]:text-sidebar-accent-foreground",
708 |       size === "sm" && "text-xs",
709 |       size === "md" && "text-sm",
710 |       "group-data-[collapsible=icon]:hidden",
711 |       className,
712 |     )}
713 |     {...props}
714 |   />
715 | );
716 | });
717 | SidebarMenuSubButton.displayName = "SidebarMenuSubButton";
718 |
719 | export {
720 |   Sidebar,
721 |   SidebarContent,
722 |   SidebarFooter,
723 |   SidebarGroup,
724 |   SidebarGroupAction,
725 |   SidebarGroupContent,
726 |   SidebarGroupLabel,
727 |   SidebarHeader,
728 |   SidebarInput,
729 |   SidebarInset,
730 |   SidebarMenu,
731 |   SidebarMenuAction,
732 |   SidebarMenuBadge,
733 |   SidebarMenuButton,
734 |   SidebarMenuItem,
735 |   SidebarMenuSkeleton,
736 |   SidebarMenuSub,
737 |   SidebarMenuSubButton,
738 |   SidebarMenuSubItem,
739 |   SidebarProvider,
740 |   SidebarRail,
741 |   SidebarSeparator,
742 |   SidebarTrigger,
743 |   useSidebar,
744 | };
```

FILE #89: src\components\utils\skeleton.tsx

```
1 | import { cn } from "@lib/utils";
2 |
3 | function Skeleton({ className, ...props }: React.HTMLAttributes<HTMLDivElement>) {
4 |   return <div className={cn("animate-pulse rounded-md bg-primary/10", className)} {...props} />;
5 | }
6 |
7 | export { Skeleton };
```

FILE #90: src\components\utils\slider.tsx

```
1 | import * as React from "react";
2 | import * as SliderPrimitive from "@radix-ui/react-slider";
3 |
4 | import { cn } from "@lib/utils";
5 |
```

```
6 | const Slider = React.forwardRef<
7 |   React.ElementRef<typeof SliderPrimitive.Root>,
8 |   React.ComponentPropsWithoutRef<typeof SliderPrimitive.Root>
9 | >(({ className, ...props }, ref) => (
10 |   <SliderPrimitive.Root
11 |     ref={ref}
12 |     className={cn("relative flex w-full touch-none select-none items-center", className)}
13 |     {...props}
14 |   >
15 |     <SliderPrimitive.Track className="relative h-1.5 w-full grow overflow-hidden rounded-full bg-primary/..."
16 |       <SliderPrimitive.Range className="absolute h-full bg-primary" />
17 |     </SliderPrimitive.Track>
18 |     <SliderPrimitive.Thumb className="block h-4 w-4 rounded-full border border-primary/50 bg-background s..."
19 |   </SliderPrimitive.Root>
20 | ));
21 | Slider.displayName = SliderPrimitive.Root.displayName;
22 |
23 | export { Slider };
```

FILE #91: src\components\uilsonner.tsx

```
1 | import { Toaster as Sonner } from "sonner";
2 |
3 | type ToasterProps = React.ComponentProps<typeof Sonner>;
4 |
5 | const Toaster = ({ ...props }: ToasterProps) => {
6 |   return (
7 |     <Sonner
8 |       className="toaster group"
9 |       toastOptions={{
10 |         classNames: {
11 |           toast:
12 |             "group toast group-[.toaster]:bg-background group-[.toaster]:text-foreground group-[.toaster]:..."
13 |           description: "group-[.toast]:text-muted-foreground",
14 |           actionButton: "group-[.toast]:bg-primary group-[.toast]:text-primary-foreground",
15 |           cancelButton: "group-[.toast]:bg-muted group-[.toast]:text-muted-foreground",
16 |         },
17 |       }}
18 |       {...props}
19 |     </>
20 |   );
21 | };
22 |
23 | export { Toaster };
```

FILE #92: src\components\uilswitch.tsx

```
1 | import * as React from "react";
2 | import * as SwitchPrimitives from "@radix-ui/react-switch";
3 |
4 | import { cn } from "@lib/utlis";
5 |
6 | const Switch = React.forwardRef<
7 |   React.ElementRef<typeof SwitchPrimitives.Root>,
8 |   React.ComponentPropsWithoutRef<typeof SwitchPrimitives.Root>
9 | >(({ className, ...props }, ref) => (
10 |   <SwitchPrimitives.Root
11 |     className={cn(
12 |       "peer inline-flex h-5 w-9 shrink-0 cursor-pointer items-center rounded-full border-2 border-transpa..."
13 |       className,
14 |     )}
```

```
15 |     {...props}
16 |     ref={ref}
17 |   >
18 |     <SwitchPrimitives.Thumb
19 |       className={cn(
20 |         "pointer-events-none block h-4 w-4 rounded-full bg-background shadow-lg ring-0 transition-transfo...
21 |       )}
22 |     />
23 |   </SwitchPrimitives.Root>
24 | ));
25 | Switch.displayName = SwitchPrimitives.Root.displayName;
26 |
27 | export { Switch };
```

FILE #93: src\components\uitable.tsx

```
1 | import * as React from "react";
2 |
3 | import { cn } from "@lib/utils";
4 |
5 | const Table = React.forwardRef<HTMLTableElement, React.HTMLAttributes<HTMLTableElement>>(>
6 |   ({ className, ...props }, ref) => (
7 |     <div className="relative w-full overflow-auto">
8 |       <table ref={ref} className={cn("w-full caption-bottom text-sm", className)} {...props} />
9 |     </div>
10 |   ),
11 | );
12 | Table.displayName = "Table";
13 |
14 | const TableHeader = React.forwardRef<
15 |   HTMLTableSectionElement,
16 |   React.HTMLAttributes<HTMLTableSectionElement>
17 | >(({ className, ...props }, ref) => (
18 |   <thead ref={ref} className={cn("&_tr:border-b", className)} {...props} />
19 | ));
20 | TableHeader.displayName = "TableHeader";
21 |
22 | const TableBody = React.forwardRef<
23 |   HTMLTableSectionElement,
24 |   React.HTMLAttributes<HTMLTableSectionElement>
25 | >(({ className, ...props }, ref) => (
26 |   <tbody ref={ref} className={cn("&_tr:last-child:border-0", className)} {...props} />
27 | ));
28 | TableBody.displayName = "TableBody";
29 |
30 | const TableFooter = React.forwardRef<
31 |   HTMLTableSectionElement,
32 |   React.HTMLAttributes<HTMLTableSectionElement>
33 | >(({ className, ...props }, ref) => (
34 |   <tfoot
35 |     ref={ref}
36 |     className={cn("border-t bg-muted/50 font-medium [&tr]:last:border-b-0", className)}
37 |     {...props}
38 |   />
39 | ));
40 | TableFooter.displayName = "TableFooter";
41 |
42 | const TableRow = React.forwardRef<HTMLTableRowElement, React.HTMLAttributes<HTMLTableRowElement>>(>
43 |   ({ className, ...props }, ref) => (
44 |     <tr
45 |       ref={ref}
46 |       className={cn(
47 |         "border-b transition-colors hover:bg-muted/50 data-[state=selected]:bg-muted",
```

```
48 |         className,
49 |       )}
50 |     {...props}
51 |   />
52 | ),
53 | );
54 | TableRow.displayName = "TableRow";
55 |
56 | const TableHead = React.forwardRef<
57 |   HTMLTableCellElement,
58 |   React.ThHTMLAttributes<HTMLTableCellElement>
59 | >(({ className, ...props }, ref) => (
60 |   <th
61 |     ref={ref}
62 |     className={cn(
63 |       "h-10 px-2 text-left align-middle font-medium text-muted-foreground [&:has([role=checkbox])]:pr-0 [...
64 |       className,
65 |     )}
66 |     {...props}
67 |   />
68 | ));
69 | TableHead.displayName = "TableHead";
70 |
71 | const TableCell = React.forwardRef<
72 |   HTMLTableCellElement,
73 |   React.TdHTMLAttributes<HTMLTableCellElement>
74 | >(({ className, ...props }, ref) => (
75 |   <td
76 |     ref={ref}
77 |     className={cn(
78 |       "p-2 align-middle [&:has([role=checkbox])]:pr-0 [&[role=checkbox]]:translate-y-[2px]",
79 |       className,
80 |     )}
81 |     {...props}
82 |   />
83 | ));
84 | TableCell.displayName = "TableCell";
85 |
86 | const TableCaption = React.forwardRef<
87 |   HTMLTableCaptionElement,
88 |   React.HTMLAttributes<HTMLTableCaptionElement>
89 | >(({ className, ...props }, ref) => (
90 |   <caption ref={ref} className={cn("mt-4 text-sm text-muted-foreground", className)} {...props} />
91 | ));
92 | TableCaption.displayName = "TableCaption";
93 |
94 | export { Table, TableHeader, TableBody, TableFooter, TableHead, TableRow, TableCell, TableCaption };
```

FILE #94: src\components\uitabs.tsx

```
1 | import * as React from "react";
2 | import * as TabsPrimitive from "@radix-ui/react-tabs";
3 |
4 | import { cn } from "@lib/utils";
5 |
6 | const Tabs = TabsPrimitive.Root;
7 |
8 | const TabsList = React.forwardRef<
9 |   React.ElementRef<typeof TabsPrimitive.List>,
10 |   React.ComponentPropsWithoutRef<typeof TabsPrimitive.List>
11 | >(({ className, ...props }, ref) => (
12 |   <TabsPrimitive.List
13 |     ref={ref}
```

```
14 |     className={cn(
15 |       "inline-flex h-9 items-center justify-center rounded-lg bg-muted p-1 text-muted-foreground",
16 |       className,
17 |     )}
18 |     {...props}
19 |   />
20 | ));
21 | TabsList.displayName = TabsPrimitive.List.displayName;
22 |
23 | const TabsTrigger = React.forwardRef<
24 |   React.ElementRef<typeof TabsPrimitive.Trigger>,
25 |   React.ComponentPropsWithoutRef<typeof TabsPrimitive.Trigger>
26 | >(({ className, ...props }, ref) => (
27 |   <TabsPrimitive.Trigger
28 |     ref={ref}
29 |     className={cn(
30 |       "inline-flex items-center justify-center whitespace-nowrap rounded-md px-3 py-1 text-sm font-medium...",
31 |       className,
32 |     )}
33 |     {...props}
34 |   />
35 | ));
36 | TabsTrigger.displayName = TabsPrimitive.Trigger.displayName;
37 |
38 | const TabsContent = React.forwardRef<
39 |   React.ElementRef<typeof TabsPrimitive.Content>,
40 |   React.ComponentPropsWithoutRef<typeof TabsPrimitive.Content>
41 | >(({ className, ...props }, ref) => (
42 |   <TabsPrimitive.Content
43 |     ref={ref}
44 |     className={cn(
45 |       "mt-2 ring-offset-background focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-rin...",
46 |       className,
47 |     )}
48 |     {...props}
49 |   />
50 | ));
51 | TabsContent.displayName = TabsPrimitive.Content.displayName;
52 |
53 | export { Tabs, TabsList, TabsTrigger, TabsContent };
```

FILE #95: src\components\uiltextarea.tsx

```
1 | import * as React from "react";
2 |
3 | import { cn } from "@/lib/utils";
4 |
5 | const Textarea = React.forwardRef<HTMLTextAreaElement, React.ComponentProps<"textarea">>(
6 |   ({ className, ...props }, ref) => {
7 |     return (
8 |       <textarea
9 |         className={cn(
10 |           "flex min-h-[60px] w-full rounded-md border border-input bg-transparent px-3 py-2 text-base sha...",
11 |           className,
12 |         )}
13 |         ref={ref}
14 |         {...props}
15 |       />
16 |     );
17 |   },
18 | );
19 | Textarea.displayName = "Textarea";
20 |
```



```
21 | export { Textarea };
```

FILE #96: src\components\uitoggle-group.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import * as ToggleGroupPrimitive from "@radix-ui/react-toggle-group";
5 | import { type VariantProps } from "class-variance-authority";
6 |
7 | import { cn } from "@lib/utils";
8 | import { toggleVariants } from "@components/ui/toggle";
9 |
10 | const ToggleGroupContext = React.createContext<VariantProps<typeof toggleVariants>>({
11 |   size: "default",
12 |   variant: "default",
13 | });
14 |
15 | const ToggleGroup = React.forwardRef<
16 |   React.ElementRef<typeof ToggleGroupPrimitive.Root>,
17 |   React.ComponentPropsWithoutRef<typeof ToggleGroupPrimitive.Root> &
18 |     VariantProps<typeof toggleVariants>
19 | >(({ className, variant, size, children, ...props }, ref) => (
20 |   <ToggleGroupPrimitive.Root
21 |     ref={ref}
22 |     className={cn("flex items-center justify-center gap-1", className)}
23 |     {...props}
24 |   >
25 |     <ToggleGroupContext.Provider value={{ variant, size }}>{children}</ToggleGroupContext.Provider>
26 |   </ToggleGroupPrimitive.Root>
27 | ));
28 |
29 | ToggleGroup.displayName = ToggleGroupPrimitive.Root.displayName;
30 |
31 | const ToggleGroupItem = React.forwardRef<
32 |   React.ElementRef<typeof ToggleGroupPrimitive.Item>,
33 |   React.ComponentPropsWithoutRef<typeof ToggleGroupPrimitive.Item> &
34 |     VariantProps<typeof toggleVariants>
35 | >(({ className, children, variant, size, ...props }, ref) => {
36 |   const context = React.useContext(ToggleGroupContext);
37 |
38 |   return (
39 |     <ToggleGroupPrimitive.Item
40 |       ref={ref}
41 |       className={cn(
42 |         toggleVariants({
43 |           variant: context.variant || variant,
44 |           size: context.size || size,
45 |         }),
46 |         className,
47 |       )}
48 |       {...props}
49 |     >
50 |       {children}
51 |     </ToggleGroupPrimitive.Item>
52 |   );
53 | });
54 |
55 | ToggleGroupItem.displayName = ToggleGroupPrimitive.Item.displayName;
56 |
57 | export { ToggleGroup, ToggleGroupItem };
```

FILE #97: src\components\uitoggle.tsx

```
1 | import * as React from "react";
2 | import * as TogglePrimitive from "@radix-ui/react-toggle";
3 | import { cva, type VariantProps } from "class-variance-authority";
4 |
5 | import { cn } from "@lib/utils";
6 |
7 | const toggleVariants = cva(
8 |   "inline-flex items-center justify-center gap-2 rounded-md text-sm font-medium cursor-pointer transition...
9 |   {
10 |     variants: {
11 |       variant: {
12 |         default: "bg-transparent",
13 |         outline:
14 |           "border border-input bg-transparent shadow-sm hover:bg-accent hover:text-accent-foreground",
15 |       },
16 |       size: {
17 |         default: "h-9 px-2 min-w-9",
18 |         sm: "h-8 px-1.5 min-w-8",
19 |         lg: "h-10 px-2.5 min-w-10",
20 |       },
21 |     },
22 |     defaultVariants: {
23 |       variant: "default",
24 |       size: "default",
25 |     },
26 |   },
27 | );
28 |
29 | const Toggle = React.forwardRef<
30 |   React.ElementRef<typeof TogglePrimitive.Root>,
31 |   React.ComponentPropsWithoutRef<typeof TogglePrimitive.Root> & VariantProps<typeof toggleVariants>
32 | >(({ className, variant, size, ...props }, ref) => (
33 |   <TogglePrimitive.Root
34 |     ref={ref}
35 |     className={cn(toggleVariants({ variant, size, className }))}
36 |     {...props}
37 |   />
38 | ));
39 |
40 | Toggle.displayName = TogglePrimitive.Root.displayName;
41 |
42 | export { Toggle, toggleVariants };
```

FILE #98: src\components\uitooltip.tsx

```
1 | "use client";
2 |
3 | import * as React from "react";
4 | import * as TooltipPrimitive from "@radix-ui/react-tooltip";
5 |
6 | import { cn } from "@lib/utils";
7 |
8 | const TooltipProvider = TooltipPrimitive.Provider;
9 |
10 | const Tooltip = TooltipPrimitive.Root;
11 |
12 | const TooltipTrigger = TooltipPrimitive.Trigger;
13 |
14 | const TooltipContent = React.forwardRef<
15 |   React.ElementRef<typeof TooltipPrimitive.Content>,
```

```
16 |   React.ComponentPropsWithoutRef<typeof TooltipPrimitive.Content>
17 | >(({ className, sideOffset = 4, ...props }, ref) => (
18 |   <TooltipPrimitive.Portal>
19 |     <TooltipPrimitive.Content
20 |       ref={ref}
21 |       sideOffset={sideOffset}
22 |       className={cn(
23 |         "z-50 overflow-hidden rounded-md bg-primary px-3 py-1.5 text-xs text-primary-foreground animate-i...
24 |         className,
25 |       )}
26 |       {...props}
27 |     />
28 |   </TooltipPrimitive.Portal>
29 | ));
30 | TooltipContent.displayName = TooltipPrimitive.Content.displayName;
31 |
32 | export { Tooltip, TooltipTrigger, TooltipContent, TooltipProvider };
```

FILE #99: src\hooks\use-mobile.tsx

```
1 | import * as React from "react";
2 |
3 | const MOBILE_BREAKPOINT = 768;
4 |
5 | export function useIsMobile() {
6 |   const [isMobile, setIsMobile] = React.useState<boolean | undefined>(undefined);
7 |
8 |   React.useEffect(() => {
9 |     const mql = window.matchMedia(`(max-width: ${MOBILE_BREAKPOINT - 1}px)`);
10 |     const onChange = () => {
11 |       setIsMobile(window.innerWidth < MOBILE_BREAKPOINT);
12 |     };
13 |     mql.addEventListener("change", onChange);
14 |     setIsMobile(window.innerWidth < MOBILE_BREAKPOINT);
15 |     return () => mql.removeEventListener("change", onChange);
16 |   }, []);
17 |
18 |   return !!isMobile;
19 | }
```

FILE #100: src\lib\error-capture.ts

```
1 | // Captures the original Error out-of-band so server.ts can recover the stack
2 | // when h3 has already swallowed the throw into a generic 500 Response.
3 |
4 | let lastCapturedError: { error: unknown; at: number } | undefined;
5 | const TTL_MS = 5_000;
6 |
7 | function record(error: unknown) {
8 |   lastCapturedError = { error, at: Date.now() };
9 | }
10 |
11 | if (typeof globalThis.addEventListener === "function") {
12 |   globalThis.addEventListener("error", (event) => record((event as ErrorEvent).error ?? event));
13 |   globalThis.addEventListener("unhandledrejection", (event) =>
14 |     record((event as PromiseRejectionEvent).reason),
15 |   );
16 | }
17 |
18 | export function consumeLastCapturedError(): unknown {
19 |   if (!lastCapturedError) return undefined;
```

```
20 |   if (Date.now() - lastCapturedError.at > TTL_MS) {
21 |     lastCapturedError = undefined;
22 |     return undefined;
23 |   }
24 |   const { error } = lastCapturedError;
25 |   lastCapturedError = undefined;
26 |   return error;
27 | }
```

FILE #101: src/lib/error-page.ts

```
1 | export function renderErrorPage(): string {
2 |   return `<!doctype html>
3 | <html lang="en">
4 |   <head>
5 |     <meta charset="utf-8" />
6 |     <title>This page didn't load</title>
7 |     <meta name="viewport" content="width=device-width, initial-scale=1" />
8 |     <style>
9 |       body { font: 15px/1.5 system-ui, -apple-system, sans-serif; background: #fafafa; color: #111; displ...
10 |       .card { max-width: 28rem; width: 100%; text-align: center; padding: 2rem; }
11 |       h1 { font-size: 1.25rem; margin: 0 0 0.5rem; }
12 |       p { color: #4b5563; margin: 0 0 1.5rem; }
13 |       .actions { display: flex; gap: 0.5rem; justify-content: center; flex-wrap: wrap; }
14 |       a, button { padding: 0.5rem 1rem; border-radius: 0.375rem; font: inherit; cursor: pointer; text-dec...
15 |       .primary { background: #111; color: #fff; }
16 |       .secondary { background: #fff; color: #111; border-color: #d1d5db; }
17 |     </style>
18 |   </head>
19 |   <body>
20 |     <div class="card">
21 |       <h1>This page didn't load</h1>
22 |       <p>Something went wrong on our end. You can try refreshing or head back home.</p>
23 |       <div class="actions">
24 |         <button class="primary" onclick="location.reload()">Try again</button>
25 |         <a class="secondary" href="/">Go home</a>
26 |       </div>
27 |     </div>
28 |   </body>
29 | </html>`;
30 | }
```

FILE #102: src/lib/lovable-error-reporting.ts

```
1 | type LovableErrorOptions = {
2 |   mechanism?: "manual" | "onerror" | "unhandledrejection" | "react_error_boundary";
3 |   handled?: boolean;
4 |   severity?: "error" | "warning" | "info";
5 | };
6 |
7 | type LovableEvents = {
8 |   captureException?: (
9 |     error: unknown,
10 |     context?: Record<string, unknown>,
11 |     options?: LovableErrorOptions,
12 |   ) => void;
13 | };
14 |
15 | declare global {
16 |   interface Window {
17 |     __lovableEvents?: LovableEvents;
```

```
18 |   }
19 | }
20 |
21 | export function reportLovableError(error: unknown, context: Record<string, unknown> = {}) {
22 |   if (typeof window === "undefined") return;
23 |   window.__lovableEvents?.captureException?.(
24 |     error,
25 |     {
26 |       source: "react_error_boundary",
27 |       route: window.location.pathname,
28 |       ...context,
29 |     },
30 |     {
31 |       mechanism: "react_error_boundary",
32 |       handled: false,
33 |       severity: "error",
34 |     },
35 |   );
36 | }
```

FILE #103: src\lib\motion.ts

```
1 | import type { Variants } from "motion/react";
2 |
3 | /** The one easing curve used across the entire site. Confident, no overshoot. */
4 | export const EASE = [0.22, 1, 0.36, 1] as const;
5 |
6 | /** Standard scroll-reveal: fade + rise, staggered by index. */
7 | export const revealVariants: Variants = {
8 |   hidden: { opacity: 0, y: 20 },
9 |   visible: (i: number = 0) => ({
10 |     opacity: 1,
11 |     y: 0,
12 |     transition: {
13 |       duration: 0.7,
14 |       ease: EASE,
15 |       delay: i * 0.07,
16 |     },
17 |   }),
18 | };
19 |
20 | /** Softer variant for large display type ? a touch of blur sells "premium". */
21 | export const revealBlurVariants: Variants = {
22 |   hidden: { opacity: 0, y: 16, filter: "blur(6px)" },
23 |   visible: (i: number = 0) => ({
24 |     opacity: 1,
25 |     y: 0,
26 |     filter: "blur(0px)",
27 |     transition: {
28 |       duration: 0.8,
29 |       ease: EASE,
30 |       delay: i * 0.07,
31 |     },
32 |   }),
33 | };
```

FILE #104: src\lib\project-details.ts

```
1 | export interface ProjectDetail {
2 |   whyBuilt: {
3 |     problem: string;
```

```
4 |     solution: string;
5 |     result: string;
6 |     scenario?: { t: string; event: string; danger?: boolean }[];
7 |     solutionSteps?: { step: string; detail?: string; file?: string }[];
8 |     resultStats?: { label: string; value: string }[];
9 | };
10 | architecture: {
11 |     id: string;
12 |     label: string;
13 |     desc: string;
14 | }[];
15 | features: string[];
16 | challenges: {
17 |     title: string;
18 |     problem: string;
19 |     difficulty: string;
20 |     solution: string;
21 |     fixSteps?: { step: string; detail?: string; file?: string }[];
22 |     learned: string;
23 | }[];
24 | journey: {
25 |     day: string;
26 |     milestone: string;
27 |     details: string;
28 | }[];
29 | lighthouse?: {
30 |     performance: number;
31 |     accessibility: number;
32 |     bestPractices: number;
33 |     seo: number;
34 | };
35 | reliability?: {
36 |     label: string;
37 |     value: string;
38 |     detail: string;
39 | }[];
40 | decisions: {
41 |     tech: string;
42 |     title: string;
43 |     explanation: string;
44 | }[];
45 | lessons: string[];
46 | gallery: {
47 |     label: string;
48 |     img: string;
49 | }[];
50 | theme: {
51 |     primary: string; // Tailwind class like "from-cyan-400"
52 |     secondary: string; // Tailwind class like "to-violet-500"
53 |     glow: string; // CSS color string or class like "rgba(34, 211, 238, 0.15)"
54 | };
55 | }
56 |
57 | export const PROJECT_DETAILS: Record<string, ProjectDetail> = {
58 |     Vurlo: {
59 |         theme: {
60 |             primary: "from-cyan-400",
61 |             secondary: "to-violet-500",
62 |             glow: "rgba(34, 211, 238, 0.15)",
63 |         },
64 |         whyBuilt: {
65 |             problem:
66 |                 'Firestore has no "hold this unit while someone pays" primitive ? two people can buy the last lam...
67 |             solution:
68 |                 "Run atomic stock check-and-decrement transactions for each item before payment, with rollbacks o...
```

```
69 |     scenario: [  
70 |       {  
71 |         t: "T+0s",  
72 |         event: "Customer A adds the last Saturn Lamp (stock: 1) to cart, hits checkout.",  
73 |       },  
74 |       { t: "T+1s", event: "Customer B does the exact same thing, same product, same second." },  
75 |       {  
76 |         t: "T+2s",  
77 |         event:  
78 |           'Both clients read products/{id}.stock ? both see 1. Both pass the "is there enough stock" ch...  
79 |         danger: true,  
80 |       },  
81 |       {  
82 |         t: "T+3s",  
83 |         event:  
84 |           "Both writes land. Stock goes 1 ? 0 ? then to -1 if nothing stops the second write.",  
85 |         danger: true,  
86 |       },  
87 |       {  
88 |         t: "T+9s",  
89 |         event: "Razorpay confirms both payments. Vurlo just sold a lamp it doesn't have.",  
90 |       },  
91 |     ],  
92 |     solutionSteps: [  
93 |       {  
94 |         step: "Reserve inside a transaction, not before it",  
95 |         detail:  
96 |           "placeOrder() opens one runTransaction per cart line item and re-reads stock at commit time ?...  
97 |         file: "use-cart.tsx",  
98 |       },  
99 |       {  
100 |        step: "Throw before any write lands",  
101 |        detail:  
102 |          "If stock < quantity inside the transaction body, it throws. Firestore retries the whole read...  
103 |        file: "use-cart.tsx",  
104 |      },  
105 |      {  
106 |        step: "Batch the order creation",  
107 |        detail:  
108 |          "One writeBatch creates the order doc, clears the cart subcollection, and writes a notificati...  
109 |        file: "use-cart.tsx",  
110 |      },  
111 |      {  
112 |        step: "Roll back on every failure exit",  
113 |        detail:  
114 |          "Signature mismatch, modal dismiss, SDK constructor throw, raw catch ? all four delete the pe...  
115 |        file: "checkout-component.tsx",  
116 |      },  
117 |    ],  
118 |     result:  
119 |       "Zero oversells since launch ? COD and Razorpay both run through the same reserve ? commit ? veri...  
120 |     resultStats: [  
121 |       { label: "Stock leaks in prod", value: "0" },  
122 |       { label: "Failed checkout recoveries", value: "100%" },  
123 |       { label: "Lighthouse perf", value: "98 / 100" },  
124 |       { label: "Initial bundle size", value: "-45%" },  
125 |     ],  
126 |   },  
127 |   architecture: [  
128 |     {  
129 |       id: "routes",  
130 |       label: "src/routes/ ? routeTree.gen.ts",  
131 |       desc: "automatic page routes matched directly to your project folders ? uses @tanstack/router-plu...  
132 |     },  
133 |     {
```

```
134 |         id: "cart-hook",
135 |         label: "use-cart.tsx",
136 |         desc: "keeps client cart state fully synced across open tabs and guest sessions ? uses a live onS...
137 |     },
138 |     {
139 |         id: "checkout-ui",
140 |         label: "checkout-component.tsx",
141 |         desc: "manages shipping validations, real-time coupon calculations, and checkout branching ? lazy...
142 |     },
143 |     {
144 |         id: "reserve",
145 |         label: "placeOrder()",
146 |         desc: "reserves stock atomically inside a database transaction so two people never buy the same l...
147 |     },
148 |     {
149 |         id: "razorpay-order",
150 |         label: "api/create-razorpay-order.ts",
151 |         desc: "a secure backend helper to initiate online credit/debit card orders ? instantiates the Raz...
152 |     },
153 |     {
154 |         id: "razorpay-verify",
155 |         label: "api/verify-razorpay-payment.ts",
156 |         desc: "the check that stops someone faking a 'payment successful' callback and getting a free ord...
157 |     },
158 |     {
159 |         id: "rules",
160 |         label: "firestore.rules",
161 |         desc: "in plain terms: a customer can cancel their own pending order and change nothing else abou...
162 |     },
163 |     {
164 |         id: "email",
165 |         label: "api/send-order-email.ts",
166 |         desc: "sends confirmation emails asynchronously without slowing down checkout ? fires a backgroun...
167 |     },
168 |     {
169 |         id: "ai-recommend",
170 |         label: "api/ai-recommend.ts",
171 |         desc: "uses AI to suggest complementary products directly on the cart page ? sends current cart c...
172 |     },
173 | ],
174 | features: [
175 |     "Complete Firebase Authentication system ? secure email/password login/signup workflows with instan...
176 |     "Granular Firestore security rules ? locks down user data, admin portals, and orders, preventing ba...
177 |     "Interactive customer AI chatbot assistant ? answers catalog questions, recommends items, and handl...
178 |     "Dynamic product 'Quick View' preview overlays ? lets customers check product specs, variants, and ...
179 |     "Optimized client-side lazy-loading bundles ? routes, modals, and admin panels are loaded on-demand...
180 |     "Complete SEO & GEO optimization ? server-side generated tags, geographic-aware configuration, and ...
181 |     "Secure stock reservation before payment ? reserves items using a database transaction before the p...
182 |     "Automatic stock recovery on aborted payments ? a compensating rollback deletes pending orders and ...
183 |     "Seamless guest-to-account cart merging ? preserves local items in browser storage and combines the...
184 |     "Flexible promo coupons ? handles percentage/fixed discount rules, expiry dates, minimum spending t...
185 |     "Tamper-proof payment integrations ? runs online card payments verified via server-side HMAC-SHA256...
186 |     "Secure merchant dashboard ? role-gated access control for admins to manage product listings, coupo...
187 | ],
188 | challenges: [
189 |     {
190 |         title: "Stopping Double-Sells in the Cart ? Payment Gap",
191 |         problem:
192 |             "Between 'add to cart' and Razorpay's async confirmation, nothing stopped the same lamp being b...
193 |         difficulty:
194 |             "Two customers could both end up thinking they'd bought the last lamp. Firestore has no built-i...
195 |         solution:
196 |             "placeOrder() and reserveStockAndGetTotal() both wrap each cart line item in its own runTransac...
197 |         fixSteps: [
198 |             { step: "Wrap each cart line item in its own runTransaction", file: "use-cart.tsx" },
```



```
199 |         { step: "Re-read stock inside the transaction body, not before it" },
200 |         { step: "Throw before any write ? Firestore retries the whole pair on conflict" },
201 |     ],
202 |     learned:
203 |       "Firestore transactions are scoped per-document, not per-request. Protecting a multi-item order...
204 | },
205 | {
206 |   title: "Not Leaking Stock on Every Way Razorpay Can Fail",
207 |   problem:
208 |     "Once stock is decremented for a pending order, at least four separate paths can still kill the...
209 |   difficulty:
210 |     "If checkout fails at any point after stock is reserved, inventory leaks. Signature mismatch, c...
211 |   solution:
212 |     "The same rollback block ? deleteDoc the order, then runTransaction per cart item to restore st...
213 |   fixSteps: [
214 |     {
215 |       step: "deleteDoc(orders/{orderId}) on any failed exit",
216 |       file: "checkout-component.tsx",
217 |     },
218 |     { step: "runTransaction per cart item to add stock back" },
219 |     {
220 |       step: "Wire identically into all 4 exits: !verifyRes.ok, ondismiss, SDK throw, outer catch",
221 |     },
222 |   ],
223 |   learned:
224 |     "Payment integrations fail in more places than the happy path suggests. Every exit needs the sa...
225 | },
226 | {
227 |   title: "Preventing Customers from Promoting Themselves to Admin",
228 |   problem:
229 |     "Customers need to cancel their own pending orders; admins need full access; nobody should self...
230 |   difficulty:
231 |     "Standard database rules are often too broad. If you allow users to edit their profile, they ca...
232 |   solution:
233 |     "users/{userId} rejects role, emailVerified, and admin on create/update. The orders/{orderId} r...
234 |   fixSteps: [
235 |     {
236 |       step: "Make it impossible for a user to grant themselves admin",
237 |       detail:
238 |         "Firestore's hasNone/hasAny checks block role, emailVerified, and admin from ever appearing...
239 |       file: "firestore.rules",
240 |     },
241 |     {
242 |       step: "Let a cancel change status and nothing else",
243 |       detail:
244 |         "diff().affectedKeys().hasOnly(['status']) ? the write only goes through if status is the o...
245 |     },
246 |     {
247 |       step: "Require the order is currently 'pending' and the new value is exactly 'cancelled'",
248 |     },
249 |   ],
250 |   learned:
251 |     "diff().affectedKeys() is the actual mechanism that stops someone smuggling an extra field chan...
252 | },
253 | {
254 |   title: "Making the Guest ? Account Cart Merge Safe to Run Twice",
255 |   problem:
256 |     "Guest carts sit in localStorage; logging in has to fold them into Firestore without duplicatin...
257 |   difficulty:
258 |     "If the merge fires twice (due to network retries, fast double-taps, or double-renders), a naiv...
259 |   solution:
260 |     "performMerge() reads the existing Firestore quantity, clamps the merged total against current ...
261 |   fixSteps: [
262 |     { step: "Read existing Firestore itemSnap for each local item", file: "use-cart.tsx" },
263 |     { step: "Clamp: Math.min(stock, dbQty + item.quantity)" },
```

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264 |         { step: "Only clear vurlo_local_cart after batch.commit() resolves" },
265 |     ],
266 |     learned:
267 |       "Write merge logic assuming it'll run twice. Clamping against server state instead of trusting ...
268 |   },
269 |   {
270 |     title: "Keeping the Admin Panel Out of the Customer's Bundle",
271 |     problem:
272 |       "Vurlo ships a full admin console inside the same route tree as the customer storefront.",
273 |     difficulty:
274 |       "Every visitor who just wanted to buy a lamp was also downloading the admin dashboard's code fo...
275 |     solution:
276 |       "checkout.tsx and admin routes lazy-load via React.lazy + Suspense; vite.config.ts splits vendo...
277 |     fixSteps: [
278 |       { step: "React.lazy + Suspense on checkout.tsx and every admin route" },
279 |       {
280 |         step: "manualChunks split: vendor / firebase / router / query / ui",
281 |         file: "vite.config.ts",
282 |       },
283 |       { step: "sharp resize pass caps product images at 760px with matching webp" },
284 |     ],
285 |     learned:
286 |       "Route-level code splitting plus an actual image size budget moves real-world load time more th...
287 |   },
288 | ],
289 | journey: [
290 |   {
291 |     day: "Day 1",
292 |     milestone: "Scaffold & File-based Routing",
293 |     details: "TanStack Start scaffold, Tailwind/shadcn config, src/routes/ tree established.",
294 |   },
295 |   {
296 |     day: "Day 2",
297 |     milestone: "Firebase Wiring",
298 |     details:
299 |       "firebase.json set up; Auth + Firestore SDK wired into use-auth.tsx and lib/firebase.ts.",
300 |   },
301 |   {
302 |     day: "Day 3",
303 |     milestone: "Cart & Guest Persistence",
304 |     details:
305 |       "use-cart.tsx built out ? onSnapshot cart, localStorage guest fallback, merge-on-login batch.",
306 |   },
307 |   {
308 |     day: "Day 5",
309 |     milestone: "Firestore Security Rules",
310 |     details:
311 |       "firestore.rules written ? admin role gating, field-diff protections, nested orders lockdown.",
312 |   },
313 |   {
314 |     day: "Day 6",
315 |     milestone: "Checkout & Coupons",
316 |     details:
317 |       "checkout-component.tsx multi-step flow shipped, coupon validation against the coupons collecti...
318 |   },
319 |   {
320 |     day: "Day 7",
321 |     milestone: "Razorpay Integration",
322 |     details:
323 |       "create-razorpay-order.ts and verify-razorpay-payment.ts built, HMAC verification wired into ev...
324 |   },
325 |   {
326 |     day: "Day 8",
327 |     milestone: "Admin Console",
328 |     details:
```

```
329 |         "admin.tsx role-gated layout shipped with products, orders, coupons, stock-request panels.",
330 |     },
331 |     {
332 |         day: "Day 10",
333 |         milestone: "AI, Email & Performance Pass",
334 |         details:
335 |             "Gemini recommendations, Resend emails, image pipeline, manualChunks split, sitemap gen before ...
336 |     },
337 | ],
338 | lighthouse: {
339 |     performance: 98,
340 |     accessibility: 100,
341 |     bestPractices: 96,
342 |     seo: 100,
343 | },
344 | decisions: [
345 |     {
346 |         tech: "TanStack Start / Router",
347 |         title: "File-based Routes, Compiled at Build Time",
348 |         explanation:
349 |             "Routes are plain files under src/routes/; @tanstack/router-plugin compiles them into routeTree...
350 |     },
351 |     {
352 |         tech: "React 19",
353 |         title: "Suspense-Driven Code Splitting",
354 |         explanation:
355 |             "checkout-component.tsx (906 lines) and the admin dashboard are both React.lazy imports behind ...
356 |     },
357 |     {
358 |         tech: "Firestore Transactions",
359 |         title: "Per-Document Atomic Stock Decrements",
360 |         explanation:
361 |             "runTransaction runs once per product doc in both placeOrder() and reserveStockAndGetTotal() (u...
362 |     },
363 |     {
364 |         tech: "Vite manualChunks",
365 |         title: "Deliberate Vendor Splitting",
366 |         explanation:
367 |             "rollupOptions.output.manualChunks explicitly separates vendor, firebase, router, query, and ui...
368 |     },
369 |     {
370 |         tech: "Razorpay + firebase-admin",
371 |         title: "Server-Only Payment Verification",
372 |         explanation:
373 |             "verify-razorpay-payment.ts is the only place RAZORPAY_KEY_SECRET recomputes the HMAC signature...
374 |     },
375 | ],
376 | lessons: [
377 |     "Extract the four copy-pasted rollback blocks in checkout-component.tsx into one shared function ? ...
378 |     "Batch the per-item runTransaction calls in placeOrder() into a single transaction that reads all p...
379 |     "Add a scheduled job to expire orders stuck at paymentStatus: 'upi_pending' when someone abandons t...
380 |     "Move ai-recommend.ts's in-memory Map cache into Firestore or Redis ? it resets on every serverless...
381 |     "Wrap coupon usageCount increments in the same transaction as the usageLimit check ? right now two ...
382 | ],
383 | gallery: [
384 |     { label: "Homepage Preview", img: "/vurlo-preview.webp" },
385 |     { label: "Product Collections", img: "/vurlo-shop.webp" },
386 |     { label: "Product Quick View", img: "/vurlo-checkout-view.webp" },
387 |     { label: "Order Tracking", img: "/vurlo-orders.webp" },
388 |     { label: "Inventory Console", img: "/vurlo-admin.webp" },
389 | ],
390 | },
391 | Veltrix: {
392 |     theme: {
393 |         primary: "from-emerald-400",
```

```
394 |     secondary: "to-teal-500",
395 |     glow: "rgba(16, 185, 129, 0.15)",
396 | },
397 | whyBuilt: {
398 |     problem:
399 |         "Running a growth Instagram account solo means someone has to write, fact-check, design, and publ...
400 |     solution:
401 |         "Built an autonomous pipeline where a caption only ships after two independent AI auditors agree ...
402 |     scenario: [
403 |         {
404 |             t: "T+0s",
405 |             event:
406 |                 "GitHub Actions cron fires at 18:00 IST ? a brand-new, stateless runner boots with zero memor...
407 |         },
408 |         {
409 |             t: "T+1s",
410 |             event:
411 |                 "auto_post.py checks pipeline_checkpoint.json first, in case a post from the last run is stil...
412 |         },
413 |         {
414 |             t: "T+3s",
415 |             event:
416 |                 "No pending checkpoint found. config.json's WEEKLY_SCHEDULE resolves today's evening slot to ...
417 |         },
418 |         {
419 |             t: "T+5s",
420 |             event:
421 |                 "The topic's embedding is compared against the last 30 days via cosine similarity ? adaptive...
422 |         },
423 |         {
424 |             t: "T+6s",
425 |             event:
426 |                 "Similarity lands between the low and high threshold ? the ambiguous middle band Vurlo-style ...
427 |             danger: true,
428 |         },
429 |         {
430 |             t: "T+7s",
431 |             event: "is_semantic_duplicate_llm() breaks the tie: not a duplicate. Pipeline proceeds.",
432 |         },
433 |         {
434 |             t: "T+9s",
435 |             event:
436 |                 "verify_text_content() sends the caption to Groq and Cerebras independently ? both must retur...
437 |         },
438 |         {
439 |             t: "T+11s",
440 |             event:
441 |                 "Both approve. Jinja2 renders the slide templates; Playwright screenshots them at 1080x1350, ...
442 |         },
443 |         {
444 |             t: "T+14s",
445 |             event:
446 |                 "save_checkpoint() base64-encodes the slide images into pipeline_checkpoint.json with status:...
447 |         },
448 |         {
449 |             t: "T+1,800s",
450 |             event:
451 |                 "30 minutes later, the next cron tick's load_checkpoint() finds the same checkpoint, sees age...
452 |         },
453 |     ],
454 |     solutionSteps: [
455 |         {
456 |             step: "Persist state across two runners that never coexist",
457 |             detail:
458 |                 "save_checkpoint()/load_checkpoint() serialize the topic, caption, and base64-encoded media b...
```

```

459 |         file: "database.py",
460 |     },
461 |     {
462 |         step: "Score duplicates against a moving baseline, not a fixed cutoff",
463 |         detail:
464 |             "adaptive_thresholds() derives low/high similarity cutoffs each run from ? + ? and ? + 1.5? o...
465 |         file: "database.py",
466 |     },
467 |     {
468 |         step: "Never let one model grade its own homework",
469 |         detail:
470 |             "verify_text_content() routes the caption to Groq (llama-3.3-70b-versatile) and Cerebras (gpt...
471 |         file: "text_auditor.py",
472 |     },
473 |     {
474 |         step: "Force a human decision point before anything goes live",
475 |         detail:
476 |             "A generated post sits at status: pending_review for a 30-minute window on the dashboard; the...
477 |         file: "auto_post.py",
478 |     },
479 | ],
480 | result:
481 |     "Every post clears two independent auditors and a statistically-adaptive duplicate filter before ...
482 | resultStats: [
483 |     { label: "Inference cost / post", value: "~$0.0002" },
484 |     { label: "Independent audits required", value: "2 of 2" },
485 |     { label: "Human review window", value: "30 min" },
486 |     { label: "Output formats", value: "4 (photo/carousel/reel/listicle)" },
487 | ],
488 | },
489 | architecture: [
490 |     {
491 |         id: "cron",
492 |         label: "GitHub Actions cron ? auto_post.py",
493 |         desc: "fires at fixed IST slot hours and resolves the day's format (photo/carousel/reel/listicle)...
494 |     },
495 |     {
496 |         id: "checkpoint",
497 |         label: "database.py ? save_checkpoint() / load_checkpoint()",
498 |         desc: "the only state that survives between two entirely separate ephemeral runners ? serializes ...
499 |     },
500 |     {
501 |         id: "content-gen",
502 |         label: "content_generator.py",
503 |         desc: "drafts a topic and caption using Gemini, pulling from a weighted voice pool (Punchy, Listi...
504 |     },
505 |     {
506 |         id: "dedupe",
507 |         label: "database.py ? cosine_similarity() / adaptive_thresholds()",
508 |         desc: "a three-tier duplicate filter: fast-reject below a statistically adaptive low threshold, f...
509 |     },
510 |     {
511 |         id: "auditor",
512 |         label: "text_auditor.py ? verify_text_content()",
513 |         desc: "local regex/heuristic pre-checks (emoji density, disclaimer placement, shadowban phrase pa...
514 |     },
515 |     {
516 |         id: "renderer",
517 |         label: "listicle_pipeline.py ? render_pages()",
518 |         desc: "fills Jinja2 HTML templates with the approved copy, then screenshots them via headless Pla...
519 |     },
520 |     {
521 |         id: "publisher",
522 |         label: "instagram_publisher.py / threads_publisher.py",
523 |         desc: "publishes the approved asset set through the Instagram Graph API and cross-posts the same ...

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524 |     },
525 |     {
526 |         id: "dashboard",
527 |         label: "database.py ? update_dashboard() ? dashboard.html",
528 |         desc: "renders a static control panel showing the pending-review card with Approve/Reject buttons...",
529 |     },
530 | ],
531 | features: [
532 |     "Zero-touch daily publishing ? a GitHub Actions cron resolves each day's format from a weekly sched...",
533 |     "Adversarial fact-checking before anything ships ? every caption is drafted by Gemini, then indepen...",
534 |     "Statistically-adaptive duplicate detection ? cosine similarity against the account's post history, ...",
535 |     "Free local pre-screening before any API spend ? regex and heuristic checks (emoji count, disclaime...",
536 |     "State that survives an ephemeral runner dying mid-post ? a self-expiring JSON checkpoint carries t...",
537 |     "Built-in human veto window ? every generated post sits for 30 minutes as pending_review on a live ...",
538 |     "Headless-render pipeline for branded slides ? Jinja2 templates screenshot through Playwright Chrom...",
539 |     "Perceptual duplicate guards for media, separate from text ? dhash/pHash image comparisons stop the...",
540 |     "Cross-platform publishing from one pipeline ? the same approved asset set republishes to Threads w...",
541 |     "Per-run cost telemetry ? an APITracker tallies Gemini token costs and per-provider fallback usage ...",
542 | ],
543 | challenges: [
544 |     {
545 |         title: "Making State Survive Between Two Runners That Never Coexist",
546 |         problem:
547 |             "auto_post.py runs as a brand-new GitHub Actions job every invocation ? no memory, no disk, no ...",
548 |         difficulty:
549 |             "A GitHub Actions runner is torn down completely when the job ends. Adding a human review windo...",
550 |         solution:
551 |             "save_checkpoint() serializes the topic, caption, and base64-encoded asset bytes to a local pip...",
552 |         fixSteps: [
553 |             {
554 |                 step: "Base64-encode every binary asset ? image bytes, video slide frames ? into the checkpoi...",
555 |                 file: "auto_post.py",
556 |             },
557 |             {
558 |                 step: "Stamp every checkpoint with an ISO timestamp and self-expire it after 24 hours",
559 |                 file: "database.py",
560 |             },
561 |             {
562 |                 step: "Check checkpoint.status === 'pending_review' before drafting anything new, so a cron t...",
563 |             },
564 |         ],
565 |         learned:
566 |             "Idempotency isn't optional on ephemeral infrastructure ? it's the only way a stateless schedul...",
567 |     },
568 |     {
569 |         title: "One Model Grading Its Own Homework",
570 |         problem:
571 |             "Having Gemini draft a caption and then having Gemini check that same caption for factual accur...",
572 |         difficulty:
573 |             "Self-review only catches errors the model already 'knows' are errors. Hallucinated facts and s...",
574 |         solution:
575 |             "verify_text_content() runs cheap deterministic checks first, then routes the caption to Groq a...",
576 |         fixSteps: [
577 |             {
578 |                 step: "Run local regex/heuristic pre-checks first ? emoji density over 4, disclaimer placed a...",
579 |                 file: "text_auditor.py",
580 |             },
581 |             {
582 |                 step: "Query Groq and Cerebras with the identical audit prompt and brand system instruction, ...",
583 |             },
584 |             {
585 |                 step: "AND the two decisions together ? one dissenting auditor rejects the whole caption, wit...",
586 |                 file: "text_auditor.py",
587 |             },
588 |         ],
589 |     },
590 | ],
```

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589 |         learned:
590 |             "Cross-vendor disagreement is a signal, not noise. Two different model families are far less li...
591 |     },
592 |     {
593 |         title: "A Fixed Similarity Cutoff Couldn't Tell 'Repeat Topic' From 'Same Niche'",
594 |         problem:
595 |             "Embedding similarity is supposed to flag duplicate topics before they get posted twice, but a ...
596 |         difficulty:
597 |             "The account posts within one niche (AI/fintech), so baseline similarity between any two unrela...
598 |         solution:
599 |             "adaptive_thresholds() computes the mean and standard deviation across every pairwise similarit...
600 |         fixSteps: [
601 |             {
602 |                 step: "Compute ? and ? across all recent embedding pairs on every run, not once at setup",
603 |                 file: "database.py",
604 |             },
605 |             {
606 |                 step: "Derive low = min(0.78, ?+?) and high = min(0.88, ?+1.5?) instead of one constant for e...
607 |             },
608 |             {
609 |                 step: "Route only the ambiguous middle band to a slower LLM tiebreaker (is_semantic_duplicate...
610 |             },
611 |         ],
612 |         learned:
613 |             "A threshold tuned once against a snapshot of data quietly goes stale as the data distribution ...
614 |     },
615 |     {
616 |         title: "Headless Chromium Was Screenshotting Slides Before Fonts Finished Loading",
617 |         problem:
618 |             "Playwright screenshots taken on a clean GitHub Actions runner occasionally captured branded li...
619 |         difficulty:
620 |             "A fresh CI container has no font cache the way a local dev machine does, and Playwright's netw...
621 |         solution:
622 |             "render_pages() loads each rendered Jinja2 template from a local file:// URI, waits for network...
623 |         fixSteps: [
624 |             {
625 |                 step: "page.goto() the rendered HTML from a local temp file so relative image assets resolve ...
626 |                 file: "listicle_pipeline.py",
627 |             },
628 |             {
629 |                 step: "page.wait_for_load_state('networkidle', timeout=5000), then page.wait_for_timeout(1500...
630 |                 file: "listicle_pipeline.py",
631 |             },
632 |             {
633 |                 step: "Launch the browser context with viewport 1080x1350 and device_scale_factor: 2 so slide...
634 |             },
635 |         ],
636 |         learned:
637 |             "networkidle is a proxy for 'probably done,' not a guarantee ? it doesn't know about a font swa...
638 |     },
639 | ],
640 | journey: [
641 |     {
642 |         day: "Day 1",
643 |         milestone: "Core Pipeline & LLM Hooks",
644 |         details:
645 |             "content_generator.py drafting flow wired to Gemini; config.json's weekly schedule map and voic...
646 |     },
647 |     {
648 |         day: "Day 3",
649 |         milestone: "Dual-Brain Auditor",
650 |         details:
651 |             "text_auditor.py built out ? local heuristic pre-checks, then independent Groq and Cerebras aud...
652 |     },
653 |     {
```

```
654 |         day: "Day 5",
655 |         milestone: "Adaptive Duplicate Detection",
656 |         details:
657 |             "Embedding cache and adaptive_thresholds() written in database.py, replacing an earlier fixed-c...
658 |     },
659 |     {
660 |         day: "Day 7",
661 |         milestone: "Playwright + Jinja2 Render Pipeline",
662 |         details:
663 |             "listicle_pipeline.py template rendering and headless screenshot capture stood up; chased down ...
664 |     },
665 |     {
666 |         day: "Day 9",
667 |         milestone: "Checkpoint & Review Gate",
668 |         details:
669 |             "save_checkpoint()/load_checkpoint() shipped in database.py; auto_post.py wired to the 30-minut...
670 |     },
671 |     {
672 |         day: "Day 10",
673 |         milestone: "Publishing & Dashboard",
674 |         details:
675 |             "instagram_publisher.py and threads_publisher.py cross-posting wired up; dashboard.html control...
676 |     },
677 | ],
678 | reliability: [
679 |     {
680 |         label: "Orchestrated APIs",
681 |         value: "18+ Keys",
682 |         detail:
683 |             "Integrates content channels, competitor scrapers, database logs, alerting systems, and deploym...
684 |     },
685 |     {
686 |         label: "Paid Gateway Gate",
687 |         value: "Gemini API",
688 |         detail:
689 |             "Paid Gemini API invoked ONLY after the free/low-cost Llama audits approve the draft",
690 |     },
691 |     {
692 |         label: "Audit Fallback Chain",
693 |         value: "4 Layers",
694 |         detail:
695 |             "Audits chain dynamically from Groq ?? Cerebras ?? OpenRouter ?? Raw Gemini to avoid blockages",
696 |     },
697 |     {
698 |         label: "Visual Generation",
699 |         value: "3 Layers",
700 |         detail:
701 |             "Fails back from SiliconFlow (Flux/SD3) ?? Hugging Face ?? Gemini internal image generator",
702 |     },
703 |     {
704 |         label: "Checkpoint TTL",
705 |         value: "24 hrs",
706 |         detail: "self-expires so a stuck runner never resurrects a stale, half-finished post",
707 |     },
708 |     {
709 |         label: "Duplicate-check tiers",
710 |         value: "3",
711 |         detail:
712 |             "embedding similarity ?? adaptive threshold band ?? LLM tiebreaker for the ambiguous zone",
713 |     },
714 | ],
715 | decisions: [
716 |     {
717 |         tech: "Gemini 2.5 Flash + gemini-embedding-2",
718 |         title: "One Provider for Both Drafting and Embeddings",
```



```

719 |         explanation:
720 |             "Gemini writes the initial topic/caption and also generates the embedding vectors used for dupl...
721 |     },
722 |     {
723 |         tech: "Groq + Cerebras",
724 |         title: "Independent Adversarial Audit, Not a Second Opinion From the Same Vendor",
725 |         explanation:
726 |             "verify_text_content() requires both providers ? running different base models (llama-3.3-70b-v...
727 |     },
728 |     {
729 |         tech: "Playwright + Jinja2",
730 |         title: "Headless Card Generation",
731 |         explanation:
732 |             "Branded listicle slides are real HTML/CSS templates screenshotted at 1080x1350, 2x scale ? lay...
733 |     },
734 |     {
735 |         tech: "Local JSON checkpoint + SQLite",
736 |         title: "Zero-Infra State for a Scheduler With No Server",
737 |         explanation:
738 |             "No process stays alive between runs. pipeline_checkpoint.json and veltrix.db are the entire st...
739 |     },
740 |     {
741 |         tech: "GitHub Actions cron + workflow_dispatch",
742 |         title: "Free Scheduler That Doubles as the Approval Queue",
743 |         explanation:
744 |             "The same GitHub Actions infra that runs the cron also handles the manual approve/reject action...
745 |     },
746 | ],
747 | lessons: [
748 |     "Move the checkpoint's binary asset payload out of a committed JSON file and into object storage (S...
749 |     "Cache Groq/Cerebras audit responses so a topic that gets re-evaluated after a checkpoint recovery ...
750 |     "Extend perceptual image-hash duplicate checking to Reel source frames, not just PHOTO/CAROUSEL sti...
751 |     "Replace the fixed 1.5s Playwright settle buffer with an actual document.fonts.ready wait inside th...
752 | ],
753 | gallery: [
754 |     { label: "Live Control Panel", img: "/veltrix-preview.webp" },
755 |     { label: "Credential Health Board", img: "/veltrix-logs-view.webp" },
756 | ],
757 | },
758 | Vcentre: {
759 |     theme: {
760 |         primary: "from-indigo-400",
761 |         secondary: "to-amber-500",
762 |         glow: "rgba(99, 102, 241, 0.15)",
763 |     },
764 |     whyBuilt: {
765 |         problem:
766 |             "Scraping every competitor post nightly buries the handful of genuinely viral posts inside routin...
767 |         solution:
768 |             "Score every post against its own account's rolling median, isolated by format ? Reels vs. Photos...
769 |         scenario: [
770 |             {
771 |                 t: "T+0s",
772 |                 event:
773 |                     "GitHub Actions cron fires main.py at 2AM UTC ? config.load() validates niches.yaml/agents.ya...
774 |             },
775 |             {
776 |                 t: "T+2s",
777 |                 event:
778 |                     "db.set_pipeline_start() anchors the Gemini daily budget window, then OurPostsSync pulls the ...
779 |             },
780 |             {
781 |                 t: "T+5s",
782 |                 event:
783 |                     "scraper.scrape_batch() fires one parallel Apify call across every competitor + owned-account...

```

```
784 |     },
785 |     {
786 |         t: "T+40s",
787 |         event:
788 |             "The Apify dataset resolves; posts are matched back to handles by ownerUsername, keyword-filt...
789 |     },
790 |     {
791 |         t: "T+42s",
792 |         event:
793 |             "outlier.py splits each handle's posts into Reels vs. Photos cohorts, computes the rolling me...
794 |         danger: true,
795 |     },
796 |     {
797 |         t: "T+45s",
798 |         event:
799 |             "Flagged outliers route through router.run() for vision audit, caption audit, hook scoring, a...
800 |     },
801 |     {
802 |         t: "T+50s",
803 |         event:
804 |             "Gemini synthesizes the structured JSON brief only if the daily call budget has room left; th...
805 |     },
806 |     {
807 |         t: "T+52s",
808 |         event:
809 |             "HealthChecker scans for zero-post niches, zero-outlier runs, and 14-day zero-brief streaks, ...
810 |     },
811 | ],
812 | solutionSteps: [
813 |     {
814 |         step: "Batch every handle into one parallel Apify call",
815 |         detail:
816 |             "scrape_batch() sends all competitor + owned-account profile URLs into a single Apify actor r...
817 |         file: "scraper.py",
818 |     },
819 |     {
820 |         step: "Gate every scrape behind a cooldown and a circuit breaker",
821 |         detail:
822 |             "_is_cooldown_active() skips any handle successfully scraped in the last 12 hours; _is_circui...
823 |         file: "scraper.py",
824 |     },
825 |     {
826 |         step: "Isolate cohorts before a median is ever computed",
827 |         detail:
828 |             "detect() splits posts into is_reel==1 vs is_reel==0 before statistics.median() runs, so a Re...
829 |         file: "outlier.py",
830 |     },
831 |     {
832 |         step: "Require three signals to agree before calling something viral",
833 |         detail:
834 |             "is_outlier only trips when a post clears the median multiplier, an absolute floor (3,000 vie...
835 |         file: "outlier.py",
836 |     },
837 |     {
838 |         step: "Route every AI task through a 10-provider fallback chain",
839 |         detail:
840 |             "AgentRouter.run() tries the task's preferred provider first, then walks a fixed fallback ord...
841 |         file: "router.py",
842 |     },
843 |     {
844 |         step: "Feed real post performance back into pattern scoring",
845 |         detail:
846 |             "process_closed_loop_feedback() compares each newly-synced post's real likes against a 30-day...
847 |         file: "db.py",
848 |     },

```

```

849 |     ],
850 |     result:
851 |         "Nightly runs scan every configured niche, isolate true format-relative outliers instead of raw v...
852 |     resultStats: [
853 |         { label: "Monthly infra cost", value: "$0" },
854 |         { label: "AI fallback depth", value: "10 providers" },
855 |         { label: "Outlier confirmation gates", value: "3 (median + floor + ER)" },
856 |         { label: "Scrape credit guards", value: "2 (cooldown + circuit breaker)" },
857 |     ],
858 | },
859 | architecture: [
860 |     {
861 |         id: "scraper",
862 |         label: "core/scraper.py",
863 |         desc: "Batches every competitor + owned handle into one parallel Apify actor call, gated by a 12-...
864 |     },
865 |     {
866 |         id: "outlier",
867 |         label: "core/outlier.py",
868 |         desc: "Splits each handle's posts into Reels vs. Photos cohorts, computes a rolling median per co...
869 |     },
870 |     {
871 |         id: "router",
872 |         label: "core/router.py",
873 |         desc: "Routes every AI task ? vision audits, caption audits, hook scoring, final synthesis ? thro...
874 |     },
875 |     {
876 |         id: "brief",
877 |         label: "core/brief.py",
878 |         desc: "Runs vision, caption, hook, and strategy-contrast audits per outlier, then synthesizes eve...
879 |     },
880 |     {
881 |         id: "pattern-engine",
882 |         label: "core/pattern_engine.py",
883 |         desc: "Clusters outliers into hook, caption, and visual-theme patterns, deduplicating via Supabas...
884 |     },
885 |     {
886 |         id: "health",
887 |         label: "core/health.py",
888 |         desc: "Runs five layered checks after every run ? zero posts, low scrape yield, zero outliers, bu...
889 |     },
890 | ],
891 | features: [
892 |     "Format-isolated median outlier detection (Reels vs. Photos scored separately)",
893 |     "Composite viral confirmation (median multiplier + absolute floor + engagement-rate gate)",
894 |     "12-hour cooldown + 2-hit/24h circuit breaker to protect scrape credits",
895 |     "Single parallel Apify batch call across every competitor + owned account",
896 |     "10-provider AI fallback chain with per-task routing and rate-limit cooldowns",
897 |     "Semantic pattern deduplication via vector similarity + LLM tiebreaker fallback",
898 |     "Closed-loop feedback scoring real post performance back into pattern weights",
899 |     "Layered health diagnostics distinguishing quiet days from silent failures",
900 | ],
901 | challenges: [
902 |     {
903 |         title: "Keeping One Viral Reel From Poisoning the Whole Account's Baseline",
904 |         problem:
905 |             "A single 500k-view Reel sitting in the same post list as an account's usual 2k-like photos wou...
906 |         difficulty:
907 |             "Reels and photos measure success on totally different scales ? views vs. likes ? and mixing th...
908 |         solution:
909 |             "_calculate_cohort() runs independently per format: OutlierEngine.detect() splits every handle'...
910 |         fixSteps: [
911 |             { step: "Split posts by is_reel flag before any statistics call", file: "outlier.py" },
912 |             { step: "Use views as the Reel metric, likes as the Photo metric ? never mix them" },
913 |         ]

```

```
914 |         step: "Require a minimum cohort size of 3 valid posts before computing a median at all",
915 |     },
916 | ],
917 | learned:
918 |     "Median resistance to outliers only holds if it's computed over a genuinely comparable populati...
919 | },
920 | {
921 |     title: "Not Burning Apify Credits on Accounts That Haven't Changed",
922 |     problem:
923 |         "Every handle scraped costs real Apify credits, and a naive nightly loop would re-scrape all 18...
924 |     difficulty:
925 |         "A free-tier scraping budget doesn't survive a scheduler that can't tell the difference between...
926 |     solution:
927 |         "Two independent gates run before scrape_batch() includes a handle: _is_cooldown_active() check...
928 |     fixSteps: [
929 |         {
930 |             step: "Query scrape_history for the last successful scrape timestamp per handle+niche",
931 |             file: "scraper.py",
932 |         },
933 |         { step: "Skip the handle if that timestamp is under 12 hours old" },
934 |         { step: "Independently cap attempts ? success or failure ? at 2 per rolling 24h window" },
935 |     ],
936 |     learned:
937 |         "Rate limiting a scraper isn't just about avoiding a block from the target platform ? it's a bu...
938 | },
939 | {
940 |     title: "One AI Provider Going Down Shouldn't Stall the Whole Brief",
941 |     problem:
942 |         "Free-tier LLM providers rate-limit or go down independently, and a brief pipeline hard-wired t...
943 |     difficulty:
944 |         "Different tasks ? vision audits, caption audits, hook scoring, final JSON synthesis ? have dif...
945 |     solution:
946 |         "AgentRouter.run() resolves a task's preferred agent from routing_rules in agents.yaml, then wa...
947 |     fixSteps: [
948 |         {
949 |             step: "Read the task's preferred provider from routing_rules, then insert it first in the sha...
950 |             file: "router.py",
951 |         },
952 |         {
953 |             step: "Catch rate-limit-specific errors and place that agent on a 5-minute cooldown instead o...
954 |         },
955 |         {
956 |             step: "Skip any agent that doesn't support vision when a base64 image is part of the payload",
957 |         },
958 |     ],
959 |     learned:
960 |         "A fallback chain is only as good as its failure classification ? treating a genuine rate limit...
961 | },
962 | {
963 |     title: "Telling 'No Outliers Today' Apart From 'The Pipeline Is Silently Broken'",
964 |     problem:
965 |         "Zero briefs generated on a given day is completely normal on a quiet news day, but it's also e...
966 |     difficulty:
967 |         "A single-run alert would either fire too often, crying wolf on quiet days, or too rarely, miss...
968 |     solution:
969 |         "HealthChecker.check() runs five layered checks ? zero-post detection, low scrape yield vs. exp...
970 |     fixSteps: [
971 |         {
972 |             step: "Distinguish 'Gemini budget exhausted' from 'brief generator actually failed' by checki...
973 |             file: "health.py",
974 |         },
975 |         {
976 |             step: "Track consecutive zero-brief days via a 14-day rolling window query against creative_b...
977 |             file: "health.py",
978 |         },
979 |     ],
980 | }
```

```
979 |         {
980 |             step: "Write findings to GITHUB_STEP_SUMMARY and an optional Discord webhook so failures surf...",
981 |         },
982 |     ],
983 |     learned:
984 |         "A health check that can't tell 'nothing happened because there was nothing to find' from 'noth...",
985 | },
986 | ],
987 | journey: [
988 |     {
989 |         day: "Day 1",
990 |         milestone: "Skeleton, Config & Schema",
991 |         details:
992 |             "config.py's lazy singleton loader, the niches.yaml/agents.yaml schema, and the initial SQLite ...
993 |     },
994 |     {
995 |         day: "Day 3",
996 |         milestone: "Apify Scraper + Rate Limit Gates",
997 |         details:
998 |             "scraper.py built out against Apify's instagram-scraper actor, with the 12-hour cooldown and 2-...
999 |     },
1000 |     {
1001 |         day: "Day 5",
1002 |         milestone: "Format-Isolated Outlier Engine",
1003 |         details:
1004 |             "outlier.py's cohort-splitting median logic and composite engagement-rate gate replaced an earl...
1005 |     },
1006 |     {
1007 |         day: "Day 7",
1008 |         milestone: "10-Provider Router + Brief Generator",
1009 |         details:
1010 |             "router.py's fallback order and brief.py's vision/caption/hook/contrast pipeline shipped togeth...
1011 |     },
1012 |     {
1013 |         day: "Day 9",
1014 |         milestone: "Pattern Engine, Closed-Loop Feedback & Health Checks",
1015 |         details:
1016 |             "pattern_engine.py's semantic pattern matching, db.py's closed-loop performance scoring, and he...
1017 |     },
1018 | ],
1019 | reliability: [
1020 |     {
1021 |         label: "Scrape cooldown",
1022 |         value: "12 hrs",
1023 |         detail:
1024 |             "minimum gap between successful scrapes of the same handle in the same niche, checked before an...
1025 |     },
1026 |     {
1027 |         label: "Circuit breaker",
1028 |         value: "2 hits / 24h",
1029 |         detail:
1030 |             "a handle stops being scraped mid-day once it's already been attempted twice, protecting the cr...
1031 |     },
1032 |     {
1033 |         label: "Outlier confirmation gates",
1034 |         value: "3",
1035 |         detail:
1036 |             "median multiplier, absolute floor, and minimum engagement rate must all pass together before a...
1037 |     },
1038 |     {
1039 |         label: "AI fallback depth",
1040 |         value: "10 providers",
1041 |         detail:
1042 |             "Groq ? Cerebras ? NVIDIA ? OpenRouter ? GitHub Models ? Cohere ? SambaNova ? Mistral ? Gemini ...
1043 |     },
```

```
1044 |     ],
1045 |     decisions: [
1046 |       {
1047 |         tech: "Apify (apify-client)",
1048 |         title: "Reliable Scraping Without Maintaining Cookies",
1049 |         explanation:
1050 |           "Offloads the actual scraping to a managed actor instead of hand-rolling Instagram session/cook...",
1051 |       },
1052 |       {
1053 |         tech: "Groq + Cerebras",
1054 |         title: "Free, Fast Inference for the Bulk of the Analysis",
1055 |         explanation:
1056 |           "Vision audits, caption audits, and hook scoring all run on Groq/Cerebras's free tiers first ? ...",
1057 |       },
1058 |       {
1059 |         tech: "SQLite + Supabase",
1060 |         title: "Local-First With an Optional Sync Layer",
1061 |         explanation:
1062 |           "SQLite is the source of truth for a single run; Supabase sync (profiles, patterns, competitor ...",
1063 |       },
1064 |       {
1065 |         tech: "FastAPI dashboard",
1066 |         title: "A Thin Control Layer, Not the Pipeline Itself",
1067 |         explanation:
1068 |           "dashboard_server.py exposes read/trigger endpoints over the same SQLite data the cron job writ...",
1069 |       },
1070 |     ],
1071 |     lessons: [
1072 |       "Move the semantic pattern-matching fallback off a full LLM call and onto a cheaper local embedding...",
1073 |       "Make the circuit breaker's 2-hit/24h window niche-configurable instead of hardcoded, since some ni...",
1074 |       "Add TikTok as a second scrape source so outlier detection isn't blind to a competitor's cross-plat...",
1075 |       "Surface closed-loop feedback scores directly on the dashboard so a declining pattern's success rat...",
1076 |     ],
1077 |     gallery: [
1078 |       { label: "Competitor Outliers", img: "/vcentre-preview.webp" },
1079 |       { label: "AI Creative Briefs", img: "/vcentre-briefs-view.webp" },
1080 |     ],
1081 |   },
1082 | };
```

FILE #105: src/lib/site-data.ts

```
1 | /**
2 |  * Content model.
3 |  * Kept separate from presentation so components stay pure/reusable
4 |  * and copy can be edited without touching layout code.
5 |  */
6 |
7 | export const PROFILE = {
8 |   name: "Ayush Agarwal",
9 |   role: "Systems & Agent Engineer",
10 |   location: "Vellore, IN",
11 |   status: "17-year-old developer crafting high-fidelity products & interfaces",
12 |   tagline: "Autonomous pipelines. Commerce engines. Interfaces that feel alive.",
13 |   headline: {
14 |     line1: "Building systems that ship.",
15 |     line2: "Agents that work. Products",
16 |     accent: "that scale.",
17 |   },
18 |   pitch:
19 |     "I'm a developer and designer specialized in high-performance e-commerce engines, multi-model AI agen...",
20 |   bio: [
21 |     "I'm 17, based in India, and I've spent the last year building things that were probably too ambitiou..."
```

```
22 |     "I care more about whether a system actually works than whether it looks impressive on a slide. Every...
23 | ],
24 | currently: "Shipped 3 production AI/SaaS projects in 30 days. Starting VIT CSE in August.",
25 | previously: "Freelance Designer · Two open-source teams.",
26 | email: "theayush.codes@gmail.com",
27 | socials: {
28 |   github: "https://github.com/theayushagarwal",
29 |   linkedin: "https://linkedin.com/in/ayushagarwal17",
30 |   twitter: "https://x.com/theayushcodes",
31 |   resume: "#",
32 | },
33 | } as const;
34 |
35 | export const HERO_STATS = [
36 |   { k: "Infra cost", v: "?0" },
37 |   { k: "Lighthouse Score", v: "100" },
38 |   { k: "Shipped products", v: "3" },
39 |   { k: "Build time", v: "30 days" },
40 | ] as const;
41 |
42 | export type Project = {
43 |   index: string;
44 |   name: string;
45 |   tagline: string;
46 |   summary: string;
47 |   role: string;
48 |   year: string;
49 |   stack: string[];
50 |   metrics: { k: string; v: string }[];
51 |   href: string;
52 |   visual: "agent-trace" | "terminal-evals" | "waveform" | "graph-network";
53 |   visualLabel: string;
54 |   image?: string;
55 |   secondaryImage?: string;
56 |   secondaryLabel?: string;
57 | };
58 | export const PROJECTS: Project[] = [
59 |   {
60 |     index: "01",
61 |     name: "Vurlo",
62 |     tagline:
63 |       "Full-stack e-commerce SaaS platform with Razorpay payments, Firebase rules, and custom AI chatbot",
64 |     summary:
65 |       "I built and shipped a complete e-commerce SaaS platform solo at vurlo.store. It integrates Firebas...
66 |     role: "Founder & Full-Stack Engineer",
67 |     year: "2026",
68 |     stack: [
69 |       "React 19",
70 |       "TanStack Start",
71 |       "Firebase Auth & Rules",
72 |       "Razorpay",
73 |       "Tailwind CSS",
74 |       "Gemini API",
75 |       "TypeScript",
76 |     ],
77 |     metrics: [
78 |       { k: "Build Time", v: "10 days" },
79 |       { k: "Lighthouse SEO", v: "100" },
80 |       { k: "Best Practices", v: "96" },
81 |       { k: "Accessibility", v: "86" },
82 |     ],
83 |     href: "https://vurlo.store",
84 |     visual: "agent-trace",
85 |     visualLabel: "vurlo.store",
86 |     image: "/vurlo-preview.webp",
```

```
87 |     secondaryImage: "/vurlo-admin.webp",
88 |     secondaryLabel: "admin panel",
89 | },
90 | {
91 |     index: "02",
92 |     name: "Veltrix",
93 |     tagline:
94 |         "An Instagram growth account that writes, fact-checks, and publishes itself twice a day",
95 |     summary:
96 |         "Every post ships through an adversarial pipeline before it goes live: Gemini drafts the topic and ...
97 |     role: "Solo developer",
98 |     year: "2026",
99 |     stack: [
100 |         "Python",
101 |         "Gemini API",
102 |         "Groq",
103 |         "Cerebras",
104 |         "SQLite",
105 |         "Supabase",
106 |         "Playwright",
107 |         "GitHub Actions",
108 |         "Jinja2",
109 |     ],
110 |     metrics: [
111 |         { k: "posts/day", v: "2 (auto)" },
112 |         { k: "independent audits", v: "2 of 2 required" },
113 |         { k: "formats", v: "4 (photo/carousel/reel/listicle)" },
114 |         { k: "cost/post", v: "~$0.0002" },
115 |     ],
116 |     href: "",
117 |     visual: "agent-trace",
118 |     visualLabel: "pipeline.run",
119 |     image: "/veltrix-preview.webp",
120 |     secondaryImage: "/veltrix-logs-view.webp",
121 |     secondaryLabel: "credentials board",
122 | },
123 | {
124 |     index: "03",
125 |     name: "Vcentre",
126 |     tagline: "Competitor intelligence pipeline that turns viral outliers into creative briefs",
127 |     summary:
128 |         "Vcentre scrapes competitor Instagram accounts nightly and scores Reels and Photos as separate coho...
129 |     role: "Solo developer",
130 |     year: "2026",
131 |     stack: [
132 |         "Python",
133 |         "Apify",
134 |         "SQLite",
135 |         "Supabase",
136 |         "Groq",
137 |         "Cerebras",
138 |         "GitHub Actions",
139 |         "FastAPI",
140 |     ],
141 |     metrics: [
142 |         { k: "monthly cost", v: "$0" },
143 |         { k: "outlier threshold", v: "3x median" },
144 |         { k: "AI fallback chain", v: "10 providers" },
145 |         { k: "cron cadence", v: "daily @ 2AM UTC" },
146 |     ],
147 |     href: "",
148 |     visual: "graph-network",
149 |     visualLabel: "outlier.scan",
150 |     image: "/vcentre-preview.webp",
151 | },
```



```
152 | ];
153 | export const STACK: { group: string; desc: string; items: string[] }[] = [
154 |   {
155 |     group: "AI Agents & Pipelines",
156 |     desc: "State machine routing, adversarial review loops, multi-model consensus, and autonomous automat...",
157 |     items: ["Python", "Gemini API", "Groq", "Cerebras", "LangGraph", "GitHub Actions"],
158 |   },
159 |   {
160 |     group: "E-Commerce & SaaS Systems",
161 |     desc: "Production transactional logic, payment routing, real-time sync, and granular database securit...",
162 |     items: ["React 19", "TanStack Start", "Firebase Auth", "Firestore Rules", "Razorpay Checkout"],
163 |   },
164 |   {
165 |     group: "Data & Systems Infrastructure",
166 |     desc: "Web scraping cohort analysis, embedding similarities, headless browsers, and serverless hostin...",
167 |     items: ["SQLite", "Supabase", "FastAPI", "Playwright", "Jinja2 Templates", "Cloudflare Pages"],
168 |   },
169 |   {
170 |     group: "Client & Interface Crafting",
171 |     desc: "High-fidelity layouts, dynamic 3D physics, glassmorphic rendering, and fluid animations.",
172 |     items: ["TypeScript", "Tailwind CSS", "Framer Motion", "Figma Design", "HTML5 & CSS3"],
173 |   },
174 | ];
175 |
176 | export const NAV_ITEMS = [
177 |   { label: "Work", href: "#work" },
178 |   { label: "About", href: "#about" },
179 |   { label: "Stack", href: "#stack" },
180 |   { label: "Contact", href: "#contact" },
181 | ] as const;
```

FILE #106: src/lib/sound.ts

```
1 | // Synthesized Web Audio API manager for zero-latency, asset-free tactile sounds.
2 | let audioCtx: AudioContext | null = null;
3 |
4 | // Module-level timestamp to rate-limit/debounce ticks during rapid mouse movement
5 | let lastTickTime = 0;
6 |
7 | // Idle timer to put AudioContext to sleep and free up CPU resources
8 | let suspendTimeout: NodeJS.Timeout | null = null;
9 |
10 | function initAudio() {
11 |   if (typeof window === "undefined") return;
12 |   try {
13 |     // eslint-disable-next-line @typescript-eslint/no-explicit-any
14 |     const AudioContextClass = window.AudioContext || (window as any).webkitAudioContext;
15 |     audioCtx = new AudioContextClass();
16 |   } catch (e) {
17 |     console.warn("Web Audio API not supported in this browser:", e);
18 |   }
19 | }
20 |
21 | // Reset idle suspend timer (10s of silence will suspend context to yield CPU threads)
22 | function resetSuspendTimer() {
23 |   if (typeof window === "undefined" || !audioCtx) return;
24 |
25 |   if (suspendTimeout) {
26 |     clearTimeout(suspendTimeout);
27 |   }
28 |
29 |   suspendTimeout = setTimeout(() => {
30 |     if (audioCtx && audioCtx.state === "running") {
```

```
31 |     audioCtx.suspend().catch((e) => console.warn("Failed to suspend AudioContext:", e));
32 |   }
33 | }, 10000); // 10 seconds of idle inactivity
34 | }
35 |
36 | // Helper to apply random micro-jitter to frequencies and durations
37 | function jitter(base: number, percent: number): number {
38 |   const range = base * percent;
39 |   return base + (Math.random() * 2 - 1) * range;
40 | }
41 |
42 | // Synchronously ensure audio context is active (avoids async/await microtask delay)
43 | function resumeContextSync(): boolean {
44 |   if (!audioCtx) {
45 |     initAudio();
46 |   }
47 |   if (!audioCtx) return false;
48 |
49 |   if (audioCtx.state === "suspended") {
50 |     // Kick off resume() without waiting for it. We still schedule and start
51 |     // the oscillator below; the browser resumes almost immediately in
52 |     // response to the gesture, and the already-scheduled nodes play as soon
53 |     // as it does.
54 |     audioCtx.resume().catch((e) => console.warn("Failed to resume AudioContext:", e));
55 |   }
56 |
57 |   // Refresh the suspend timer upon activity
58 |   resetSuspendTimer();
59 |
60 |   return audioCtx.state !== "closed";
61 | }
62 |
63 | export function resumeAudio(): Promise<void> {
64 |   if (!audioCtx) {
65 |     initAudio();
66 |   }
67 |   if (!audioCtx) return Promise.resolve();
68 |
69 |   // Refresh the suspend timer
70 |   resetSuspendTimer();
71 |
72 |   return audioCtx.resume();
73 | }
74 |
75 | // Set up a one-time listener to initialize/resume AudioContext on first gesture
76 | export function initAudioOnFirstInteraction() {
77 |   if (typeof window === "undefined" || typeof document === "undefined") return;
78 |
79 |   const handleInteraction = () => {
80 |     resumeAudio().catch((e) => console.warn("Failed to resume context on interaction:", e));
81 |     cleanup();
82 |   };
83 |
84 |   const cleanup = () => {
85 |     document.removeEventListener("pointerdown", handleInteraction);
86 |     document.removeEventListener("keydown", handleInteraction);
87 |   };
88 |
89 |   document.addEventListener("pointerdown", handleInteraction, { once: true });
90 |   document.addEventListener("keydown", handleInteraction, { once: true });
91 | }
92 |
93 | /**
94 |  * 1. playTick: Crisp, mechanical hover click (10/10 liquid pop).
95 |  * Debounced at 45ms and memory-optimized via explicit node graph de-allocation on finish.
```

```
96 | */
97 | export function playTick(pan = 0) {
98 |   // Rate-limiting: Prevent multiple ticks triggering too close together (under 45ms)
99 |   const nowMs = performance.now();
100 |   if (nowMs - lastTickTime < 45) {
101 |     return;
102 |   }
103 |
104 |   if (!resumeContextSync() || !audioCtx) return;
105 |
106 |   // Set the start time of the playback
107 |   lastTickTime = nowMs;
108 |
109 |   const now = audioCtx.currentTime;
110 |
111 |   // Flavor selection: 3 distinct timbre flavors picked randomly each hover
112 |   const flavors = [
113 |     { type: "sine" as OscillatorType, startFreq: 2000, endFreq: 1000, baseDuration: 0.025 }, // Flavor A:...
114 |     { type: "triangle" as OscillatorType, startFreq: 1400, endFreq: 700, baseDuration: 0.025 }, // Flavor...
115 |     { type: "sine" as OscillatorType, startFreq: 2800, endFreq: 1800, baseDuration: 0.018 }, // Flavor C:...
116 |   ];
117 |   const selectedFlavor = flavors[Math.floor(Math.random() * flavors.length)];
118 |
119 |   // Timing jitter: +/-10% variation on chosen flavor's base duration (clamped to 12ms minimum)
120 |   const duration = Math.max(0.012, jitter(selectedFlavor.baseDuration, 0.1));
121 |
122 |   // Stereo panner shared by both noise transient and tone layers
123 |   const panner = audioCtx.createStereoPanner();
124 |   const panValue = typeof pan === "number" ? pan : 0;
125 |   panner.pan.setValueAtTime(panValue, now);
126 |   panner.connect(audioCtx.destination);
127 |
128 |   // 1. Synthesize percussive transient click (highly resonant liquid pop)
129 |   const sampleCount = Math.floor(audioCtx.sampleRate * 0.007); // ~7ms duration
130 |   const noiseBuffer = audioCtx.createBuffer(1, sampleCount, audioCtx.sampleRate);
131 |   const bufferData = noiseBuffer.getChannelData(0);
132 |
133 |   // Fill buffer with random noise samples
134 |   for (let i = 0; i < sampleCount; i++) {
135 |     bufferData[i] = (Math.random() * 2 - 1) * 0.5;
136 |   }
137 |
138 |   const noiseSource = audioCtx.createBufferSource();
139 |   noiseSource.buffer = noiseBuffer;
140 |
141 |   const noiseGain = audioCtx.createGain();
142 |   const noiseFilter = audioCtx.createBiquadFilter();
143 |
144 |   noiseGain.gain.setValueAtTime(0.17, now);
145 |   noiseGain.gain.exponentialRampToValueAtTime(0.0001, now + 0.007);
146 |
147 |   // Resonant bandpass filter: centers the noise burst to behave like a tiny physical chamber
148 |   noiseFilter.type = "bandpass";
149 |   const bandpassFreq = selectedFlavor.type === "triangle" ? 1800 : 2800;
150 |   noiseFilter.frequency.setValueAtTime(bandpassFreq, now);
151 |   noiseFilter.Q.setValueAtTime(8.0, now); // High Q factor creates the wooden click/pop character
152 |
153 |   noiseSource.connect(noiseGain);
154 |   noiseGain.connect(noiseFilter);
155 |   noiseFilter.connect(panner); // Route through panner
156 |
157 |   // 2. Synthesize base tone sweep
158 |   const osc = audioCtx.createOscillator();
159 |   const gainNode = audioCtx.createGain();
160 |   const filter = audioCtx.createBiquadFilter();
```

```
161 |
162 |   const startFreq = jitter(selectedFlavor.startFreq, 0.06);
163 |   const endFreq = jitter(selectedFlavor.endFreq, 0.06);
164 |
165 |   osc.type = selectedFlavor.type;
166 |   osc.frequency.setValueAtTime(startFreq, now);
167 |   osc.frequency.exponentialRampToValueAtTime(endFreq, now + duration);
168 |
169 |   gainNode.gain.setValueAtTime(0.045, now);
170 |   gainNode.gain.exponentialRampToValueAtTime(0.0001, now + duration * 0.65);
171 |
172 |   filter.type = "highpass";
173 |   filter.frequency.setValueAtTime(1700, now);
174 |
175 |   osc.connect(gainNode);
176 |   gainNode.connect(filter);
177 |   filter.connect(panner); // Route through panner
178 |
179 |   // Memory Optimization: Disconnect and sever the nodes immediately after playback ends to bypass GC pre...
180 |   osc.onended = () => {
181 |     osc.disconnect();
182 |     gainNode.disconnect();
183 |     filter.disconnect();
184 |     panner.disconnect();
185 |   };
186 |
187 |   noiseSource.onended = () => {
188 |     noiseSource.disconnect();
189 |     noiseGain.disconnect();
190 |     noiseFilter.disconnect();
191 |   };
192 |
193 |   // Trigger both components simultaneously
194 |   noiseSource.start(now);
195 |   osc.start(now);
196 |
197 |   osc.stop(now + duration);
198 | }
199 |
200 | /**
201 |  * 2. playSwell: Cinematic, low-frequency ascending system swell (1.8s) (10/10 with gleam shimmer).
202 |  * Memory-optimized via explicit node graph de-allocation on finish.
203 |  */
204 | export function playSwell() {
205 |   if (!resumeContextSync() || !audioCtx) return;
206 |
207 |   const now = audioCtx.currentTime;
208 |   const duration = 1.8;
209 |
210 |   const subOsc = audioCtx.createOscillator();
211 |   const subGain = audioCtx.createGain();
212 |
213 |   const midOsc = audioCtx.createOscillator();
214 |   const midGain = audioCtx.createGain();
215 |
216 |   const filter = audioCtx.createBiquadFilter();
217 |
218 |   subOsc.type = "sine";
219 |   subOsc.frequency.setValueAtTime(45, now);
220 |   subOsc.frequency.linearRampToValueAtTime(95, now + duration);
221 |
222 |   midOsc.type = "triangle";
223 |   midOsc.frequency.setValueAtTime(180, now);
224 |   midOsc.frequency.linearRampToValueAtTime(280, now + duration);
225 |
```

```
226 | subGain.gain.setValueAtTime(0, now);
227 | subGain.gain.linearRampToValueAtTime(0.22, now + 1.2);
228 | subGain.gain.setValueAtTime(0.22, now + 1.5);
229 | subGain.gain.exponentialRampToValueAtTime(0.0001, now + duration);
230 |
231 | midGain.gain.setValueAtTime(0, now);
232 | midGain.gain.linearRampToValueAtTime(0.035, now + 1.2);
233 | midGain.gain.setValueAtTime(0.035, now + 1.5);
234 | midGain.gain.exponentialRampToValueAtTime(0.0001, now + duration);
235 |
236 | filter.type = "lowpass";
237 | filter.frequency.setValueAtTime(100, now);
238 | filter.frequency.exponentialRampToValueAtTime(600, now + duration);
239 |
240 | subOsc.connect(subGain);
241 | midOsc.connect(midGain);
242 |
243 | subGain.connect(filter);
244 | midGain.connect(filter);
245 |
246 | filter.connect(audioCtx.destination);
247 |
248 | // 3. Shimmer/Gleam layer: quiet, high-frequency tremolo sine sweep at the end of the swell (1.4s to 1.8s)
249 | const shimmerOsc = audioCtx.createOscillator();
250 | const shimmerGain = audioCtx.createGain();
251 | const lfo = audioCtx.createOscillator();
252 | const lfoGain = audioCtx.createGain();
253 |
254 | shimmerOsc.type = "sine";
255 | shimmerOsc.frequency.setValueAtTime(3500, now + 1.4);
256 | shimmerOsc.frequency.exponentialRampToValueAtTime(4200, now + 1.8);
257 |
258 | // LFO modulates the shimmer gain at 12Hz to create a crystalline sparkle
259 | lfo.type = "sine";
260 | lfo.frequency.setValueAtTime(12, now + 1.4);
261 | lfoGain.gain.setValueAtTime(0.012, now + 1.4); // Subtle amplitude modulation depth
262 |
263 | shimmerGain.gain.setValueAtTime(0, now);
264 | shimmerGain.gain.setValueAtTime(0, now + 1.4);
265 | shimmerGain.gain.linearRampToValueAtTime(0.02, now + 1.68);
266 | shimmerGain.gain.exponentialRampToValueAtTime(0.0001, now + 1.8);
267 |
268 | lfo.connect(lfoGain);
269 | lfoGain.connect(shimmerGain.gain);
270 |
271 | shimmerOsc.connect(shimmerGain);
272 | shimmerGain.connect(audioCtx.destination);
273 |
274 | // Memory Optimization: Disconnect nodes upon completion
275 | subOsc.onended = () => {
276 |   subOsc.disconnect();
277 |   subGain.disconnect();
278 |   midOsc.disconnect();
279 |   midGain.disconnect();
280 |   filter.disconnect();
281 | };
282 |
283 | shimmerOsc.onended = () => {
284 |   shimmerOsc.disconnect();
285 |   shimmerGain.disconnect();
286 |   lfo.disconnect();
287 |   lfoGain.disconnect();
288 | };
289 |
290 | // Start oscillators
```

```
291 | subOsc.start(now);
292 | midOsc.start(now);
293 | lfo.start(now + 1.4);
294 | shimmerOsc.start(now + 1.4);
295 |
296 | // Stop oscillators
297 | subOsc.stop(now + duration);
298 | midOsc.stop(now + duration);
299 | lfo.stop(now + duration);
300 | shimmerOsc.stop(now + duration);
301 | }
302 |
303 | /**
304 |  * 3. playExit: Textured suction dissolve (10/10 air whoosh).
305 |  * Replaced the arcade-style sine glide with a premium lowpass white-noise whoosh sweep.
306 |  */
307 | export function playExit() {
308 |   if (!resumeContextSync() || !audioCtx) return;
309 |
310 |   const now = audioCtx.currentTime;
311 |   const duration = 0.3; // Quick 300ms suction whoosh
312 |
313 |   // Generate white noise for the air sweep
314 |   const sampleCount = Math.floor(audioCtx.sampleRate * duration);
315 |   const noiseBuffer = audioCtx.createBuffer(1, sampleCount, audioCtx.sampleRate);
316 |   const bufferData = noiseBuffer.getChannelData(0);
317 |
318 |   for (let i = 0; i < sampleCount; i++) {
319 |     bufferData[i] = (Math.random() * 2 - 1) * 0.45;
320 |   }
321 |
322 |   const noiseSource = audioCtx.createBufferSource();
323 |   noiseSource.buffer = noiseBuffer;
324 |
325 |   const gainNode = audioCtx.createGain();
326 |   const filter = audioCtx.createBiquadFilter();
327 |
328 |   // Volume: smooth fade-out over 300ms
329 |   gainNode.gain.setValueAtTime(0.18, now);
330 |   gainNode.gain.exponentialRampToValueAtTime(0.0001, now + duration);
331 |
332 |   // Lowpass filter sweep: sweeps downward from 3.5kHz to 80Hz to create the organic suction dissolve
333 |   filter.type = "lowpass";
334 |   filter.frequency.setValueAtTime(3500, now);
335 |   filter.frequency.exponentialRampToValueAtTime(80, now + duration);
336 |
337 |   noiseSource.connect(gainNode);
338 |   gainNode.connect(filter);
339 |   filter.connect(audioCtx.destination);
340 |
341 |   // Memory Optimization: Disconnect node graph on ended
342 |   noiseSource.onended = () => {
343 |     noiseSource.disconnect();
344 |     gainNode.disconnect();
345 |     filter.disconnect();
346 |   };
347 |
348 |   noiseSource.start(now);
349 |   noiseSource.stop(now + duration);
350 | }
```

FILE #107: src/lib/utils.ts

```
1 | import { clsx, type ClassValue } from "clsx";
2 | import { twMerge } from "tailwind-merge";
3 |
4 | export function cn(...inputs: ClassValue[]) {
5 |   return twMerge(clsx(inputs));
6 | }
```

FILE #108: src\$routeTree.gen.ts

```
1 | /* eslint-disable */
2 |
3 | // @ts-nocheck
4 |
5 | // noinspection JSUnusedGlobalSymbols
6 |
7 | // This file was automatically generated by TanStack Router.
8 | // You should NOT make any changes in this file as it will be overwritten.
9 | // Additionally, you should also exclude this file from your linter and/or formatter to prevent it from b...
10 |
11 | import { Route as rootRouteImport } from './routes/__root'
12 | import { Route as IndexRouteImport } from './routes/index'
13 | import { Route as ProjectsVurloRouteImport } from './routes/projects/vurlo'
14 | import { Route as ProjectsVeltrixRouteImport } from './routes/projects/veltrix'
15 | import { Route as ProjectsVcentreRouteImport } from './routes/projects/vcentre'
16 |
17 | const IndexRoute = IndexRouteImport.update({
18 |   id: '/',
19 |   path: '/',
20 |   getParentRoute: () => rootRouteImport,
21 | } as any)
22 | const ProjectsVurloRoute = ProjectsVurloRouteImport.update({
23 |   id: '/projects/vurlo',
24 |   path: '/projects/vurlo',
25 |   getParentRoute: () => rootRouteImport,
26 | } as any)
27 | const ProjectsVeltrixRoute = ProjectsVeltrixRouteImport.update({
28 |   id: '/projects/veltrix',
29 |   path: '/projects/veltrix',
30 |   getParentRoute: () => rootRouteImport,
31 | } as any)
32 | const ProjectsVcentreRoute = ProjectsVcentreRouteImport.update({
33 |   id: '/projects/vcentre',
34 |   path: '/projects/vcentre',
35 |   getParentRoute: () => rootRouteImport,
36 | } as any)
37 |
38 | export interface FileRoutesByFullPath {
39 |   '/': typeof IndexRoute
40 |   '/projects/vcentre': typeof ProjectsVcentreRoute
41 |   '/projects/veltrix': typeof ProjectsVeltrixRoute
42 |   '/projects/vurlo': typeof ProjectsVurloRoute
43 | }
44 | export interface FileRoutesByTo {
45 |   '/': typeof IndexRoute
46 |   '/projects/vcentre': typeof ProjectsVcentreRoute
47 |   '/projects/veltrix': typeof ProjectsVeltrixRoute
48 |   '/projects/vurlo': typeof ProjectsVurloRoute
49 | }
50 | export interface FileRoutesById {
51 |   __root__: typeof rootRouteImport
52 |   '/': typeof IndexRoute
53 |   '/projects/vcentre': typeof ProjectsVcentreRoute
54 |   '/projects/veltrix': typeof ProjectsVeltrixRoute
```

```
55 |   '/projects/vurlo': typeof ProjectsVurloRoute
56 | }
57 | export interface FileRouteTypes {
58 |   fileRoutesByFullPath: FileRoutesByFullPath
59 |   fullPaths: '/' | '/projects/vcentre' | '/projects/veltrix' | '/projects/vurlo'
60 |   fileRoutesByTo: FileRoutesByTo
61 |   to: '/' | '/projects/vcentre' | '/projects/veltrix' | '/projects/vurlo'
62 |   id:
63 |     | '__root__'
64 |     | '/'
65 |     | '/projects/vcentre'
66 |     | '/projects/veltrix'
67 |     | '/projects/vurlo'
68 |   fileRoutesById: FileRoutesById
69 | }
70 | export interface RootRouteChildren {
71 |   IndexRoute: typeof IndexRoute
72 |   ProjectsVcentreRoute: typeof ProjectsVcentreRoute
73 |   ProjectsVeltrixRoute: typeof ProjectsVeltrixRoute
74 |   ProjectsVurloRoute: typeof ProjectsVurloRoute
75 | }
76 |
77 | declare module '@tanstack/react-router' {
78 |   interface FileRoutesByPath {
79 |     '/': {
80 |       id: '/'
81 |       path: '/'
82 |       fullPath: '/'
83 |       preLoaderRoute: typeof IndexRouteImport
84 |       parentRoute: typeof rootRouteImport
85 |     }
86 |     '/projects/vurlo': {
87 |       id: '/projects/vurlo'
88 |       path: '/projects/vurlo'
89 |       fullPath: '/projects/vurlo'
90 |       preLoaderRoute: typeof ProjectsVurloRouteImport
91 |       parentRoute: typeof rootRouteImport
92 |     }
93 |     '/projects/veltrix': {
94 |       id: '/projects/veltrix'
95 |       path: '/projects/veltrix'
96 |       fullPath: '/projects/veltrix'
97 |       preLoaderRoute: typeof ProjectsVeltrixRouteImport
98 |       parentRoute: typeof rootRouteImport
99 |     }
100 |     '/projects/vcentre': {
101 |       id: '/projects/vcentre'
102 |       path: '/projects/vcentre'
103 |       fullPath: '/projects/vcentre'
104 |       preLoaderRoute: typeof ProjectsVcentreRouteImport
105 |       parentRoute: typeof rootRouteImport
106 |     }
107 |   }
108 | }
109 |
110 | const rootRouteChildren: RootRouteChildren = {
111 |   IndexRoute: IndexRoute,
112 |   ProjectsVcentreRoute: ProjectsVcentreRoute,
113 |   ProjectsVeltrixRoute: ProjectsVeltrixRoute,
114 |   ProjectsVurloRoute: ProjectsVurloRoute,
115 | }
116 | export const routeTree = rootRouteImport
117 |   ._addFileChildren(rootRouteChildren)
118 |   ._addFileTypes<FileRouteTypes>()
119 |
```



```
120 | import type { getRouter } from './router.tsx'
121 | import type { startInstance } from './start.ts'
122 | declare module '@tanstack/react-start' {
123 |   interface Register {
124 |     ssr: true
125 |     router: Awaited<ReturnType<typeof getRouter>>
126 |     config: Awaited<ReturnType<typeof startInstance.getOptions>>
127 |   }
128 | }
```

FILE #109: src\router.tsx

```
1 | import { QueryClient } from "@tanstack/react-query";
2 | import { createRouter } from "@tanstack/react-router";
3 | import { routeTree } from "../routeTree.gen";
4 |
5 | export const getRouter = () => {
6 |   const queryClient = new QueryClient();
7 |
8 |   const router = createRouter({
9 |     routeTree,
10 |     context: { queryClient },
11 |     scrollRestoration: true,
12 |     defaultPreloadStaleTime: 0,
13 |   });
14 |
15 |   return router;
16 | };
```

FILE #110: src\routes\README.md

```
1 | # Routes
2 |
3 | TanStack Start uses file-based routing. Every .tsx file in this directory
4 | defines a route. Do not create src/pages/, src/routes/_app/index.tsx, or
5 | app/layout.tsx ? those are Next.js / Remix conventions. The only root layout
6 | is src/routes/__root.tsx.
7 |
8 | ## Conventions
9 |
10 | | File | URL |
11 | | ----- | ----- |
12 | | index.tsx | / |
13 | | about.tsx | /about |
14 | | users/index.tsx | /users |
15 | | users/$id.tsx | /users/:id (dynamic ? bare $, no curly braces) |
16 | | posts/{-$category}.tsx | /posts/:category? (optional segment) |
17 | | files/$.tsx | /files/* (splat ? read via _splat param, never **) |
18 | | _layout.tsx | layout route (renders children via <Outlet />) |
19 | | __root.tsx | app shell ? wraps every page; preserve <Outlet /> |
20 |
21 | routeTree.gen.ts is auto-generated. Don't edit it by hand.
```

FILE #111: src\routes__root.tsx

```
1 | import { QueryClient, QueryClientProvider } from "@tanstack/react-query";
2 | import {
3 |   Outlet,
```

```
4 |   Link,
5 |   createRootRouteWithContext,
6 |   useRouter,
7 |   HeadContent,
8 |   Scripts,
9 | } from "@tanstack/react-router";
10 | import { useEffect, type ReactNode } from "react";
11 |
12 | import appCss from "../styles.css?url";
13 | import { reportLovableError } from "../lib/lovable-error-reporting";
14 | import { initAudioOnFirstInteraction } from "../lib/sound";
15 |
16 | function NotFoundComponent() {
17 |   return (
18 |     <div className="flex min-h-screen items-center justify-center bg-background px-4">
19 |       <div className="max-w-md text-center">
20 |         <p className="eyebrow mb-4">404 ? not found</p>
21 |         <h1 className="text-4xl font-semibold text-foreground">
22 |           This page slipped through the cracks
23 |         </h1>
24 |         <p className="mt-3 text-sm text-muted-foreground">
25 |           The URL you followed doesn't match anything here.
26 |         </p>
27 |         <div className="mt-8">
28 |           <Link
29 |             to="/"
30 |             className="btn-primary inline-flex items-center justify-center rounded-md px-5 py-2.5 text-sm...
31 |           >
32 |             Back to home
33 |           </Link>
34 |         </div>
35 |       </div>
36 |     </div>
37 |   );
38 | }
39 |
40 | function ErrorComponent({ error, reset }: { error: Error; reset: () => void }) {
41 |   console.error(error);
42 |   const router = useRouter();
43 |   useEffect(() => {
44 |     reportLovableError(error, { boundary: "tanstack_root_error_component" });
45 |   }, [error]);
46 |
47 |   return (
48 |     <div className="flex min-h-screen items-center justify-center bg-background px-4">
49 |       <div className="max-w-md text-center">
50 |         <p className="eyebrow mb-4">Something broke</p>
51 |         <h1 className="text-2xl font-semibold text-foreground">This page didn't load</h1>
52 |         <p className="mt-2 text-sm text-muted-foreground">Try refreshing, or head back home.</p>
53 |         <div className="mt-6 flex flex-wrap justify-center gap-2">
54 |           <button
55 |             onClick={() => {
56 |               router.invalidate();
57 |               reset();
58 |             }}
59 |             className="btn-primary inline-flex items-center justify-center rounded-md px-5 py-2.5 text-sm...
60 |           >
61 |             Try again
62 |         </button>
63 |         <a
64 |           href="/"
65 |           className="btn-ghost inline-flex items-center justify-center rounded-md px-5 py-2.5 text-sm f...
66 |         >
67 |           Go home
68 |         </a>
```

```
69 |         </div>
70 |     </div>
71 | </div>
72 | );
73 | }
74 |
75 | export const Route = createRootRouteWithContext<{ queryClient: QueryClient }>()({
76 |   head: () => ({
77 |     title: "Ayush Agarwal | Systems & Agent Engineer",
78 |     meta: [
79 |       { charSet: "utf-8" },
80 |       { name: "viewport", content: "width=device-width, initial-scale=1" },
81 |       {
82 |         name: "description",
83 |         content:
84 |           "Systems & Agent Engineer designing and shipping production LLM systems, agents, and inference ...
85 |       },
86 |       { name: "author", content: "Ayush Agarwal" },
87 |       { name: "theme-color", content: "#09090B" },
88 |       { name: "apple-mobile-web-app-capable", content: "yes" },
89 |       { name: "apple-mobile-web-app-status-bar-style", content: "black-translucent" },
90 |       { name: "apple-mobile-web-app-title", content: "Ayush Portfolio" },
91 |       {
92 |         property: "og:title",
93 |         content: "Ayush Agarwal | Systems & Agent Engineer",
94 |       },
95 |       {
96 |         property: "og:description",
97 |         content:
98 |           "Systems & Agent Engineer designing and shipping production LLM systems, agents, and inference ...
99 |       },
100 |       { property: "og:type", content: "website" },
101 |       { name: "twitter:card", content: "summary_large_image" },
102 |       { name: "twitter:title", content: "Ayush Agarwal | Systems & Agent Engineer" },
103 |       {
104 |         name: "twitter:description",
105 |         content:
106 |           "Systems & Agent Engineer designing and shipping production LLM systems, agents, and inference ...
107 |       },
108 |       { name: "color-scheme", content: "dark" },
109 |       { property: "og:url", content: "https://theayush.pages.dev/" },
110 |       { property: "og:image", content: "https://theayush.pages.dev/vurlo-preview.webp" },
111 |       { property: "og:image:width", content: "1024" },
112 |       { property: "og:image:height", content: "553" },
113 |       {
114 |         property: "og:image:alt",
115 |         content:
116 |           "Screenshot mockup of Vurlo e-commerce storefront dashboard featuring premium aesthetic room de...
117 |       },
118 |       { name: "twitter:image", content: "https://theayush.pages.dev/vurlo-preview.webp" },
119 |       {
120 |         name: "robots",
121 |         content: "index, follow, max-image-preview:large, max-snippet:-1, max-video-preview:-1",
122 |       },
123 |       { property: "og:locale", content: "en_US" },
124 |       { property: "og:site_name", content: "Ayush Agarwal Portfolio" },
125 |       { name: "geo.region", content: "IN-TN" },
126 |       { name: "geo.placename", content: "Vellore" },
127 |       { name: "geo.position", content: "12.9692;79.1559" },
128 |       { name: "ICBM", content: "12.9692, 79.1559" },
129 |     ],
130 |     links: [
131 |       { rel: "stylesheet", href: appCss },
132 |       { rel: "canonical", href: "https://theayush.pages.dev/" },
133 |       { rel: "icon", href: "/favicon.png", type: "image/png" },
```

```
134 |     { rel: "apple-touch-icon", href: "/favicon.png" },
135 |     { rel: "preload", href: "/vurlo-preview.webp?v=1.2", as: "image", type: "image/webp" },
136 |     { rel: "preconnect", href: "https://fonts.googleapis.com" },
137 |     {
138 |       rel: "preconnect",
139 |       href: "https://fonts.gstatic.com",
140 |       crossOrigin: "anonymous",
141 |     },
142 |     {
143 |       rel: "stylesheet",
144 |       href: "https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600&family=Inter+Tight:wght@50...",
145 |     },
146 |   ],
147 | }),
148 | shellComponent: RootShell,
149 | component: RootComponent,
150 | notFoundComponent: NotFoundComponent,
151 | errorComponent: ErrorComponent,
152 | });
153 |
154 | const PERSON_JSON_LD = JSON.stringify({
155 |   "@context": "https://schema.org",
156 |   "@type": "Person",
157 |   "@id": "https://theayush.pages.dev/#person",
158 |   name: "Ayush Agarwal",
159 |   jobTitle: "Systems & Agent Engineer",
160 |   contactPoint: {
161 |     "@type": "ContactPoint",
162 |     email: "theayush.codes@gmail.com",
163 |     contactType: "professional",
164 |   },
165 |   knowsAbout: [
166 |     "AI Agents",
167 |     "Systems Engineering",
168 |     "Large Language Models",
169 |     "Python",
170 |     "TypeScript",
171 |     "React",
172 |     "TanStack Start",
173 |     "Database Security",
174 |     "Web Scraping",
175 |   ],
176 |   hasOccupation: {
177 |     "@type": "Occupation",
178 |     name: "Systems & Agent Engineer",
179 |     description:
180 |       "Designing and deploying production-grade AI systems, e-commerce engines, and web architectures glo...",
181 |     skills: "Python, TypeScript, React, Go, SQL",
182 |     occupationLocation: {
183 |       "@type": "AdministrativeArea",
184 |       name: "Remote / Global",
185 |     },
186 |   },
187 |   alumniOf: {
188 |     "@type": "EducationalOrganization",
189 |     name: "Vellore Institute of Technology",
190 |     alternateName: "VIT",
191 |   },
192 |   url: "https://theayush.pages.dev",
193 |   sameAs: ["https://github.com/theayushagarwal", "https://linkedin.com/in/ayushagarwal17"],
194 | });
195 |
196 | const WEBSITE_JSON_LD = JSON.stringify({
197 |   "@context": "https://schema.org",
198 |   "@type": "WebSite",
```

```
199 |   "@id": "https://theayush.pages.dev/#website",
200 |   name: "Ayush Agarwal Portfolio",
201 |   alternateName: ["Ayush Agarwal", "Ayush Agarwal AI Engineer", "Ayush Agarwal Systems Engineer"],
202 |   url: "https://theayush.pages.dev",
203 |   description: "Portfolio of Ayush Agarwal, Systems & Agent Engineer.",
204 |   inLanguage: "en-US",
205 | });
206 |
207 | function RootShell({ children }: { children: ReactNode }) {
208 |   return (
209 |     <html lang="en">
210 |       <head>
211 |         <title>Ayush Agarwal | Systems & Agent Engineer</title>
212 |         <HeadContent />
213 |         <script type="application/ld+json" dangerouslySetInnerHTML={{ __html: PERSON_JSON_LD }} />
214 |         <script type="application/ld+json" dangerouslySetInnerHTML={{ __html: WEBSITE_JSON_LD }} />
215 |       </head>
216 |       <body>
217 |         <div className="noise-overlay" />
218 |         {children}
219 |         <Scripts />
220 |       </body>
221 |     </html>
222 |   );
223 | }
224 |
225 | import { CommandPalette } from "../components/portfolio/CommandPalette";
226 |
227 | function RootComponent() {
228 |   const { queryClient } = Route.useRouteContext();
229 |
230 |   useEffect(() => {
231 |     initAudioOnFirstInteraction();
232 |   }, []);
233 |
234 |   return (
235 |     <QueryClientProvider client={queryClient}>
236 |       <Outlet />
237 |       <CommandPalette />
238 |     </QueryClientProvider>
239 |   );
240 | }
```

FILE #112: src/routes/index.tsx

```
1 | import { createFileRoute } from "@tanstack/react-router";
2 | import { useState, useEffect, useRef } from "react";
3 | import { AnimatePresence } from "motion/react";
4 |
5 | import { Intro } from "@components/portfolio/Intro";
6 | import { Nav } from "@components/portfolio/Nav";
7 | import { Hero } from "@components/portfolio/Hero";
8 | import { Projects } from "@components/portfolio/Projects";
9 | import { About } from "@components/portfolio/About";
10 | import { Stack } from "@components/portfolio/Stack";
11 | import { Faq } from "@components/portfolio/Faq";
12 | import { Contact } from "@components/portfolio/Contact";
13 | import { Footer } from "@components/portfolio/Footer";
14 | import { ScrollProgress } from "@components/portfolio/ScrollProgress";
15 | import { CursorFollower } from "@components/portfolio/CursorFollower";
16 | import { Chatbox } from "@components/portfolio/Chatbox";
17 | import { StatsBar } from "@components/portfolio/StatsBar";
18 |
```

```
19 | export const Route = createFileRoute("/")({
20 |   head: () => ({
21 |     title: "Ayush Agarwal | Systems & Agent Engineer",
22 |     meta: [
23 |       {
24 |         name: "description",
25 |         content:
26 |           "Systems & Agent Engineer designing and shipping production LLM systems, agents, and inference ...
27 |       },
28 |       { property: "og:title", content: "Ayush Agarwal | Systems & Agent Engineer" },
29 |       {
30 |         property: "og:description",
31 |         content:
32 |           "Systems & Agent Engineer designing and shipping production LLM systems, agents, and inference ...
33 |       },
34 |       { property: "og:url", content: "https://theayush.pages.dev/" },
35 |       { property: "og:image", content: "https://theayush.pages.dev/portfolio-preview.webp" },
36 |       { name: "twitter:card", content: "summary_large_image" },
37 |       { name: "twitter:title", content: "Ayush Agarwal | Systems & Agent Engineer" },
38 |       {
39 |         name: "twitter:description",
40 |         content:
41 |           "Systems & Agent Engineer designing and shipping production LLM systems, agents, and inference ...
42 |       },
43 |       { name: "twitter:image", content: "https://theayush.pages.dev/portfolio-preview.webp" },
44 |     ],
45 |     links: [{ rel: "canonical", href: "https://theayush.pages.dev/" }],
46 |   }),
47 |   component: Portfolio,
48 | });
49 |
50 | const FAQ_JSON_LD = JSON.stringify({
51 |   "@context": "https://schema.org",
52 |   "@type": "FAQPage",
53 |   mainEntity: [
54 |     {
55 |       "@type": "Question",
56 |       name: "How does Veltrix's adversarial AI consensus loop work?",
57 |       acceptedAnswer: {
58 |         "@type": "Answer",
59 |         text: "Veltrix is an autonomous pipeline that orchestrates 18+ active APIs (content channels, com...
60 |       },
61 |     },
62 |     {
63 |       "@type": "Question",
64 |       name: "What is Vcentre and how does it analyze competitor engagement?",
65 |       acceptedAnswer: {
66 |         "@type": "Answer",
67 |         text: "Vcentre scrapes competitor accounts nightly using isolated Reels and Photos baseline calcula...
68 |       },
69 |     },
70 |     {
71 |       "@type": "Question",
72 |       name: "What technology stack does Ayush Agarwal specialize in?",
73 |       acceptedAnswer: {
74 |         "@type": "Answer",
75 |         text: "For AI and pipelines, the primary languages are Python and TypeScript, deploying models wi...
76 |       },
77 |     },
78 |     {
79 |       "@type": "Question",
80 |       name: "Is Ayush available for remote internships, freelance, or contract work?",
81 |       acceptedAnswer: {
82 |         "@type": "Answer",
83 |         text: "Yes. Ayush starts his Computer Science Engineering (CSE) degree at VIT in August 2026 and ...
```

```
84 |     },
85 |   },
86 |   {
87 |     "@type": "Question",
88 |     name: "How was the Vurlo e-commerce SaaS platform optimized for performance?",
89 |     acceptedAnswer: {
90 |       "@type": "Answer",
91 |       text: "Vurlo was built solo in 10 days and achieved a 100/100 Lighthouse SEO score. It integrates...",
92 |     },
93 |   },
94 | ],
95 | });
96 |
97 | function Portfolio() {
98 |   const [showIntro, setShowIntro] = useState(true);
99 |   const containerRef = useRef<HTMLDivElement>(null);
100 |
101 |   useEffect(() => {
102 |     const container = containerRef.current;
103 |     if (!container) return;
104 |
105 |     const handleMouseMove = (e: MouseEvent) => {
106 |       container.style.setProperty("--mouse-x", `${e.clientX}px`);
107 |       container.style.setProperty("--mouse-y", `${e.clientY}px`);
108 |     };
109 |
110 |     container.addEventListener("mousemove", handleMouseMove);
111 |     return () => {
112 |       container.removeEventListener("mousemove", handleMouseMove);
113 |     };
114 |   }, []);
115 |
116 |   return (
117 |     <div
118 |       ref={containerRef}
119 |       className="relative min-h-screen bg-background text-foreground overflow-hidden"
120 |     >
121 |       <script type="application/ld+json" dangerouslySetInnerHTML={{ __html: FAQ_JSON_LD }} />
122 |       <div className="global-spotlight" />
123 |       <a href="#top" className="skip-link">
124 |         Skip to content
125 |       </a>
126 |
127 |       <AnimatePresence>{showIntro && <Intro onDone={() => setShowIntro(false)} />}</AnimatePresence>
128 |
129 |       <ScrollProgress />
130 |       <CursorFollower />
131 |       <Chatbox />
132 |       <Nav />
133 |
134 |       <main>
135 |         <Hero revealed={!showIntro} />
136 |         <StatsBar />
137 |         <Projects />
138 |         <div className="hairline mx-auto max-w-6xl px-6" />
139 |         <About />
140 |         <div className="hairline mx-auto max-w-6xl px-6" />
141 |         <Stack />
142 |         <div className="hairline mx-auto max-w-6xl px-6" />
143 |         <Faq />
144 |         <div className="hairline mx-auto max-w-6xl px-6" />
145 |         <Contact />
146 |       </main>
147 |
148 |       <Footer />
```

```
149 |     </div>
150 |   );
151 | }
```

FILE #113: src\routes\projects\vcentre.tsx

```
1 | import { createFileRoute } from "@tanstack/react-router";
2 | import { PROJECTS } from "@/lib/site-data";
3 | import { WorkspacePanel } from "@/components/portfolio/WorkspacePanel";
4 |
5 | export const Route = createFileRoute("/projects/vcentre")({
6 |   head: () => ({
7 |     title: "Vcentre Case Study | Ayush Agarwal",
8 |     meta: [
9 |       {
10 |         name: "description",
11 |         content:
12 |           "Detailed case study of Vcentre: a competitor intelligence crawler that flags viral outlier pos...
13 |       },
14 |       { property: "og:title", content: "Vcentre Case Study | Ayush Agarwal" },
15 |       {
16 |         property: "og:description",
17 |         content:
18 |           "Detailed case study of Vcentre: a competitor intelligence crawler that flags viral outlier pos...
19 |       },
20 |       { property: "og:type", content: "article" },
21 |       { property: "og:url", content: "https://theayush.pages.dev/projects/vcentre" },
22 |       { property: "og:image", content: "https://theayush.pages.dev/vcentre-preview.webp" },
23 |       { name: "twitter:card", content: "summary_large_image" },
24 |       { name: "twitter:title", content: "Vcentre Case Study | Ayush Agarwal" },
25 |       {
26 |         name: "twitter:description",
27 |         content:
28 |           "Detailed case study of Vcentre: a competitor intelligence crawler that flags viral outlier pos...
29 |       },
30 |       { name: "twitter:image", content: "https://theayush.pages.dev/vcentre-preview.webp" },
31 |     ],
32 |     links: [{ rel: "canonical", href: "https://theayush.pages.dev/projects/vcentre" }],
33 |   }),
34 |   component: VcentreCaseStudy,
35 | });
36 |
37 | const VCENTRE_PROJECT_JSON_LD = JSON.stringify({
38 |   "@context": "https://schema.org",
39 |   "@type": "SoftwareSourceCode",
40 |   name: "Vcentre",
41 |   url: "https://theayush.pages.dev/projects/vcentre",
42 |   description:
43 |     "Competitor intelligence scraper that detects cohort-specific outliers (Reels vs. Photos medians) and...
44 |   programmingLanguage: ["Python", "FastAPI"],
45 |   codeRepository: "https://github.com/theayushagarwal/PORTFOLIO",
46 |   author: {
47 |     "@type": "Person",
48 |     "@id": "https://theayush.pages.dev/#person",
49 |   },
50 | });
51 |
52 | function VcentreCaseStudy() {
53 |   const project = PROJECTS.find((p) => p.name.toLowerCase() === "vcentre");
54 |   if (!project) return null;
55 |   return (
56 |     <>
57 |     <script
```



```

58 |         type="application/ld+json"
59 |         dangerouslySetInnerHTML={{ __html: VCENTRE_PROJECT_JSON_LD }}
60 |     />
61 |     <WorkspacePanel project={project} />
62 | </>
63 | );
64 | }

```

FILE #114: src/routes/projects/veltrix.tsx

```

1 | import { createFileRoute } from "@tanstack/react-router";
2 | import { PROJECTS } from "@lib/site-data";
3 | import { WorkspacePanel } from "@components/portfolio/WorkspacePanel";
4 |
5 | export const Route = createFileRoute("/projects/veltrix")({
6 |   head: () => ({
7 |     title: "Veltrix Case Study | Ayush Agarwal",
8 |     meta: [
9 |       {
10 |         name: "description",
11 |         content:
12 |           "Deep dive into Veltrix: an autonomous Instagram publishing engine orchestrating 18+ active API...
13 |       },
14 |       { property: "og:title", content: "Veltrix Case Study | Ayush Agarwal" },
15 |       {
16 |         property: "og:description",
17 |         content:
18 |           "Deep dive into Veltrix: an autonomous Instagram publishing engine orchestrating 18+ active API...
19 |       },
20 |       { property: "og:type", content: "article" },
21 |       { property: "og:url", content: "https://theayush.pages.dev/projects/veltrix" },
22 |       { property: "og:image", content: "https://theayush.pages.dev/veltrix-preview.webp" },
23 |       { name: "twitter:card", content: "summary_large_image" },
24 |       { name: "twitter:title", content: "Veltrix Case Study | Ayush Agarwal" },
25 |       {
26 |         name: "twitter:description",
27 |         content:
28 |           "Deep dive into Veltrix: an autonomous Instagram publishing engine orchestrating 18+ active API...
29 |       },
30 |       { name: "twitter:image", content: "https://theayush.pages.dev/veltrix-preview.webp" },
31 |     ],
32 |     links: [{ rel: "canonical", href: "https://theayush.pages.dev/projects/veltrix" }],
33 |   }),
34 |   component: VeltrixCaseStudy,
35 | });
36 |
37 | const VELTRIX_PROJECT_JSON_LD = JSON.stringify({
38 |   "@context": "https://schema.org",
39 |   "@type": "SoftwareSourceCode",
40 |   name: "Veltrix",
41 |   url: "https://theayush.pages.dev/projects/veltrix",
42 |   description:
43 |     "An autonomous publishing engine orchestrating 18+ APIs. Employs a multi-model adversarial consensus ...
44 |   programmingLanguage: ["Python"],
45 |   codeRepository: "https://github.com/theayushagarwal/PORTFOLIO",
46 |   author: {
47 |     "@type": "Person",
48 |     "@id": "https://theayush.pages.dev/#person",
49 |   },
50 | });
51 |
52 | function VeltrixCaseStudy() {
53 |   const project = PROJECTS.find((p) => p.name.toLowerCase() === "veltrix");

```

```
54 |   if (!project) return null;
55 |   return (
56 |     <>
57 |     <script
58 |       type="application/ld+json"
59 |       dangerouslySetInnerHTML={{ __html: VELTRIX_PROJECT_JSON_LD }}
60 |     />
61 |     <WorkspacePanel project={project} />
62 |   </>
63 | );
64 | }
```

FILE #115: src\routes\projects\vurlo.tsx

```
1 | import { createFileRoute } from "@tanstack/react-router";
2 | import { PROJECTS } from "@lib/site-data";
3 | import { WorkspacePanel } from "@components/portfolio/WorkspacePanel";
4 |
5 | export const Route = createFileRoute("/projects/vurlo")({
6 |   head: () => ({
7 |     title: "Vurlo Case Study | Ayush Agarwal",
8 |     meta: [
9 |       {
10 |         name: "description",
11 |         content:
12 |           "Detailed case study of Vurlo, a full-stack e-commerce SaaS platform engineered in 10 days with...",
13 |       },
14 |       { property: "og:title", content: "Vurlo Case Study | Ayush Agarwal" },
15 |       {
16 |         property: "og:description",
17 |         content:
18 |           "Detailed case study of Vurlo, a full-stack e-commerce SaaS platform engineered in 10 days with...",
19 |       },
20 |       { property: "og:type", content: "article" },
21 |       { property: "og:url", content: "https://theayush.pages.dev/projects/vurlo" },
22 |       { property: "og:image", content: "https://theayush.pages.dev/vurlo-preview.webp" },
23 |       { name: "twitter:card", content: "summary_large_image" },
24 |       { name: "twitter:title", content: "Vurlo Case Study | Ayush Agarwal" },
25 |       {
26 |         name: "twitter:description",
27 |         content:
28 |           "Detailed case study of Vurlo, a full-stack e-commerce SaaS platform engineered in 10 days with...",
29 |       },
30 |       { name: "twitter:image", content: "https://theayush.pages.dev/vurlo-preview.webp" },
31 |     ],
32 |     links: [{ rel: "canonical", href: "https://theayush.pages.dev/projects/vurlo" }],
33 |   }),
34 |   component: VurloCaseStudy,
35 | });
36 |
37 | const VURLO_PROJECT_JSON_LD = JSON.stringify({
38 |   "@context": "https://schema.org",
39 |   "@type": "SoftwareSourceCode",
40 |   name: "Vurlo",
41 |   url: "https://theayush.pages.dev/projects/vurlo",
42 |   description:
43 |     "A production-grade e-commerce SaaS platform solo-built in 10 days. Features real-time stock-locking ...",
44 |   programmingLanguage: ["TypeScript", "React", "CSS"],
45 |   codeRepository: "https://github.com/theayushagarwal/vurlo-ecommerce",
46 |   author: {
47 |     "@type": "Person",
48 |     "@id": "https://theayush.pages.dev/#person",
49 |   },
```

```
50 | });
51 |
52 | function VurloCaseStudy() {
53 |   const project = PROJECTS.find((p) => p.name.toLowerCase() === "vurlo");
54 |   if (!project) return null;
55 |   return (
56 |     <>
57 |       <script
58 |         type="application/ld+json"
59 |         dangerouslySetInnerHTML={{ __html: VURLO_PROJECT_JSON_LD }}
60 |       />
61 |       <WorkspacePanel project={project} />
62 |     </>
63 |   );
64 | }
```

FILE #116: src\server.ts

```
1 | import "../lib/error-capture";
2 |
3 | import { consumeLastCapturedError } from "../lib/error-capture";
4 | import { renderErrorPage } from "../lib/error-page";
5 |
6 | type ServerEntry = {
7 |   fetch: (request: Request, env: unknown, ctx: unknown) => Promise<Response> | Response;
8 | };
9 |
10 | let serverEntryPromise: Promise<ServerEntry> | undefined;
11 |
12 | async function getServerEntry(): Promise<ServerEntry> {
13 |   if (!serverEntryPromise) {
14 |     serverEntryPromise = import("@tanstack/react-start/server-entry").then(
15 |       (m) => (m.default ?? m) as ServerEntry,
16 |     );
17 |   }
18 |   return serverEntryPromise;
19 | }
20 |
21 | // h3 swallows in-handler throws into a normal 500 Response with body
22 | // {"unhandled":true,"message":"HTTPError"} ? try/catch alone never fires for those.
23 | async function normalizeCatastrophicSsrResponse(response: Response): Promise<Response> {
24 |   if (response.status < 500) return response;
25 |   const contentType = response.headers.get("content-type") ?? "";
26 |   if (!contentType.includes("application/json")) return response;
27 |
28 |   const body = await response.clone().text();
29 |   if (!isH3SwallowedErrorBody(body)) return response;
30 |
31 |   console.error(consumeLastCapturedError() ?? new Error(`h3 swallowed SSR error: ${body}`));
32 |   return new Response(renderErrorPage(), {
33 |     status: 500,
34 |     headers: { "content-type": "text/html; charset=utf-8" },
35 |   });
36 | }
37 |
38 | function isH3SwallowedErrorBody(body: string): boolean {
39 |   try {
40 |     const payload = JSON.parse(body) as { unhandled?: unknown; message?: unknown };
41 |     return payload.unhandled === true && payload.message === "HTTPError";
42 |   } catch {
43 |     return false;
44 |   }
45 | }
```

```
46 |  
47 | export default {  
48 |   async fetch(request: Request, env: unknown, ctx: unknown) {  
49 |     try {  
50 |       const handler = await getServerEntry();  
51 |       const response = await handler.fetch(request, env, ctx);  
52 |       return await normalizeCatastrophicSsrResponse(response);  
53 |     } catch (error) {  
54 |       console.error(error);  
55 |       return new Response(renderErrorPage(), {  
56 |         status: 500,  
57 |         headers: { "content-type": "text/html; charset=utf-8" },  
58 |       });  
59 |     }  
60 |   },  
61 | };
```

FILE #117: src\start.ts

```
1 | import { createStart, createMiddleware } from "@tanstack/react-start";  
2 |  
3 | import { renderErrorPage } from "../lib/error-page";  
4 |  
5 | const errorMiddleware = createMiddleware().server(async ({ next }) => {  
6 |   try {  
7 |     return await next();  
8 |   } catch (error) {  
9 |     if (error != null && typeof error === "object" && "statusCode" in error) {  
10 |       throw error;  
11 |     }  
12 |     console.error(error);  
13 |     return new Response(renderErrorPage(), {  
14 |       status: 500,  
15 |       headers: { "content-type": "text/html; charset=utf-8" },  
16 |     });  
17 |   }  
18 | });  
19 |  
20 | export const startInstance = createStart(() => ({  
21 |   requestMiddleware: [errorMiddleware],  
22 | }));
```

FILE #118: src\styles.css

```
1 | @import "tailwindcss" source(none);  
2 | @source "../src";  
3 | @import "tw-animate-css";  
4 |  
5 | @custom-variant dark (&:is(.dark *));  
6 |  
7 | /*  
8 |  * Portfolio design system ? premium dark.  
9 |  * Background #09090B · Surface #111113 · Card #18181B  
10 |  * Primary #8B5CF6 (violet) · Secondary #06B6D4 (cyan)  
11 |  * Text: FAFAFA / A1A1AA / 71717A  
12 |  */  
13 |  
14 | @theme inline {  
15 |   --radius-sm: calc(var(--radius) - 4px);  
16 |   --radius-md: calc(var(--radius) - 2px);  
17 |   --radius-lg: var(--radius);
```

```
18 | --radius-xl: calc(var(--radius) + 4px);
19 | --radius-2xl: calc(var(--radius) + 8px);
20 |
21 | --color-background: var(--background);
22 | --color-foreground: var(--foreground);
23 | --color-surface: var(--surface);
24 | --color-card: var(--card);
25 | --color-card-foreground: var(--card-foreground);
26 | --color-popover: var(--popover);
27 | --color-popover-foreground: var(--popover-foreground);
28 | --color-primary: var(--primary);
29 | --color-primary-foreground: var(--primary-foreground);
30 | --color-secondary: var(--secondary);
31 | --color-secondary-foreground: var(--secondary-foreground);
32 | --color-muted: var(--muted);
33 | --color-muted-foreground: var(--muted-foreground);
34 | --color-subtle: var(--subtle);
35 | --color-accent: var(--accent);
36 | --color-accent-foreground: var(--accent-foreground);
37 | --color-destructive: var(--destructive);
38 | --color-destructive-foreground: var(--destructive-foreground);
39 | --color-border: var(--border);
40 | --color-input: var(--input);
41 | --color-ring: var(--ring);
42 |
43 | --font-sans: "Inter", ui-sans-serif, system-ui, -apple-system, "Segoe UI", sans-serif;
44 | --font-display: "Inter Tight", "Inter", ui-sans-serif, system-ui, sans-serif;
45 | --font-mono: "JetBrains Mono", ui-monospace, SFMono-Regular, Menlo, monospace;
46 | /* Editorial accent for a handful of emphasis words ? never body copy. */
47 | --font-serif-accent: "Instrument Serif", ui-serif, Georgia, serif;
48 | }
49 |
50 | :root {
51 |   --radius: 0.75rem;
52 |
53 |   /* Core surfaces */
54 |   --background: #07050f;
55 |   --surface: #0d0a18;
56 |   --card: #131020;
57 |   --card-foreground: #fafafa;
58 |   --foreground: #fafafa;
59 |   --popover: #131020;
60 |   --popover-foreground: #fafafa;
61 |
62 |   /* Text scale */
63 |   --muted-foreground: #a1a1aa;
64 |   --subtle: #8a8a94;
65 |
66 |   /* Accents */
67 |   --primary: #8b5cf6;
68 |   --primary-foreground: #ffffff;
69 |   --secondary: #06b6d4;
70 |   --secondary-foreground: #07050f;
71 |
72 |   --muted: #131020;
73 |   --accent: #1b162a;
74 |   --accent-foreground: #fafafa;
75 |
76 |   --destructive: oklch(0.65 0.22 25);
77 |   --destructive-foreground: #fafafa;
78 |
79 |   --border: rgba(255, 255, 255, 0.08);
80 |   --input: rgba(255, 255, 255, 0.1);
81 |   --ring: #8b5cf6;
82 | }
```

```
83 |   color-scheme: dark;
84 | }
85 |
86 | @layer base {
87 |   * {
88 |     border-color: var(--color-border);
89 |   }
90 |
91 |   html {
92 |     scroll-behavior: smooth;
93 |     background-color: var(--color-background);
94 |   }
95 |
96 |   body {
97 |     background-color: var(--color-background);
98 |     color: var(--color-foreground);
99 |     font-family: var(--font-sans);
100 |     -webkit-font-smoothing: antialiased;
101 |     -moz-osx-font-smoothing: grayscale;
102 |     text-rendering: optimizeLegibility;
103 |     font-feature-settings: "cv11", "ss01", "ss03";
104 |   }
105 |
106 |   ::selection {
107 |     background: color-mix(in oklab, var(--primary) 40%, transparent);
108 |     color: #fff;
109 |   }
110 |
111 |   h1,
112 |   h2,
113 |   h3,
114 |   h4 {
115 |     font-family: var(--font-display);
116 |     letter-spacing: -0.02em;
117 |     text-wrap: balance;
118 |   }
119 |
120 |   a,
121 |   button {
122 |     outline-offset: 3px;
123 |   }
124 |
125 |   a:focus-visible,
126 |   button:focus-visible {
127 |     outline: 2px solid var(--color-ring);
128 |     outline-offset: 3px;
129 |     border-radius: 2px;
130 |   }
131 |
132 |   @media (prefers-reduced-motion: reduce) {
133 |     html {
134 |       scroll-behavior: auto;
135 |     }
136 |     *,
137 |     *::before,
138 |     *::after {
139 |       animation-duration: 0.01ms !important;
140 |       animation-iteration-count: 1 !important;
141 |       transition-duration: 0.01ms !important;
142 |     }
143 |   }
144 | }
145 |
146 | @utility font-serif-accent {
147 |   font-family: var(--font-serif-accent);
```

```
148 | background: linear-gradient(to right, #a78bfa, #22d3ee, #f472b6, #a78bfa);
149 | background-size: 300% auto;
150 | -webkit-background-clip: text;
151 | background-clip: text;
152 | color: transparent;
153 | animation: text-gradient-shift 8s linear infinite;
154 | display: inline-block;
155 | }
156 |
157 | @keyframes text-gradient-shift {
158 |   0% {
159 |     background-position: 0% 50%;
160 |   }
161 |   50% {
162 |     background-position: 100% 50%;
163 |   }
164 |   100% {
165 |     background-position: 0% 50%;
166 |   }
167 | }
168 |
169 | /* Fine structural dot-grid used behind the hero ? no glow, just texture. */
170 | @utility dot-grid {
171 |   background-image: radial-gradient(rgba(255, 255, 255, 0.09) 1px, transparent 1px);
172 |   background-size: 28px 28px;
173 | }
174 |
175 | /*
176 |  * A slow, single light-band sweeping across the text ? the "shiny text"
177 |  * treatment used sparingly on hero names/badges. Deliberately monochrome
178 |  * (foreground ? white ? foreground), not a rainbow gradient, and slow
179 |  * enough to read as premium rather than a loading shimmer.
180 |  */
181 | @utility shiny-text {
182 |   background-image: linear-gradient(
183 |     100deg,
184 |     var(--color-foreground) 38%,
185 |     #ffffff 50%,
186 |     var(--color-foreground) 62%
187 |   );
188 |   background-size: 220% auto;
189 |   background-clip: text;
190 |   -webkit-background-clip: text;
191 |   color: transparent;
192 |   animation: shine 5s linear infinite;
193 |   animation-delay: 1.4s;
194 |
195 |   @media (prefers-reduced-motion: reduce) {
196 |     animation: none;
197 |     color: var(--color-foreground);
198 |     background: none;
199 |   }
200 | }
201 |
202 | @keyframes shine {
203 |   to {
204 |     background-position: -220% center;
205 |   }
206 | }
207 |
208 | /* Visible only when focused via keyboard ? standard skip-navigation link. */
209 | @utility skip-link {
210 |   position: fixed;
211 |   top: -100%;
212 |   left: 1rem;
```

```
213 | z-index: 100;
214 | background: var(--color-primary);
215 | color: var(--color-primary-foreground);
216 | padding: 0.6rem 1rem;
217 | border-radius: var(--radius-md);
218 | font-size: 0.8rem;
219 | font-weight: 500;
220 | transition: top 0.2s ease;
221 | &:focus-visible {
222 |   top: 1rem;
223 | }
224 | }
225 |
226 | /* Subtle grain + noise for premium feel */
227 | @utility grain {
228 |   position: relative;
229 |   &::after {
230 |     content: "";
231 |     position: absolute;
232 |     inset: 0;
233 |     pointer-events: none;
234 |     opacity: 0.035;
235 |     mix-blend-mode: overlay;
236 |     background-image: url("data:image/svg+xml;utf8,<svg xmlns='http://www.w3.org/2000/svg' width='160' he...
237 |   }
238 | }
239 |
240 | /* Hairline divider used site-wide */
241 | @utility hairline {
242 |   background: linear-gradient(
243 |     90deg,
244 |     transparent,
245 |     rgba(255, 255, 255, 0.08) 20%,
246 |     rgba(255, 255, 255, 0.08) 80%,
247 |     transparent
248 |   );
249 |   height: 1px;
250 |   width: 100%;
251 | }
252 |
253 | /* Underline hover for links */
254 | @utility link-underline {
255 |   position: relative;
256 |   &::after {
257 |     content: "";
258 |     position: absolute;
259 |     left: 0;
260 |     bottom: -2px;
261 |     width: 100%;
262 |     height: 1px;
263 |     background: currentColor;
264 |     transform: scaleX(0);
265 |     transform-origin: right;
266 |     transition: transform 0.4s cubic-bezier(0.22, 1, 0.36, 1);
267 |   }
268 |   &:hover::after {
269 |     transform: scaleX(1);
270 |     transform-origin: left;
271 |   }
272 | }
273 |
274 | /* Card treatment used across projects and stack */
275 | @utility surface-card {
276 |   background:
277 |     linear-gradient(180deg, rgba(255, 255, 255, 0.02), rgba(255, 255, 255, 0) 60%),
```



```
278 |     var(--color-card);
279 |     border: 1px solid var(--color-border);
280 |     border-radius: var(--radius-xl);
281 |     box-shadow: 0 4px 20px rgba(0, 0, 0, 0.15);
282 |     transition:
283 |       border-color 0.4s ease,
284 |       transform 0.6s cubic-bezier(0.22, 1, 0.36, 1),
285 |       background 0.4s ease,
286 |       box-shadow 0.4s ease;
287 |     &:hover {
288 |       border-color: rgba(6, 182, 212, 0.35);
289 |       box-shadow: 0 20px 40px rgba(0, 0, 0, 0.45);
290 |       background:
291 |         linear-gradient(180deg, rgba(6, 182, 212, 0.04), rgba(255, 255, 255, 0) 70%),
292 |         var(--color-card);
293 |     }
294 | }
295 |
296 | /* Violet CTA */
297 | @utility btn-primary {
298 |   background: var(--primary);
299 |   color: var(--primary-foreground);
300 |   border: 1px solid color-mix(in oklab, var(--primary) 70%, white 10%);
301 |   box-shadow:
302 |     inset 0 1px 0 rgba(255, 255, 255, 0.14),
303 |     0 1px 0 rgba(0, 0, 0, 0.4);
304 |   transition:
305 |     background 0.25s ease,
306 |     transform 0.25s ease,
307 |     box-shadow 0.25s ease;
308 |   &:hover {
309 |     background: color-mix(in oklab, var(--primary) 88%, white 4%);
310 |     box-shadow:
311 |       inset 0 1px 0 rgba(255, 255, 255, 0.2),
312 |       0 0 0 4px rgba(139, 92, 246, 0.15);
313 |   }
314 | }
315 |
316 | /* Ghost / secondary button */
317 | @utility btn-ghost {
318 |   background: transparent;
319 |   color: var(--foreground);
320 |   border: 1px solid var(--color-border);
321 |   transition:
322 |     background 0.25s ease,
323 |     border-color 0.25s ease,
324 |     color 0.25s ease;
325 |   &:hover {
326 |     color: var(--secondary);
327 |     border-color: color-mix(in oklab, var(--secondary) 45%, transparent);
328 |     background: color-mix(in oklab, var(--secondary) 6%, transparent);
329 |   }
330 | }
331 |
332 | /* Section label eyebrow */
333 | @utility eyebrow {
334 |   font-family: var(--font-mono);
335 |   font-size: 0.72rem;
336 |   letter-spacing: 0.18em;
337 |   text-transform: uppercase;
338 |   color: var(--subtle);
339 | }
340 |
341 | /* =====
342 | * Workspace Portal ? case-study overlay
```

```
343 | * A dev-tool console rather than a generic modal: glass panels,
344 | * bracket-cornered frames, a monospace boot sequence, and a
345 | * violet?cyan signal running through progress/metric accents.
346 | * ===== */
347 |
348 | /* Fine blueprint-style grid behind the portal, faded toward the edges. */
349 | @utility workspace-grid {
350 |   background-image:
351 |     linear-gradient(rgba(255, 255, 255, 0.035) 1px, transparent 1px),
352 |     linear-gradient(90deg, rgba(255, 255, 255, 0.035) 1px, transparent 1px);
353 |   background-size: 34px 34px;
354 |   mask-image: radial-gradient(ellipse 75% 60% at 50% 35%, black 35%, transparent 88%);
355 |   -webkit-mask-image: radial-gradient(ellipse 75% 60% at 50% 35%, black 35%, transparent 88%);
356 | }
357 |
358 | /* Portal backdrop fade ? opacity only. */
359 | @keyframes portalFadeIn {
360 |   from {
361 |     opacity: 0;
362 |   }
363 |   to {
364 |     opacity: 1;
365 |   }
366 | }
367 | .animate-portal-in {
368 |   animation: portalFadeIn 240ms ease-out both;
369 | }
370 |
371 | /* Boot-sequence card entrance. */
372 | @keyframes bootCardIn {
373 |   from {
374 |     opacity: 0;
375 |     transform: translateY(6px) scale(0.98);
376 |   }
377 |   to {
378 |     opacity: 1;
379 |     transform: translateY(0) scale(1);
380 |   }
381 | }
382 | .animate-boot-in {
383 |   animation: bootCardIn 320ms cubic-bezier(0.16, 1, 0.3, 1) both;
384 | }
385 |
386 | /* Segmented, glowing boot progress bar. */
387 | .boot-progress-track {
388 |   position: relative;
389 |   height: 3px;
390 |   border-radius: 999px;
391 |   overflow: hidden;
392 |   background: rgba(255, 255, 255, 0.06);
393 |   box-shadow: inset 0 0 0 1px rgba(255, 255, 255, 0.04);
394 | }
395 | .boot-progress-fill {
396 |   height: 100%;
397 |   border-radius: 999px;
398 |   background: linear-gradient(90deg, var(--color-primary), var(--color-secondary));
399 |   box-shadow:
400 |     0 0 14px 1px color-mix(in oklab, var(--color-secondary) 55%, transparent),
401 |     0 0 5px color-mix(in oklab, var(--color-primary) 65%, transparent);
402 |   transition: width 260ms cubic-bezier(0.16, 1, 0.3, 1);
403 | }
404 | /* Thin dark dividers on top of the fill read as terminal segments/ticks. */
405 | .boot-progress-ticks {
406 |   position: absolute;
407 |   inset: 0;
```

```
408 | background-image: repeating-linear-gradient(
409 |     90deg,
410 |     transparent 0 6px,
411 |     rgba(9, 9, 11, 0.6) 6px 7px
412 | );
413 | pointer-events: none;
414 | }
415 |
416 | /* Blinking terminal cursor next to the status text. */
417 | .boot-cursor {
418 |     display: inline-block;
419 |     width: 6px;
420 |     height: 12px;
421 |     margin-left: 5px;
422 |     vertical-align: -2px;
423 |     background: var(--color-secondary);
424 |     box-shadow: 0 0 6px color-mix(in oklab, var(--color-secondary) 80%, transparent);
425 |     animation: bootCursorBlink 1s steps(1) infinite;
426 | }
427 | @keyframes bootCursorBlink {
428 |     0%,
429 |     49% {
430 |         opacity: 1;
431 |     }
432 |     50%,
433 |     100% {
434 |         opacity: 0;
435 |     }
436 | }
437 |
438 | /* Metric cell ? gradient hairline top edge + soft corner glow. */
439 | @utility metric-cell {
440 |     &::before {
441 |         content: "";
442 |         position: absolute;
443 |         inset: 0 0 auto 0;
444 |         height: 1px;
445 |         background: linear-gradient(
446 |             90deg,
447 |             transparent,
448 |             color-mix(in oklab, var(--color-primary) 75%, transparent),
449 |             color-mix(in oklab, var(--color-secondary) 75%, transparent),
450 |             transparent
451 |         );
452 |         opacity: 0.75;
453 |     }
454 |     &::after {
455 |         content: "";
456 |         position: absolute;
457 |         right: -25%;
458 |         bottom: -45%;
459 |         width: 65%;
460 |         height: 65%;
461 |         background: radial-gradient(
462 |             circle,
463 |             color-mix(in oklab, var(--color-secondary) 16%, transparent),
464 |             transparent 70%
465 |         );
466 |         pointer-events: none;
467 |     }
468 | }
469 |
470 | /* Workspace panel spring entrance ? a slight overshoot for a springy,
471 |  * organic settle rather than a linear ease. opacity + transform only. */
472 | @keyframes workspaceSpringIn {
```

```
473 | 0% {
474 |     opacity: 0;
475 |     transform: translateY(18px) scale(0.97);
476 |     filter: blur(12px);
477 | }
478 | 55% {
479 |     opacity: 1;
480 |     transform: translateY(-3px) scale(1.01);
481 |     filter: blur(0px);
482 | }
483 | 80% {
484 |     transform: translateY(1px) scale(0.998);
485 | }
486 | 100% {
487 |     opacity: 1;
488 |     transform: translateY(0) scale(1);
489 |     filter: blur(0px);
490 | }
491 | }
492 | .animate-workspace-in {
493 |     animation: workspaceSpringIn 620ms cubic-bezier(0.16, 1, 0.3, 1) both;
494 | }
495 |
496 | @keyframes workspaceSpringOut {
497 |     from {
498 |         opacity: 1;
499 |         transform: translateY(0) scale(1);
500 |         filter: blur(0px);
501 |     }
502 |     to {
503 |         opacity: 0;
504 |         transform: translateY(-14px) scale(1.015);
505 |         filter: blur(18px);
506 |     }
507 | }
508 | .animate-workspace-out {
509 |     animation: workspaceSpringOut 260ms cubic-bezier(0.4, 0, 1, 1) both;
510 | }
511 |
512 | /* scroll-reactive edge vignette on the workspace scroll container */
513 | .workspace-scroll-mask {
514 |     mask-image: linear-gradient(
515 |         to bottom,
516 |         transparent 0,
517 |         #000 80px,
518 |         #000 calc(100% - 80px),
519 |         transparent 100%
520 |     );
521 |     -webkit-mask-image: linear-gradient(
522 |         to bottom,
523 |         transparent 0,
524 |         #000 80px,
525 |         #000 calc(100% - 80px),
526 |         transparent 100%
527 |     );
528 | }
529 |
530 | /* Premium Visual Effects */
531 |
532 | /* 1. Global noise overlay for high fidelity texture */
533 | .noise-overlay {
534 |     position: fixed;
535 |     inset: 0;
536 |     width: 100%;
537 |     height: 100%;
```

```
538 |     z-index: 50;
539 |     pointer-events: none;
540 |     opacity: 0.015; /* Extremely subtle premium grain */
541 |     background: url("data:image/svg+xml,%3Csvg viewBox='0 0 200 200' xmlns='http://www.w3.org/2000/svg'%3E...
542 | }
543 |
544 | /* 2. Global cursor spotlight */
545 | .global-spotlight {
546 |     pointer-events: none;
547 |     position: fixed;
548 |     inset: 0;
549 |     z-index: 0;
550 |     background: radial-gradient(
551 |         500px circle at var(--mouse-x, -999px) var(--mouse-y, -999px),
552 |         rgba(139, 92, 246, 0.06) 0%,
553 |         transparent 70%
554 |     );
555 |     will-change: background;
556 | }
557 |
558 | /* 3. GPU-Accelerated Shifting Background Mesh */
559 | @keyframes mesh-float-1 {
560 |     0%,
561 |     100% {
562 |         transform: translate3d(0px, 0px, 0px) scale(1);
563 |     }
564 |     33% {
565 |         transform: translate3d(40px, -60px, 0px) scale(1.15);
566 |     }
567 |     66% {
568 |         transform: translate3d(-30px, 30px, 0px) scale(0.9);
569 |     }
570 | }
571 | @keyframes mesh-float-2 {
572 |     0%,
573 |     100% {
574 |         transform: translate3d(0px, 0px, 0px) scale(1);
575 |     }
576 |     50% {
577 |         transform: translate3d(-50px, 50px, 0px) scale(1.1);
578 |     }
579 | }
580 | @keyframes mesh-float-3 {
581 |     0%,
582 |     100% {
583 |         transform: translate3d(0px, 0px, 0px) scale(1);
584 |     }
585 |     40% {
586 |         transform: translate3d(60px, 30px, 0px) scale(0.85);
587 |     }
588 | }
589 |
590 | .mesh-blob {
591 |     position: absolute;
592 |     border-radius: 50%;
593 |     filter: blur(140px);
594 |     opacity: 0.25;
595 |     will-change: transform;
596 |     pointer-events: none;
597 |     backface-visibility: hidden;
598 |     transform: translate3d(0, 0, 0);
599 | }
600 | .mesh-blob-1 {
601 |     width: 450px;
602 |     height: 450px;
```

```
603 | background: radial-gradient(circle, rgba(6, 182, 212, 0.45) 0%, transparent 80%);
604 | top: -10%;
605 | left: 15%;
606 | animation: mesh-float-1 25s infinite ease-in-out;
607 | }
608 | .mesh-blob-2 {
609 | width: 550px;
610 | height: 550px;
611 | background: radial-gradient(circle, rgba(139, 92, 246, 0.35) 0%, transparent 80%);
612 | top: 15%;
613 | right: 10%;
614 | animation: mesh-float-2 30s infinite ease-in-out;
615 | }
616 | .mesh-blob-3 {
617 | width: 400px;
618 | height: 400px;
619 | background: radial-gradient(circle, rgba(236, 72, 153, 0.2) 0%, transparent 80%);
620 | top: 5%;
621 | left: 40%;
622 | animation: mesh-float-3 22s infinite ease-in-out;
623 | }
624 |
625 | .mesh-container {
626 | animation: hue-shift 30s linear infinite;
627 | position: absolute;
628 | inset: 0;
629 | pointer-events: none;
630 | will-change: filter, transform;
631 | transform: translate3d(0, 0, 0);
632 | backface-visibility: hidden;
633 | }
634 |
635 | @keyframes hue-shift {
636 | 0% {
637 | filter: hue-rotate(0deg);
638 | }
639 | 100% {
640 | filter: hue-rotate(360deg);
641 | }
642 | }
643 |
644 | /* 4. Elegant Premium Scrollbar */
645 | ::-webkit-scrollbar {
646 | width: 5px;
647 | height: 5px;
648 | }
649 | ::-webkit-scrollbar-track {
650 | background: var(--color-background);
651 | }
652 | ::-webkit-scrollbar-thumb {
653 | background: rgba(139, 92, 246, 0.25);
654 | border-radius: 999px;
655 | border: 1px solid var(--color-background);
656 | }
657 | ::-webkit-scrollbar-thumb:hover {
658 | background: rgba(139, 92, 246, 0.45);
659 | }
660 |
661 | /* 5. Hide Lovable AI watermark badge */
662 | #lovable-badge,
663 | [id*="lovable-badge"],
664 | [class*="lovable-badge"] {
665 | display: none !important;
666 | opacity: 0 !important;
667 | pointer-events: none !important;
```

```
668 |   visibility: hidden !important;
669 | }
670 |
671 | /* =====
672 |  * Premium 3D Scroll Reveal
673 |  * Uses perspective, rotateX, and translateZ to create a deep
674 |  * 3D pop-out effect. Driven entirely by CSS compositor thread.
675 |  * ===== */
676 | .reveal-3d {
677 |   opacity: 0;
678 |   transform: perspective(1200px) translateY(50px) translateZ(-80px) rotateX(10deg);
679 |   filter: blur(8px) brightness(0.7);
680 |   transition:
681 |     opacity 0.9s cubic-bezier(0.16, 1, 0.3, 1),
682 |     transform 1.1s cubic-bezier(0.16, 1, 0.3, 1),
683 |     filter 0.9s ease;
684 |   will-change: transform, opacity, filter;
685 |   transform-origin: center center;
686 | }
687 |
688 | .reveal-3d.in-view {
689 |   opacity: 1;
690 |   transform: perspective(1200px) translateY(0) translateZ(0) rotateX(0deg);
691 |   filter: blur(0px) brightness(1);
692 | }
693 |
694 | @media (prefers-reduced-motion: reduce) {
695 |   .reveal-3d,
696 |   .reveal-light {
697 |     opacity: 1 !important;
698 |     transform: none !important;
699 |     filter: none !important;
700 |     transition: none !important;
701 |   }
702 | }
703 |
704 | /* =====
705 |  * Light 2D Scroll/Load Reveal (For text & hero elements)
706 |  * ===== */
707 | .reveal-light {
708 |   opacity: 0;
709 |   transform: translateY(15px);
710 |   filter: blur(4px);
711 |   transition:
712 |     opacity 0.8s cubic-bezier(0.16, 1, 0.3, 1),
713 |     transform 0.8s cubic-bezier(0.16, 1, 0.3, 1),
714 |     filter 0.8s ease;
715 |   will-change: transform, opacity, filter;
716 | }
717 |
718 | .reveal-light.in-view {
719 |   opacity: 1;
720 |   transform: translateY(0);
721 |   filter: blur(0px);
722 | }
723 |
724 | /* =====
725 |  * Directional 3D Reveals (Method A)
726 |  * Cards rotate in from different sides based on their position.
727 |  * ===== */
728 | .reveal-3d-left {
729 |   opacity: 0;
730 |   transform: perspective(1200px) translateX(-60px) translateZ(-80px) rotateY(15deg);
731 |   filter: blur(8px) brightness(0.7);
732 |   transition:
```

```
733 |     opacity 0.9s cubic-bezier(0.16, 1, 0.3, 1),
734 |     transform 1.1s cubic-bezier(0.16, 1, 0.3, 1),
735 |     filter 0.9s ease;
736 |     will-change: transform, opacity, filter;
737 |     transform-origin: left center;
738 | }
739 | .reveal-3d-left.in-view {
740 |     opacity: 1;
741 |     transform: perspective(1200px) translateX(0) translateZ(0) rotateY(0deg);
742 |     filter: blur(0px) brightness(1);
743 | }
744 |
745 | .reveal-3d-right {
746 |     opacity: 0;
747 |     transform: perspective(1200px) translateX(60px) translateZ(-80px) rotateY(-15deg);
748 |     filter: blur(8px) brightness(0.7);
749 |     transition:
750 |         opacity 0.9s cubic-bezier(0.16, 1, 0.3, 1),
751 |         transform 1.1s cubic-bezier(0.16, 1, 0.3, 1),
752 |         filter 0.9s ease;
753 |     will-change: transform, opacity, filter;
754 |     transform-origin: right center;
755 | }
756 | .reveal-3d-right.in-view {
757 |     opacity: 1;
758 |     transform: perspective(1200px) translateX(0) translateZ(0) rotateY(0deg);
759 |     filter: blur(0px) brightness(1);
760 | }
761 |
762 | /* =====
763 | * Native CSS Scroll-Driven Parallax (Method B)
764 | * Uses zero JS. Runs 100% on the browser's compositor thread.
765 | * Gracefully degrades in older browsers (stays static).
766 | * ===== */
767 | @supports (animation-timeline: view()) {
768 |     .scroll-parallax-y {
769 |         animation: parallax-y linear both;
770 |         animation-timeline: view();
771 |         animation-range: entry exit; /* Animates from when it enters until it exits */
772 |         will-change: transform;
773 |     }
774 |
775 |     @keyframes parallax-y {
776 |         from {
777 |             transform: translateY(-15%);
778 |         }
779 |         to {
780 |             transform: translateY(15%);
781 |         }
782 |     }
783 | }
784 |
785 | @media (prefers-reduced-motion: reduce) {
786 |     .reveal-3d-left,
787 |     .reveal-3d-right {
788 |         opacity: 1 !important;
789 |         transform: none !important;
790 |         filter: none !important;
791 |         transition: none !important;
792 |     }
793 |     .scroll-parallax-y {
794 |         animation: none !important;
795 |     }
796 | }
797 |
```



```
798 | /* =====
799 | * Workspace navigation responsiveness custom override
800 | * ===== */
801 | .workspace-nav-mobile {
802 |   display: block !important;
803 | }
804 | .workspace-nav-desktop {
805 |   display: none !important;
806 | }
807 |
808 | @media (min-width: 768px) {
809 |   .workspace-nav-mobile {
810 |     display: none !important;
811 |   }
812 |   .workspace-nav-desktop {
813 |     display: flex !important;
814 |   }
815 | }
816 |
817 | /* =====
818 | * Hide scrollbars utility
819 | * ===== */
820 | .scrollbar-none {
821 |   -ms-overflow-style: none; /* IE and Edge */
822 |   scrollbar-width: none; /* Firefox */
823 | }
824 | .scrollbar-none::-webkit-scrollbar {
825 |   display: none; /* Chrome, Safari and Opera */
826 | }
827 |
828 | /* =====
829 | * Carousel Scroll Performance Optimizations
830 | * ===== */
831 | /* Only disable pointer events on scroll for hover-capable devices (desktop/mouse) to prevent mobile tap ...
832 | @media (hover: hover) {
833 |   .carousel-scrolling .snap-center {
834 |     pointer-events: none !important;
835 |   }
836 | }
837 |
838 | /* Instantly hide spotlight glows during scroll to free browser repaint resource loops */
839 | .carousel-scrolling .project-glow-element {
840 |   opacity: 0 !important;
841 |   visibility: hidden !important;
842 |   display: none !important;
843 |   transition: none !important;
844 | }
845 |
846 | /* Instantly disable all transitions inside cards during scroll to prevent exit hover lag */
847 | .carousel-scrolling .surface-card,
848 | .carousel-scrolling .surface-card img,
849 | .carousel-scrolling .surface-card * {
850 |   transition: none !important;
851 | }
852 |
853 | /* Force permanent GPU layer promotion on card images to prevent rasterization/demotion lag when scroll s...
854 | .surface-card img {
855 |   will-change: transform;
856 |   transform: translateZ(0);
857 | }
```

FILE #119: tsconfig.json

```
1 | {
2 |   "include": ["src/**/*.ts", "src/**/*.tsx", "vite.config.ts", "eslint.config.js"],
3 |   "compilerOptions": {
4 |     "target": "ES2022",
5 |     "jsx": "react-jsx",
6 |     "module": "ESNext",
7 |     "lib": ["ES2022", "DOM", "DOM.Iterable"],
8 |     "types": ["vite/client"],
9 |
10 |    /* Bundler mode */
11 |    "moduleResolution": "Bundler",
12 |    "allowImportingTsExtensions": true,
13 |    "verbatimModuleSyntax": false,
14 |    "noEmit": true,
15 |
16 |    /* Linting */
17 |    "skipLibCheck": true,
18 |    "strict": true,
19 |    "noUnusedLocals": false,
20 |    "noUnusedParameters": false,
21 |    "noFallthroughCasesInSwitch": true,
22 |    "noUncheckedSideEffectImports": true,
23 |    "paths": {
24 |      "@/*": [".src/*"]
25 |    }
26 |  }
27 | }
```

FILE #120: tunnel.log

FILE #121: vite.config.ts

```
1 | // @lovable.dev/vite-tanstack-config already includes the following ? do NOT add them manually
2 | // or the app will break with duplicate plugins:
3 | //   - tanstackStart, viteReact, tailwindcss, tsConfigPaths, nitro (build-only using cloudflare as a defa...
4 | //     componentTagger (dev-only), VITE_* env injection, @ path alias, React/TanStack dedupe,
5 | //     error logger plugins, and sandbox detection (port/host/strictPort).
6 | // You can pass additional config via defineConfig({ vite: { ... }, etc... }) if needed.
7 | import { defineConfig } from "@lovable.dev/vite-tanstack-config";
8 |
9 | export default defineConfig({
10 |   tanstackStart: {
11 |     // Redirect TanStack Start's bundled server entry to src/server.ts (our SSR error wrapper).
12 |     // nitro/vite builds from this
13 |     server: { entry: "server" },
14 |   },
15 | });
```